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DI & LR REVIEW

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Data Interpretation & Logical Reasoning

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1. INTRODUCTION

It is widely accepted that people with high and advanced level of reasoning are highly successful and adapt very well to their surrounding environment. This is exactly opposite the case for those with low level of reasoning. For this very reason, reasoning tests are employed by colleges in their admission process and also by companies to establish the suitability of a person for a job. Logical reasoning is the process of thinking which aims to arrive at certain valid conclusions. This test is used for assessing the ability of the student to comprehend, dissect and evaluate the given information.

As a student it is important for you to follow a few standard steps while solving logical questions:
1. Take a complete 'preview of the situation' clearly understanding the context. Remember to accept the situation as it is given.
2. Read each and every part of the question carefully. You should concentrate hard and focus fully while reading the question. This is a very important prerequisite for solving questions on logic since very often in long sentences there will be individual single words which will transform the meaning of the sentences. If you fail to take into account these words, the end result will be errors in logic & deduction.

Let us now proceed to understand how all this applies to real life problem solving through examining questions which have been asked in different competitive exams and CAT.

Example 1) A party is held at the house of the Mehtas. There were five other couples present (besides Mr. and Mrs. Mehta), and many, but not all, pairs of people shook hands. Nobody shook hands with anyone twice, and nobody shook hands with his/her spouse. Both the host and hostess shook some hands. At the end of the party, Mr. Mehta polls each person present to see how many hands each person (other than himself) shook. Each person gives a different answer. Determine how many hands Mrs. Mehta must have shaken. (Can we prove that it was not Mrs. Mehta who shook 10 hands?)

Solution 1) Let there be 5 couples:
A — A
B — B
C — C
D — D
E — E
& M — M

Deduction 1 From the condition that nobody shook hands with his/her spouse, it is clear that none of the twelve people in the party shook more than 10 hands. (Since, nobody shakes hands with himself or his/her spouse; it leaves a maximum of 10 people to shake hands with).

Deduction 2 Mr. Mehta has asked the question to eleven different people and each of them has given a different answer. Also, the highest answer anyone could have possibly given is 10. Hence, the only way to distribute different numbers of handshakes amongst the 11 people is: 0,1,2,3,4,5,6,7,8,9,10. [Note: somebody shook 0 hands and somebody shook 10].

Deduction 3 Since the host & hostess have both shaken some hands, the person who shook '0' hands cannot be either M or M. It has to be one of the other 10 people in the party. [At this point you need to realize that in the context of this problem A,A B,B,C, C, D, D, E & E are alike, i e. there is no logical difference amongst these 10 and you have exactly the same information about each of these 10 people. However, Mrs. Mehta is different because she stands out as the hostess as well as the wife of the person who has asked the question].

Since all ten guests are the same, assume 'A' shook no hands. This leads us to the following deduction.

Deduction 4 Take any one person apart from A & A; say B. B will not shake hands with himself & his wife. Besides B will also not shake hands with A (who has shaken no hands). Thus, B can shake a maximum of 9 hands and will thus not be the person to shake 10 hands. What applies to B, applies to B, C, C, D, D, E, E and M. Hence, A is the only person who could have shaken 10 hands. Hence, amongst the couple A & A, if we suppose that A had shaken 0 hands, then A must have shaken 10 hands.

Note: The main result here is that, out of the people to whom M has asked the question, and amongst whom we have to distribute the numbers 0 to 10 there has to be a couple who has had 0 & 10 handshakes. It could be any of the five couples, but it cannot be M who has either 0 or 10 handshakes.

We now proceed, using the same line of reason as follows.

Deduction 5 Suppose B has 1 handshake — he must have shaken hands with A (who has shaken everybody's hands she can). Then, B wouldn't have shaken hands with anyone out of A, B, C, C, D, D, E, E & M. At this point the following picture emerges:

A	— 0	A	— 10
B	— 1	B	—
C	—	C	—
D	—	D	—
E		E	
M			

Numbers left to be allocated — 2, 3, 4, 5, 6, 7, 8, 9.

Considering C, as a general case, he cannot shake hands with C, A (Who shook no hands) & B (Who shook hands only with A. This is mandatory since A has shaken hands with 10 people).

Thus, C can shake hands with a maximum of 8 people and this deduction will be true for C, D, D ,E , E & M too. Hence, the only person who could get 9 handshakes is B. Thus, we conclude that just like 0 and 10 handshakes were in one pair, similarly 1 and 9 handshakes too have to be part of one pair of husband and wife. Similar deductions, will lead to the realization that 2 & 8, 3 & 7 and 4 & 6 handshakes will also occur for couples amongst the 11 people questioned.

Hence, M must have shaken 5 hands. The above question was solved on the basis of a series of deductions, which were based on a series of Logical Form (LF), then logical structures.

Example 2) Consider the following grid:

L		P
M	O	Q
N		R

Each letter in the above grid represents a different digit from 0 to 9, such that $L \times M \times N = M \times O \times Q = P \times Q \times R$. Find the value of 'O'.

Solution 2 In order to solve such a question, one need to proceed systematically, making one deduction at a time.

Deduction 1 There are seven alphabets and ten digits. We need to somehow eliminate 3 of these 10 to define the 7 digits required to be allocated. It is obvious, that '0' cannot be used since if we make any alphabet '0' we will end up with a product that '0' in one or a maximum of two of the three cases.

Deduction 2 You need to find a product which can be made in three different ways.

Deduction 3 Two of these three ways have to be independent of each other with no matching digits and the third way has to be drawn out of one digit each from the first two ways and one independent digit that has not been used. Also, at this point there are 9 digits and 2 more to be eliminated.

Now we move to the process of Trial & Error.

Notes on Interpretation

Trial & Error is one of the most useful processes for solving questions based on reasoning. Principally there are three ways for carrying out trial & error search.

(1) Complete trial & error.

(2) Directed trial & error.

(3) Blind trial & error.

The blind trial and error is what most students practice and hence are unable to solve logical questions. Since they do not use their deductive logic to do a more focused search, they end up going round in circles while trying to solve such questions. Instead directed trial & error and comprehensive trial & error are superior problem solving processes and therefore score above the complete trial and error method.

Application of Directed Trial & Error The question above is a classical situation warranting a directed trial & error. Hence, this is best illustrated through the example. At this stage we are at a situation where we know: 1, 2, 3, 4, 5, 6, 7, 8 and 9 have to be allocated to L, M, N, O, P, Q and R. At this point, take a call as to whether you want to use 9 or not? 9 and 1 are different from the other numbers primarily because they are the highest and lowest numbers respectively and also because 9 gives us "maximum room for maneuvering" the question as compared to the other

numbers, while 1 is a useful tool for a third multiplier if we do not want to change the value of the multiplication.

Note: The student should analogize the thinking process applied here to the thinking process used to unravel a ball of wool which has got entangled.

To disentangle an entangled ball of wool, we need to search the end of the ball. Once you identify either end, the remaining process of disentangling the ball requires very elementary logic coupled with patience and perseverance.

The logic for picking up the end point of the ball of wool is that it is different from all other points in the ball. Similarly, in reasoning questions, we need to identify things/objects/people which are different from other things/objects/ people and start our solution from there. After that, the whole process becomes one of use of elementary logic for elementary deductions coupled with patience and perseverance. In the question under consideration, if we take a call on using '9' and decide to do so, we can then deduce that the required product has to be a multiple of 9. [Remember that at this point of time we have ignored the line of thinking that neglects 9. We will have to consider it, if we do not get an answer by using 9].

Since the product has to be a multiple of 9, assume that the product is 36. But, this product eliminates the use of 8, 7 & 5 and leaves only 6 digits. Going below 36 as a product for further trial and error will further reduce the number of possibilities. Hence, let us try to go to the higher extreme & try to experiment with the number 72. We see that 72 = $3^2 \times 2^3$ and can be formed by $9\times8\times1$, $6\times4\times3$ or $9\times4\times2$.

This satisfies our deduction that 72 gives three ways of solving the question. Also, the second requirement that there should be two ways which are independent of each other and a third way which uses one term each from the two independent ways and one unique term is also satisfied. Since $6\times3\times4$ is independent of $9\times8\times1$ in all its digits. Also, $9\times4\times2$ uses the no. 9 from $9\times8\times1$ & 4 from $6\times4\times3$.

Hence, the following possibility emerges:

L = 6	P = 8	
M = 4	O = 2	Q = 9
N = 3	R = 1	

You need to understand here that, the digit '2' is the only one which is fixed to its place in the grid. All the other digits can be changed. Thus, we can have alternative arrangements like:

L = 3		P = 1
M = 4	O = 2	Q = 9
N = 6		R = 8

OR

L = 6		P = 6
M = 9	O = 2	Q = 4
N = 1		R = 3

The only thing that is fixed is the number 2 for O.

Example 3 Let us now consider another question, which is a classic case of **complete** or **comprehensive trial & error method**.

Two people A & B are playing a game. Both A & B are logical people. There are two boxes on the table. One of them contains 9 balls, the other contains 4 balls. In this game, the players are supposed to take alternate turns of picking up balls according to the following rules:

(a) Pick up as many balls as you want to from any box.

(b) Pick up an equal number of balls from both boxes. (If you pick from both boxes)

The person who picks up the last ball wins the game. In his/her turn it is mandatory to pick up at least one ball. A has to play first. What should he do to ensure a win?

Solution 3

Deduction 1 The rules of the game define that there are two legal moves: Picking up an equal number from each box or picking up any number from either box (at least 1).

Deduction 2 A has 17 possible moves to make and since the question asks for one particular move that will ensure a win, one of these 17 must be the winning move. It is at this stage that you should realize that the question calls for a **comprehensive trial & error** which should result in the elimination of 16 possibilities. The starting position is 9, 4.

A's options at the start of the game can be basically divided into three options:

I Pick up balls from the first box.
II Pick up balls from the second box.
III Pick up balls from both the boxes.

The position after A plays his move can be documented as follows (in terms of the number of balls B has in front of him.):

[There are 9 moves in option I.] (Balls from first box are picked by A.) If
(a) A picks up 1 ball, B will be left with 8, 4.
(b) A picks up 2 balls, B will be left with 7, 4
(c) A picks up 3 balls, B will be left with 6, 4
(d) A picks up 4 balls, B will be left with 5, 4
(e) A picks up 5 balls, B will be left with 4, 4
(f) A picks up 6 balls, B will be left with 3, 4
(g) A picks up 7 balls, B will be left with 2, 4
(h) A picks up 8 balls, B will be left with 1, 4
(i) A picks up 9 balls, B will be left with 0, 4

Similarly, if he picks up balls from the second box (option II) he will have an end result of: If
(a) A picks up 1 ball, B will be left with 9, 3
(b) A picks up 2 balls, B will be left with 9, 2
(c) A picks up 3 balls, B will be left with 9, 1
(d) A picks up 4 balls, B will be left with 9, 0

And for option III, If:
(a) A picks up one ball each from both boxes, B is left with 8, 3
(b) A picks up two balls each from both boxes, B is left with 7, 2
(c) A picks up three balls each from both boxes, B is left with 6, 1
(d) A picks up four balls each from both boxes, B is left with 5, 0

Out of these 17 options, the options of leaving (0, 4), (9, 0) and (5, 0) are infeasible since it will result in an immediate win for B, who can clean up the board in one move. Similarly leaving (4, 4) will also cause an immediate win for B. This leaves A with 13 options which he needs to consider. These are:

I	II	III
8, 4	9, 3	8, 3
7, 4	9, 2	7, 2
6, 4	9, 1	6, 1
5, 4		
3, 4		
2, 4		
1, 4		

From these 13, the easiest option to check is 1, 4. If B gets 1, 4 he has the following options:

I	II	III
0, 4	1, 3	0, 3
	1, 2	
	1, 1	
	1, 0	

Obviously, he cannot play (0, 4), (1, 0), (1, 1) or (0, 3). This means he can leave A with ((1, 2) or (1, 3). These need to be further investigated.
If B leaves (1, 2):
A has the following options to leave for B

I	II	III
0, 2	1, 1	0, 1
	1, 0	

Obviously, if he plays any of this, A will lose. Hence, if B leaves A with (1, 2) A will definitely lose.

Deduction 3 A cannot leave B with a situation in which B can make into (1, 2) or (2, 1). Evaluating the 13 options left for A, we get that (1, 4), (2, 4), (3, 4), (5, 4), (9, 1) , (9, 2), (6, 1) & (6, 2) are situations from which B can reach (2, 1) or (1, 2) in one move. This means B will win if A leaves him with any of these eight situations.
Thus, A will eliminate these eight options from his list of 13 and come down to five options which need further checking. These are: (6, 4), (7, 4), (8, 4), (9, 3) and (8, 3). At this stage we further need to eliminate four of these five options to come to the correct answer.
Now let us start our investigation with (6, 4). If B is left with (6, 4) he has the following options:

I	II	III
5, 4	6, 3	5, 3
4, 4	6, 2	4, 2
3, 4	6, 1	3, 1
2, 4	6, 0	2, 0
1, 4		
0, 4		

Of this, B cannot play (0,4), (1, 4), (2, 4), (3, 4), (4, 4), (5, 4), (6, 0), (6, 1), (6, 2), (2, 0), (3, 1) & (4, 2) [Since these lead to a (2, 1) or a clean win in one move.] Thus, B can either, leave (6, 3) or (5, 3).
If B leaves (5, 3) for A, A's options are:

I	II	III
4, 3	5, 2	4, 2
3, 3	5, 1	3, 1
2, 3	5, 0	2,0
1, 3		
0, 3		

In each of these cases, B will either win in one move or get to (2, 1) in one move. Thus, A cannot do anything if B leaves (5, 3) in return for A leaving (6, 4).
Deduction 4 Just like being left with (2, 1) means a definite loss, so does being left with (5, 3). Hence, A cannot allow a situation in which B can make it (5, 3) in one move.
Thus, A cannot leave B with (6, 4), (9, 3) or (8,3) since, each of these will result in B giving back (5, 3) in one move.
Thus A's options are down to (7, 4) or (8, 4). These need to be further investigated now:
Let us consider A leaving (7, 4) for B.

B's options:

I	II	III
6, 4		
5, 4	7, 3	6, 3
4, 4	7 2	5, 2
4, 4	7, 1	4, 1
3, 4	7, 0	3, 0
2, 4		
(1, 4) or (0, 4)		

After closely evaluating each of these, all possible options will get eliminated because in one move they lead to either a straight win or a (2, 1) situation or a (5, 3) situation.
If A does any of this, then B cannot do anything to stop from losing the game.
The above question obviously is extremely long and complicated. However, what needs to be noticed is that even for this extremely long question, at no point of time is there any big logical jump i.e. from one logic emerges the next one and so on.

Example 4 The year was 2006. All six teams in Pool A of World Cup Hockey play each other exactly once. Each win earns a team three points, a draw earns one point and a loss earns zero points. The two teams with the highest points qualify for the semifinals. In case of a tie, the team with the highest goal difference (goals for–goals against) qualifies. In the opening match, Spain lost to Germany. After the second round (after each team played two matches), the (pool A) table looked as shown on the next page.
In the third round, Spain played Pakistan, Argentina played Germany, and New Zealand played South Africa. All the third round matches were drawn. The following are some results from the fourth and fifth round matches.
(a) Spain won both the fourth and fifth round matches.
(b) Both Argentina and Germany won their fifth round matches by 3 goals to 0.
(c) Pakistan won both the fourth and fifth round matches by 1 goal to 0.

Solution 4 For solving the above question, one requires tremendous alacrity, logical consistently and above all a cool head.
Also, the solution of the above question is dependent on your ability to interpret the table and find out the appropriate linkages (something that might not be as easily done as said).
Let us look at the table and create our interpretations:
Deduction 1 Germany and Argentina have both won their matches while New Zealand and South Africa have lost both theirs.
Let us start by making tables for the possible result of the first two rounds for each of the six teams.

Germany

Round one Opponent	Result
1. Spain	Won 1–0/2–1
2. Pak/New Zealand/South Africa	Won 2–1/1–0
3. Argentina	

It is given that Germany has played Spain in the opening match (and obviously won that game). Looking at the goals for & goals against columns and the fact that Germany has won both the first and second round matches, we deduce that its two wins must have been in one of the following combinations:

(Pool A)

Team	Games Played	Won	Drawn	Lost	Goals For	Goals Against	Points
Germany	2	2	0	0	3	1	6
Argentina	2	2	0	0	2	0	6
Spain	2	1	0	1	5	2	3
Pakistan	2	1	0	1	2	1	3
New Zealand	2	0	0	2	1	6	0
South Africa	2	0	0	2	1	4	0

Won 1 – 0 & Won 2 – 1
Or
Won 1 – 0 & Won 2 – 1

Notes on Interpretation Germany cannot win either match by 2–0 or 3–0 or 3–1 margins since it will not be able to win the other match and maintain a 3–1 goal for goals against situation.
Deduction If Germany had won the first game 2–1 against Spain, Spain would have won its second round match by 4–0, while if Germany won by 1–0, then Spain would have won its second round match 5–1 (since Spain has Goals For = 5 and Goals Against = 2).
Further, since only two teams — New Zealand and South Africa have conceded 4 or more goals, Spain must have played one of them. Looking into South Africa's G.F/G.A columns, if South Africa had conceded 4 goals in the second round, then it should have won the first round (1,0). But, South Africa has lost both rounds.
Hence, Spain played its second round against New Zealand. Further, if this is true, no other team can play New Zealand in round two.

At this stage, the following possibilities emerge.

Team Germany				
Round 1	vs.	Spain	Won	1–0 or 2–1
Round 2	vs.	Pak/S.A.	Won	2–1 or 1–0
Round 3	vs.	Argentina	Draw	
Team Spain				
Round 1	vs.	Germany	Lost	0–1 or 2–1
Round 2	vs.	New Zealand	Won	5–1 or 4–0
Round 3	vs.	Pakistan	Draw	
Team New Zealand				
Round 1	vs.	Arg/Pakistan	Lost	0–1 or 1–2
Round 2	vs.	Spain	Lost	1–5 or 0–4
Round 3	vs.	South Africa	Draw	

Deduction Team Pakistan won one round and lost one and GF/GA 2/1. Hence, won 2–0 and lost 0–1.

Now, since New Zealand played its first round against Pakistan or Argentina it could not have lost 1–2. This is because in the case of Pakistan, if Pakistan had won 2–1 against NZ in round 1, its round 2 would have been a draw.

Further, Argentina has conceded no goals. Hence, it could not have won 2–1 against N.Z. This means that N.Z. must have lost 0–1 in its first match to Argentina (that cannot happen against Pakistan, because Pakistan cannot win 1–0 in the first round since it will result in a 1–1 draw in round 2).

Consequently N.Z. lost 1–5 in its second match to Spain and hence Spain must have lost 0–1 to Germany.

The following scenario emerges from these deductions:

Team Germany				
Round 1	vs.	Spain	Won	1–0
Round 2	vs.	S.A.	Won	2–1
Note: Here Germany's round 2 has to be vs. S.A., since Pakistan cannot lose 2–1				
Team Spain				
Round 1	vs.	Germany	Lost	1–0
Round 2	vs.	N.Z.	Won	5–1
		Team N.Z.		

Round 1	vs.	Argentina	Lost	0–1
Round 2	vs.	Spain	Lost	1–5
Team Pakistan				
Round 1	vs.	S.A.	Won	2–0
Round 2	vs.	Argentina	Lost	0–1
The first three rounds are as under:				
Round 1 matches:				
Germany	beat	Spain	1–0	
Argentina	beat	N.Z.	1–0	
Pakistan	beat	S.A.	2–0	
Round 2 matches:				
Spain	Beat	N.Z.	5–1	
Argentina	beat	Pak	1–0	
Germany	beat	S.A.	2–1	

Putting all deductions into one table, the following picture emerges:

	Germany	Argentina	Spain	Pak	New Zealand	S. Africa
Germany	–	D#3	W(1–0)#1			W(2–1)#
Argentina	D#3	–		W(1–0)#2	W(1–0)#1	
Spain	L(0–1)#1		–	D#3	W(9–1)#2	
Pakistan		L(0–1)#2	D#3	–		W(2–0)#1
New Zealand		L(0–1)#1	L(1–5)#2	–	D#3	
S. Africa	L(1–2)#2			L(0–2)#1	D#3	–

According to the information available about the fourth and fifth round of matches:

Germany — Pakistan, Loss (0–1) & N.Z. won (3–0)

Argentina — Spain, Loss by ‘x’ goals & S.A. won (3,0)
Spain — Argentina won by ‘x’ goals & S.A. won by ‘y’ goal
Pakistan — Germany won (1–0) & N.Z. won 1–0
N.Z. — Germany loss (0–3) & Pakistan lost (0–1)
S.A. — Argentina lost (0–3) & Spain lost by ‘y’ goals

And the goal differences for the six teams are:

Germany	+1+1+0–1+3=+4
Argentina	+1+1+0–x+3=5–x=Max. 4 or less
Spain	–1+4+0+x+y=3+x+y=Min. 5 or more
Pakistan	2–1+0+1+1=+3
N.Z.	–1–4+0–3–y=–6–y

Based on these deductions, the following questions can be answered.

1. Which one of the following statements is true about the matches played in the first two rounds?
(a) Pakistan beat South Africa by 2 goals to 1.
(b) Argentina beat Pakistan by 1 goal to 0.
(c) Germany beat Pakistan by 2 goals to 1.
(d) Germany best Spain by 2 goals to 1.

2. Which one of the following statements is true about the matches played in the first two rounds?
(a) Germany beat New Zealand by 1 goal to 0.
(b) Spain beat New Zealand by 4 goals to 0.
(c) Spain beat South Africa by 2 goals to 0.
(d) Germany beat South Africa by 2 goals to 1.

3. Which team finished at the top of the pool after five rounds of matches?
(a) Argentina
(b) Germany
(c) Spain
(d) Cannot be determined
Spain must be top of the pool since it has the best goal difference even in its worst case scenario.

4. If Pakistan qualified as one of the two teams from Pool A, which was the other team that qualified?
(a) Argentina
(b) Germany
(c) Spain
(d) Cannot be determined

Notes on Interpretation- This question has an ambiguity since according to the deductions, Spain and Germany both should be above Pakistan in terms of goal difference and hence Pakistan cannot qualify. However, if Pakistan qualifies, so do both Spain and Germany.

Example 5 Prof. Singh has been tracking the number of visitors to his homepage. His service provider has provided him with the following data on the country of origin of the visitors and the university they belong to:

	Number of Visitors/Day		
University	1	2	3
University 1	1	0	0
University 2	2	0	0
University 3	0	1	0
University 4	0	0	2
University 5	1	0	0
University 6	1	0	1
University 7	2	0	0
University 8	0	2	0

	Number of Visitors/Day		
Country	1	2	3
Canada	2	0	0
Netherlands	1	1	0
India	1	2	0
UK	2	0	2
USA	1	0	1

Deduction1 Looking at Day 3, University 4 must belong to the UK and University 6 must belong to the USA.

Deduction 2 From Day 2 it is clear that University 8 has to be an Indian University while University 3 has to be from Netherlands.

Deduction 3 From the analysis of Day 1 data , University 2 and University 7 should be distributed amongst UK and Canada in either order, i.e. 2 belongs to UK and 7 to Canada or 2 belongs to Canada and 7 to UK. [Symbolically, (2UK + 7 Canada) vs (2Canada + 7 UK)]

Deduction 4 The visitor from USA on Day 1 must come from University 6. Hence, University 1 and University 5 should be distributed between India and Netherlands.

With this set of deductions, we get the following table. Using this table the answers to the following questions become quite elementary.

Number of Visitors/Day

University	1
University 1	I v N
University 2	UK v C
University 3	N
University 4	UK
University 5	N v I
University 6	USK
University 7	C v UK
University 8	I

1. To which country does University 5 belong?
(a) India or Netherlands but not USA
(b) India or USA but not Netherlands
(c) Netherlands or USA but not India
(d) India or USA but not UK

2. University 1 can belong to:
(a) UK (b) Canada
(c) Netherlands (d) USA

3. Which among the listed countries can possibly host three of the eight listed universities?
(a) None
(b) Only UK
(c) Only India
(d) Both India and UK

4. Visitors from how many universities from UK visited Prof. Singh's homepage in three days?
(a) 1
(b) 2
(c) 3
(d) 4

1.1 TYPES OF QUESTIONS ASKED

Below is a list of various types of questions have been asked already in CAT, and some new ones as it is always probable to find new question types on the exam.

Arrangement and Ranking
Team formation
Quantitative reasoning
Puzzle test
Syllogisms
Assumptions
Statements
Set Theory
Binary Logic
Venn Diagrams
Clocks & Calendars
Blood Relations
Direction Sense
Miscellaneous topics (new)

2. ARRANGEMENTS AND RANKING

I. ARRANGEMENTS

Questions on arrangements typically involve arranging people or objects in straight lines or around circles/squares or other geometrical shapes. The key skills involved in solving questions on arrangements include but are not limited to:
(i) The ability to visualize the geometrical shape of the arrangement situation.
(ii) The ability to order the clues in the correct order of usage (as explained in the theory of logical reasoning).
(iii) The ability to perceive what indirect clues are talking about – and how to use them.
(iv) The ability to convert clues written in language form into visual cues so that you do not need to read the text again and again. Also, converting the language clues to visual cues is critical for the purpose of being able to 'see' all the clues at one go.

II. RANKINGS

Reasoning questions on Rankings involve an ordering of people objects based on their heights/weights/performance in an exam, etc. As the name suggests, in questions on rankings you are supposed to place people/objects in a decreasing or increasing order based on an attribute being measured. Key skills required in solving Logical Reasoning questions based on rankings:
(i) The ability to visualize the structure in which the rankings have to be created.
(ii) The ability to order the clues in the correct order of usage (as explained in the theory of logical reasoning).
(iii) The ability to perceive what indirect clues are talking about, and find the appropriate point in the solving process about how to use them.
(iv) The ability to convert clues written in language form into visual cues so that you do not need to read the text again and again, and also are able to 'see' all the clues at one go.

Q1)
The eight employees of a company are G, H, I, J, K, M, N, and O. In each of the years 2001 through 2008, exactly one of the employees joined the company.

i. H joined the company before N.

ii. K joined the company before J.

iii. N and J joined the company before G.

iv. N joined the company before O.

v. J joined the company before M.

vi. G joined the company before I.

1. Which one of the following CANNOT be true?

A) H joined the company in 2001.
B) H joined the company in 2003.
C) G joined the company in 2004.
D) M joined the company in 2004.
E) O joined the company in 2004.

2. If J joined the company in 2002, which one of the following CANNOT be true?

A) H joined the company in 2003.
B) M joined the company in 2003.
C) H joined the company in 2004.
D) N joined the company in 2004.
E) O joined the company in 2004.

3. Of the following, which one is the latest year in which J could have joined the company?

A) 2002
B) 2003
C) 2004
D) 2005
E) 2006

4. If O joined the company in 2005 and M joined it in 2007, one can determine the years in which exactly how many of the other employees joined the company?

A) 1
B) 2
C) 3
D) 4
E) 5

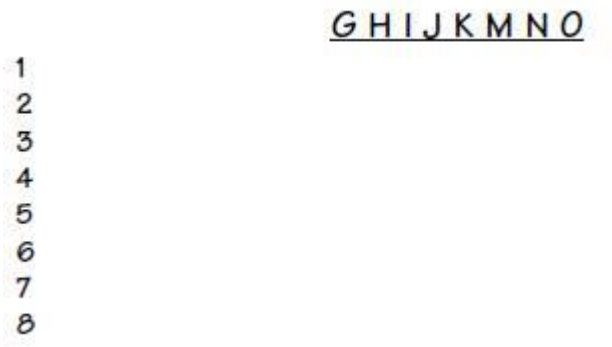
5. Assume that O joined the company before M. Of the following, which one is the earliest year in which M could have joined it?

A) 2003
B) 2004
C) 2005
D) 2006
E) 2007

S1)

The Initial Setup:

With most “free-floating” sequence sets, the most natural way to work is vertically. Put the years along the side and list the employees. Notice that, as in Set 2, we’ve shortened the years to just 1 through 8. (You could also have worked horizontally by listing the years left to right like a timeline. Whichever way you choose, just make sure that your sketch is neat and not misleading.)



The Rules:

A good way to start is to locate an employee or employees who might come at the top of the sketch; that is, anyone who doesn’t seem to have joined the company after other employees. If we scan the rules, we see that H in Rule 1 and K in Rule 2 both fit that description. Let’s therefore start by putting an “H” and a “K” at the top of the sketch.

1. H joined before N. From the H at the top of the sketch, add a line down to an N.
2. K joined the company before J. Next to the H at the top of the sketch, we already have a K. From that K draw a line down to a J.
3. N and J both joined the company before G. There’s already an N and a J in the sketch, so draw lines down from each of them to a G.
4. Gives more information about N. N joined the company before O. From the N in the sketch, draw another line down to an O.
5. Here’s more information about J. J joined the company before M. From the J in the sketch, draw another line down to an M.
6. G joined the company before I. From the G in the sketch, draw a line down to an I.

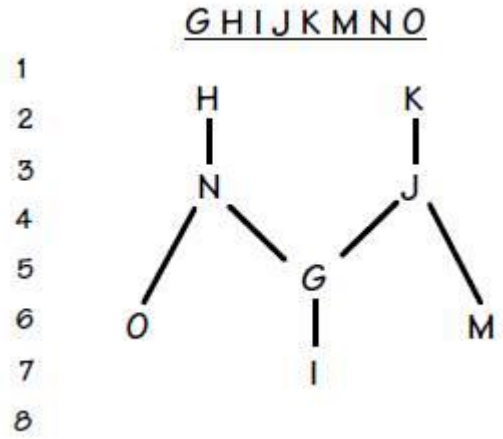
Key Deductions:

The first thing to do is check and make sure that all of the entities are included in the master sketch. If one isn’t, that’s a “floater” that can be plugged in anywhere. Fortunately, all of the entities are included in the sketch, so we can depend solely on our master sketch to answer every question we’ll face.

Before moving on, take a moment and review the sketch. Some things should be obvious. For instance, K joined the company before G; I joined the company after N, etc. Also take note of the possibilities regarding the specific years; what entities could have joined the company in 2001? What entities could have joined in 08? H and K are the only entities that aren't below an entity, so either H or K must have joined in 01. I, O, and M are the only employees with no entities below them, so they're the only ones who could have joined in 08. While these aren't earth-shattering deductions, it's a good idea to take a moment and review what the completed sketch can tell you. This sets up the set's parameters in your mind.

It's also wise to take a moment and recognize what the sketch can't tell us. Be careful not to assume any relationship between employees that aren't connected by lines. For example, I, in our sketch, is slightly lower than M and O. But we can't assume that M and O joined the company before I. We don't know that. Be very careful when dealing with the two "arms" of the sketch. The "K—J—M" arm is not connected to the "H—N—O" arm. Therefore, we can't deduce any relationship between the two. M, for instance, is quite low on the sketch, but since it's not connected to the "H—N—O" arm, M could have joined the company even before H. Likewise, O could have joined before K.

The Final Visualization: Be sure you've kept all of your notations neat and easily accessible:



The Questions:

1. (C)

In a "free-floating" sequence set like this (when all the entities are in the master sketch), just check each choice against the sketch, using it to figure out the choice that is impossible.

(A) H (or K) could have joined the company in 01. Answer choice (A) can be true and isn't the answer.

(B) If K joined in 01 and J in 02, then H could have joined in 03. (B) can be true and isn't the answer.

(C) Four entities, H, K, J, and N are all above G in the sketch, so at least four employees joined the company before G. The earliest that G could have joined the company is 05. Answer choice (C) can't be true and is the answer. Quickly, here is why choice (D) and choice (E) can be true.

(D) M could have joined the company anytime from 03 (K and J joined earlier) to 08.

(E) O could have joined the company anytime from 03 (H and N joined earlier) to 08.

2. (E)

Again, we can solely depend on the sketch. If J joined in 02, exactly one employee joined earlier (the one that joined in 01). Look at the sketch, K must have joined the company earlier than J, so K joined in 01. That's all we know for sure. Either H or M could have joined in 03, either H or M or N could have joined in 04, and so on. Check each choice look for the one that can't be true. (A) and (B) Either H or M could have joined in 03 after K (01) and J (02).

(C) Could H have joined in 04? Sure, here is the ordering: K, J, M, H, N, G, I, O.

(D) Could N have joined in 04? Sure, here is the ordering: K, J, H, N, M, G, I, O.

(E) K joined in 01 and J joined in 02. At least H and N joined before O, so the earliest O could have joined is in 05. (E) can't be true and is the answer.

3. (D)

We're asked for the latest year that J could have joined the company. There are two ways to go about this: We can put as many employees as possible before J, or we can see which employees MUST be after J. Since the latter deals with what is DEFINITELY true while the former deals with what is POSSIBLE only, the latter is most likely going to be simpler and, more importantly, quicker. Look at the sketch, only G, I, and M are definitely placed below J. Therefore if G, I, and M joined the company in 06, 07, and 08, the latest that J could have joined the company is 05, choice (D).

4. (B)

From the master sketch, we're able to deduce that either, I, M, or O must have joined the company last, in 08. This question stem assigns O to 05 and M to 07. Therefore, I must have joined the company in 08. Now check the master sketch. H and N joined before O in the 01-04 slots. K and J joined before M and also before G. Since H and N must occupy two of the four 01-04 slots, K and J will take the other two slots which leaves G (which must go AFTER K and J) to take the remaining 06 slot. G and I are the only other entities whose slots can be determined, choice (B).

5. (D)

Here we're to assume that O joined the company before M, and we're looking for the earliest that M could have joined. Since H and N joined the company before O, all three of these must have joined the company before M. K and J joined the company before M (as always), so that makes a total of five employees that must have joined the company before M. These five will at least take up the 01-05 slots, so the earliest that M could have joined the company is 06, choice (D).

Q2)

The Mammoth Corporation has just completed hiring nine new workers: B, C, D, E, F, G, H, I, and J.

(a) F and I were hired on the same day as each other, and no one else was hired that day.
(b) C and G were hired on the same day as each other, and no one else was hired that day.
(c) On each of the other days of hiring, exactly one worker was hired.
(d) E was hired before B.
(e) H was hired before D.
(f) D was hired after I but before E
(g) G was hired after both J and B.
(h) B was hired before J.

1. Who were the last two workers to be hired?

A) E and J
B) B and G
C) B and C
D) G and C
E) J and B

2. Who was hired on the fourth day of hiring?

A) E
B) B
C) I
D) G
E) J

3. Exactly how many workers were hired before J?

A) 6
B) 5
C) 4
D) 3
E) 2

4. Which one of the following must be true?

A) D was the first worker to be hired.
B) H was the first worker to be hired.
C) F and I were the first two workers to be hired.
D) H and F were the first two workers to be hired.
E) Either H was the first worker to be hired or F and I were the first two workers to be hired.

5. If E was hired on a Monday, what is the earliest day on which G could have been hired?

A) Monday
B) Tuesday
C) Wednesday
D) Thursday
E) Friday

S2)

The Action:

In no time at all you should see that this set asks you to arrange the entities in order of hiring—a sequence set. Moreover, it's a special type of sequencing set, one we call a "free-floating" sequencing set because all of the entities are described relative to each other, as opposed to in a standard sequencing set in which the entities are slated for definite slots (Monday to Friday, 1 to 7 in a line, etc.). In contrast, in "free-floating" sets, the important aspect is where each entity is relative to the others—H is before D, D is after I, etc. On the basis of these types of rules, there are no specific slots we're asked to fill in, although sometimes, as in this case, we're able to deduce much of the relative ordering. That is, in easy sets of this type (and this is one of the easier ones), we get so much relative information that we're able to fit entities into specific time slots, but that's not always the case.

Each of the nine new workers—B, C, D, E, F, G, H, I, and J—was recently hired. The Key Issues will be:

1) What worker can, must, or cannot have been hired before or after what other worker?

2) What worker can, must, or cannot be hired on the same day as what other worker?

The Initial Setup:

Many people find it helpful to visualize a “free-floating” sequencing set in a vertical fashion. Let’s put the workers who are hired earlier above the workers who were hired later. Remember to list the workers’ letters off to the side, and we’re ready to hit the rules: B C D E F G H I J The Rules: A good way to start is to find a worker or workers who might come at the top of the sketch; that is, anyone who doesn’t seem to be hired after other workers. If you scan the rules, you’ll see that H fits that description, as does the FI pair from the first rule. Let’s therefore start by putting an “FI” and an “H” at the top of the sketch.



The Rules:

A good way to start is to find a worker or workers who might come at the top of the sketch; that is, anyone who doesn’t seem to be hired after other workers. If you scan the rules, you’ll see that H fits that description, as does the FI pair from the first rule. Let’s therefore start by putting an “FI” and an “H” at the top of the sketch.

1) F and I were the only two workers hired on their day, so whatever is true of one is true of the other. That means they’re an inseparable pair. We’ve already put our “FI” pair at the top of the sketch, so let’s move on. Where to now? We may as well find entities related to the ones we’ve already put on the page. Rule 6, containing I, links up nicely:

6) Break this information into parts. First of all, D was hired after I. Draw a line down from the “FI” pair to a “D.” Secondly, D was hired before E, so draw a line down from the “D” and add an “E.”

5) We now continue to incorporate the rules that link up with the entities already on the page: “H was hired before D,” so we can draw another line up from D connecting H we drew earlier.

4) Now we can work down from E: This rule says that E was hired before B, so draw a line down from the “E,” and add a “B.”

8) This seems like the logical rule to incorporate next: B was hired before J, so draw a line down from the “B,” and add a “J.”

7) Now that J and B are placed, deal with Rule 7: G was hired after J and B. Add a “G” below “J.”

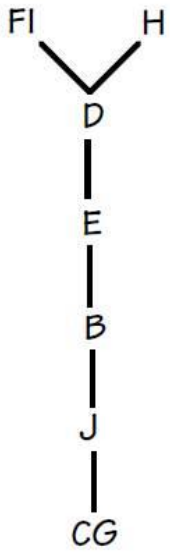
2) Like F and I, C and G were hired on the same day (along with no one else that day), so add C to the G we’ve just placed below J.

3) We’re left with one rule: Except for the two pairs of workers in Rules 1 and 2 (FI and CG), on every other day exactly one employee was hired. Basically a loophole closer, no need to draw anything. (Note that this does confirm total of seven hiring days, not necessarily consecutive.)

Key Deductions:

Much should be clear just by reviewing the master sketch: H and FI must have been hired on days 1 and 2, in either order. D must have been hired on day 3; E on day 4; B on day 5; J on day 6; and GC on the last day, day 7. All that's ambiguous, in fact, is the H vs. FI relationship. Talk about concrete information! Small wonder that there are so many "non-if" questions.

The Final Visualization: And here is that neat, accessible sketch:



The Questions:

1. (D)

The master sketch clearly shows that the last two workers hired were G and C, choice (D).

2. (A)

Again, if your sketch is complete and accurate, you're rewarded with another ten-second question. The FI pair and H were hired on days 1 and 2 (although we don't know in which order), D was hired on day 3, and E on day 4, choice (A).

3. (A)

Here you're looking for the workers who must have been hired before J. Again, let your master sketch do the work for you. Just count the entities, who must have come before J: F, I, H, D, E, and B, which makes a total of six, choice (A).

4. (E)

Thanks to the master sketch, you know nearly everything about the entities. It's an easy (and quick) matter of checking each choice looking for the one that must be true.

(A) No, D must have been fourth, not first.

(B) and (C) H could have been hired first, but so could the FI pair. Each of these choices is possible only. One of them is true, but we can't figure out which.

(D) No—clearly the first two workers hired could be I and F, with H third. (D) therefore need not be true.

(E) is the only choice left, but for the record: The sketch clearly shows something we've known from the very beginning: that either H was hired first, or F and I were. (E) must be true and is the answer.

5. (D)

If E was hired on Monday, the earliest day on which B could have been hired is Tuesday. If B was hired on Tuesday, the earliest J could have been hired is Wednesday. And if J was hired on Wednesday, the earliest G (as well as C, of course) could have been hired is Thursday, answer choice (D).

Q3)
There are exactly seven houses on a street. Each house is occupied by exactly one of seven families: the Kahns, Lowes, Muirs, Newmans, Owens, Piatts, Rutans. All the houses are on the same side of the street, which runs from west to east.

(a) The Rutans do not live in the first or the last house on the street.
(b) The Kahns live in the fourth house from the west end of the street.
(c) The Muirs live next to the Kahns.
(d) The Piatts live east of both the Kahns and the Muirs but west of the Lowes.

1. Which one of the following families could live in the house that is the farthest east?

A) the Kahns
B) the Muirs
C) the Newmans
D) the Piatts
E) the Rutans

2. Which one of the following families CANNOT live next to the Kahns?

A) the Lowes
B) the Newmans
C) the Owens
D) the Piatts
E) the Rutans

3. If the Muirs live west of the Kahns, then the Rutans CANNOT live next to both

A) The Kahns and the Piatts
B) The Lowes and the Piatts
C) The Muirs and the Piatts
D) The Muirs and the Owens
E) The Muirs and the Newmans

4. If the Newmans live immediately west of the Kahns, which one of the following statements must be false?

A) The Owens lives next to the Newmans.
B) The Owens lives next to the Rutans.
C) The Piatts lives next to the Lowes.
D) The Piatts lives next to the Muirs.
E) The Rutans lives next to the Newmans.

5. If the Owens live east of the Muirs, which one of the following statements must be true?

A) The Kahns lives east of the Muirs.
B) The Kahns lives west of the Rutans.
C) The Owens lives west of the Lowes.
D) The Owens lives east of the Piatts.
E) The Owens lives west of the Piatts.

6. If the Owens live east of the Kahns, which one of the following pairs of families must live next to each other?

A) The Kahns and the Piatts
B) The Lowes and the Owens
C) The Muirs and the Newmans
D) The Newmans and the Rutans
E) The Owens and the Piatts

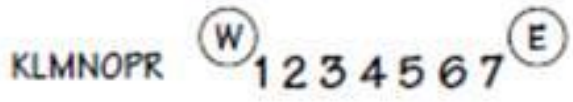
S3)

The Action: A fairly straightforward sequencing set, with a very minor wrinkle: the use of the terms “west” and “east” to refer to the different sides of the sequence. We’re asked to order seven families (K, L, M, N, O, P, and R) in seven houses lined up west (left) to east (right) on one side of a street. The Key Issues are:

1) Which families can, must, or cannot occupy which house?

2) Which family can, must, or cannot occupy a house adjacent to what other family?

The Initial Setup: Keep this setup simple; seven dashes or the numbers 1 to 7 from left to right on your page will suffice. You may wish to jot down a "W" to the left and an "E" to the right of the sketch, just for good measure, to remind you which side is which. List the roster of families off to the side, and you're ready to fill the Rules into this basic setup:



The Rules:

2) Once again, the most concrete rule first: The Kahns live in the fourth house from the west. Don't let the awkward wording throw you. This simply means that the Kahns live in house 4, so put that directly into the sketch.

1) "No R" over houses 1 and 7 should help us keep this in mind.

3) Since the Muirs live next to the Kahns, who live in house 4, the only possible houses for the Muirs are 3 and 5. Place an "M" with arrows pointing to houses 3 and 5 into your sketch.

4) Since the Kahns live in house 4, this rule tells us that the Piatts can only live in houses 5, 6, or 7. But the Piatts can't live in 7, because the Lowes must live to the east (right) of them. So the "P . . . L" chunk will have to fall somewhere within houses 5, 6, and 7. Jot that down, and we're ready to move on.

Key Deductions:

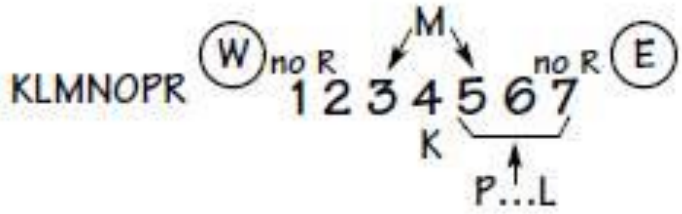
There's no major deduction here, but there are only a limited number of ways the sequence could play out. It's therefore worth quickly exploring the possibilities: M lives in either 3 or 5. Let's say M occupies the 5th house. Since the P. . . L pair must be east of K, we get this:

1	2	3	4	5	6	7
O/N			K	M	P	L

R, O, and N are left to fill in the 1, 2 and 3 houses. R, you remember, can't be on either end, so that family would be in either 2 or 3 in this situation, leaving O and N for the other spots, in either order.

The second scenario is a little more flexible, and arises if we instead place M 3rd. P and L still need to be to the right of K, and must still maintain that K..P..L ordering. The other 3—N, O, and R—would fill the remaining spots, so long as R isn't first or last. This ordering isn't as determined as the other, and naturally doesn't help us as much, but it's still good to consider the possibilities in advance, which we've done.

The Final Visualization: Here's what we're armed with to tackle the questions:



The Questions:

1. (C)

The Lowes lives in house 7 in our first possibility above, but that's not a choice. (A), (B), (D), and (E) directly violate Rules 2, 3, 4, and 1, respectively, which leaves only (C). A glance at our sketch would quickly confirm the same result.

2. (A)

We can easily infer from Rule 4 that the Piatts lives between the Kahns and the Lowes. Therefore, there's no way that the Kahns can live next to the Piatts. The rest of the families in (B) through (E) can live next to the Kahns.

3. (C)

The info in the stem puts M in house 3, which conjures up the second scenario we discussed earlier. This was the less helpful scenario, so let's check out the choices.

(A) R can live next to both K and P: K in 4, R in 5, P in 6, and L in 7, with N and O floating between 1 and 2.

(B) is possible too: M, K, P, R, and L in 3 through 7, again with N and O floating at the beginning.

(C) M is in house 3, K is in 4, and P is either in 5 or 6. It's therefore not possible for the Rutans to live between the Muirs and the Piatts, making (C) our answer.

(D) and (E) Either O or N can live in house 1, and with R in 2 and M in 3, so both are possible.

4. (A)

N in 3 points us to our first scenario:

1	2	3	4	5	6	7
		N	K	M	P	L

We're left with R and O for houses 1 and 2, and Rule 1 forces R into house 2 and O into house 1. The entire setup of families to houses is complete: O, R, N, K, M, P, L. The Owens clearly do not live next to the Newmans. Choice (A) it is.

5. (A)

Here we can deduce that the second scenario above is operative: The only way for the Owens to live to the east, or right, of the Muirs, is for M to live in 3, with O, P and L fitting somewhere in 5, 6, and 7. The two remaining entities, N and R, will therefore take houses 1 and 2, respectively (R can't take 1 from Rule 1). The choice that corresponds to this is (A): K must be east of M. (B) is false, and (C) through (E) are possible only.

6. (D)

The test makers liked question 12 so much, here it is again (nearly). So we should consult the same sketch as the one for number 12: M is in 3, K is in 4, O, P, and L are relatively free to float between spaces 5, 6, and 7 (as long as P is before L), N is in house 1, and R is in house 2. Scan down the list for adjacent families; N and R must live next to one another, specifically in houses 1 and 2, respectively. (A), (B), and (E) contain pairs that are possible neighbors, but need not be, whereas the pair in (C) are definitely separated by the Rutans.

Q4)

Seven consecutive time slots for a broadcast, numbered in chronological order 1 through 7, will be filled by six song tapes—G, H, L, O, P, S—and exactly one news tape. Each tape is to be assigned to a different time slot, and no tape is longer than any other tape. The broadcast is subject to the following restrictions:

(a) L must be played immediately before O.
(b) The news tape must be played at some time after L.
(c) There must be exactly two time slots between G and P, regardless of whether G comes before P or whether G comes after P.

1. If G is played second, which one of the following tapes must be played third?

A) the news
B) H
C) L
D) O
E) S

2. The news tape can be played in any one of the following time slots EXCEPT the

A) second
B) third
C) fourth
D) fifth

E) sixth

3. If H and S are to be scheduled as far from each other as possible, then the first, the second, and the third time slots could be filled, respectively, by

A) G, H, and L
B) S, G, and the news
C) H, G, and L
D) H, L, and O
E) L, O, and S

4. If P is played fifth, L must be played

A) first
B) second
C) third
D) fourth
E) sixth

5. What is the maximum number of tapes that can separate S from the news?

A) 1
B) 2
C) 3
D) 4
E) 5

6. Which one of the following is the latest time slot in which L can be played?

A) the third
B) the fourth
C) the fifth
D) the sixth
E) the seventh

7. The time slot in which O must be played is completely determined if G is assigned to which one of the following time slots?

A) the first
B) the third
C) the fourth
D) the fifth
E) the sixth

S4)

The Action:

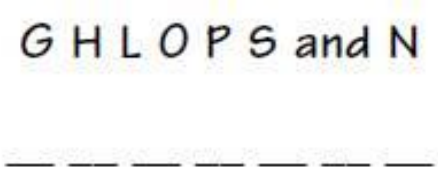
"Seven consecutive time slots" immediately suggests a sequencing game, and that's exactly what we get in Game 1. A slight oddity is introduced, but it turns out to be inconsequential: Six of the seven entities are song tapes with letter names, while the last entity is referred to as a "news tape." Usually such a differentiation between entities means something, but here it doesn't—if we call the news tape N, we simply have seven lettered entities to fit into seven time slots, which is as familiar a game action as any you'll see on the Logic Games section. We're told that no tapes share a time slot, and no tape is longer than any other, which are just loophole closers. So far then, any tape can be slotted anywhere. The Key Issues will be our normal sequencing concerns:

1) Which tapes must, can, or cannot be played in which time slots?

2) Which tapes must, can, or cannot be played consecutively?

The Initial Setup:

A roster of the entities and seven dashes on your page should do it for an initial setup—there's no need to get any fancier than that. You can include the news tape, N, alphabetically between L and O, or keep N separate for now, just in case this tape takes on some special significance in the game. (As we'll see, it doesn't—N will be regarded as just another entity.)



The Rules:

1) Rule 1 is a traditional sequencing rule that pairs up a couple of entities in a specific order. The upshot is that we'll always have LO somewhere in the list. Simply denote this as ALWAYS LO.

2) Rule 2 links up nicely with Rule 1, and you probably didn't have to wait until you got to Step 4 of the Kaplan Method to combine these rules to create an even bigger cluster: LO....N. Song tape O comes immediately after L, and then N must come sometime after that. The dots remind us that it's okay for other tapes to come between O and N; the rule is satisfied as long as N is anywhere to the right of LO.

3) Don't let the wordiness of Rule 3 intimidate you: All it says is that two tapes must come between G and P, but G and P can appear in either order. This allows for two possibilities: G _ _ P or P _ _ G.

Key Deductions:

In a game so basic, there's often not much in the way of Key Deductions. However, there are usually some implications of the rules that are worth noting. Take, for example, our main LO...N bloc. What does this tell us about L? Well, L can't be played in slots six or seven, right? If it was, there wouldn't be enough room for N at the end, and possibly O as well. Similarly, N can't be played in slots one or two, because that would shove L, O, or both off the map at the beginning of the sequence. These may not seem like major deductions, but these limitations directly help us bag a few questions, notably Qs. 2 and 6.

The Final Visualization: Here's what we have to work with heading into the questions:

G H L O P S and N

_ _ _ _ _ _ _

Always LO...N

G _ _ P or P _ _ G

The Questions:

1. (C)

The first question launches right into a solid hypothetical, so it's best to get the information into a new sketch on the page and see where it leads. Placing G into the second slot kicks Rule 3 into effect, forcing P into slot five; that's the only way that exactly two time slots will fit between G and P as mandated by Rule 3. That leaves the LO...N bloc as our next major concern. Well, the LO pair physically fits into slots three and four or slots six and seven, but the latter case leaves no room for N. So L must be played third and O fourth. The stem asks which tape must be third, and there we have it: L, choice (C).

2. (A)

We've already given N all the consideration that Q. 2 requires. Recall N's limitations, discussed in the Key Deductions section above: N cannot be played first, because then there would be no room for the LO pair before it and N can't be played second, because even if we squeeze O into slot one, L would be left out in the cold. We're looking for the time slot that N can't take, so it's either one or two. The test makers chose "second" for choice (A).

3. (C)

Here, our two free agents, H and S, are to be slotted as far from each other as possible. How far is that? — is the question that should come to mind. We may as well try placing them on opposite sides of the ordering, that is, in slots one and seven. And it doesn't matter which is on which side, since neither H nor S has any relationship to any other entity, thus making them functionally identical. Trying it out, we find that it is possible to place H and S on opposite ends of the ordering, as long as L and O take slots three and four, respectively. If this pair is in any other consecutive slots between two and six, there's no way to satisfy Rule 3. A little trial and error shows that L and O must be in slots three and four, surrounded by G and P, in either order, in two and five. N is left for slot six.

Now, we're asked about the first three slots, and while there's some variability in slots one and two (slot one can be H or S, and slot two can be G or P), slot three is definitely reserved for L, so our best bet is to check the choices for that. This narrows the possibilities down to (A) and (C), and (A) is out for two reasons: First, G H L is an impossible way to start the sequence under any circumstances: Rule 1 would force O into slot four, while Rule 3 would have it going to P. That's enough to eliminate (A). Also, (A) doesn't fulfill the requirement in the stem: H, in slot two, is not the furthest it could be from S. That leaves (C): H G L is a possible beginning for the ordering under these circumstances, as long as we flesh it out with O P N S, in that order.

4. (C)

What happens when P is fifth? G is second, and we're right back at Q. 1 again. Lucky for us, since we've already worked through this scenario: As we saw there, when G is second and P is fifth, L must be third, choice (C).

5. (E)

What's the greatest number of tapes we can get between S and N? As mentioned just above, we can use our previous orderings to help us out here. We've seen the same partial ordering in Qs. 1 and 4:

__ G L O P __ __

Can we get all the tapes between S and N? Sure; if our familiar GLOP sequence occupies slots two through five, we can place S into slot one and N into slot seven (which leaves H for slot six), and we're done. It is possible to fit all five of the other tapes between S and N, choice (E).

6. (C)

Another quick point for work already done: We thought through L's limitations up front, so this one shouldn't take much more than a few seconds. Since both O and N must come after L, L

cannot be played sixth or seventh, which means the earliest slot possible for L is fifth, choice (C). And placing L fifth causes no other violations, as the ordering “G H S P L O N” readily proves.

7. (D)

We’re looking for a time slot for G that will decisively fix the time slot for O, and once again our previous work in Qs. 1 and 4 can help us out. In both of those questions, G was second, which forced L to be third and O to be fourth. “Second,” which would therefore be a viable answer, is conspicuously absent; notice how the choices jump right from “first” to “third.” This is not a problem, though, once we remember that G and P are interchangeable in this game. We can switch

__ G L O P __ __

from Qs. 1 and 4 to

__ P L O G __ __

without altering the logic in any way. P in two and G in five would thus force the same situation we’ve seen a number of times—L in 3 and O in 4. So placing G in slot five fixes O in slot four, and (D) is correct for the final question of this first game.

For the record:

(A) If G is first, then P is fourth, and O can be slotted third or sixth.

(B) If G is third, then P is sixth, and O can be slotted second or fifth.

(C) If G is fourth, then P could be first, in which case O can be slotted third or sixth.

(E) If G is sixth, then P is third, and O can be slotted second or fifth.

Q5)

A gymnastics instructor is planning a weekly schedule, Monday through Friday, of individual coaching sessions for each of six students—H, I, K, O, U, and Z. The instructor will coach exactly one student each day, except for one day when the instructor will coach two students in separate but consecutive sessions. The following restrictions apply:

(a) H’s session must take place at some time before Z’s session.
(b) I’s session is on Thursday.
(c) K’s session is always scheduled for the day immediately before or the day immediately after the day for which O’s session is scheduled.
(d) Neither Monday nor Wednesday can be a day for which two students are scheduled.

1. Which one of the following is a pair of students whose sessions can both be scheduled for Tuesday, not necessarily in the order given?

A) H and U
B) H and Z
C) K and O
D) O and U
E) U and Z

2. If K's session is scheduled for Tuesday, then which one of the following is the earliest day for which Z's session can be scheduled?

A) Monday
B) Tuesday
C) Wednesday
D) Thursday
E) Friday

3. Which one of the following must be true?

A) If U's session is scheduled for Monday, H's session is scheduled for Tuesday.
B) If U's session is scheduled for Tuesday, O's session is scheduled for Wednesday.
C) If U's session is scheduled for Wednesday, Z's session is scheduled for Tuesday.
D) If U's session is scheduled for Thursday, Z's session is scheduled for Friday.
E) If U's session is scheduled for Friday, Z's session is scheduled for Thursday.

4. Scheduling Z's session for which one of the following days determines the day for which U's session must be scheduled?

A) Monday
B) Tuesday
C) Wednesday
D) Thursday
E) Friday

5. If H's session is scheduled as the next session after U's session, which one of the following could be true about H's session and U's session?

A) U's session is scheduled for Monday, and H's session is scheduled for Tuesday.
B) U's session is scheduled for Thursday, and H's session is scheduled for Friday.
C) They are both scheduled for Tuesday.
D) They are both scheduled for Thursday.
E) They are both scheduled for Friday

S5)

The Action:

"Weekly schedule" is a sure sign of sequencing, and indeed, we are mainly concerned with ordering students throughout a five-day period from Monday to Friday. One small wrinkle, though: There are six students for the five days, so exactly one day will get two students, whose sessions will be "separate but consecutive." This two-student day lends a very small grouping element to the set, but so small in fact that we'll just consider this to be a standard sequence with a minor twist. If you took a peek at the rules, you'd notice that all but the last deal with our typical sequencing concerns—or as we like to call them, the Key Issues—namely:

1) Which students must, can, or cannot be scheduled for which days of the week?

2) Which students must, can, or cannot be scheduled consecutively with which other students?

To which we'll simply add the one additional issue here:

3) Which day of the week gets two students, and which students are they?

The Initial Setup:

No great difficulty picturing this setup; simply get the days of the week and students down on the page and you're ready to see what the rules have in store for our aspiring gymnasts: ("No days free" is a good reminder of the stipulation in the intro paragraph that aside from the double-student day, the coach meets with exactly one student each day.)



The Rules:

2) No reason not to jump right to the most concrete rule of the bunch: Student I's on Thursday, so build that right into the sketch.

1) Standard sequencing fare: H . . . Z should help us to remember this. You should make a mental note, however, that due to the slight wrinkle (the two-student day), this rule does not necessarily imply that H's session will be on an earlier day of the week than Z's. It must come "some time before," which leaves open the possibility that H can be the first student of the two-student day with Z's lesson later that day.

3) Here's the more standard adjacency type of sequencing rule that we're used to: K and O must fall on consecutive days of the week, in either order. KO or OK will remind us of this.

4) Monday and Wednesday will have one student, so the two-student day will be either Tuesday, Thursday, or Friday.

Key Deductions: Not much to combine here, is there? Notice how no entity appears in more than one rule, a fact that makes it less likely that a big deduction will exist. Trying out a few scenarios leads to the conclusion that the possibilities are not very limited thus far, which means that our best bet is to move right on to the questions.

The Final Visualization contains all that we'll take with us into the questions:

H I K O U Z

			I	
M (1)	T	W (1)	Th	F

H...Z
KO or OK
NO DAYS FREE

The Questions:

1. (D)

The set gets off to somewhat of a nasty start with this "could be true" question—with no big deductions at our disposal, there's not much to do but simply try out the choices. We could postpone this one for later, hoping to see a pair among the choices show up together on Tuesday in another question, but that's not all that likely, so we may as well simply get to it. First, we can check to see if any choices directly violate a rule, and in fact (C) bites the dust on account of Rule 3: K and O, scheduled for immediately consecutive days, can never be on the same day. It would nice if a choice contained I who must be on Thursday, but no such luck, so from here we'll just have to test the remaining choices:

(A) Can H and U pair up on Tuesday? Nope—that leaves no room for the KO bloc. With H and U taking up the double day, there's no way to get K and O on consecutive days without doubling up somewhere else. And that means that *any* pair that doesn't include either K or O for Tuesday will result in the same violation. This insight effectively eliminates (B) and (E), and we've already axed (C) for the reason stated above, so (D) must be correct. If you test it out—and by all means in this situation, you probably should—you'll see that

M	T	W	Th	F
K	O	H	I	Z
	U			

is a valid ordering, confirming (D) as the pair that could be scheduled for Tuesday.

2. (B)

K's on Tuesday, so get that into a new sketch. Not much to work with, but the question's specificity tells us where to go next—we need to find the earliest spot for Z. The earliest day is Monday, and while Rule 1 would allow H and Z on Monday (as long as H went first), Rule 4 forbids it—the double-student day can't be Monday. What about Tuesday for Z? No problem there:

M	T	W	Th	F
H	K	O	I	U
	Z			

This ordering takes care of the stem and satisfies all the rules, proving that Tuesday, choice (B), is the earliest day for Z under these circumstances.

3. (D)

Yuck. A non-if "must be true" question is perfectly welcome when there's a big deduction up front, but what must be true here given no new information? Nothing, really, except maybe something negative like "Z cannot be scheduled for Monday"—but the choices here are all themselves hypothetical. However, the task is made a little easier by using work from the previous two questions. The schedule we formed in Q. 2 allows us to kill (E)—there we see that U can be on Friday while Z need not be on Thursday. And the schedule we created in Q. 1 shows that U and O could both be on Tuesday, effectively killing choice (B). The rest, however, we'll have to work out:

(A) If U is on Monday, does H have to be on Tuesday? The test is to see whether we can place H on some other day. And we can: If K and O are on Tuesday and Wednesday (in either order), then H can be on Thursday with Z on Friday. So (A) need not be a true statement.

(C) With U on Wednesday, K and O can take Monday and Tuesday, leaving H and Z for Thursday and Friday, respectively. So U on Wednesday does not force Z into Tuesday, and we can cross off (C).

(D) is all that's left, so it must be correct. For the record: With U on Thursday, the KO pair must be scheduled during two consecutive days between Monday and Wednesday. To get H before Z, H must take the leftover day before Thursday while Z is slotted for Friday.

4. (D)

Q. 4 requires some trial and error, but a little cleverness makes the job easier. We're asked to find the day for Z that would fix U's position in the schedule. The shortcut here is to recognize that the end of the week will probably pose more restrictions given the fact that I is already placed on Thursday. Testing out Z on those days at the end, you'll find that Thursday, choice (D),

is indeed the spot for Z that would fix U's time slot as well: If Z is on Thursday with I, then our double-student day is used up. The adjacent KO bloc will have to fall on consecutive days between Monday and Wednesday (in either order), and H's session must also be before Thursday to satisfy Rule 1. Either way, U is the only student left for Friday. All of the other choices leave U an option of at least two days in the ordering. (Note: If you started testing from the beginning, you would have noticed, as we pointed out above in Q. 2, that Z can never be scheduled for Monday. If Z's on Monday, then H would have to come before Z on Monday (Rule 1), setting up a situation that would violate Rule 4.)

5. (A)

We get some info to work with here, but once again we're faced with the prospect of testing out various scenarios. The schedule from Q. 1 applies, as H's session could be immediately after U's in that one, but unfortunately none of the choices conform to that schedule. And there's not even a likely candidate here as in the previous question, so we really have no choice but to try them out. Luckily, (A) comes through as the winner, as illustrated by the following schedule:

M	T	W	Th	F
U	H	Z	I	O
			K	

There are other ways to space out Z, K, and O from the middle to end of the week, but all we need is one instance that works with respect to H and U to choose (A) for this "could be true" question. As for the rest:

(B) With U doubling up with I on Thursday, it's not possible to put H on Friday and still have H's session before Z's (Rule 1). Sure, we could put Z later on Friday, but then we'd have two two-student days, in violation of the numbers rule in the intro.

(C) Scheduling both H and U on Tuesday causes the same problem we saw back in Q. 1—there would be no way to place the consecutive KO pair without doubling up somewhere else.

(D) With I already on Thursday (Rule 2), placing this pair there would result in three Thursday students. No good.

(E) If H and U were on Friday, then Rule 1 would force Z onto Friday's roster as well, overburdening the poor coach with three Friday lessons. Again, no good.

Q6)
Exactly six trade representatives negotiate a treaty: K, L, M, N, O, and P. There are exactly six chairs evenly spaced around a circular table. The chairs are numbered 1 through 6, with

successively numbered chairs next to each other and chair number 1 next to chair number 6. Each chair is occupied by exactly one of the representatives. The following conditions apply:

(a) P sits immediately next to N.

(b) L sits immediately next to M, N, or both.

(c) K does not sit immediately next to M.

(d) If O sits immediately next to P, O does not sit immediately next to M.

1. Which one of the following seating arrangements of the six representatives in chairs 1 through 6 would NOT violate the stated conditions?

A) K, P, N,M, O, L

B) K, L,M, P, N, O

C) K, L,M, O, P, N

D) K, O, P, N, L,M

E) K, N, L, O,M, P

2. If L sits immediately next to P, which one of the following is a pair of representatives who must sit immediately next to each other?

A) K and O

B) L and N

C) L and O

D) M and N

E) M and P

3. If K sits directly between L and P, then M must sit directly between

A) L and N

B) L and O

C) N and O

D) N and P

E) O and P

4. If N sits immediately next to M, then K can sit directly between

A) L and M

B) L and P

C) N and O

D) N and P

E) P and O

5. If L sits immediately next to M, then which one of the following is a complete and accurate list of representatives any one of whom could also sit immediately next to L?

A) K

B) K, N

C) N, P

D) K, O, P

E) K, N, O, P

6. If L sits immediately next to N, which one of the following statements must be false?

A) K sits immediately next to O.

B) L sits immediately next to M.

C) O sits immediately next to P.

D) N sits directly between L and P.

E) O sits directly between K and M.

7. If K sits immediately next to O, then L CANNOT sit directly between

A) K and M

B) K and N

C) M and N

D) M and P

E) N and O

S6)

The Key Issues are:

1) Which representatives can, must, or cannot occupy adjacent chairs?

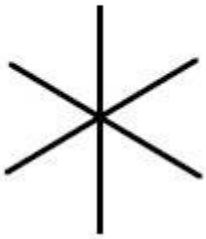
2) Which representatives can, must, or cannot sit between which other representatives?

In the majority of circle sequencing sets, another key issue is who is across from whom. How do we know that this set is not concerned with who's across from whom? A quick scan of the questions tells us. None of the question stems or answer choices include the phrase "across from" or "opposite from." Therefore, we know that we needn't concern ourselves with this element here.

Another important issue to look for in a circle sequencing set is the left/right issue. We need to recognize whether the set is concerned with who sits to the left and right of whom around the circle. How can we determine if this is an issue? A quick scan of the set will tell us. Nowhere in the intro, rules, or questions do the words "left" or "right" occur. As indicated clearly in the rules and questions, the set is concerned only with who is next to and between whom.

The Initial Setup: here are two ways to lay out a circle sequencing set. We can either, draw a circle (and place the entities in chairs around it) or draw intersecting lines like the spokes of a wheel (and place the entities at the ends of those spokes). The number of spokes will depend upon the number of entities we're placing around the table. The "spoke" method is often preferable because it clearly shows not only which chairs are next to each other, but also which chairs are across from each other. As we mentioned above, most circle sequencing sets are concerned with who's across from whom, but scanning the questions, it's evident that this set isn't. From the scan of the questions, it's also evident that even though the set numbers the chairs 1 through 6, none of the questions but the first mentions a chair number. So we won't bother numbering the chairs in our sketch:

K L M N O P



The Rules:

1) Pairs of entities are quite useful. Two entities that must always be together (or adjacent in this set) give us a solid point from which to begin work on each question. Rule 1 provides us with P and N as such a pair. Under Key issues, we already recognized that this set isn't concerned with who's to the left or right of whom, so we don't need to take that into account when symbolizing the rules. "ALWAYS PN" will serve as a reminder that these two must always sit together. (If this set did involve the left/right issue, we'd have to write "ALWAYS PN OR NP.")

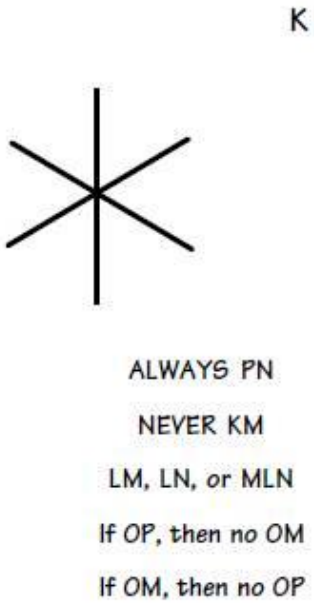
2) This rule may have looked complex and confusing at first, but remember to ask yourself what this rule means, not just what it says. After interrogating the rule in this way, it hopefully appeared neither complex nor confusing. It simply says that L must always sit next to M or N or both of them. Again, this set isn't concerned with left/right, so "LM, LN, or MLN" is a good way to keep this rule in mind.

3) Here we're given a pair of representatives who can't sit next to each other. Knowing who can't sit together isn't as useful as knowing who must sit together (as in Rule 1), but this type of rule does serve to limit the possibilities. K and M must never be next to each other around the table. Write "NEVER KM" to serve as a reminder.

4) is an if-then rule, so be sure to take your time and carefully think through the rule and its contra-positive. If O is sitting next to P, then O isn't also next to M. We can get the contra-positive simply by reversing these terms and negating them. If O is sitting next to M, then O isn't also next to P. "If OP, then no OM" and "If OM, then no OP" should keep the rule and its contra-positive in our minds.

Key Deductions: The most important deduction here is recognizing that this set isn't concerned with who's across from whom and who's to the right or left of whom. These realizations may not seem like enormous deductions, but they give us quite a leg up on this set. Now, we don't have to waste time worrying about those issues, and this also leaves us greater flexibility to toss entities into the circle at the beginning of each question. Furthermore, we can simply throw the P/N pair from Rule 1 into the circle as the starting point when there seems to be no other logical place to begin (since we need not worry about the order they take as long as they're adjacent).

The Final Visualization: Here's what we have as we go on to the questions:



The Questions:

1. (B)

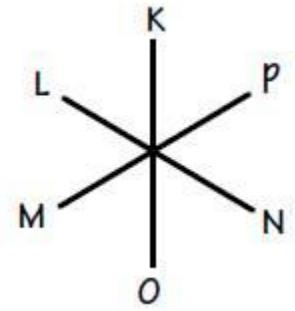
Oh good, here's an acceptability question. Normally we'd just check each rule against the choices and eliminate the choices that violate a rule. However, there's something we need to notice about this acceptability question. Thanks to the circular design of the table, the first person of each list (in chair 1) is next to the last (in chair 6). For example, we need to realize that, in choice (A), K sits next to L. With that in mind, we can treat this just like any other acceptability question; just grab each rule and cross off any choice that violates that rule. Rule 1 is violated by answer choice (E) which doesn't have P next to N. Rule 2 requires that L be next to M, N, or both. Answer choice (A) has L next to O and K (L in seat 6 and K in seat 1) and so can be eliminated. Rule 3 forbids K to be next to M which answer choice (D) tries (M in seat 6 wraps around and is next to K in seat 1). Finally, the conditional Rule 4 is violated by answer choice (C) with has O next to both P and M. Choice (B) remains and is the credited response.

2. (A)

The stem says that L sits next to P. Get this new information down on the page. The set isn't concerned with who's to the left and right, so just make a new sketch and put L and P next to each other. Where to now? Look for rules that have L or P in them. Rule 1 says that P must sit next to N. L is on one side of P, so N must be on the other. Rule 2 says that L must sit next to M or N or both. Here N is on one side of P with L on the other side, so M must be next to L. Who's left? K and O are left to occupy the remaining adjacent chairs in between M and N. Rule 3 says that K can't sit next to M, so K must sit next to N while O sits next to M. The complete ordering is now set, so it's a simple matter of skimming the choices for the pair that must sit together. K and O must sit next to each other and (A) is the answer.

3. (B)

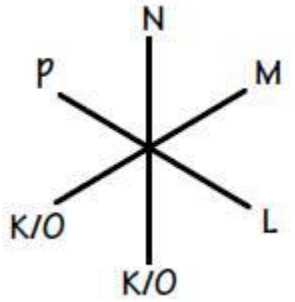
Once again, get the new information down and check the rules. We're not concerned with the "left/right" issue, so just draw a new sketch and put K between L and P. Now look for rules with K, L, or P. Rule 1 says that P and N must always sit next to each other. K is next to P on one side, so N must be on the other side of P. Our new sketch clearly shows that L and N aren't next to each other, which means it's time to enact Rule 2: L must have M beside her. Only one entity and one slot remain, so O will fill in the chair between M and N. The complete sketch looks like this:



The question asks for the reps that M sits between, and that's L and O, choice (B).

4. (E)

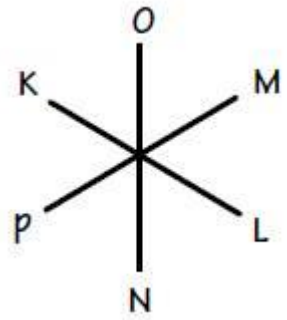
Once again, all you have to do is create a new sketch and add the new information: N is next to M. By now you should be used to putting P and N together. M is on one side of N, so P must be on the other. You should also be used to placing L next to M and/or N. There's no open chair next to N, so L must go next to M. O and K are left to fill in the two chairs between P and L, and this time nothing forbids either of these people from taking either of the remaining chairs. The sketch therefore looks like this:



The question asks which reps K can sit between. Depending on where K and O sit, K can either sit between P and O or O and L. The test makers chose the former in choice (E).

5. (E)

We're helped immensely in this question by referring to our previous work. M is on one side of L, and the question asks for a complete list of all of the reps who could be on the other side. In the previous question, Q. 4, L was next to M on one side, and could be next to either K or O on the other. This immediately allows us to eliminate (A), (B), and (C), none of which includes O. (Incidentally, (C) is doubly wrong by not including K.) (D) and (E) are left. How are these two choices different? (E) includes N while (D) does not. If we can prove that N can be on the other side of L, then (E) is the answer; if not, we go with (D). Is it possible for N to sit on the other side of L? Sure, here is the complete ordering:



N must be included, so (E) gets the point for Q. 5.

6. (C)

Once again, don't hesitate to quickly redraw the master sketch including the new information from the stem (L next to N). L is on one side of N, so Rule 1 forces P to sit on the other side of N. Where to now? We're left with K, M, and O to fill in the remaining three adjacent slots. We know from Rule 3 that K and M can't sit next to each other, so they will have to be separated by O. Either K or M will sit next to L while the other sits next to P— and don't forget O in between K and M, no matter which way K and M sit. With the exception of K and M, this ordering is completely set. Now look for the choice that must be false.

(A) must be true. K must sit between either O and L or O and P.

(B) could be true. M could sit between L and O with K sitting between P and O.

(C) No way. O can't sit next to P. She must sit between K and M, as we deduced. (C) must be false and is the answer.

(D) must be true. We deduced from the information in the stem that N must sit between L and P.

(E) must be true. As we saw in our original analysis and again in the context of correct choice (C), O must sit between K and M.

7. (E)

This stem tells us that K sits next to O. And we can't take it any further. It looks like we'll just have to check each choice. But don't despair! Use your previous work. Look for instances where K and O sat next to each other and see which reps L was sitting between in those situations. This question asks who L can't sit between, so we can eliminate any choice that is disproved by those previous instances. In the previous question, K was next to O, and in that setup, L could sit between either, K and N or M and N. Eliminate choices (B) and (C). Where else was K next to O? In Q. 4, K and O sat together, and in that scenario we saw it was possible for L to sit between K and M—eliminate choice (A). Finally, in question 2, K and O sat together, and L sat between M and P. Eliminate choice (D). Choice (E) remains and is the answer.

Q7)

The organisms W, X, Y, and Z respond to the antibiotics ferromycin, ganocyclene, and heptocillin in a manner consistent with the following:
(1)Each of the organisms responds to at least one of the antibiotics.

(2)No organism responds to all three antibiotics.

(3)At least two but not all four or the organisms respond to ferromycin.

(4)If W responds to any antibiotic, then X responds to that antibiotic.

(5)If an organism responds to ferromycin, then it responds to ganocyclene.

(6)Y responds to ferromycin.

1. Each of the following can be true EXCEPT:

A) W responds to heptocillin.

B) X responds to ganocyclene.

C) X responds to heptocillin.

D) Y responds to heptocillin.

E) Z responds to ganocyclene.

2. Which one of the following could be true?

A) W, X, and Z all respond to ferromycin.

B) W, X, and Z all respond to ganocyclene.

C) W and exactly one other organism respond to ganocyclene.

D) W responds to more of the antibiotics than X does.

E) More of the organisms respond to ferromycin than to ganocyclene.

3. Which one of the following could be true?

A) Exactly one of the organisms responds to ferromycin.

B) All four of the organisms respond to heptocillin.

C) At least one of the organisms responds both to ferromycin and to heptocillin.

D) At least one of the organisms responds neither to ganocyclene nor to heptocillin.

E) At least one of the organisms responds to ganocyclene but does not respond to ferromycin.

4. If X does not respond to ferromycin, then which one of the following must be true?

A) W responds to ganocyclene.

B) X responds to ganocyclene.

C) X responds to heptocillin.

D) Z responds to ferromycin.

E) Z responds to heptocillin.

5. If any of the organisms responds to two of the antibiotics, then which one of the following is true about such an organism?

A) It must respond to ferromycin.

B) It must respond to ganocyclene.

C) It must respond to heptocillin.

D) It cannot respond to ferromycin.

E) It cannot respond to ganocyclene.

6. If none of the organisms responds to heptocillin, then which one of the following must be true?

A) W responds to ferromycin.

B) X responds to ferromycin.

C) Z responds to ferromycin.

D) Exactly three of the organisms respond to ganocyclene.

E) Exactly four of the organisms respond to ganocyclene.

7. If three of the organisms respond to exactly the same set of antibiotics as each other, and if Z does not respond to ferromycin, then each of the following must be true EXCEPT:

A) W responds to ferromycin.

B) X responds to ganocyclene.

C) Z responds to ganocyclene.

D) W responds to exactly the same set of antibiotics as Y.

E) X responds to exactly the same set of antibiotics as Y.

S7)

The Action: The key verb phrase in the opening paragraph here is “respond to”; each organism responds to the three antibiotics in some manner, so we’re dealing with a matching set. Does W get F, or G, or H—or some combination thereof? No doubt that’s what the rules are going to help us to decide, and the same goes for the other three organisms. No tricks here—the **Key Issue** is very straightforward:
1) Which antibiotics does each organism respond to? (All other issues are simply subsets of that. For example, knowing exactly which antibiotics an organism responds to implies which ones it doesn’t, which ones it has in common with the other organisms, which ones it doesn’t have in common, etc.)

The Initial Setup: The simplest way to match up the entities in two groups is to list the definite entities horizontally in columns across your page (here, the four organisms) and place the choices for those entities off to the side, to be filled in as needed. A simple list or grid like so will suffice, and certainly won’t take long to draw or recopy when necessary:

	W	X	Y	Z
F G H				

The Rules:

6) Keeping with the Kaplan strategy of beginning with the most concrete information, a quick scan of the rules reveals that Y responds to F, so draw it right in. Having incorporated this, you may have been drawn to the other rules about F.

5) Anywhere we place an F, we’ll also have to place a G, and so we may as well place a G under organism Y since we just placed an F there from Rule 6. Off to the side, you’ll want to indicate “If F, then G.” The contra positive should be readily apparent, and you may have jotted it down or simply taken notice of it here, never to lose sight of it throughout the set: “If no G, then no F.”

3) Rule 3 is about F too, but this one we’ll have to shorthand off to the side. Did you simply make the rule shorter, by writing something to the effect of “2 but not 4 F’s”? If you did, you need to remind yourself to ferret out the implications of the rules: that is, to turn negatives into positives and state the meaning of the rules as clearly as possible. “2 or 3 F’s” is much better. That’s what the rule really means in the context of this set.

There’s no indication that any of the remaining rules should be given priority at this point, so let’s jump back to the beginning. As it turns out, Rules 1 and 2 can be combined on the fly:

1) and 2) Each organism responds to at least one antibiotic, but no organism responds to all three. We can and should perform the same simple arithmetic as above to deduce that each organism will respond to either 1 or 2 antibiotics. “Each gets 1 or 2” is one of many ways to remind us of this.

4) Lastly, Rule 4: Whatever antibiotics W gets, X gets too. Make sure you take a few seconds to work out the implications of this. Whatever goes in the W column must appear also in the X column. But does it work the other way around? No. If W responds to two antibiotics, then X must have the same two. But if W gets only one—for example, G—then X must get G but can also get one more antibiotic, F or H.

Key Deductions: We’ve already begun our deductive work by piecing together the number information above. We saw that **each organism will respond to either, 1 or 2 antibiotics**, and also that **exactly 2 or 3 organisms will respond to F**. We also went right ahead and put FG into our Y column, and now, factoring in Rule 2, we can safely conclude that **organism Y, responding to F and G, is done and fully accounted for**. What else? We can combine Rules 2 and 5 to deduce that **no organism responds to both F and H**. F requires G, and adding H as well would violate Rule 2. And here’s one more thing you may have noticed, (but if you didn’t, is not a huge drawback): What happens if W responds to F? Then so does X (Rule 4). And we know that Y responds to F (Rule 6), so Z cannot because then all four would be responding to F, in violation of Rule 3, stated simply: **If W responds to F, then Z doesn’t respond to F.** If you saw this, great, if not, no biggie; you probably had the opportunity to work through this at some point during the questions. We won’t even bother including it in our master sketch. And speaking of which:

The Final Visualization: Here is all the information we have at our disposal to dispose of the seven questions hanging in the balance:

	W	X	Y	Z
F G H			FG	
	1 or 2	1 or 2	done	1 or 2

2 or 3 F’s
If F, then G - - - If no G, then no F
Whatever W gets, X gets

The Questions:

1. (D)

May as well begin by checking our key deductions, looking for the odd man out here. And our previous work with Y takes us where we need to go. We saw that since Y gets F (Rule 6), and therefore gets G (Rule 5), it cannot get H also because that would violate Rule 2. (D) is the impossibility here; all the rest work just fine.

2. (B)

We've most likely deduced enough up front to knock these choices out quickly, so take 'em one at a time. We're looking for something that's possible.

(A) Since Y already has F, linking W, X, and Z with F would result in four F's—a direct violation of Rule 3.

(B) No problem here, nothing prohibits four G's across the board. We need to get one more F in there to take care of Rule 3, but other than that, smooth sailing.

(C) If W responds to G, then so does X (Rule 4), and we already know that Y responds to G. So (C) is impossible: If W responds to G, and then at least two more that we know of must respond to G as well.

(D) directly contradicts Rule 4: Whatever W gets, X gets, so there's no way W can respond to more antibiotics than X.

(E) Same thought as (D), only stemming from Rule 5: Whatever gets F, gets G; so there can't be more organisms with F than G.

3. (E)

Same drill as 19. We're simply looking for what could be true, so our job is to test the choices until something possible appears. Unfortunately, we have to wait until the bottom of the list in choice (E): Rule 5 states that wherever there's F, there's G. This doesn't work the other way around. While Y has both F and G, it's perfectly possible for at least one of the other organisms to have G without F. In fact, consulting the partial arrangement worked out for correct choice (B) in Q. 2 confirms this: it's possible that G is linked to all four while only two organisms respond to F. (E) could be true. As for the others:

(A) directly contradicts Rule 3—we need 2 or 3 F's here.

(B) As we saw earlier, Y cannot respond to H because it already responds to F and G and no organism is allowed to respond to all three.

(C) Here's another deduction we worked out above: Since whatever responds to F must also respond to G (Rule 5), no organism that responds to F may respond to H because that would lead to a prohibited threesome.

(D) is the trickiest of the bunch because it's stated in the negative. Is it possible for any organisms to be without G and H? Well, Rule 1 reminds us that each organism must match up with at least one antibiotic, and if G and H are out, then F is the only one left. But a solo F is against the rules—Rule 5 in particular, which states that wherever F appears, G appears.

4. (D)

Our first hypothetical of the set appears here in 21: No F for X. The contra positive of Rule 4 tells us that whatever X doesn't get, W doesn't get, so no F for X also means no F for W. We still need another F to satisfy Rule 3, so Z must respond to F, as well as to G thanks to Rule 5. Do we have our answer yet? Yup—(D) must be true, and gets the point for 21. (A), (B), and (C) can, but need not, be true, while (E) is flat-out impossible: If Z responds to H, then Z's got the trifecta, all three antibiotics—and we know of course that that's not allowed.

5. (B)

There are many ways to go about this one. One quick way is to simply plot out the possibilities and check them against the choices. The only pairs of antibiotics an organism can respond to are FG and GH. The only other pair that can be formed from the three antibiotics is FH, and we've seen a number of times why this pair doesn't work (where F goes, G goes, therefore no H—all that Rule 2 and Rule 5 business). So let's call FG option 1, and GH option 2. Checking the choices:

(A) Not in option 2 it doesn't.

(B) Yup. Either way, G's called into service. That's our answer, and naturally we'd move on immediately. But for the record:

(C) No H in option 1.

(D) Sure it can—right there in option 1.

(E) Au-contraire! Such an organism must respond to G.

6. (E)

H is off the market for the sake of this question, which means that this scenario is going to be dominated by F's and G's. But wait a minute: Wherever F is, G is. And wherever F isn't, G's going to have to be there, too, since H is temporarily out of the picture. Bottom line: G's ubiquitous; omnipresent so to speak. Or, another way to put it is choice (E). (D) is obviously wrong if (E) is right; there must be four G's, not three. (A) and (B) bite the dust if we give an FG pair to Z and solo G's to W and X. However, we can also place FG pairs under W and X, with a solo G in Z's slot, contrary to the assertion in (C).

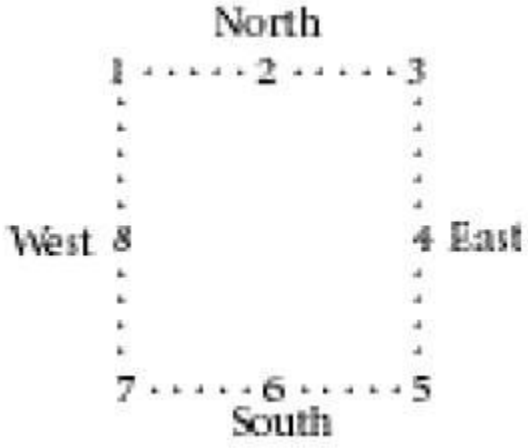
7. (C)

The second if-clause is more concrete than the first, so it's wise to begin there. If Z doesn't get F, then Z gets G or H or both. So far so good. But that plays into the more abstract hypothetical which tells us that three of the four organisms respond to the same set of antibiotics. Since Y responds F, and Z doesn't, and we still need at least two F's listed, Z must be the odd organism out: W, X, and Y must therefore be the three identical ones. Each of those responds to both F

and G, while Z responds to G or H or both, as noted above. That tells us almost everything we could possibly know about the situation, and all the choices match this scenario except for (C): it's possible that Z responds to only H.

Q8)

A square parking lot has exactly eight lights—numbered 1 through 8—situated along its perimeter as diagrammed below.



The lot must always be illuminated in such a way that the following specifications are met:

1. At least one of any three consecutively numbered lights is off.

2. Light 8 is on.

3. Neither light 2 nor light 7 is on when light 1 is on.

4. At least one of the three lights on each side is on.

5. If any side has exactly one of its three lights on, then that light is its center light.

6. Two of the lights on the north side are on.

1. Which one of the following could be a complete and accurate list of lights that are on together?

A) 1, 3, 5, 7

B) 2, 4, 6, 8

C) 2, 3, 5, 6, 8

D) 3, 4, 6, 7, 8

E) 1, 2, 4, 5, 6, 8

2. Which one of the following lights must be on?

A) light 2

B) light 3

C) light 4

D) light 5

E) light 6

3. If light 1 is off, which one of the following is a light that must also be off?

A) light 3

B) light 4

C) light 5

D) light 6

E) light 7

4. Which one of the following statements must be true?

A) If light 2 is on, then light 6 is off.

B) If light 3 is on, then light 2 is on.

C) If light 4 is on, then light 3 is off.

D) If light 5 is off, then light 4 is on.

E) If light 6 is off, then light 1 is on.

5. If light 5 is on, which one of the following could be true?

A) Light 1 is off and light 6 is off.

B) Light 1 is on and light 7 is on.

C) Light 2 is off and light 4 is on.

D) Light 2 is off and light 6 is off.

E) Light 6 is on and light 7 is on.

6. If light 4 is on, each of the following statements must be true EXCEPT:

A) Light 1 is on.

B) Light 2 is on.

C) Light 5 is off.

D) Light 6 is on.

E) Light 7 is off.

7. Suppose that it is no longer part of the specifications that two lights on the north side be on. If all of the other original specifications remain the same, and if exactly one light on the north side is on, which one of the following statements could be false?

A) Light 1 is off.

B) Light 2 is on.

C) Light 3 is off.

D) Light 4 is on.

E) Light 5 is on.

S8)

The Action: Lighting the parking lot—that's our job here. A picture is provided, illustrating the location of eight lights around a square lot. And what exactly are we called on to do? It's nothing fancy; we're simply to decide whether each light is on or off. Some understood this set action to be similar to that of a grouping set of selection—each light being either in or out, on or off. If that conception helped you, great! We can just as easily group this one in the "oddball" category—a spatial arrangement set with a selection element. A glance at the questions confirms that there's really only one **Key Issue** here:

1) Which lights are on, and which are off given various circumstances?

The Initial Setup: There's nothing initially to draw; your best bet is to simply work off the picture on the page. There's plenty of room on the bottom of the page to recreate the square when necessary. But it's worth deciding at this point how you're going to indicate whether a light's on or off. A simple but effective solution is to circle a number whose light is on and put an X through a number whose light is off.

The Rules:

2) No reason not to jump to the simplest and most concrete rule of the bunch: Light 8 is on. Circle it.

1) This one needs some interpretation. "At least one of any three consecutively numbered lights is off." First of all, what are consecutively numbered lights? Well, lights 1, 2, and 3, for example, or 6, 7, 8. All these rules tells us that if any consecutive trio of lights like these are all on, that's a violation. In other words, if lights 2 and 3, for example, are on, then 1 and 4 must be off. But wait

a minute! What about the trio 7, 8, 1, or 8, 1, 2? The lights in each of these trios are physically adjacent to one another in the picture, but certainly wouldn't be considered "consecutively numbered." It's a good question, but this possible slight ambiguity is taken care of by Rule 3. In the meantime, your task is to represent the rule on the page. Perhaps you came up with "NO 3 CONSECUTIVE ON" or something to that effect that will enable to you to quickly access this restriction.

3) The safest way to handle Rule 3 is to first translate it into if-then form: "IF 1 IS ON, THEN 2 IS OFF AND 7 IS OFF." Did you work through the contra positive? "IF 2 OR 7 OR BOTH ARE ON, THEN 1 IS OFF." Get both down on your page.

As just mentioned, this rule turns the possible ambiguity alluded to above into a moot point: Since 1 and 2 can never be on at the same time, and 1 and 7 can never be on at the same time, we don't have to worry about the trios 8, 1, 2 and 7, 8, 1. We don't think the test makers meant for these to qualify as "consecutively numbered" trios, but Rule 3 makes the whole issue irrelevant.

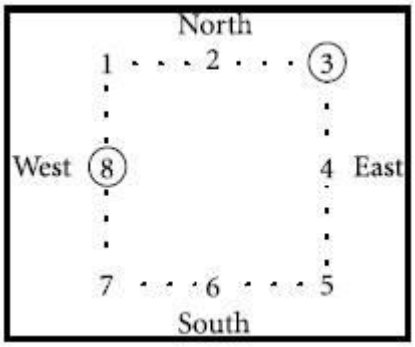
4) "AT LEAST ONE ON EACH SIDE" captures the gist, although as shorthand it's not much shorter than the original. You may have simply chosen to remember this rule, or circle it on the page. If you were unsure of the meaning, again, try out an example. Lights 5, 6, and 7, for example—the three south side lights—cannot all be off. If two of them are off, then the other one must be on. The west side, of course, automatically satisfies this rule thanks to Rule 2's illumination of light 8.

5) This one requires care as well: If a side has exactly one light on, then it has to be the middle light. Perhaps another quick example is in order to clarify this: Light 5 cannot be the only illuminated light on the south side; that would violate this rule. Again, the shorthand is a little problematic; there's just no really concise way to say this. Come up with some written way to express this, or commit it to memory, or circle it on the page. Here, we'll just say "IF EX 1 ONE ON, THEN MIDDLE." As long as you know what the rule means, you'll be okay.

6) Finally, a rule that's a little more specific than some of the previous ones—two northern lights are on, which means two out of the trio 1, 2, 3. And that, thankfully, leads us right to a big deduction. Did you see it?

Key Deductions: Hopefully upon registering the final rule you asked yourself "which two can they be?" 1 and 2? Nope—Rule 3 forbids it, which means we're talking either, 1 and 3, or 2 and 3. One of those pairs of northern lights must be on to satisfy Rule 6. In either case, **light 3 must be on**, and one of the questions, Q. 2, is answered already. Circle light 3 in your master sketch, and you're ready to roll.

The Final Visualization: You can redraw the sketch as needed, when new information arrives. But expect to be able to get some quick points simply by referring to the master sketch created around the picture on the page:



No 3 Consecutive On
If 1 is On 2 is off and 7 is off
If 2 or 7 both On 1 is off
At least one on each side
If Ex 1 On Middle

The Questions:

1. (C)

You shouldn't have much trouble here if you use the Kaplan Method for this Acceptability question. Rule 2 is the easiest to check—light 8 must be included in any "complete and accurate" list of illuminated lights, but (A) begs to differ, and must be axed. (A) also violates Rule 3 by having lights 1 and 7 on together, and (E) violates the same rule by lighting up both 1 and 2. No good. Rule 1 gets rid of (D); 6, 7, and 8 are consecutive and therefore can't all be on at the same time. (D) also violates Rule 6, a rule which is also not met by (B), which fails to light two lights on the north side. (You may have chucked (B) and (E) right off the bat by noticing that they don't reflect the big deduction above, namely that 3 is always on.) The losers are chock full of violations here. (C) remains, and gets the point for Q. 1.

2. (B)

Here's the reward for those of you who worked out the implications of Rule 6 as we did above: Light 3 must be on because of a combination of Rules 3 and 6 (see Key Deductions above).

3. (B)

If you're on top of the rules, a glance at the master sketch should be enough to answer this one. Light 1 is up north, and we know something special about the northern lights—two of them must be on. If 1 is off, then 2 and 3 are on, this has direct implication for light 4: Rule 1 insists

that under these conditions, light 4 must be off. Luckily, that's exactly what we're looking for. If 1 is off, then 4 must be off—choice (B).

4. (D)

No new info to add here, so we have a choice. We can either hold off on Q. 4 until we have more possible scenarios worked out in the final three questions (although Q. 7 doesn't help in this case because it involves a rule change), or we can just plow through the choices. If you were feeling good with the set to this point, and were having no problem using the sketch to eyeball the situations posed by the choices, then tackling this one now wouldn't be a terrible mistake. For the sake of continuity, we'll work through each choice here, but for the record, (A) can be eliminated from the correct answer to Q. 1 (which you should have noticed in either case), and (B) can be axed thanks to the arrangement worked out in Q. 6, and (E) thanks to the correct choice of Q. 5. Choice (C) falls to our big deduction (a.k.a. the answer to Q. 2), as light 3 can never be off. So the method of using other work allows us to narrow the choices all the way down to correct choice (D). Here's how it would have gone if you handled the question in the order presented:

(A) Correct choice (C) in acceptability Q. 1 offers us a valid collection of lights that are on: 2, 3, 5, 6, 8. Clearly, since this is acceptable, choice (A) is not—if light 2 is on, light 6 can be on, too.

(B) If 3 is on (and we know by now it's always on), must 2 be on? No; we can satisfy Rule 6 by turning on light 1 instead, which is fine as long as 7 is off.

(C) Light 3 is never off. A quick kill! Thanks to our big deduction.

(D) Yup: If 5 is off, then we're left with only light 3 blazing on the east side. That's a no-no, according to Rule 5—better send some juice over to light 4, pronto. So (D) must be true. If 5 is off, then 4 must be on. For the record:

(E) is out-and-out false: If southern middle light 6 is off, then 5 and 7 must be on (Rules 4 and 5). And if 7's on, then 1's off, contrary to (E).

5. (A)

You certainly may have benefited from jotting down a new sketch to incorporate Q. 5's hypothetical—circle light 5 along with 8 and 3 and you're ready to go. The first thing to notice is that light 4 must be off; otherwise Rule 1 is violated with 3, 4, and 5 on simultaneously. We're looking for a choice that could be true under these circumstances, so it's on to the choices.

(A) And we don't have to look far: It's quite possible for lights 1 and 6 to be off here, provided that 2 and 7 are on. So (A) it is. (Note how this eliminates (E) in Q. 4). On test day, you would circle (A) and move on, but for the record, here's how the others pan out:

(B) directly contradicts Rule 3—1 and 7 can never be on together.

(C) denies what we've just deduced, namely that 4 must be off any time 5 is on.

(D) is perhaps the trickiest one to eliminate, because it forces many rules into play. But ultimately it fails, too, and here's why: If both 2 and 6 are off, then 1 must be on (Rule 6), and 7 must be on (Rule 5). But firing up both 1 and 7 violates Rule 3, so (D) isn't possible.

(E) would create a consecutive 5, 6, 7 trio, not to mention a 6, 7, 8 trio, both of which are expressly forbidden by Rule 1.

6. (B)

You may or may not have created a new sketch for Q. 5, but it certainly helps here to plot out the chain of deductions. A circle around light 4 demands an X through lights 2 and 5 (lest we end up with three consecutive lights illuminated). Light 1 must therefore be the other light on up north (Rule 6). And with 1 on, 7's off, thanks to Rule 3. That leaves 6 as the only light left to illuminate the south side, and Rule 4 tells us that the south side can't go dark. So there we have it, the entire lighting design, so to speak: 1, 3, 4, 6, and 8 on; 2, 5, and 7 off. Every choice matches up except (B).

7. (E)

Uh-oh, they're shaking things up. No sooner is the "two northern lights" rule lifted than we're told that exactly one light's on up there. No problem, we know which one it is—must be the middle one, light 2, because of Rule 5. For the first time, we get to go back on our big deduction and cross out light 3 along with light 1. We need at least one light on for the east side, and with 3 off, it's down to 4 and 5. We've seen this issue before: If only the end light, light 5, were on, then Rule 5 would be violated. So at least middle light 4 must be on, and light 5 could, but need not, be on as well. The choice that could be false, therefore, is (E); maybe yes, maybe no. All of the others are definite.

Q9)

A small software firm has four offices, numbered 1, 2, 3, and 4. Each of its offices has exactly one computer and exactly one printer. Each of these eight machines was bought in either 1987, 1988, or 1989. The eight machines were bought in a manner consistent with the following conditions:

1.The computer in each office was bought either in an earlier year than or in the same year as the printer in that office.

2.The computer in office 2 and the printer in office 1 were bought in the same year.

3.The computer in office 3 and the printer in office 4 were bought in the same year.

4.The computer in office 2 and the computer in office 3 were bought in different years.

5.The computer in office 1 and the printer in office 3 were bought in 1988.

1. If the computer in office 3 was bought in an earlier year than the printer in office 3 was, then which one of the following statements could be true?

A) The computer in office 2 was bought in 1987.

B) The computer in office 2 was bought in 1988.

C) The computer in office 4 was bought in 1988.

D) The printer in office 4 was bought in 1988.

E) The printer in office 4 was bought in 1989.

2. Which one of the following statements could be true?

A) The printer in office 1 was bought in 1987.

B) The computer in office 2 was bought in 1987.

C) The computer in office 3 was bought in 1989.

D) The printer in office 4 was bought in 1988.

E) The printer in office 4 was bought in 1989.

3. If as few of the eight machines as possible were bought in 1987, then what is the exact number of machines that were bought in 1987?

A) 0

B) 1

C) 2

D) 3

E) 4

4. If the computer in office 4 was bought in 1988, then which one of the following statements must be true?

A) The printer in office 1 was bought in 1988.

B) The printer in office 1 was bought in 1989.

C) The computer in office 2 was bought in 1988.

D) The computer in office 3 was bought in 1987.

E) The printer in office 4 was bought in 1989.

5. If the computer in office 3 was bought in 1988, then which one of the following statements could be true?

A) The printer in office 1 was bought in 1988.

B) The computer in office 2 was bought in 1987.

C) The printer in office 2 was bought in 1988.

D) The computer in office 4 was bought in 1987.

E) The printer in office 4 was bought in 1989.

6. Suppose that the computer in office 2 and the computer in office 3 had been bought in the same year as each other. If all of the other conditions remained the same, then which one of the following machines could have been bought in 1989?

A) The printer in office 1

B) The computer in office 2

C) The printer in office 2

D) The computer in office 4

E) The printer in office 4

S9)

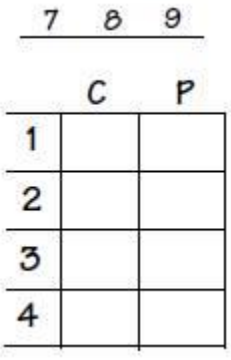
The **Key Issues** will deal with:
1.In what year was each computer and printed bought?

2.What computer or printer can, must, or cannot have been bought earlier or later than what other computer or printer?

3.What computer or printer can, must, or cannot have been bought the same year as what other computer or printer?

The Initial Setup: One of the most important choices we have to make with this set is deciding how to organize its information. As with most matching sets, you can either use a grid or lists to keep track of the info. Looking at this set, a grid might be slightly more useful than the lists. Why? Well, look at the information. When deciding how to organize a set, look for what is variable and what is definitely set. Here, the four offices are definitely set, and one computer and one printer in each are definitely set.

The only thing that is variable is the date when each machine was purchased. So we have two fixed points and one variable. That is a hint that a grid might be the best way to go. One fixed point will go along the top of the grid and one will go along the side. We'll fill the grid in with the variable aspect, in this case, the date:



Note that we shortened the dates from 1987, 1988, 1989 to just 7, 8, 9. This is a time saving trick. The dates are all identical except for the last digit, so we'll only concern ourselves with that digit.

The Rules:

1) In each office, the computer was bought either earlier than or the same year as the printer in that office. In other words, if the computer in 1 was bought in 1988, then the printer in 1 must have been bought in 1988 or 89. Build this directly into the sketch. In between each office's C and P, draw a "£."

2) Here's information that connects the machines in two different offices. The computer in office 2 and the printer in office 1 were bought the same year. Build this directly into the sketch by drawing arrows between the boxes representing these two machines.

3) More inter-office information—the computer in office 3 and the printer in office 4 were bought in the same year. Again, build this directly into your sketch by drawing arrows connecting these two squares of the grid.

4) This time we're given two machines bought in different years: the computer in office 2 and the computer in office 3. Build this information directly into the sketch by placing a "X" on the line between the corresponding squares in the grid.

5) gives the only concrete information of the bunch. The computer in office 1 and the printer in office 3 were both bought in 88. Build this right into the sketch.

Key Deductions:

This set has a ton of entities and a ton of rules. Both of these facts are hints that there's some major deducing to be done here. So let's get started.

Begin with the concrete info in Rule 5. The computer in office 1 was bought in 88. Rule 1 says that each office's computer was bought earlier than or the same year as that office's printer. So the printer in office 1 must have been bought in 88 or 89. Add this to the sketch.

Rule 2 says that the printer in office 1 was bought the same year as the computer in office 2. Since we just saw that the printer in office 1 was bought in 88 or 89, so was the computer in office 2. Add this to the sketch.

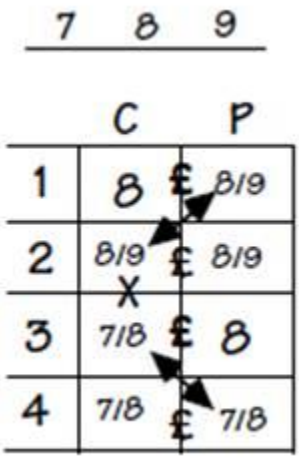
From there Rule 1 comes back into play. Since the computer in office 2 was bought in 88 or 89, the printer in office 2 must also have been bought in 88 or 89. Add this to the sketch.

Back to Rule 5. Since the printer in office 3 was bought in 88, the computer in office 3 (bought earlier than or the same year as the printer—Rule 1) must have been bought in 87 or 88. Add this to the sketch.

Since the computer in office 3 was bought in 87 or 88, we can combine this with Rule 3 to deduce that the printer in office 4 must also have been bought in 87 or 88. Add this as well.

Finally, since the printer in office 4 was bought in 87 or 88, Rule 1 means that the computer in office 4 must also have been bought in 87 or 88. Add this to the sketch.

The Final Visualization: Here's that ton of information added to the sketch:



The Questions:

1. (B)

The printer in office 3 was bought in 88 (Rule 5). The only way for the computer in 3 to be bought earlier is if it was bought in 87. Rule 3 says that the computer in 3 and the printer in 4 were bought the same year, so the printer in office 4 must have also been bought in 87. Eliminate (D) and (E). Each office's computer was bought in the same year as or earlier than the printer in that office (Rule 1), so since the printer in 4 was bought in 87, the only year that the computer in 4 could have been bought is also 87. Eliminate (C). (A) and (B) are all that's left. We deduced in the setup that the computer in office 2 must have been bought in 88 or 89. Cross off (A), which leaves (B), the answer.

2. (D)

Here's a could be true question with no new information. Normally this type of question might take a lot of time to answer. Here, however, with all the good work we did up front, we should be able to eliminate the wrong answer choices quickly. Just compare each choice against our master sketch.

(A) No, the printer in 1 must have been bought in 88 or 89.

(B) No, the computer in 2 must have been bought in 88 or 89.

(C) No, the computer in 3 must have been bought in 87 or 88.

(D) Yes, the printer in 4 could have been bought in 88 (or 87). (D) could be true and is the answer.

(E) No, the printer in 4 must have been bought in 87 or 88.

3. (A)

We want as few machines bought in 87 as possible. Check our master sketch. There are only three machines that could have been bought in 87. The computer in 3, the computer in 4, and the printer in 4 all were bought in 87 or 88. The stem asks for the fewest number of machines bought in 87, so could all three of these have been bought in 88? Sure. Rule 3 says that the computer in 3 was bought the same year as the printer in 4, so both of these could have been bought in 88. Rule 1 says that the computer in each office must have been bought the same year as or before the printer in that office, so the computer in 4 could have been bought in 88 as well. All of the other machines couldn't have been purchased in 87 (see Key Deductions), so it turns out that we need not have any 87 machines. In other words, the fewest possible number of machines that were bought in 87 is 0, choice (A).

4. (B)

Remember that the printer in each office was bought either the same year as or later than the computer in that office (Rule 1). If the computer in 4 was bought in 88, the printer in 4 must also have been bought in 88 (we ruled out 89 as a possibility for the printer in 4 under Key Deductions). We've just made a new deduction, so stop and see if that is an answer choice. It's not (although we can eliminate (E) which has the printer in 4 bought in 89), so keep looking.

Rule 3 says that the printer in 4 and the computer in 3 were bought the same year, so the computer in 3 must have been bought in 88. Is that a choice? No, but we can cross off (D) which has it bought in 87.

Rule 4 says that the computer in 2 can't have been purchased the same year as the computer in 3, so here the computer in 2 must have been bought in 89 (again check Key Deductions to see how 87 was eliminated for the computer in 2). Is "computer in 2 bought in 89" an answer choice? No, but we now know that (C) must be false, so cross it off.

Moving along, Rule 2 says that the computer in 2 and the printer in 1 were bought the same year, so the printer in 1 was bought in 89, and finally that's the answer, choice (B).

5. (D)

If the computer in 3 was bought in 88, Rule 3 dictates that the printer in 4 was bought in 88 as well. Rule 4 (computer in 3 and computer in 2 bought in different years) means that the computer in 2 was bought in 89. From there, Rule 2 (computer in 2 and printer in 1 bought in the same year) insists that the printer in 1 was bought in 89. Also, since the computer in 2 was bought in 89, Rule 1 means that the printer in 2 was also bought in 89. The stem asks for what could be true. The only machine that still has more than one option is the computer in 4 which could have been bought in 87 or 88. Look first at any choice that includes the computer in 4. Sure enough, choice (D), the computer in 4 being bought in 87, is the only possible choice.

6. (C)

And here's the one question that absolutely must have a new sketch. Now we're to consider what happens when the opposite of Rule 4 is true. The computer in 2 and the computer in 3 now are bought in the same year. Take a moment and create a new master sketch including this new information. Look at the original sketch and note what information still holds. The deduction that the printer in 1 was bought in 88 or 89 still holds, as does Rule 2 which means that the computer in 2 was also bought in 88 or 89. The stem says that the computers in 2 and 3 were bought in the same year, so the computer in 3 must also have been bought in 88 or 89. But wait. The printer in 3 was bought in 88, so Rule 1 means that the computer in 3 can only have been bought in 88. This also means that the computer in 2 was also bought in 88 (thanks to the new information in the stem). From there, Rule 2 means that the printer in 1 was bought in 88 as

well. Since the computer in 3 was bought in 88, Rule 3 forces the printer in 4 to have been bought in 88. The only machines that possibly weren't bought in 88 are the printer in 2 (88 or 89) and the computer in 4 (87 or 88). The printer in 2 is the only machine that could have been bought in 89, choice (C).

Q10)

A railway company has exactly three lines: line 1, line 2, and line 3. The company prints three sets of tickets for January and three sets of tickets for February: one set for each of its lines for each of the two months. The company's tickets are printed in a manner consistent with the following conditions:

1.Each of the six sets of tickets is exactly one of the following colors: green, purple, red, yellow.

2.For each line, the January tickets are a different color than the February tickets.

3.For each month, tickets for different lines are in different colors.

4.Exactly one set of January tickets is red.

5.For line 3, either the January tickets or the February tickets, but not both, are green.

6.The January tickets for line 2 are purple.

7.No February tickets are purple.

1. If the line 3 tickets for January are red, then which one of the following statements must be true?

A) The line 1 tickets for January are green.

B) The line 1 tickets for January are yellow.

C) The line 1 tickets for February are red.

D) The line 2 tickets for February are yellow.

E) The line 3 tickets for February are green.

2. If one set of the line 2 tickets is green, then which one of the following statements must be true?

A) The line 1 tickets for January are red.

B) The line 3 tickets for January are red.

C) The line 1 tickets for February are red.

D) The line 3 tickets for February are green.

E) The line 3 tickets for February are yellow.

3. Which one of the following statements could be true?

A) No January ticket is green.

B) No February ticket is green.

C) Only line 2 tickets are red.

D) One set of January tickets is green and one set of January tickets is yellow.

E) The line 2 tickets for January are the same color as the line 1 tickets for February.

4. Which one of the following statements could be true?

A) Both the line 1 tickets for January and the line 2 tickets for February are green.

B) Both the line 1 tickets for January and the line 2 tickets for February are yellow.

C) Both the line 1 tickets for January and the line 3 tickets for February are yellow.

D) The line 1 tickets for January are green, and the line 3 tickets for February are red.

E) The line 3 tickets for January are yellow, and the line 1 tickets for February are red.

5. If the line 3 tickets for February are yellow, then each of the following statements must be true EXCEPT:

A) One set of January tickets is green.

B) One set of line 1 tickets is red.

C) One set of line 2 tickets is red.

D) The tickets in two of the six sets are red.

E) The tickets in two of the six sets are yellow.

6. Suppose that none of the ticket sets are purple. If all of the other conditions remain the same, then which one of the following statements could be true?

A) None of the January tickets are green.

B) None of the February tickets are green.

C) None of the line 2 tickets are green.

D) No line 1 or line 2 tickets are yellow.

E) No line 2 or line 3 tickets are red.

S11)

The Action:

After reading the opening paragraph and rules, we see that this set requires that we match up the January and February tickets for three railway lines with the color of each—a matching set. The action doesn't become fully apparent until Rule 1 which introduces the concept of the "color" of each set of tickets. Some students were thrown by the notion of "sets" of tickets in the opening paragraph. They were worried that this implied a whole lot of tickets. If we read carefully, though, we know that that's not a problem. There is one "set" of tickets per month, as simple as that.

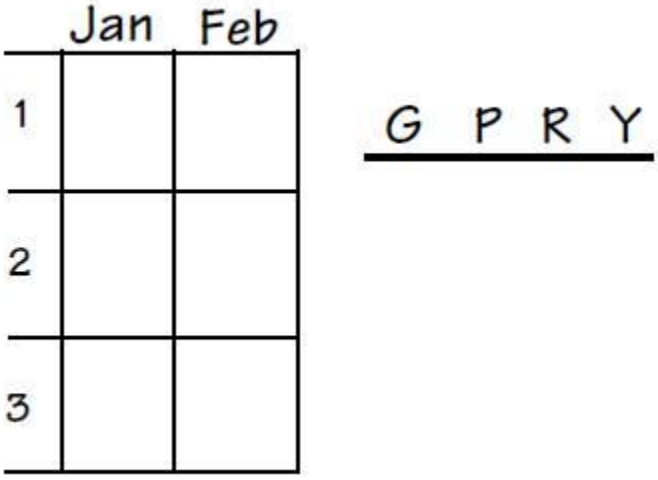
The Key Issues are:

1) What color is each of the six sets of tickets?

2) What sets of tickets can, must, or cannot be the same color as what other sets of tickets?

The Initial Setup:

As with Set 2 and other matching sets, a grid or lists both work well. In Set 2, we talked about identifying the variable and fixed aspects when using a grid. What are the variable and fixed points in this set? The two months, January and February, are fixed. They don't change. The three railway lines are also fixed, not going anywhere. What can change from line to line and month to month is the color of the sets of tickets. This is the variable aspect. So when creating our grid, the two fixed points (months and lines) will go along the top and side, and we will fill in the grid squares with the variable (color):



The Rules:

1) gives the information that we've already included in the sketch above, namely that each set of tickets will be green, purple, red, or yellow.

2) Let's take our time and make sure that we fully understand this rule. The January set of tickets of each line (1, 2, and 3) will be a different color from the February set of tickets of that line. Let's do a "what if" to make sure we have a handle on this rule. If the Jan 1 (January line 1) set of tickets is red, then the Feb 1 (February line 1) set of tickets cannot be red. Draw a big "X" between each line's January and February squares in our grid.

3) Again, we need to take our time when decoding this rule. Within each month, the set of tickets of each line must be a different color than the set of tickets of the other two lines. Let's do another "what if." If the Jan 1 (January line 1) set of tickets is red, then the Jan 2 (January line 2) set of tickets can't be red. Between each month's 1 and 2 squares draw a big "X," and between each month's 2 and 3 squares draw a big "X."

4) Of the three sets of January tickets, exactly one must be red. Build this directly into our grid. Write "ex. 1 R" at the bottom of the January column.

5) Once more, take care when translating Rule 5. This rule is quite similar to Rule 4. In Rule 4, exactly one of the sets of January tickets (either line 1, 2, or 3) was red. Here, exactly one of the sets of line 3 tickets (either January or February) must be green. At the end of the line 3 row, write "ex. 1 G."

6) Here is a nice, concrete piece of information. The Jan 2 set of tickets is purple. Build this right into the master sketch.

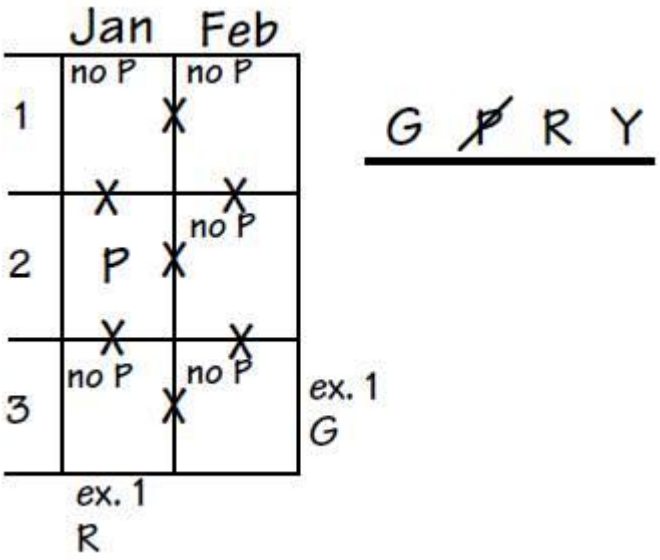
7) Finally, Rule 7 states that none of the sets of February tickets are purple. Write "NO P" in each of the February squares in our grid.

Key Deductions:

Rule 6 said that Jan 2 is purple. Rule 3 said that within each month, the sets of tickets for the three lines must be of different colors. So we can deduce that Jan 1 and Jan 3 can't be purple. Rule 7 already said that NO sets of February tickets are purple, so we definitely know that Jan 2 is the only set of tickets that is purple.

We can deduce something else from Rule 7 which says that none of the February tickets are purple. Rule 3 says that each month's set of tickets must be different colors. Since none of the February tickets are purple, that means that all three of the other colors must be used exactly once for the three sets of February tickets, one green, one red, and one yellow.

The Final Visualization: Here's our master sketch:



The Questions:

1. (E)

Rule 5 says that one of the sets of line 3 tickets (either January or February) must be green. The stem says that Jan 3 is red, so Feb 3 must be green, choice (E).

2. (A)

What must be true if one of the sets of line 2 tickets is green? Jan 2 is purple (Rule 6), so the only 2 line left to be green is Feb 2. Is Feb 2 green an answer choice? Nope, keep going. Rule 3 says that no two sets of tickets in the same month can be the same color. Since Feb 2 is green, Feb 3 can't be green. Rule 5 (either Jan 3 or Feb 3 must be green) comes into effect again and Jan 3 must be green. Is Jan 3 green a choice? No, so keep looking. Rule 4 says that one of the January tickets is red, and since Jan 2 is purple and Jan 3 is green, Jan 1 must be red. And that's choice (A) and the answer.

3. (A)

There's not much to do with this "could be true" question with no new information but try out each choice.

(A) Could none of the January tickets be green? If none are, we're left with purple, red, and yellow as the colors of the three sets of January tickets. Jan 2 is purple (Rule 6). Jan 1 could be red, and as long as Feb 3 is green (to fulfill Rule 5), Jan 3 could be yellow. Answer choice (A) could be true and is the answer. Quickly here is why the remaining choices can't be true:

(B) Since none of the sets of February tickets are purple (Rule 7), exactly one set of the February tickets MUST be green.

(C) The Jan 2 tickets are purple (Rule 6) and one set of the January tickets is red (Rule 4), so either Jan 1 or Jan 3 must be red.

(D) Jan 2 is purple, so if one set of January tickets was green and the other set of January tickets was yellow, there wouldn't be any set of January tickets that was red which would violate Rule 4.

(E) Jan 2 is purple, and Rule 7 says that absolutely none of the sets of February tickets can be purple.

4. (B)

Another "could be true" question with no new information (but with positive choices, at least—see Bullet Point to Q. 21 above). Again, there is not much to do but to try out each of the choices.

(A) If Jan 1 and Feb 2 are both green, Rule 3 means that none of the remaining sets of tickets can be green. But Rule 5 insists that either Jan 3 or Feb 3 be green. (A) can't be true and it isn't the answer.

(B) Can Jan 1 and Feb 2 both be yellow? Sure! Jan 1—Y, Jan 2—P, Jan 3—R; Feb 1—R, Feb 2—Y, Feb 3—G. (B) can be true and is the answer. Quickly, here is why the remaining choices can't be true:

(C) If Jan 1 is yellow, the only set of January tickets left to be red (as Rule 4 requires) is Jan 3. Rule 5 then requires that Feb 3 be green, not yellow.

(D) If Jan 1 is green, Rule 3 means that none of the other sets of January tickets are green, so Rule 5 again requires that Feb 3 be green, not red.

(E) If Jan 3 is yellow, the only set of January tickets left to be red (as Rule 4 requires) is Jan 1. Since Jan 1 is red, Rule 2 doesn't permit Feb 1 to also be red.

5. (E)

If Feb 3 is yellow, Rule 5 forces Jan 3 to be green. Since Jan 3 is green, Jan 1 is left to be red (as Rule 4 requires). Since none of the sets of February tickets are purple, Feb 1 and Feb 2 must be green and red. Since Jan 1 is red, Rule 2 means that Feb 1 can't be red and must be green. Feb 2 is left to be red. The entire setup is fixed: Jan 1—R, Jan 2—P, Jan 3—G; Feb 1— G, Feb 2—R, Feb 3—Y. The only choice that doesn't agree with this setup is (E)—only one of the sets of tickets is yellow.

6. (C)

For this question, none of the sets of tickets are purple. This doesn't affect the sets of February tickets since none were purple anyway (Rule 7). But since none of the sets of January tickets are

purple (sorry Jan 2, you're not purple anymore), one will be green, one red, and one yellow. These three colors are the only options for both months' tickets.

Now check the choices looking for the one that can be true. (A) and (B) We can kill both of these choices immediately. Since none of the sets of tickets are purple, all of the other colors must be used exactly once per month. Exactly one set of January tickets and exactly one set of February tickets must be green. (C) Could neither of the sets of line 2 tickets be green? Sure, here is the complete setup: Jan 1—Y, Jan 2—R, Jan 3—G; Feb 1—G, Feb 2—Y, Feb 3—R. (C) could be true and is the answer. (D) If none of the sets of line 1 or line 2 tickets are yellow, then both Jan 3 and Feb 3 must be yellow which Rule 2 forbids. (E) If none of the sets of line 2 or line 3 tickets are red, then both Jan 1 and Feb 1 must be red which Rule 2 forbids.

Q11)

A committee ranks five towns—Pune, Qulloo, Rampur, Saharanpur, Tilak Nagar—from first (best) to fifth (worst) on each of three criteria: climate, location, friendliness.

1. For each of the three criteria, none of the five towns receives the same ranking as any other town does.

2. In climate, Tilak Nagar is ranked third, and Saharanpur fourth.

3. In location, Qulloo is ranked second, Rampur third, Pune fourth.

4. In friendliness, Tilak Nagar's ranking is better than Pune's, Qulloo is ranked fourth and Saharanpur fifth.

5. Rampur receives a better ranking in climate than in friendliness.

6. Qulloo's three ranking are all different from each other.

1. Which of the following is a complete and accurate list of the rankings any one of which could be the ranking on climate given to Rampur?

A) first

B) first, second

C) first, fifth

D) second, fifth

E) first, second, fifth

2. Which of the following is a town that CANNOT be ranked fifth on any one of the three criteria?

A) Pune

B) Qulloo

C) Rampur

D) Saharanpur

E) Tilak Nagar

3. Which of the following could be true?

A) Pune is ranked first in both climate and friendliness.

B) Qulloo is ranked second in both climate and location.

C) Rampur is ranked first in climate and third in both location and friendliness.

D) Saharanpur is ranked fifth in friendliness and fourth in both climate and location.

E) Tilak Nagar is ranked third in both climate and friendliness.

4. If Qulloo is ranked first in climate, then it must be true that

A) Pune is ranked second in climate

B) Pune is ranked third in friendliness

C) Rampur is ranked second in friendliness

D) Rampur is ranked third in friendliness

E) Tilak Nagar is ranked fifth in location

5. If Pune is ranked second in climate, then which one of the following can be true?

A) Pune is ranked second in friendliness.

B) Qulloo is ranked first in climate.

C) Rampur is ranked first in friendliness.

D) Rampur is ranked fifth in climate.

E) Tilak Nagar is ranked third in friendliness.

6. If Tilak Nagar is ranked first in location and Rampur is ranked second in friendliness, then it is possible to deduce with certainty all three rankings for exactly how many of the towns?

A) One

B) Two

C) Three

D) Four

E) Five

7. Which one of the following statements CANNOT be true?

A) Pune is ranked first in climate.

B) Qulloo is ranked fifth in climate.

C) Rampur is ranked third in friendliness.

D) Saharanpur is ranked first in location.

E) Tilak Nagar is ranked second in friendliness.

S11)

The Action: Something of a rarity—a triple sequencing set. But that doesn't necessarily make it problematic. Check how much data we're given. We're ranking five towns—Pune, Qulloo, Rampur, Saharanpur, and Tilak Nagar (abbreviated as P, Q, R, S, T)—from 1 to 5, best to worst, in three different categories—climate, location, and friendliness. Our job will be to put the towns in order based on the rankings they receive in each of the three categories. The **Key Issues** are:
1) How are the towns ranked in each category?

2) How are the towns ranked in each category based on their rankings in the other categories?

3) What town is ranked higher and lower than which other towns in each category?

The Initial Setup: As with our previous ranking set—Set 1—working vertically seems to make the most sense. It's very intuitive and easy to handle:

	CLI	LOC	FRIEND	P Q R S T
1				
2				
3				
4				
5				

Remember that each town has a rank in each category.

The Rules:

1) The first rule is simply a loophole closer. There are no ties is what it's saying.

2) In the climate category, enter T 3rd and S 4th. Simple.

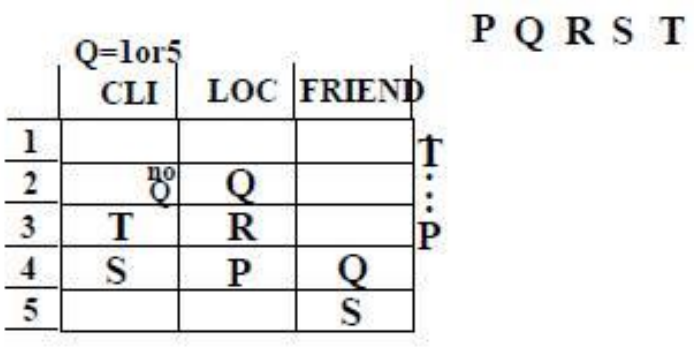
3) Here three of the five location rankings are given to us: Q 2nd, R 3rd, P 4th.

4) As in question stems with more than one piece of information, start with the concrete and then go to the abstract. Under friendliness, rank Q 4th and S 5th. We're also told that T is better than P in this category. Jot down a physical reminder nearby that T is ranked higher than P in friendliness (see figure, below).

5) R is ranked higher in climate than friendliness. Just make a note of this rule or circle it for now. We'll come back to it under **Key Deductions.**

6) Q's rankings are already set for location and friendliness, so we can deduce quite a lot from this requirement that Q's rankings all be different. Since Q takes 2nd in one category and 4th in another, Q's climate ranking—the one remaining—cannot be 2nd or 4th. And the 3rd slot under climate is already taken. Therefore, Q must be ranked 1st or 5th in climate. Make a note of that.

Here's what we've got so far:



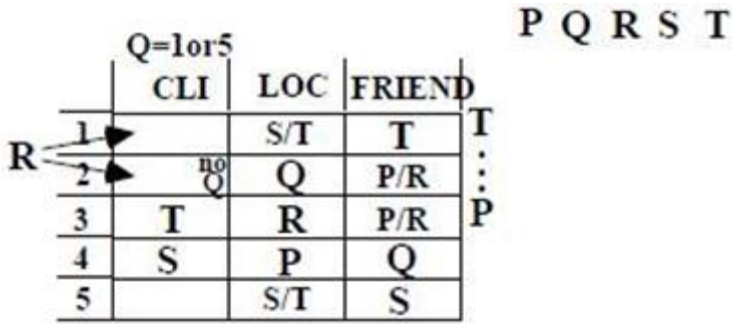
Seven out of 15 slots filled. But we can do more:

Key Deductions: Back to Rule 5 which said that R is ranked higher in climate than in friendliness. Since T and S are 3rd and 4th in climate, respectively, R must be ranked either 1st or 2nd in climate. (Look at it this way: If R were 5th in climate, there would be no way for its climate ranking to be better than its friendliness ranking.) Make a note of that, too. Meanwhile, note that R will have to be ranked either 2nd or 3rd in friendliness (since the 3rd and 4th slots are taken, and a ranking of #1 for R would violate Rule 5).

And yet we can go further. Under location, note that slots 1 and 5 are available to S and T in either order. Jot that down in the sketch.

Finally, we're rewarded if we combine Rules 4 and 5. If T's friendliness ranking is to be higher than P's, and if R can only be ranked 2nd or 3rd in friendliness, then there are no two ways about it: T will have to be ranked 1st in friendliness. (Nothing else is left!) It all looks like this:

The Final Visualization: An extremely helpful sketch:



Note that the most ambiguity exists under climate, where P, Q, and R are yet to be placed—yet we know the limitations.

The Questions:

1. (B)

The only climate rankings available for R are 1st and 2nd, choice (B).

2. (C)

R must be either 1st or 2nd in climate, 3rd in location, and either 2nd or 3rd in friendliness. So R can't be ranked 5th in any of the three criteria, answer choice (C). All of the others are open to at least one 5th place ranking in at least one category.

3. (C)

Here we simply need to test each choice.

(A) T must be ranked 1st in friendliness, so P can't be 1st there.

(B) violates Rule 6—Q can't have identical rankings in any two categories.

(C) R can be ranked 1st in climate with P 2nd and Q 5th, and we know that R is for sure ranked 3rd in location (Rule 3). Meanwhile, R can be ranked 3rd in friendliness with T 1st and P 2nd. (C) is possible and correct. On Test Day, you'd go on to the next question, but for the record:

(D) S can't be ranked 4th in location, because P was assigned that spot by Rule 3.

(E) We deduced above that T must be ranked 1st in friendliness, not 3rd.

4. (D)

We've seen that R must be ranked either 1st or 2nd in climate, so if as the stem demands, Q is ranked 1st in climate, then R is 2nd. But scanning the choices, we see that's not a choice, so keep going. Since R is 2nd in climate, Rule 5 demands that R be ranked 3rd in friendliness, and that's choice (D). The long road would be to eliminate the wrong choices:

(A) With Q 1st, R, not P, must be ranked 2nd in climate.

(B) P can't be ranked 3rd in friendliness, since we deduced that R is.

(C) If R is ranked 3rd in friendliness, then it's not ranked 2nd there.

(E) T could be ranked 5th in location, but it needn't be. T could be ranked 1st with S 5th.

5. (A)

The stem info (that P is 2nd in climate) completes the climate rankings: R must take 1st and Q is left in 5th, all of which eliminates choices (B) and (D) as statements that can be true. With R 1st in climate, R can be ranked either 2nd or 3rd in friendliness (Rule 5), which means that P can either be ranked 2nd or 3rd in friendliness as well. Choice (A) is possible, and thus the correct answer. T is still ranked 1st in friendliness, so (C) and (E) are out.

6. (E)

Here's a question where—because you're given two concrete pieces of information—you might want to create a unique sketch, with T 1st in location (leaving S 5th in that category), and R 2nd in friendliness (leaving P 3rd in that category). A new sketch quickly reveals that both of those categories are complete.

What's up with climate? Since R is 2nd in friendliness, R must be ranked 1st in climate lest we violate Rule 5. Q, we had deduced, is always either 1st or 5th in climate, so now Q is 5th. That leaves P ranked 2nd. Every town's ranking is determined for each of the three criteria—choice (E).

7. (E)

Several times so far, we've used the deduction we made upfront that T must be ranked 1st in friendliness. It's useful again now: (E), which attempts to place T 2nd, can't be true, and is the answer. All the other choices are things we've seen throughout the set; all of them could be true.

Q12)

Planes 1, 2, 3, and 4—and no others—are available to fly in an air show.
1. Pilots Anna, Bob, and Cindy are all aboard planes that are flying in the show and they are the only qualified pilots in the show.

2. Copilots Dave, Ed, and Fran are all aboard planes that are flying in the show and they are the only qualified copilots in the show.

3. No plane flies in the show without a qualified pilot aboard.

4. No one but qualified pilots and qualified copilots flies in the show.

5. Anna will only fly in either plane 1 or plane 4.

6. Dave will only fly in either plane 2 or plane 3.

1. If Anna flies in plane 4 and Dave flies in plane 2, which one of the following must be true?

A) Cindy flies in either plane 1 or plane 3.

B) If Cindy flies in plane 3, Bob flies in plane 2.

C) Bob and one other person fly in plane l.

D) If Bob is aboard plane 4, Cindy flies in plane 3.

E) If Cindy is in plane 2, Bob flies in plane 3.

2. If Bob and Anna fly on the same plane, which one of the following must be true?

A) Cindy flies with Dave and Ed.

B) Cindy flies with Ed.

C) Dave flies with Cindy.

D) Dave flies with Cindy, Ed, and Fran.

E) Fran flies with Ed.

3. If Cindy and Fran are the only people in one of the planes, which one of the following must be true?

A) Bob flies with Anna.

B) Dave flies with Ed.

C) Dave and Ed fly with Bob.

D) Dave flies with Bob.

E) Ed flies with Anna.

4. If plane 1 is used, its crew could consist of

A) Anna, Bob, Cindy, Fran

B) Anna, Bob, Ed, Fran

C) Bob, Cindy, Ed, Fran

D) Bob, Cindy, Dave, Ed

E) Bob, Dave, Ed, Fran

5. If as many of the pilots and copilots as possible fly in plane 4, that group will consist of

A) exactly two people

B) exactly three people

C) exactly four people

D) exactly five people

E) three pilots and two copilots

S12)

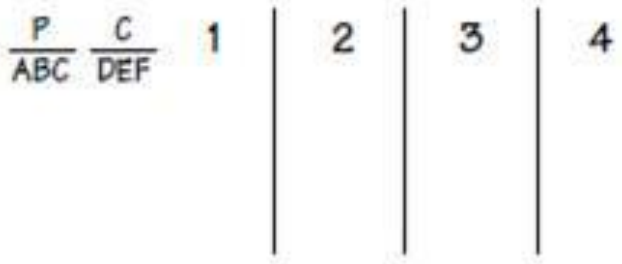
The Action:

In this grouping game, we're asked to distribute six entities—three pilots and three copilots—among four planes—planes 1, 2, 3, and 4. While the numbering of the planes may suggest sequencing, your overview should have dispelled that misconception right away: There's no mention that these planes are "in a row;" they're not numbered "from left to right;" and none of the rules say anything about people being "in adjacent planes," or anything like that. Our job is simply to distribute the pilots and copilots into the planes. The main Key Issue, therefore, is a grouping concern:

1) Who is in what plane? And by extension: Which pilots and copilots can, must, or cannot fly in the same plane as which other pilots and copilots?

The Initial Setup:

Keep this setup simple; four circles or columns, numbered 1 to 4, can represent the planes. Then list the pilots and copilots off to the side:



The Rules:

1) and 2) You most likely already used these rules to get a handle on the entities. However, some test takers overlooked a key element of these rules—that the pilots and copilots “are all aboard planes that are flying in the show.” This means that everyone flies. Selecting who flies isn’t an issue; they’re all up in the air. The only question here is which plane each person is in. As for

listing the entities, some find it helpful to use capital letters for the pilots and lowercase letters for the copilots, some don't. Do what's easiest for you.

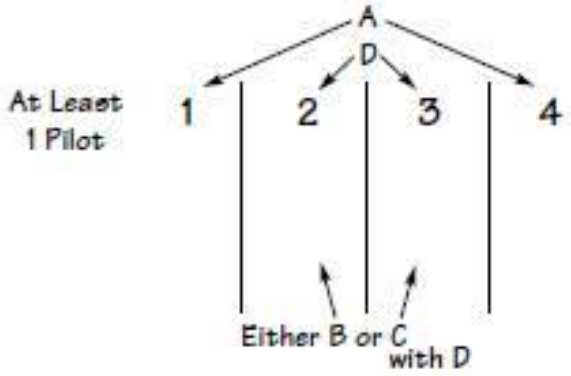
3) Translation: Every plane that's flying needs one pilot—at least one. This rule says nothing about copilots, nor does it imply that only one pilot may fly in a particular plane. So far, it's quite possible that exactly one pilot flies a plane without a copilot, just as it's possible for more than one pilot to fly in one plane.

4) This rule is basically a loophole closer to ensure that no one from the audience or anywhere else rushes out and pilots a plane.

5) Anna will only fly in plane 1 or plane 4. Since everyone is flying in the show, we know that one of these planes must be used. To build this into our master sketch, write "A" with arrows pointing to planes 1 and 4.

6) Dave only flies in plane 2 or plane 3— "D" with arrows to 2 and 3 takes care of this. Key Deductions: Not much in the way of deductions, but there are a few issues that are worth working out before hitting the questions. First, the numbers: No plane flies without a qualified pilot aboard. But we have only three qualified pilots, which means that a maximum of three of the four planes are flying; at least one is going to remain empty and on the ground. Also, since a plane can't fly without a pilot (Rule 3), we know that one of the pilots must join Dave (a copilot) in either plane 2 or plane 3. It can't be Anna, since she's in plane 1 or 4, so Dave must fly with either B or C, in plane 2 or 3.

The Final Visualization: Here's what we're armed with to reel in these five questions:



The Questions:

1. (B)

Anna is in plane 4 and Dave is in plane 2. Well, we knew we'd have to focus on the Dave situation, and here it is right off the bat. Since Bob or Cindy (or both) need to accompany Dave, Cindy in 3 forces Bob into 2 with Dave.

(A) No reason why Cindy couldn't fly in plane 2 with Dave.

(C) No, Bob could fly in plane 2.

(D) No; if Bob flies in plane 4, Cindy would have to fly with Dave in plane 2.

(E) If Cindy flies in plane 2 with Dave, then Bob could fly in any of the planes.

2. (C)

Anna can't fly with Dave (Rules 5 and 6). If Bob joins Anna, he can't fly with Dave. So Cindy would have to fly with copilot Dave, choice (C). Choices (A), (B), (D), and (E) all could be true, but none of them must be true.

3. (D)

Cindy and Fran fly alone, Anna never flies with Dave, so Bob must fly with Dave, choice (D). (A) is dead wrong, and (B), (C), and (E) are merely possible.

4. (B)

This is very much like a standard acceptability question. (D) and (E) bite the dust thanks to Rule 6. Dave can't fly in 1 but must have a pilot with him in plane 2 or 3, so any prospective plane-1 crew that included all three pilots would be impossible. That eliminates Choice (A). (C) leaves only Anna to join Dave which isn't possible. That leaves us with (B)—Anna, Bob, Ed, and Fran—a perfectly acceptable crew for plane 1, with Cindy and Dave in plane 2 or 3.

5. (C)

Dave and one pilot can't be in 4, but everybody else can be. That's the maximum, choice (C).

Q13)

Each of five illnesses—J, K, L, M, and N—is characterized by at least one of the following three symptoms: fever, headache, and sneezing. None of the illnesses has any symptom that is not one of these three.

1.Illness J is characterized by headache and sneezing.

2.Illnesses J and K have no symptoms in common.

3.Illnesses J and L have at least one symptom in common.

4.Illness L has a greater number of symptoms than illness K.

5.Illnesses L and N have no symptoms in common.

6.Illness M has more symptoms than illness J.

1. Which one of the following statements must be false?

A) Illness J has exactly two symptoms.

B) Illness K has exactly one symptom.

C) Illness L has exactly two symptoms.

D) Illness M has exactly three symptoms.

E) Illness N has exactly two symptoms.

2. In which one of the following pairs could the first member of the pair be characterized by exactly the same number and types of symptoms as the second member of the pair?

A) J and N

B) K and L

C) K and N

D) L and M

E) M and N

3. If illness L is characterized by a combination of symptoms different from any of the other illnesses, then which one of the following statements must be true?

A) Fever is a symptom of illness L.

B) Sneezing is a symptom of illness L.

C) Headache is a symptom of illness L.

D) Illnesses K and N are characterized by exactly the same symptoms.

E) Illnesses M and N are characterized by exactly the same symptoms.

4. The illnesses in which one of the following pairs must have exactly one symptom in common?

A) J and L

B) J and M

C) J and N

D) K and L

E) M and N

5. If Walter has exactly two of the three symptoms, then he cannot have all of the symptoms of

A) both illness J and illness L

B) both illness J and illness N

C) both illness K and illness L

D) both illness K and illness N

E) both illness L and illness N

S13)

The Action:
In this one we're asked to match five illnesses—J, K, L, M, and N—with their symptoms: a matching set. The Key Issues deal with the typical matching concerns:

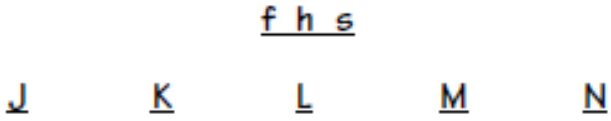
1) What symptoms does each illness have?

2) What illnesses can, must, or cannot have the same symptoms as what other illnesses?

The Initial Setup:

Either a grid or a list works well with a matching set. You could use a 5 x 3 grid with the illnesses—J, K, L, M, and N—across the top, and the symptom ms—f, h, and s—along the side. This way you'd put a "√" when you match a symptom with an illness and an "X" when you know that an illness definitely doesn't have that symptom.

However, if you were keeping track of illnesses' symptoms in real life, you'd probably just make a list of the illnesses across the page and be ready to fill in the symptoms under each, like so:



The Rules:

1) Very concrete—put an "h" for headache and an "s" for sneezing under the J.

2) Think first, don't just write "J ≠ K." Each illness has at least one symptom, and Rule 1 just said that J has headache and sneezing; therefore K must have only fever. Put an "f" for fever and "no h" and "no s" under K. Also put "no f" under J.

3) L will have at least one (or both) of headache and sneezing. Let's write "at least 1 same" and draw arrows between J and L.

4) Again, think before you draw. You know that K has exactly one symptom, so L must have two or three. Write “2 or 3” over L.

5) L and N don’t have any symptoms in common. Write an “≠” with arrows pointing to L and N to serve as a reminder.

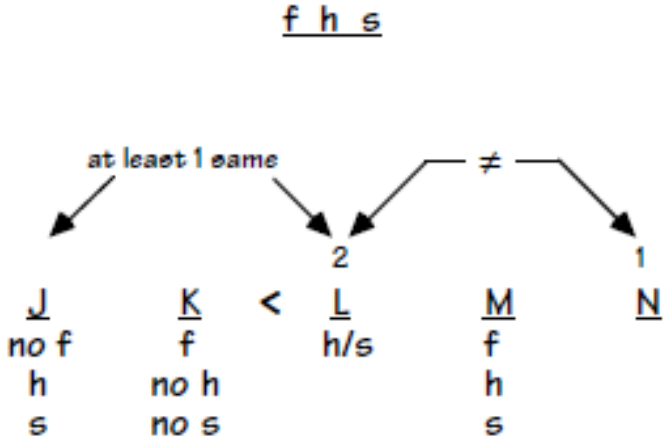
6) Don’t just write “M > J.” Think through the rule first. What did Rule 1 say about J? J has two symptoms, so M must have all three. Write “f, h, s” under M.

Key Deductions:

We made a bunch of deductions along the way as we went through the rules above. But there’s even another deduction to be made from combining Rules 4 and 5 with what we already deduced about illness L: L has either two or three symptoms (Rule 4), but N has at least one symptom, so L can’t have all three symptoms (there would be none left for N—Rule 5). L, therefore, must have exactly two symptoms, and N must have exactly one symptom. Go back and change the “2 or 3” to just “2” over the L and write “1” over N.

Three of the five illnesses (J, K, and M) are entirely filled, and you know the exact number of symptoms for the other two illnesses, L and N. This is a ton of information and should lead to some quick and easy points.

The Final Visualization: So, here’s what we have heading into the questions (and it’s quite a lot):



The Questions:

1. (E)

This question is a ten-seconder after all of our work with the setup. Scan the choices against what we’ve deduced. N has one symptom, not two as choice (E) has it.

2. (C)

Once you've decoded this question stem, it's just a matter of checking for the pair who could share exactly the same symptoms; that is, the same number and type. M is the only illness with all three symptoms so axe any choice that includes M, answer choices (D) and (E). We also know the numbers of symptoms for each of the illnesses; the only pairs that can possibly have the same number of symptoms are J and L (they each have two symptoms) or K and N (they each have one). J and L isn't a choice, but K and N is, answer choice (C).

3. (A)

Make abstract information concrete. L's symptoms aren't exactly the same as any other illness. L has two symptoms and so does J, but for this question they can't share both of them. J has headache and sneezing. Rule 3 tells us that L shares at least (and in this question, exactly) one with J. So L's other symptom must be fever, choice (A).

4. (E)

Read carefully here. You may have marked (A) thinking that it was a direct restatement of Rule 3. But the question stem says "exactly" one symptom; Rule 3 states that J and L share at least one symptom, which means they can share two. Knowing the number of symptoms in each illness helps enormously here. It may have occurred to you that M, with all three symptoms, and either K or N, with exactly one symptom, would have to share exactly one symptom. Scanning the choices we see that M and N is the pair the test makers chose in (E).

If you didn't see this, you could have answered this one fairly quickly by simply checking the choices:

(A) See discussion above.

(B) No, J with two symptoms must share exactly two with M.

(C) and (D) Both of these pairs of illnesses could share one symptom, but can also share none.

5. (E)

What combination of illnesses would force Harold to have all three symptoms? With all the deductions up front, checking each choice is a viable option, but the active way, when available, is always better. If two illnesses have all three symptoms between them, then Walter, who has only two symptoms, cannot have all the symptoms of both. J and K have all the symptoms between them, since J's symptoms are headache and sneezing, and K's symptom is fever. Unfortunately, J and K isn't a choice. However, L and N also must have all three symptoms between them, since L has two symptoms, N has one, and they have none in common. So Walter, suffering from exactly two symptoms, cannot have all the symptoms of L and N, Answer choice (E). On Tests Day, you wouldn't bother with the other choices. You'd just mark (E) and move on. But for the record, here's what's wrong with the other choices:

(A) No, J and L can share headache and sneezing, so axe (A).

(B) N's one symptom could be headache or sneezing; this kills (B).

(C) We just saw in Q. 9 that L could share one of its two symptoms with K. Walter could have these two, so axe (C).

(D) N's one symptom could be fever, just like K.

Q14)

Petworld has exactly fourteen animals (three gerbils, three hamsters, three lizards, five snakes) that are kept in four separate cages (W, X, Y, Z) according to the following conditions:
1.Each cage contains exactly two, four, or six animals.

2.Any cage containing a gerbil also contains at least one hamster; any cage containing a hamster also contains at least one gerbil.

3.Any cage containing a lizard also contains at least one snake; any cage containing a snake also contains at least one lizard.

4.Neither cage Y nor cage Z contains a gerbil.

5.Neither cage W nor cage X contains a lizard.

1. Which one of the following could be a complete and accurate list of the animals kept in cages W and Y?

A) W: one gerbil and one hamster

Y: two lizards and two snakes

B) W: one gerbil and two hamsters

Y: one lizard and three snakes

C) W: two gerbils and two hamsters

Y: one lizard and four snakes

D) W: two gerbils and two hamsters

Y: three lizards and one snake

E) W: two gerbils and two lizards

Y: two hamsters and two snakes

2. If there are exactly two hamsters in cage W and the number of gerbils in cage X is equal to the number of snakes in cage Y, then the number of snakes in cage Z must be exactly

A) one

B) two

C) three

D) four

E) five

3. If cage Z contains exactly twice as many lizards as cage Y, which one of the following can be true?

A) Cage Y contains exactly two lizards.

B) Cage Y contains exactly two snakes.

C) Cage Y contains exactly four animals.

D) Cage Z contains exactly three snakes.

E) Cage Z contains exactly two animals.

4. If the number of animals in cage W is equal to the number of animals in cage Z, then which one of the following can be true?

A) Cage W contains exactly six animals.

B) Cage X contains exactly six animals.

C) Cage Y contains exactly one snake.

D) Cage Y contains exactly three snakes.

E) Cage Z contains exactly four snakes.

5. If cage Y contains six animals, which one of the following must be true?

A) Cage W contains two gerbils.

B) Cage X contains four animals.

C) Cage Z contains two snakes.

D) The number of snakes in cage Y is equal to the number of lizards in cage Y.

E) The number of snakes in cage Z is equal to the number of lizards in cage Z.

6. At most, how many snakes can occupy cage Y at any one time?

A) one

B) two

C) three

D) four

E) five

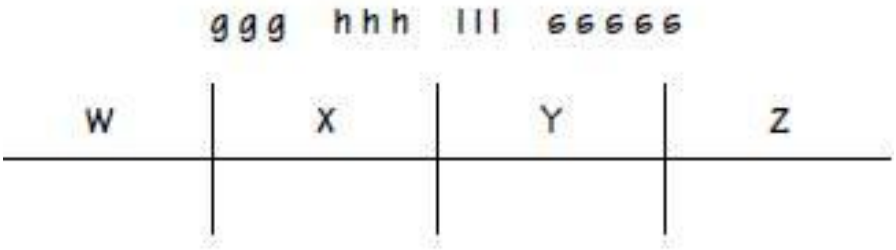
S14)

The Action: Immediately upon seeing "14 animals. . . in four separate cages," you have to figure that your job will be to place the former in the latter—to carry out, in short, a grouping set of distribution. And so it turns out. We don't at first know the impact of the information about the different types and numbers of animals, but your hunch might've been that in some way it impacts the placement. And that hunch turns out to be correct too. The **Key Issues** will be:
1) How many animals must or can be in each cage?

2) Which animals must, could, or cannot be caged with each other?

Because of the unusually large number of entities (14) and the odd number breakdown of animal types (3/3/3/5), you might have wanted to save this set till later in the 35 minutes. Set 3, and especially Set 2, might have struck you as more appealing in terms of starting off the section.

The Initial Setup: You'll likely want to picture the four cages as small boxes or circles, and label them, and make a roster of the entities. Since the cages are capitalized already, perhaps using lower-case letters for the 14 animals makes sense.



The Rules:

1) Rule 1 is best left till last. It's too vague. "Two, four, or six per cage"? Let's pass over that until we can make it more concrete.

2) This rule's two clauses make it clear that wherever you find a gerbil (g) you will find at least one hamster (h), and vice versa. So right away, we see that g's and h's are linked.

Consider the contra positive here: If a cage lacks a hamster, then it cannot contain a gerbil. And if it lacks a gerbil, then it cannot contain a hamster. This cements the link even more.

3) gives us a similar link for lizards and snakes. It works, logically speaking, exactly as Rule 2 works. The presence of a lizard requires the presence of a snake, and vice versa; and if a cage lacks one, it must lack the other.

4) Finally, some concreteness. If neither Y nor Z contains a gerbil—remember to turn negatives into positives—then all the gerbils must go into cages W and X. You can note that under your drawn cages W and X: "All g's."

5) Same as Rule 4, different terms. Since cages W and X fail to contain lizards, all the lizards will be found in cages Y and Z. So note that in your sketch of those cages: "All l's."

Key Deductions: Rule 4 links up with Rule 2, and Rule 5 with Rule 3. Let's take them step by step and in turn.

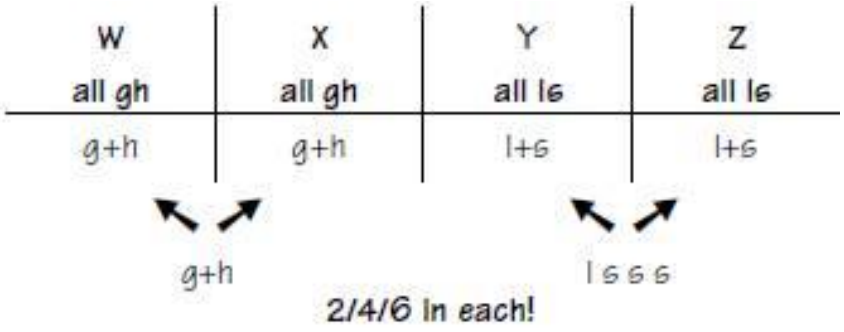
There are no gerbils in cages Y and Z, but without a gerbil in a cage we cannot place a hamster there. Conclusion: There can be no hamsters in cages Y and Z either. Therefore **all of the hamsters and gerbils will be placed in cages W and X.**

Same deal with Rules 5 and 3. Cages W and X lack lizards, which means (Rule 3) those cages must lack snakes as well. Hence **all the lizards and snakes will occupy cages Y and Z.** Now, finally, Rule 1 can help us. No cage is empty, and the minimum number of creatures per cage is two. Well, given the interrelationships of the hamster/gerbil and lizard/snake pairs, we can include this much as minimum information:

W	X	Y	Z
all gh	all gh	all ls	all ls
g+h	g+h	l+s	l+s

What's left? There's one gerbil and one hamster left for cages W and X. They can't be split between those cages (that'd make for 3 creatures per cage, a violation of Rule 1). So one of the cages will get the remaining g+h pair. The outcome is very neat: Of cages W and X, one will get just g+h, the other will get gghh. Things are slightly less neat for cages Y and Z. Four creatures (one lizard and three snakes; see above sketch) remain. They will have to be placed in cages Y and Z. But how? And in what numbers? Each cage contains its required one snake/one lizard minimum—all that's prohibited is that a cage contains snakes only or lizards only. So those four remaining creatures can be placed in either cage Y or Z, as long as Rule 1's number rules are followed. To be precise, either Y and Z will both get 4 creatures or one will get 2 and the other 6.

The Final Visualization:



The Questions:

1. (A)

As long as you consulted your sketches carefully and used the Kaplan method for this acceptability question, you should have been OK. (E) directly violates Rule 5 by giving lizards to cage W. Check the arithmetic (Rule 1): (B) is wrong to give a cage three animals, and (C) is wrong to give a cage five. Finally, look at the distribution: If cage Y gets three lizards (D), then it gets all the lizards, and that leaves cage Z with none. Can't be: (D) would make it impossible to form a valid cage Z. (A) is correct by default.

2. (D)

The first if-clause is easier to deal with: If cage W gets two hamsters, then it must get two gerbils as well (to avoid violating Rule 1). That leaves a single gerbil and hamster for cage X—which, the stem goes on to tell us, has the same number of gerbils as cage Y has snakes. One! Cage Y gets a single snake, which means that all of the remaining snakes go into cage Z; five minus one is four, choice (D).

3. (C)

The arithmetic here may be even easier than that of Q. 2. There are only three lizards to begin with, so the if-clause here must mean that Y has one lizard and Z two. (Note that that makes (A) wrong automatically.) The five snakes can be divvied up among cages Y and Z in several ways, as long as each cage contains at least one snake and an overall total of two, four, or six animals. (C) works fine: The two cages could indeed have four animals each— LSSS in Y, and LLSS in Z—so (C) is the statement that "can be true." (B) Exactly two snakes + Y's one lizard would result in three animals in cage Y, a violation of Rule 1. (D) has the same problem as (B): If we add exactly three snakes to the two lizards that we determined must be in Z (in order to satisfy the stem), we end up with an odd number of critters in both Y and Z, in violation of Rule 1. (E) Given the stem, Z has to have two lizards to Y's one, and Z needs snakes beyond that. So (E) is impossible.

4. (D)

Looking at the master sketch, if W and Z are to get the same number of animals, that number must be two or four—and if you sketch the two possibilities out briefly, it may make the work go quicker. If W and Z each get two, then each of them gets its minimum allotment of a gerbil and hamster for W and a lizard and snake for Z, leaving X and Y with all of the remains. If W and Z each get four, then W gets two gerbils and two hamsters, while Z gets some combination of lizards and snakes equaling four. Your best bet is to check each choice against these possible arrangements: (A), (B) A glance at our master sketch reveals that there are only six animals total (3g, 3h) to be split between cages W and X. If either cage got all six, then the other would be left with zero, in violation of Rule 1. (C) In option 1 above, Y gets four snakes, and in option 2, Y gets at least two snakes. Exactly one snake for Y is therefore not an option. (D) works: Y could get a lizard and three snakes, with Z getting two of each (and W getting its four animals). On test day you need look no further, but for the record: (E) in option 1, Z has just one snake, and in option 2, Z gets at most three snakes.

5. (E)

Cage Y's six animals would have to include the LS pair that's always there, and the four remaining animals, giving us LLSSSS in Y, and LS in Z. (E) is therefore correct—exactly one of each type is present in Z. (D) is flat wrong, and so is (C): Z gets only one snake. All of this distribution, of course, has no impact on cages W and X. In point of fact either (A) or (B) has to be true and the other false, but we can't tell which is which, and the Y/Z situation doesn't help any.

6. (D)

As we just saw in the previous question, Y can contain four snakes and two lizards, leaving one snake and one lizard for Z. The maximum number of snakes for Y is 4, choice (D).

Q15)

Each of seven judges voted for or else against granting Datalog Corporation's petition. Each judge is categorized as conservative, moderate, or liberal, and no judge is assigned more than one of those labels. Two judges are conservatives, two are moderates, and three are liberals. The following is known about how the judges voted:

1.If the two conservatives and at least one liberal voted the same way as each other, then both moderates voted that way.

2.If the three liberals voted the same way as each other, then no conservative voted that way.

3.At least two of the judges voted for Datalog, and at least two voted against Datalog.

4.At least one conservative voted against Datalog.

1. If the two moderates did not vote the same way as each other, then which one of the following could be true?

A) No conservative and exactly two liberals voted for Datalog.

B) Exactly one conservative and exactly one liberal voted for Datalog.

C) Exactly one conservative and all three liberals voted for Datalog.

D) Exactly two conservatives and exactly one liberal voted for Datalog.

E) Exactly two conservatives and exactly two liberals voted for Datalog.

2. Which one of the following must be true?

A) At least one conservative voted for Datalog.

B) At least one liberal voted against Datalog.

C) At least one liberal voted for Datalog.

D) At least one moderate voted against Datalog.

E) At least one moderate voted for Datalog.

3. If the three liberals all voted the same way as each other, which one of the following must be true?

A) Both moderates voted for Datalog.

B) Both moderates voted against Datalog.

C) One conservative voted for Datalog and one conservative voted against Datalog.

D) One moderate voted for Datalog and one moderate voted against Datalog.

E) All three liberals voted for Datalog.

4. If exactly two judges voted against Datalog, then which one of the following must be true?

A) Both moderates voted for Datalog.

B) Exactly one conservative voted for Datalog.

C) No conservative voted for Datalog.

D) Exactly two liberals voted for Datalog.

E) Exactly three liberals voted for Datalog.

5. Each of the following could be a complete and accurate list of those judges who voted for Datalog EXCEPT

A) two liberals

B) one conservative, one liberal

C) two moderates, three liberals

D) one conservative, two moderates, two liberals

E) one conservative, two moderates, three liberals

6. If the two conservatives voted the same way as each other, but the liberals did not all vote the same way as each other, then each of the following must be true EXCEPT:

A) Both conservatives voted against Datalog.

B) Both moderates voted for Datalog.

C) At least one liberal voted against Datalog.

D) Exactly two liberals voted for Datalog.

E) Exactly five of the judges voted against Datalog.

S15)

The Action:
We're presented with seven judges, each of which voted either for or against a petition. One way to look at this is that each entity is either in or out, which strongly suggests grouping; specifically, a grouping game of selection. The entities are two conservatives, two moderates, and three liberals. The Key Issues appear to be:

1) How did each judge vote—for or against the petition?

2) How many judges voted for, and how many voted against the petition?

It's nothing more complicated than that, and the rules, of course, will help us to narrow down the possible voting scenarios.

The Initial Setup:

First, get the basics down on the page. A simple but effective way to proceed is to list the seven entities, under which we'll endeavor to place a ✓ for "for" and an ✗ for "against."

C C M M L L L

The Rules:

4) A quick scan of the rules tells us that Rule 4 is the most concrete of the bunch, and we can put this information right into the sketch: At least one conservative gets an ✗.

3) We may as well take care of Rule 3, and save the two conditional rules for last.

at least 2 ✗'s
at least 2 ✓'s

1) Now the if-thens: If both conservatives and at least one liberal voted the same way, then both moderates voted that way as well. Thanks to Rule 4, we can make this a little more specific, since we know from that rule that at least one conservative said "nay." If both conservatives and at least one liberal voted the same way, then that way must be "against." The rule then tells us that under these circumstances, the two moderates will vote against as well:

C	C	M	M	L	L	L
✗	✗	✗	✗	✗		

Now you may have taken this rule even further, or you may have waited until Step 4 of the Kaplan Method. If you handled the rules in order from 1 to 4, that's probably what you did, but since we've already incorporated Rules 3 and 4, let's continue. Since we still need two votes in favor of the petition (Rule 3), in this situation the two remaining liberals must have voted for it. So Rule 1, if enacted (remember, it's conditional based on the if-then) really boils down to this:

If both C's ✗ and 1 L ✗, then:

C	C	M	M	L	L	L
✗	✗	✗	✗	✗	✓	✓

If you thought this through in advance, a little sketch off to the side to reflect the implications of this rule is in order. If not, that's okay—you no doubt had the opportunity to work through all this at some point in the course of answering the questions.

2) If the three liberals voted the same, then neither conservative voted that way, which is to say that both conservatives voted the other way. You can shorthand this any way that makes sense to you. Here's one possibility:

If L's the same, then C's voted the other way.

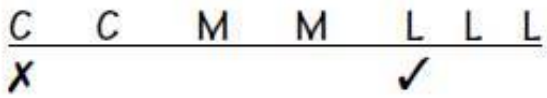
Let's take a further look at the implications of this rule in "Key Deductions."

Key Deductions:

We've already worked through a few implications of the rules by combining the more concrete Rules 3 and 4 with the first two if-then rules. But a little extra thought will reveal one more major thing—did you see it? It's derived from a combination of Rules 2 and 4. If the three liberals voted the same, then both conservatives voted the opposite, and we know from Rule 4 that at least one conservative voted against. So the situation works if all three liberals voted for the petition, in which case both conservatives must have voted against. But the other option is impossible: If all the liberals voted against the petition, then Rule 2 would require that both conservatives vote for it.

But that would violate Rule 4, so it's impossible for all three liberals to vote against the petition. Which means, driving this thought to completion, that at least one liberal must have voted for the petition. This insight bags Qs. 14 and 15 directly, and is just useful knowledge throughout. Write it into your final sketch (and we can make our shorthand for Rule 2 more specific thanks to this deduction; see shorthand below). Now we're ready to attack the questions.

The Final Visualization: Here's what a good final sketch might look like:

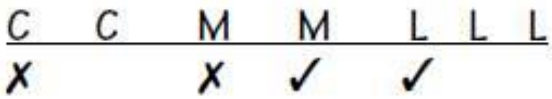


at least 2 ✗'s
at least 2 ✓'s
if both C's ✗ and 1 L ✗, then both M's ✗ and 2 L's ✓
if L's all ✓'s, then C's both ✗'s

The Questions:

1. (B)

If the moderates split their votes, then one voted for and the other voted against. Quickly jot this down in a new sketch:



The contra positive of Rule 1, which you may have noted earlier in the setup stage, is that if the moderates split their votes, then it cannot be true that both conservatives and at least one liberal voted the same way. But following this train of thought gets a little messy, because there are a few different ways to ensure this doesn't happen. The better method here, since we're dealing with a "could be true" question, is to test the choices against the sketch just created, until we find one that works.

(A) The voting arrangement that would result from (A) would violate Rule 1: We would have both conservatives and one liberal voting against the petition, which, according to Rule 1, would force both moderates to vote against as well. But the stem has it otherwise.

(B) works. If you have trouble seeing it in your mind, get the info down on your page:

C	C	M	M	L	L	L
✗	✓	✗	✓	✓	✗	✗

No problem with this scenario, so (B) could be true and is correct.

(C) violates Rule 2—if all three liberals voted for the petition, then both conservatives would have to vote against, but (C) has one of the conservatives voting in favor of the petition.

(D) and (E) both violate Rule 4, which requires at least one "nay" among the conservatives.

2. (C)

Here's the reward for the Key Deduction described above—we deduced that at least one liberal must have voted for the petition. That's because if all the liberals voted no, Rule 2 would have both conservatives voting yes. But Rule 4 disallows that, which means that it cannot be true that all the liberals voted against, which in turn means that at least one liberal must have voted in favor of the petition, choice (C).

(A), (B) No, all the liberals could vote in favor of the petition, in which case both conservatives would vote against.

(D), (E) The moderates can swing all possible ways: As we saw in Q. 13, they can split their votes. On the other hand, both can vote for, or both can vote against, so (D) and (E) both could be true, but need not be.

3. (E)

Now, whether you made the Key Deduction regarding the liberals early on, or just now in the previous question, you can and should use this information to bag Q. 15. If all three liberals voted the same way . . . yup, that's right, they must all vote for, because it's impossible for all of them to vote against, for reasons elaborated just above and in the Key Deductions section. And there's the point in choice (E).

With the liberals all voting for the petition, Rule 2 mandates that both conservatives vote against. So here's what we know of the voting so far:

C	C	M	M	L	L	L
✗	✗			✓	✓	✓

(A), (B) and (D) As Rule 1 is not in effect here, the moderates are totally free to vote any old way: both for, both against, or one each way. So these three choices all could be true, but need not be.

(C) is impossible—as shown above, both conservatives have to vote "nay" in this situation.

4. (A)

Our job in this one is to figure out how we can fit exactly two ✗'s (which means, of course, five ✓'s) into the sketch. We already know one definite ✗, a conservative thanks to Rule 4. And we've seen, most recently in Q. 15, the case in which both conservatives vote against the petition. Filling out the rest in line with the stem's mandate results in this arrangement:

C	C	M	M	L	L	L
✗	✗	✓	✓	✓	✓	✓

No problems there; but is this the only way to get a two/five breakdown of "nays" to "yeas"? Giving the second "against" to one of the moderates doesn't work—then all three liberals vote for, which according to Rule 2 means that both conservatives would have to vote against. But if the second (and therefore last) ✗ goes to a moderate, the other conservative would have to vote in favor, which cannot be. So it's impossible for a moderate here to get the second ✗, which means both moderates must vote for the petition, choice (A).

You may have waited until you worked out the other possible scenario before moving to the choices, in which case you would have worked out this arrangement:

C	C	M	M	L	L	L
✗	✓	✓	✓	✓	✓	✗

This is the only other way to satisfy the stem, which again confirms that (A) must be true. As for the wrong choices, they all could be true, but need not be: (C) and (E) are true in the first possible scenario above, but not in the second. Conversely, (B) and (D) are true in the second, but not in the first.

5. (E)

We're looking for the choice that contains the list of judges that cannot be a complete and accurate list of judges in favor of the petition. Before jumping into complicated scenarios, let's see if the answer can be derived by checking the choices against the concrete rules, Rules 3 and 4. Rule 4 is easy to check—there must always be at least one conservative voting against the petition, so if any choice contains "two conservatives," that must be the answer (remember, the entities in the choices represent judges voting for the petition). No luck. But Rule 3 does the trick. Are there any choices that would result in a lack of two judges on either side? Well, yeah: (E)

consumes every judge but one conservative, which would result in six “yea’s” and only one “nay.” Just add ‘em up—six judges voting in favor of the proposal is a no-no, a direct violation of Rule 3.

6. (B)

Take it one step at a time: The conservatives voted the same way, which means, thanks to Rule 4, that they both voted no. Next we’re told that the liberals didn’t all vote the same way. We deduced (and saw later in Q. 14) that at least one liberal must vote for the petition, so now we know that at least one must vote against. So far then, this is how the votes went:

C	C	M	M	L	L	L
✗	✗			✓	✗	

This situation triggers Rule 1—both conservatives and at least one liberal voted the same way, against—which means that both moderates must vote against the petition as well. But wait—we may as well complete the voting: With only one “for” vote out of six so far, we must satisfy Rule 3 by placing a ✓ under the remaining liberal, resulting in:

C	C	M	M	L	L	L
✗	✗	✗	✗	✓	✗	✓

Every choice conforms to this arrangement except for (B), which is impossible (and therefore correct): Under these circumstances, both moderates must have voted against the petition.

Q16)

Seven children are to be seated in seven chairs arranged in a row that runs from west to east. All seven children will face north. Four of the children are boys: Frank, Harry, Ivan, and Joel. Three are girls: Ruby, Sylvia, and Thelma. The children are assigned to chairs according to the following conditions:

1. Exactly one child sits in each chair.

2. No boy sits next to another boy.

3. Ivan sits next to and east of the fourth child in the row.

4. Sylvia sits east of Ivan.

5. Frank sits next to Ruby.

1. What is the maximum possible number of different pairs of chairs in which Frank and Ruby could sit?

A) one

B) two

C) three

D) four

E) five

2. Which one of the following statements must be false?

A) Both Harry and Joel sit east of Frank.

B) Both Harry and Ruby sit east of Frank.

C) Both Harry and Joel sit west of Frank.

D) Both Harry and Ruby sit west of Frank.

E) Both Joel and Ruby sit east of Frank.

3. If Thelma sits next to Ivan, and if Frank sits next to Thelma, which one of the following statements could be false?

A) Both Frank and Ivan sit east of Ruby.

B) Both Frank and Ruby sit west of Thelma.

C) Both Frank and Sylvia sit east of Ruby.

D) Both Frank and Thelma sit west of Sylvia.

E) Both Frank and Ruby sit west of Joel.

4. If Frank does not sit next to any child who sits next to Ivan, which one of the following statements could be true?

A) Harry sits west of Frank.

B) Joel sits west of Ivan.

C) Ruby sits west of Frank.

D) Thelma sits west of Frank.

E) Thelma sits west of Ruby.

5. If Frank sits east of Ruby, which one of the following pairs of children CANNOT sit next to each other?

A) Frank and Thelma

B) Harry and Ruby

C) Harry and Sylvia

D) Ivan and Ruby

E) Joel and Ruby

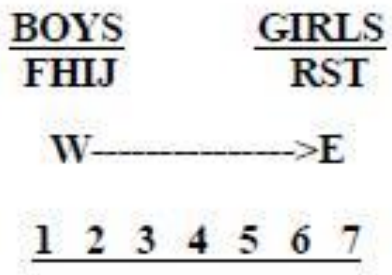
S16)

The Action: Back to the basics here in Set 3: “Seven chairs arranged in a row” immediately denotes sequencing as the set action. The entities, children, are split into boys and girls, but that’s not so tough to incorporate. And that west to east business, with all children facing north? Simply a different way of describing the kids’ relationships to one another. The reference to “facing north” suggests that there may be a left-right element to the set, which turns out not to be the case. So, it simply boils down to seven chairs, seven kids, putting them in an order—you’ve seen this (or will, before we’re through with you) dozens of times. The **Key Issue** in sequencing sets is always the same—who’s where in the ordering. A quick glance through the answer choices on the page reveals that we’ll mainly be concerned with figuring out:

1) Which children must, can, or cannot sit east and west of which other children?

2) Which children sit next to which other children?

The Initial Setup: Since the east-west thing seems to play a big part in the questions, there’s no harm in jotting down a W and E to remind yourself at a glance which children are east of and west of each other. Get the boys and girls down on the page, throw in seven chairs, and you’re ready to see how the seating arrangement will be determined.



The Rules:

1) Talk about strict. No fun for these kids; one kid per chair. Just a loophole closer, that you have most likely assumed anyway.

2) It's hard to wait for Step 4 of the Kaplan Method to consider the implications of Rule 2: There are four boys, and none of them sits next to any other one. Well, that's not so easy to do in a sequence of only seven kids, is it? Your reaction to each rule is supposed to be "what does this mean" in the context of this set—and you needn't postpone a deduction until Step 4 if it jumps right out at you. Hopefully you saw that the only way to space out four non-adjacent entities in a sequence of seven is to place them in spots 1, 3, 5, and 7—there's simply no other way. At this point, just put a b for boy over those slots in your sketch, and a g for girl over the others. (Boy, girl, boy, girl . . . how quaint.)

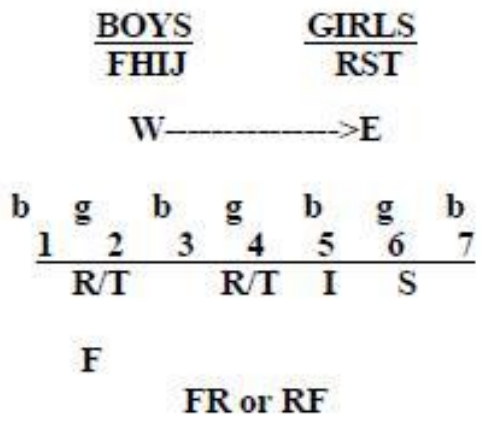
3) Here's the most concrete rule of the bunch, one that you may have jumped to right off the bat: It requires a touch of interpretation, but essentially means that Ivan is 5th. Chairs 5, 6, and 7 are the chairs east of chair 4, and the only one of those next to 4 is 5.

4) "East of" in this set means to the right, so this boils down to our usual sequencing notation I . . . S. Perhaps you thought this through in accordance with the deduction stated above. We'll hold off until Step 4 and deal with it in Key Deductions below.

5) FR or RF must appear somewhere in the ordering.

Key Deductions: We jumped all over Rule 2 above; it seemed natural to work through the implications of the rule at that point. If you didn't think it through then, you probably did so now. As you recall, we deduced that **boys must sit in chairs 1, 3, 5, and 7, which leaves the girls in chairs 2, 4, and 6**. Now we can combine that with Rules 3 and 4. Since Ivan is in chair 5, and Sylvia must be east of that, Sylvia must be in chair 6 or chair 7. But 7 is a boy's chair, so **Sylvia must sit in chair 6**. Is that it? Don't stop there; what else can we deduce? With Sylvia in 6, the other girls, **Ruby and Thelma, must split chairs 2 and 4**. And **Frank**, who must sit next to Ruby, **must therefore sit in either chair 1 or chair 3**. This latter piece of information comes in especially handy throughout the questions. (You could take it one step further by noticing that chair 7 must be occupied by either Harry or Joel, but the set isn't won or lost on this tidbit.)

The Final Visualization reflects the rules and all the deductions discussed above:



The Questions:

1. (C)

The major challenge of Q. 1 is figuring out exactly what they're looking for, so a careful reading is in order. First of all, what is a "pair of chairs in which Frank and Ruby can sit"? Let's check it out: We see from our sketch that Frank can be in chair 1 with Ruby in chair 2, so chairs 1 and 2 together count as one acceptable pair for this couple linked by Rule 5. Keeping Ruby in 2, Frank can move to 3 without a hitch, so there's a second pair, chairs 2 and 3. But Ruby can also be in chair 4 when Frank's in 3, so 3 and 4 together counts as our third pair. Any others? Nope, so three it is, choice (C).

2. (C)

Next we're to find a statement that must be false, which means that each wrong choice will either be true or possibly true. Notice how all the choices involve Frank, which is nice because we know plenty about Frank's prospects. In fact, simply recognizing that Frank can be in the first chair allows us to kill choices (A), (B), and (E), since if Frank were to take chair 1, everyone would be east of him. Upon analysis, we see that (C) is the impossibility we seek: We deduced up front that Frank must sit in either chair 1 or 3. But even if we slide Frank down as far east as possible—that is, chair 3—there's no way to fit both free agents Harry and Joel west of him, considering that either Ruby or Thelma must take chair 2. (D) is possible: One of our boy floaters, in this case Harry, can certainly hang out with Ruby on the west side of Frank—H in 1, R in 2, and F in 3 would do the trick. But we can't get both Harry and Joel west of Frank. As mentioned above, one of those guys has to be in the easternmost chair 7.

3. (E)

Good solid info is offered in this one, so here's a perfect time to use your pencil—get it all down in a new sketch. Ivan's in chair 5, so if Thelma's next to him, she must be in chair 4, considering that we've already slotted Sylvia for chair 6. The stem also tells us that Frank is next to Thelma, so put F in chair 3, which leaves R for chair 2. Free agents Harry and Joel will float between chairs 1 and 7. We're looking for something that could be false, and we can shorten our work by scanning the choices for Harry and Joel; after all, they're the only ones not fully restricted here. Only Joel appears, and way down in choice (E). Lo and behold, Joel's elusive nature does in fact make (E) the correct choice. (E) would be true if Joel's in chair 7, but would be false if Joel's in chair 1.

4. (B)

We already have Frank's location narrowed down to either chair 1 or 3, and the information in the stem allows us to deduce it must be 1. How? Because chair 3 is next to chair 4 which does seat a child sitting next to Ivan (who we know is in 5). The stem says Frank can't sit in

such a seat, so Frank goes to 1, which forces Ruby into 2 (Rule 5) and Thelma into chair 4. Once again, we're left with our two free agents, H and J, this time floating between chairs 3 and 7. Checking the choices, we see that only (B) could be true—if Joel's in 3 and Harry's in 7, then Joel indeed would be sitting west of Ivan. All of the others are impossible. (You may have eliminated (A), (C), and (D) quickly on account of the fact that with Frank in chair 1, ain't nobody west of him.)

5. (D)

If you're not convinced yet as to the importance of the Frank deduction, here's another question that falls pretty easily thanks to the fact that Frank must be in 1 or 3. If Frank took chair 1, he wouldn't be east of Ruby as the stem demands; so Frank must sit in chair 3 with Ruby in chair 2. That again lands Thelma in chair 4 (familiar territory for her), which again leaves 1 and 7 for Harry and Joel to fight over. Again, everyone's placed but those two. From this ordering, we can see quite readily that the only pair that cannot be next to each other is the duo of Ivan and Ruby—Frank and Thelma come between them. Choice (A) contains folks that are next to each other, while the others contain pairs that could be, depending on where Harry and Joel sit.

Q17)

Mr. Bankatlal acted as a judge for the beauty contest. There were four participants, viz. Ms. Andhra Pradesh, Ms. Uttar Pradesh, Ms. West Bangal and Ms. Maharashtra. Mrs. Bankatlal, who was very anxious about the result asked about it as soon as he was back home. Mr. Bankatlal told her just that the one who was wearing the yellow saree won the contest. When Mrs. Bankatlal pressed for further details, he elaborated as follows:

i. All of them were sitting in a row.
ii. All of them wore sarees of different colours, viz. green, yellow, white, red.
iii. There was only one runner-up and she was sitting beside Ms Maharashtra.
iv. The runner-up was wearing the green saree.
v. Ms West Bengal was not sitting at the end and was not the runner-up.
vi. The winner and the runner up are not sitting adjacent to each other.
vii. Ms Maharashtra was wearing a white saree.
viii. Ms Andhra Pradesh was wearing a green saree.
ix. Participants wearing yellow saree and white saree were at the ends.

1. Who wore the red saree?

A) Ms Andhra Pradesh
B) Ms West Bangal
C) Ms Uttar Pradesh
D) Ms Maharashtra

2. Ms West Bengal was sitting adjacent to

A) Ms Andhra Pradesh and Ms Maharashtra
B) Ms Uttar Pradesh and Ms Maharashtra
C) Ms Andhra Pradesh and Ms Uttar Pradesh
D) Ms Uttar Pradesh only

3. Which saree was worn by Ms Andhra Pradesh?

A) Yellow
B) Red
C) Green
D) White

4. Who was the runner-up?

A) Ms Andhra Pradesh
B) Ms West Bengal
C) Ms Uttar Pradesh
D) Ms Maharashtra

5. Who was the winner of the beauty contest?

A) Ms Andhra Pradesh
B) Ms West Bengal
C) Ms Uttar Pradesh
D) Ms Maharashtra

S17)

Based on the information given, we can come up with the following alternatives:
Possibility 1: Runner Up

Miss UP	Miss WB	Miss AP	Miss Maharashtra
Yellow	Red	Green	White

Possibility 2: Runner Up

Miss Maharashtra	Miss AP	Miss WB	Miss UP
White	Green	Red	Yellow

In either case the answers are the same:
1. Miss West Bengal wore red. Option (b) is correct.
2. Miss West Bengal was adjacent to Miss Andhra Pradesh and Miss Uttar Pradesh. Option (c) is correct.
3. Miss Andhra Pradesh wore the green saree. Option (c) is correct.
4. Miss Andhra Pradesh was the runner up as she wears green and sits next to Miss Maharashtra in both cases. Option (a) is correct.
5. Miss Uttar Pradesh was the winner. Option (c) is correct.

Q18)
Sprinters in Olympics running on the track are carrying numbers from 1 to 7 at the back of their vests. They complete the race when their feet touches the finish line at 9.10, 9.20, 9.30 seconds till 9.7 seconds i.e. at an interval of one millisecond each. Each athlete is ranked from 1 (for the fastest) to 7 (for the slowest). The following additional information is known:
(i) The square of the highest total of the vest number and rank is 169 which occur only once.
(ii) The square of the lowest total of vest number and rank is 16 and occurs only once.
(iii) The square of the total of vest number and rank = 81, occurs thrice.
(iv) The winners' vest no. exceeds that of the first runner up.

1. The vest number of the sprinter coming first is:
A) 3
B) 6
C) 1
D) 5

2. The sprinter number finishing second is:
A) 7
B) 6
C) 4
D) 3

3. The sprinter number finishing third is:
A) 6
B) 3
C) 1
D) 2

4. The sprinter number finishing last is:
A) 7

B) 2
C) 5
D) 1

5. The sprinter number finishing last but one is:
A) 7
B) 2
C) 5
D) 1

S18)

There are essentially two logical forks in this question:
The first one comes from Clue (i), according to which the highest total of vest number and time must be 13 (as the square is 169). This could mean two scenarios:
Vest number 7 with time=6; or vest number 6 with time=7.

The two scenarios can be shown as:

Possibility 1		Possibility 2	
Vest Number	Rank	Vest Number	Rank
1		1	
2		2	
3		3	
4		4	
5		5	
6		6	7
7	6	7	

Inside these two scenarios we need to merge the other three way fork: The square of the least total of vest number and rank is 16 and occurs only once. It gives us 3 scenarios on the surface: However in this scenario we can reject Vest 3 ranking 1 because then the second ranker's vest number would exceed the winner's vest number (Clue (iv))—as we cannot put the 2nd rank for either 1 or 2 as the total of 4 is the least sum and occurs only once. This leaves us with 4 principal situations 1A, 1B, 2A and 2B. Let us evaluate each of these separately:

Possibility 1A (starting thought)	
Vest Number	Rank
1	3
2	4/5/7
3	2/4/5/7
4	1/2/4/5/7
5	1/2/4/5/7
6	1/2/4/5
7	6

In the above situation, the total of vest number + rank has to occur thrice. There are only 3 places where this can actually occur given the possibilities listed above. These are 2-7; 4-5 and 5-4. Hence, the following conclusions can be made:

Possibility 1A (starting thought)	
Vest Number	Rank
1	3
2	7
3	2
4	5
5	4
6	1
7	6

3-2 and 6-1 get automatically selected. As for the vest number 3, only rank possibility left is 2nd rank and after than rank 1 must go to vest number 6. If we go for a similar analysis with the other possibilities you can see that they do not work out. For instance in Possibility 1B, if we put down what is possible to be placed against vest numbers 1 and 3 respectively we get the following figure.

Possibility 1B	
Vest Number	Rank
1	4/5/7
2	2
3	3/4/5/7
4	
5	
6	
7	6

From this point we know that there have to be exactly 3 totals of 9, which can only be achieved by matching 4-5; 5-4 and 6-3. Then, 3rd vest must be 7th rank but that leaves us with no rank possibility for vest 1 and also rank 1 not getting allotted anywhere. Thus, this possibility does not exist. Similarly we can reject possibilities 2A and 2B:

Possibility 2A	
Vest Number	Rank
1	3
2	4/5/6
3	2/4/5/6
4	1/2/4/5/6
5	1/2/4/5/6
6	7
7	1/2/4/5/6

In the above table we need three 9s. There are 4 possibilities for the sum of vest number and rank to be 9. These are 3-6, 4-5, 5-4 and 7-2. Only 3 of these 4 have to make up 9, hence we will have to leave out one of the 4. If we leave out 3-6, 4-5 or 5-4 then vest number 7 must take Position 2 (runner up). This will contradict Clue (iv) as the winner's vest number > the runner-ups vest number. If we leave out 7-2 and place 3-6, 4-5 and 5-4 there will be no place to put rank 2. Thus, this possibility gets rejected. Possibility 2B also gets rejected as follows:

Possibility 2B	
Vest Number	Rank
1	4/5/6
2	2
3	3/4/5/6
4	1/3/4/5/6
5	1/3/4/5/6
6	7
7	1/3/4/5/6

In the table above, there are only 3 cases of total of vest number + rank = 9. Thus, the table should become:

Possibility 2B	
Vest Number	Rank
1	4/5/6 no place for 3rd rank??
2	2
3	6
4	5
5	4
6	7
7	1

Thus, this possibility is also rejected. The answers are:

1. Vest number 6. Option (B)
2. Vest number 3. Option (D)
3. Vest number 2. Option (D)
4. Vest number 2. Option (B)
5. Vest number 7. Option (A)

3. TEAM FORMATION

The questions consist typically of a group of people/objects that have to be put together in teams, albeit with a set of constraints/conditions about certain people who need to be put together/ cannot be put together in a team. The key skills involved in solving team formation questions would involve the following:
(i) The ability to visualize the structure of the teams to be formed including the number of people/objects in each team.
(ii) The ability to order the clues in the correct order of usage (as explained in the theory of logical reasoning).
(iii) The ability to create symbolic representations of the various clues provided so that you can bring together each of the relevant clues while creating the teams.
(iv) The ability to wait for and reach the appropriate time in the problem solving situation from where the indirect clues provided in the question can be used.

Q1)
Vijayshree wants to take four courses this trimester. She is interested in only three marketing courses – Distribution, Advertising and PR; and four Finance courses— International Finance, Accounting, Corporate Finance and Financial Services. To meet college requirements she must take two marketing courses. There are some scheduling problems: International Finance overlaps both Advertising and Corporate Finance but she can choose Advertising and Corporate Finance as two different courses. Distribution is given at the same time as Accounting.

1. If Vijayshree decides she will take International Finance, what will her other three courses be?
(a) Distribution, PR and Advertising
(b) Distribution, PR and Financial Services
(c) Distribution, PR and Accounting
(d) PR, Advertising and Financial Services

2. If Vijayshree takes four courses this trimester, then which of the following statement is not possible?
I. She takes Accounting and does not take Advertising.
II. She takes Corporate Finance and does not take Advertising.
III. She takes International Finance and does not take PR.
(a) I only
(b) II only
(c) III only
(d) I and III only

3. Which of the following must always be true?
I. Vijayshree must take PR if she takes Corporate Finance.
II. Vijayshree must take Advertising if she takes Accounting.
III. Vijayshree must take Accounting if she takes Advertising.

(a) I, II, and III
(b) II, and III only
(c) I and II only
(d) II only

S1)

(1) It can be seen that if she takes International Finance, she would not be able to take the Advertising and Corporate Finance course. This means that she has to choose Distribution and PR as her two marketing courses and because Distribution overlaps with Accounting, her second finance course must be Financial Services. Option (b) is correct.

(2) Statement I is definitely not possible because if she takes Accounting she cannot take Distribution and she must have to take Advertising. Statement II is possible, while statement III can be seen to not be possible as if she takes International Finance she has to take PR as Advertising would not be available. Option (d) is correct.

(3) Statement I is not necessarily true as if she takes Corporate Finance she can take up Distribution and Advertising and skip PR. i.e. PR is not necessary if she has taken Corporate Finance. Statement II is mandatorily true because if she takes Accounting, she cannot take Distribution and hence she would be forced to take up Advertising as one of her two compulsory marketing courses. Statement III is not necessarily true. Thus option (d) is correct.

Q2)

M, N, P, Q, S, and T are six members of a group in which there are three female members. Females work in three departments – Accounts, Administration and Personnel, and sit on three different floors – I^{st} , IInd , IIIrd . Persons working in the same department are not on the same floor. Two persons work on each floor.
ii. No two females work in the same department or on the same floor. N and S work in the same department but not in Personnel. Q works in Administration. S and M are on the I^{st} and IIIrd floors respectively and work in the same department. Q, a female, does not work on the IInd floor, P, a male, works on I^{st} floor.

1. Which of the following groups of persons are females?
(a) SQT
(b) QMT
(c) QPT
(d) Data inadequate

2. Which of the following pairs of persons work in Administration?
(a) QP
(b) QN
(c). SP
(d) Data inadequate

3. T works in which department?
(a) Accounts
(b) Administration
(c). Personnel
(d) Accounts or Personnel

4. Which of the following pairs work on the IInd floor?
(a) PT
(b) SM
(c) QN
(d) NT

5. If T is transferred to Accounts and S is transferred to Administration, who is to be transferred to Personnel to maintain the original distribution of females on each floor?
(a) P
(b) N
(c) Q
(d) Data inadequate

S2)
From the first clue we know that: Six people M, N, P, Q, S & T, 3 departments for females: Accounts, Admin and Personnel. 3 floors (2 people each)

		III
		II
		I

Using the direct information in clue 2 we get:

M	Q-	III
		II
S	P+	I

Note: (-) Denotes Female and (+) denotes male
Since P is a male (denoted by '+' in the figure) S must be a female (Since no two female works in the same department or on the same floor).
If S is female, M must be male (since they are in the same department).
Also, since N and S are in the same department, N is also a male. This gives us the final figures as:

M+	Q-	III
N+	T-	II
S-	P+	I

Also, S, M and N work in either Accounts or Administration. Q—Administration. At this stage we can answer the questions:
1. S, Q, T (a)
For 2. data is inadequate since we just know that Q works in Admin, but we do not know who else works in Admin (and whether someone works in Admin or not). Hence (d).
For 3. Since Q is in Admin, S must be in accounts (as she cannot be in personnel). Thus, T is in Personnel. Hence (c).
4. N, T (d)
5. (c) Q needs to move to Personnel to have the three women in three different departments.

Q3)
Ten students— A, B, C, D, E, F, G, H, I and J are chosen to represent their college in four sports— tennis, badminton, table tennis and snooker. The badminton team has one student less than the tennis team. A, B and C are not tennis players, individually or as a group. D, E and F are not in badminton team individually or as a group. G, H and I are not in the table tennis team, individually or as a group. J is a table tennis player. None of the students is a snooker player.

1. Which of the following students could be table tennis players?

(a) J, B & G

(b) J, C & F

(c) J , D, E & F

(d) J, B, C & D

2. Of those listed, what is the largest possible number of students who could be table tennis players?

(a) 6

(b) 7

(c) 9

(d) None of these

3. If no other student except I is in the badminton team, which of the following students could be tennis players?

(a) D and H

(b) F and E

(c) E and G

(d) G and H

4. If the table tennis team has one member, who of the following must be a tennis player?

(a) A

(b) B

(c) D

(d) G

5. If G, H and C are the only students in the badminton team, which three must be the only students who are table tennis players?

(a) A, B and J

(b) A, D and J

(c) B, E and J

(d) B, F and J

6. If A and G are the only badminton players, how many students must be in the table tennis team?

(a) 2

(b) 3

(c) 5

(d) 6

S3)
The following table will emerge out of the given clues:

	Tennis	Badminton	Table tennis
A	X		
B	X		
C	X		
D		X	
E		X	
F		X	
G			X
H			X
I			X
J	X	X	
Total			

1. (b) Looking at the options, the only possible solution is option (b) (as Table Tennis should be an odd number of people as Tennis + Badminton is 2n–1)

2. (b) If Table Tennis players are 7 in number then, n + n – 1 = 3.
2n = 4.
n = 2. This is possible.

3. (d) Going through options we find that the only possibility is G and H as we would need 7 people to be Table Tennis players in such a case.

4. (c) If the Table-Tennis team has just one member J, then the Tennis team will surely have D, E and F as players.

5. (a) A, B and J would then be definitely Table Tennis players.

6. (c) If the Badminton team has 2 players, then the Tennis team will have 3 players The remaining players will be in the Table-Tennis team.

Q4)

Read the information given below and answer the questions that follow.

At a secret location in Nevada, a science lab exists where four research projects ROC-

I, ROC-II, ROC-III and ROC-IV are underway. The projects are assigned difficulty level L1 or L2 or L3. A total of six researchers who are given code names in order to protect their identity are—Ax, Bx, Cx, Dx, Ex and Fx—are chosen for working on the projects with the provision that for the project of difficulty level L1 only one research associate will be engaged while for the other projects two researchers will be engaged. There are, however, some constraints to be overcome before allotting the work.

A. Ax, Bx and Dx do not work with each other.

B. Projects ROC-II and ROC-III are of difficulty level L 2 /L 3. Remaining projects are of L 1 level.

C. Ex cannot work with anyone else.

D. Cx and Dx cannot work together.

E. Fx can work either with Dx or Cx only.

F. Fx cannot be engaged in L 3 level projects.

G. Ax and Cx cannot work together.

1. Researcher Ax can be allotted which of the following projects?

(a) ROC-I or ROC-II but not ROC-III

(b) ROC-II or ROC-III but not ROC-IV

(c) ROC-III or ROC-IV but not ROC-I

(d) ROC-IV or ROC-I but not ROC-II

2. If researcher Fx is engaged neither for ROC-I nor for ROC-II, and it is also known that Bx works in Project III, then assignments for how many of the six researchers can be decided?

(a) 2

(b) 4

(c) 3

(d) All

S4)

Refer to the clues given to us in the information, the best way to solve this question is to tabulate as well as comprehend the direct and indirect clues given to us. The first clue given to us is that L 1 difficulty projects will have single researchers and other difficulty level will have two researchers. Now it is given that ROC II and ROC III are L 2 and L 3 respectively. Thus, ROC I and ROC IV are L1 level. Direct clues given to us point that

Ex can be assigned only L 1 projects which are ROC I and ROC IV. Next set of clues given to us says that Ax cannot work with Bx, Cx, Dx and Fx. Thus, Ax will also be assigned to project requiring a single researcher hence either ROC I or ROC IV. Now, since Fx cannot be assigned to L 3 level so he must have been assigned to ROC II and he can work with either Dx or Cx, but Fx will work with Dx since Bx cannot work with Dx. Now Bx will be assigned to the remaining L 3 project i.e. ROC III and Bx has the option of working with Cx.

	L_1	L_2	L_3	L_1
	ROCI	ROCII	ROCIII	ROCIV
Ax	X/Y	X	X	Y/X
Bx	X	X	Y	X
Cx	X	X	Y	X
Dx	X	Y	X	X
Ex	Y/X	X	X	X/Y
Fx	X	Y	X	X

1. Hence Option (d) is the correct answer.

2. If Fx can neither work in ROC I and ROC II then we become uncertain about the project assignment for each of Ax, Ex and Fx. Hence Option (c) is the correct answer.

Q5)

Seven specialist persons Bebe, Meme, Keke, Pepe, Dede, Fefe and Hehe visit an office on four days— Tuesday, Wednesday, Friday and Saturday, in a week. At least one person but not more than two persons visit the office on each of these days. Each of them is a specialist in different fields—Paramedic, Business, Engineering, Genetics, Biometrics, Defense and Biotech.

(i) Pepe visits on Friday with a specialist in Defence.

(ii) The Engineer does not visit on Saturday nor with Dede and Hehe.

(iii) The Biotech specialist Fefe visits alone on Tuesday.

(iv) Meme visits on Wednesday and he is not engineer.

(v) Keke visits on Wednesday. Hehe is not a specialist in Defence.

(vi) The engineer visits with the paramedic specialist.

(vii) The genetics specialist visits on Friday.

(viii)Bebe is neither Business specialist nor a specialist in Defence.

1. What is the speciality of Bebe?

(a) Biometrics

(b) Paramedics

(c) Engineer

(d) Data inadequate

2. On which day of the week does Dede visit?

(a) Wednesday

(b) Saturday

(c) Wednesday or Saturday

(d) Friday

S5)

This question is a classic example of primary and secondary clues being provided in the information. To solve this set one must be able to analyze all the possibilities given in the clues to derive conclusive data arranged in an appropriate manner according to the given conditions.

Let us solve the problem. From clue one we get that the two persons who visit on Friday are Pepe and the person who visits with Pepe is a Defense specialist. Now from clue three Fefe is a Biotech specialist and he visits alone on Tuesday. Also from clue seven we get that the specialist in Genetics visits on Friday. This means that Pepe is a Specialist in genetics.

Moving onto clues four and five we get that Meme and Keke visit on Wednesday and from clue six the engineer visits with Paramedics. Thus they visit either on a Wednesday or on a Saturday.

Again, from clue two the engineer does not visit on Saturday. Thus, engineer and paramedics visit on Wednesday. Now using clue four we get that Meme is Paramedics and Keke is the engineer. Again from clue five we get that Hehe is not a defense specialist and Hehe does not visit on Friday. Hence Hehe visits on Saturdays. Similarly from clue eight we get that Bebe also visits on Saturday. The only person left to analyse is Dede who is the specialist in defense. Also since Bebe is not a businessman, Hehe is a businessman and Bebe is a specialist in Biometrics.

Now we can summarize our analysis in a table:

Tuesday	Fefe (Biotech)
Wednesday	Meme (Paramedic), Keke (Engineer)
Friday	Pepe (Genetics), Dede (Defence)
Saturday	Hehe (Business), Bebe (Biometrics)

Now we can answer the questions:

1. Option (a) is the correct answer.

2. Option (d) is the correct answer.

4. QUANTITATIVE REASONING

Quantitative reasoning, as the name itself suggests, is reasoning involving numbers and numerical logic. Quantitative reasoning questions are common in all kinds of aptitude exams and require the student to work out numerical relationships as defined by parameters in the problem.

The key skills involved in solving quantitative reasoning questions are the following:

(i) The ability to understand the specific quantitative logic being utilized inside the questions
(ii) The ability to order the clues in the correct order of usage (as explained in the theory of logical reasoning).
(iii) The ability to understand basic mathematical concepts like percentages, averages, ratios, multiplications, etc.
(iv) The ability to create symbolic representations of the various clues provided so that you can bring together each of the relevant clues while creating the teams
(v) The ability to wait for and reach the appropriate time in the problem solving situation from where the indirect clues provided in the question can be used.

Example 1)

A man would like to take a new health insurance. An officer taking care of these matters says to the man, “Please tell me how many children you have.” The man answers, “I have three of them.” The officer, “What are the ages of your children?” The man answers, “The product of the ages is equal to 36.” The officer replies, “This is not enough information Sir!”; the man replied, “Sorry that I was a little bit unclear, but the sum of the ages is equal to the number of shops in front of your office.” The officer: “This still isn’t enough information Sir!” The man replies, “My oldest child loves chocolate.” The officer: “Thanks for your cooperation, I now know the ages.” Are you as smart as the officer?

Then give the sum of the ages of the children.

1. 13

2. 22

3. 36

4. 38

Solution:

The product of the ages is 36. Using this one can make the following combination of ages:

1, 36, 1 sum = 38

1, 18, 2 sum = 22

1, 12, 3 sum = 16

1, 9, 4 sum = 14

1, 6, 6 sum = 13

2, 9, 2 sum = 13

2, 6, 3 sum = 11

3, 3, 4 sum = 10

After the man had said that the product of the ages is equal to 36, the officer didn't have enough information. Then he was told that the sum is equal to number of shops in front of the office. He replied by saying that this still isn't enough information. So the sum of the ages should be 13, because otherwise he would have known the ages immediately. The last statement is that that the oldest child loves chocolate. So there is an oldest child. Hence the officer concludes that the ages of the children are 2, 2 and 9 years.

Hence option (1) is the correct answer.

Q1)

Dhiman was admiring the output of her new program to generate random number. She had printed out the first ten numbers of the results. She soon noticed something interesting. Each of the 10 numbers had exactly one digit, in the proper placement, of the 5 digit code she used to open her car door without a key. In the first number 14073, for example, Dhiman's car code could not be 34170 (two digits correctly placed) or 92365 (none).
Find Dhiman's car entry code from these first 10 randomly generated numbers: 14073, 79588, 05892, 84771, 63136, 42936, 37145, 50811, 98174 and 29402?

A. 05892

B. 63136

C. 42936

D. None of these

S1) D.

In the given grid of 10 five digit numbers, since every number has exactly 1 digit matched correctly with the correct code, there must be exactly 10 instances of correct code matches amongst the 50 possible instances (10x5). If we were to look at the first digit it is clear that the first digit has 10 different values in the ten number. Thus, only 1 number can be correctly

matched for the first digit's value. The other 4 places in the 5 digit number must match 9 more times for the above grid to be correct. This can be principally done in 2 ways:

First way: 3+3+2+1 OR Second way: 2+2+2+3

If we were to observe the numbers the following deductions would come up:

First place	Second place	Third place	Fourth place	Fifth place
Each digit (from 0 to 9) happens only once	4 & 9 occur twice	1 occurs thrice & 8 occurs twice	7 occurs thrice & 3 occurs twice	1,2 & 6 occur twice each
	5,3,2,7,0 & 8 occur once each	0,5,7,9,8 occur once	8,9,4,1 & 0 occur once each	5,4,3 & 8 occur once each

Looking at the number 98174, it is clear that both 1 & 7 cannot occur in the third and fourth place digits as then the number 98174 would have 2 numbers matched with the code. Thus, the structure 1+2+3+3+1 or indeed 1+1+3+3+2 is ruled out (i.e., the use of two triplicate matches is ruled out). Thus, the 10 matches between the code and the 10 random numbers must be in one of the following structures:

	First place	Second place	Third place	Fourth place	Fifth place
Number of digits matched with the code	1	2	3	2	2
Number of digits matched with the code	1	2	2	3	2

Thus, the number would be either xx13x or xx87x. The first possibility is rejected because 63136 contain both 1 & 3 in the third and fourth places of the number. Thus, the number must be of the form xx87x. From this point thinking about the second place digit gives us that the second place has to be occupied by 9 (as only 4 & 9 give us 2 matches with the code & amongst this 4 cannot be taken for the second place because if the number is x487x then the number 84771 contradicts the basic condition of the problem.) Thus, the number must be x987x. Thinking of the last digit— this digit must be either, 1, 2 or 6. It cannot be 1 because of the presence of 84771 in the ten number. It cannot be 2 because the number 29402 would have 2 digits in the correct place. Hence, it must be 6. Thus, the number becomes x9876. Thinking of the first digit, it has to be 3, because it is only the number 37145 which lacks a single digit matching with the 2nd to 5th digit of the code x9876. Thus, the code is 39876.

Q2)

One early Monday morning, four snails—Aman, Bubbly, Charu and Devi, set off together down the garden path. Aman and Bubbly kept the same steady pace, slithering only 8 meters by the time Charu and Devi had already reached the azalea. Charu was winded and had to stop for an hour to rest. Although Devi was tired, too, she pressed on, but reduced her pace to be the same as Aman's and Bubbly's. Charu started off again just as Aman and Bubbly got even with her. She raced off at her original pace. Aman promptly sped up to the same speed as Charu and kept even with her. Bubbly just kept going at her original pace. When Aman reached the end of the path, she was 1 meter ahead of Bubbly, but she was a half hour later than Devi was.

How many meters long was the path?

A) 10

B) 15

C) 8

D) None of these

S2) B.

A few things are evident from the current situation:

Deduction 1. Since Charu rests for 1 hour before Aman and Bubbly reaches the azalea, it is clear that Devi would be exactly 1 hour ahead of Bubbly when she reaches the end point.

Deduction 2. Since Devi has already reached half an hour ago when Aman and Charu reach, Bubbly must be only half an hour from her destination. Since, Bubbly is 1 meter away from the destination, she would cover this 1 meter in 30 minutes. Thus, Bubbly's speed must be 2 m/hr. This would be the slower of the two speeds. This would also be the speed of Bubbly and Aman for the first 8 meters.

Deduction 3. Since, Bubbly and Aman take 4 hours to reach 8 meters, in 1 hour more (when Charu rests) they will cover 2 meters more. Thus, the azalea is at a distance of 10 meters from the starting point.

Deduction 4. The 10 meter distance to the azalea is covered in 4 hours by Charu and Devi. Thus, the faster speed would be 2.5 m/hr.

Deduction 5. After the azalea, when Aman and Charu start moving at 2.5 m/hr, Bubbly would be moving at 2 m/hr. When Aman and Charu reach the end, Bubbly is 1 meter behind. This gap can only be created in a 2 hour journey. Thus, the distance from the azalea to the end point is 5 meter. Total distance is 10 + 5 = 15 meters.

Q3)

During a game of five card draw poker, you are dealt a hand with the following characteristics:

(a) It contains no aces or face cards.

(b) No two cards have the same value.

(c) All four suits are present.

(d) The total value of the odd cards equals the total value of the even cards.

(e) There are no three card straights.

(f) The total value of the black cards is 10.

(g) The total value of the hearts is 14.

(h) The card with the lowest value is a spade.

Exactly what are the five cards in your hand?

A) 2 of spades, 5 and 9 of diamonds, 4 of spades and 8 of clubs

B) 2 of diamonds, 5 and 9 of spades, 4 of hearts and 8 of clubs

C) 2 of clubs, 5 and 9 of hearts, 4 of diamonds and 8 of spades

D) 2 of spades, 5 and 9 of hearts, 4 of diamonds, 8 of clubs

S3) D.

Deduction 1. Since there are no aces or face cards, we only have 2, 3, 4, 5, 6, 7, 8, 9 or 10 as the value of the cards.

Deduction 2. Since total of odd = total of evens and the minimum even total is 12 (2 + 4 + 6), there must be 2 odds and 3 evens. Only then can the two odds add up to an even total and be equal to the total of 3 even cards.

Deduction 3. There are 2 cards of 1 suit and 1 card each of the other 3 suits. The two cards belonging to 1 suit must be hearts because there is no other way for the hearts to total up to 14.

Deduction 4. There are 3 cases:

a) If the sum of the evens is 12, the odds would be 5 and 7 and the evens would be 2, 4, 6. This cannot happen because in such a case 2 cards would not add up to 14— which is the requirement for the 2 hearts.

b) If the sum of the evens is 14, the odds would be 5 and 9 and the evens would be 2,4,8. This can happen if we put the hearts as 5 and 9 and 2 goes to spades, and since the total of the blacks is 10, clubs must be the 8 of clubs and 4 would be the 4 of diamonds.

c) If the sum of the evens is 16, the evens should be 2, 4 and 10; the odds must be 7 and 9. But this is not possible because if we take the two hearts to total 14, it is not possible.

Thus, the solution is: 2 of spades, 5 and 9 of hearts, 4 of diamonds, 8 of clubs.

Q4)

In a class of 540 students, for every 9 girls there are 11 boys. The weight of students varies from 40 to 50 kg. There are as many 44 kg girls as there are 46 kg boys and as many 40 kg boys as 50 kg girls. The number of 50 kg boys is 35 more than the 44 kg girls while there are as many 44 kg boys as 46 kg girls. The ratio of 40 kg boys and girls is 4:3 while that of 50 kg girls and boys is 1:3.

1. How many boys weigh 40 kg?

(A) 22

(B) 24

(C) 28

(D) None of these

2. How many girls weigh 44 kg?

(A) 37

(B) 36

(C) 39

(D) None of these

3. How many girls weight 46 kg?

(A) 165

(B) 164

(C) 146

(D) None of these

4. The number of boys weighing 50 kg is:

(A) 72

(B) 74

(C) 76

(D) None of these

5. The number of girls weighing 40 kg is:

(A) 16

(B) 18

(C) 22

(D) None of these

6. The number of students weighing 50 kg is:

(A) 96

(B) 42

(C) 201

(D) None of these

S4)

Weight	Boys	Girls
40	(a+35)/3	(a+35)/4
44	A	a
46	B	b
50	a+35	(a+35)/3
Total	297	243

Start from the fourth line and take ‘a’ as the number of girls of 44 kgs.

1) B
2) A
3) B
4) A
5) B
6) A

Q5)

In November the answers of a prestigious test held nationwide were leaked to a group of unscrupulous people. The CBI has arrested the Don, the mastermind behind it and nine other people—P, Q, R, S, T, U, V, W and X in this matter. On interrogation, certain facts came into light:

Their modus operandi consisted of the Don initially obtaining the answer key, and then the other nine persons created their answer keys in the following manner: They obtained the answer key from one or two sources, then he/she compares the answer keys to a question from both sources. If the key to a question from both sources is identical, it is copied, otherwise it is left blank. If the person has only one source, he/she copies the source's answer into his/her copy. Finally, each person compulsorily replaces one of the answers (not a blank one) with a wrong answer in his/her answer key. The paper contained 150 questions. So the CBI has ruled out the possibility of two or more of them introducing wrong answer to the same question. The CBI has a copy of correct answer key and tabulated the following data. The data represents question numbers.

Name	Wrong Answer(s)	Blank Answer (s)
P	46	----------
Q	96	46, 90, 25
R	27, 56	17, 46, 90
S	17	----------
T	46, 90	----------
U	14, 46	92, 90
V	25	----------
W	46, 92	----------
X	27	17, 46, 90

1. Who among the following must have two sources?

A) P

B) Q

C) R

D) S

2. How many people (excluding the Don) needed to make answer keys before R could make his answer key?

A) 3

B) 4

C) 6

D) None of these

3. Both T and W were sources to:

A) U

B) X

C) Q

D) None of these

4. Which of the following is definitely true?

A) R introduced the wrong answer to question 27.

B) T introduced the wrong answer to question 46.

C) U introduced the wrong answer to question 14.

D) W introduced the wrong answer to question 46.

5. Which of the group has the same sources?

i. P, S & V ii. T and W

A) Only (i)

B) Only (ii)

C) both (i) & (ii)

D) None of these

S5)

Paragraph 1	Ten people P, Q, R, S, T, U, V, W, X and the Mastermind 3
Paragraph 2 & 3	Two ways of creating an answer key: ONE SOURCE: Copy entire answer key and introduce 1 wrong answer. Deduction: If you have one source, then you introduce only 1 wrong answer and carry over any wrong answers as well as any blanks from the answer key of the source.
	TWO SOURCES: Introduce blank if the two answer keys differ on one particular question. [This means that if one answer key has the answer correct and the other answer key has it wrong then we introduce a blank]. Note: If both answer keys are correct or if both answer keys are incorrect on a particular question, then they will give us the same answer and hence will not differ. Consequently their answer will be copied into the answer key 'under construction'.
Paragraph 4	Note also that since it is given that two or more people have not introduced a wrong answer to the same question, we can deduce that if two answer keys have the same answer wrong, they will be showing the same incorrect answer to that question. Consequently if someone has two sources who have the wrong answer to the same question both of them will show the same incorrect answer to that particular question and that answer will get replicated as it is, into the answer key which is constructed using them both. With this understanding we move to the information contained in the table.
Deductions from the table	The first thing we see when we see the table is that P, S and V have only 1 incorrect answer and no blanks. A little thought will give you that this can happen only if there is a single all-correct answer key as the source. Hence, P, S & V must have had the Don as their source and further that P must have introduced the wrong answer to Q 46. S must have introduced the wrong answer to Q 17 & V must have introduced the wrong answer to Q 25. At this point we also know that— In case of 1 source, there will be the introduction of only 1 extra wrong answer and that blanks can only be introduced if there are two sources. From this point on move ahead in the question using two main objectives— (a) Decoding the answer key patterns of the remaining six people (Q, R, T, U, W & X) and (b) Trying to decode the introduction of the remaining wrong answers and the blanks. From the table, T and W are pretty easy to decode. P must have been the source for both of them and T must have introduced the wrong answer to Q 90, while W must have introduced the wrong answer to Q 92.

At this point the table would look like:

Source	Wrong Answers	Blank answers
Don	P 46	--
	Q 96	46, 90, 25
	R 27, 56	17, 46, 90
Don	S 17	--
P	T 46, 90	
	U 14, 46	92, 90
Don	V 25	--
P	W 46, 92	--
	X 27	17, 46, 90

Q's Answer Key: He must have had T and V as his sources. In such a case his answer to Questions 25, 46 and 90 would remain blank and he would introduce the wrong answer to Q. 96.

We are now left with R, U & X and need to think how these answer keys could have been created.

It is evident that X must have been R's only source (since the blanks are just carried forward by R and he has introduced the wrong answer to Q. 56).

Further, X's answer key could have been formed if he had his sources as S and T. U's answer key could have been created only if his sources were T and W. In that case he would introduce the blank answers 90 and 92, copy the incorrect answer 46 as it is (since there would be no mismatch in that answer) and introduce the wrong answer to Q. 14. Thus the final table would look like:

Source	Wrong Answers	Blank answers
Don	P 46	--
T and V	Q 96	46, 90, 25
X	R 27, 56	17, 46, 90
Don	S 17	--
P	T 46, 90	
T and W	U 14, 46	92, 90
Don	V 25	--
P	W 46, 92	--
S and T	X 27	17, 46, 90

Thus the answers to the questions are:

1. (B)

2. P, T, S, X. Hence 4, Option (B).

3. U. Option (A)

4. (C)

5. (C)

Q6)

In a housing society, the newspaper boy keeps data of readers and notes down the sales figures of different newspapers and the gender of newspaper readers. The data is kept on a copy but unfortunately lost certain portions of the dairy. He has the following incomplete data:

	Different Newspapers			Total
	Sandesh	Gujarat Samachar	Ahmedabad mirror	
Females				16
Males			5	
Total		15		

Some more information that he remembered about the readers:

• Each reader only read one paper

• Half the readers either read Gujarat Samachar or Ahmedabad Mirror

• 40% of the total readers are female

• One third of the male readers read Sandesh. Also, there are female Sandesh readers.

1. What is the percentage of Ahmedabad Mirror readers among the overall readers?

(a) 12.5%

(b) 5%

(c) 0

(d) cannot be determined

2. How many of the readers were both female and read the Ahmedabad Mirror?

(a) 0

(b) 8

(c) 4

(d) cannot be determined

3. How many of the readers are both male and read the Gujarat Samachar?

(a) 8

(b) 5

(c) 11

(d) cannot be determined

4. Among the Sandesh readers, what is the ratio of male to female persons?

(a) 3:2

(b) 1:2

(c) 2:3

(d) cannot be determined

5. What proportion of Ahmedabad Mirror readers to the Gujarat Samachar readers?

(a) 0.33

(b) 0.5

(c) 0.25

(d) cannot be determined

S6)

We know the total number of female readers (16) and that they constitute 40% of the total readers. This means the total number of readers is 40. Also, half of the readers either read Gujarat Samachar or Ahmedabad Mirror. Number of people who read The Sandesh = Gujarat Samachar readers + Ahmedabad Mirror readers = 20 each

Therefore, the number of Ahmedabad Mirror readers = 20 -15 = 5

Now we have that 1/3 rd of the male readers read Sandesh. i.e. = 1/3 x 24 = 8

Since a total of 24 males readers are there and 13 out of these belong to Sandesh and Ahmedabad Mirror, the balance have to belong to Gujarat Samachar = 24 – 8 – 5 = 11
Now total number of readers who read Sandesh=20
Therefore, number of female readers of Sandesh=12
This means the remaining four female readers read Gujarat Samachar. Thus, we have following table for the given information:

	Different Newspapers			Total
	Sandesh	Gujarat Samachar	Ahmedabad mirror	
Females	12	4	0	16
Males	8	11	5	24
Total	20	15	5	40

Answer 1 : (a) The required percentage is:5/40 x 100=12.5%
Answer 2 : (a) We can see from the table above that the answer is 0.
Answer 3 : (c) We can see from the table above that the answer is 11.
Answer 4 : (c) We can see from the table above that the answer is 2:3.
Answer 5 : (a) The required proportion is= 5/15=0.33

Q7)
Sixteen teams have been invited to participate in the ABC Gold Cup cricket tournament. The tournament was conducted in two stages. In the first stage, teams are divided in two groups. Each group consists of eight teams, with each team playing every other team in its group exactly once. At the end of first stage, top four teams from each group advance to the second stage while the rest are eliminated. The second stage comprised several rounds. A round involves one match for each team. The winner of a match in a round advances to the next round, while the loser is eliminated. The team that remains undefeated in the second stage is declared the winner and claims the Gold Cup. The tournament rules are such that each match results in a winner and a loser with no possibility of a tie. In the first stage, a team earns one point for each win and no points for a loss. At the end of the first stage, teams in each group are ranked on the basis of total points to determine the qualifiers advancing to the next stage. Ties are resolved by a series of complex tie-breaking rules so that exactly four teams from each group advance to the next stage.

1. The minimum number of wins needed for a team in the first stage to guarantee its advancement to the next stage is:

(a) 4
(b) 6
(c) 5
(d) 7

2. What is the total number of matches played in the tournament?
(a) 55
(b) 35
(c) 63
(d) 28

3. Which of the following statements is true?
(a) The number of teams with exactly one win in the second stage of the tournament is 4.
(b) The winner will have more wins than any other team in the tournament.
(c) It is possible that the winner will have the same number of wins in the entire tournament as a team eliminated at the end of the first stage.
(d) At the end of the first stage, no team eliminated from the tournament will have more wins than any of the teams qualifying for the second stage.

4. What is the highest number of wins for a team in the first stage in spite of which it would be eliminated at the end of first stage?
(a) 2
(b) 3
(c) 1
(d) 4

5. What is the number of rounds in the second stage of the tournament?
(a) 4
(b) 1
(c) 3
(d) 2

S7)
Answer 1: (B) Since there are 8 teams in a group and each team has to play with every other team in the group exactly once, therefore the number of matches played are [${}^{8}C_{2}$] i.e. 28 matches. Every match has a winner therefore there are 28 wins and 28 losses. To advance to the next stage only 4 teams are selected from a group.

Suppose a team wins 5 matches and similarly there may be a case where 5 teams can win 5 matches each making it 25 wins and since only 4 teams could be selected, a team winning 5 matches cannot be sure of its advancement. However, if a team wins 6 matches it is not possible for more than 4 teams to win 6 matches (as max wins can be 28). Hence it would be assured of its advancement.

Alternative Answer: The following table shows the maximum number of matches won under extreme conditions. Sign (V) denotes win sign (×) denotes defeat and sign (√) denotes a combination of team with itself (no match can be played).

Team	A	B	C	D	E	F	G	H
A	*	√	√	√	√	√	×	×
B	×	*	√	√	√	√	√	×
C	×	×	*	√	√	√	√	√
D	×	×	×	*	-	-	-	-
E	×	×	×	-	*	-	-	-
F	×	×	×	-	-	*	-	-
G	√	×	×	√	√	√	*	√
H	√	√	×	√	√	√	×	*

From the table, we find that each of A, B, C, G and H can win 5 matches hence it cannot be decided that which team will qualify for the final. Therefore, the minimum number of wins that can assume a place in the second stage is 6.

Answer 2: (C) There will be eight teams in each group. Each team in a group will play with every other team. Hence, total number of matches will be in one group. Hence, in both the groups, there will be 56 matches. This is for the first stage game. Again, there are 8 teams in knockout rounds from which one winner emerges or 7 losers are identified. Hence, 7 more matches will be played. So, total number of matches played = (56 + 7) = 63.

Answer 3: (C) On the basis of the information, option (c) is the only true statement.

Answer 4: (D) The highest number of wins for a team is 4.

Answer 5: (C) Since there are 8 teams, there would be 7 matches in 3 rounds.

Q8)

Recently, Bosch Babu spent his winter vacation on Hawaai Island. During the vacation, he visited the local casino where he came across a new card game. Two athletes, using a normal deck of 52 playing cards, play this game. One athlete is called the ‘broker‘ and the other is called the ‘athlete‘. First, the athlete picks a card at random from the deck. This is called the base card. The amount in rupees equal to the face value of the base card is called the base amount. The face values of ace, king, queen and jack are ten. For other cards the face value is the number on the card. Once the ‘athlete‘ picks a card from the deck, the ‘broker‘ pays him the base amount. Then, the ‘broker‘ picks a card from the deck and this card is called the top card. If the top card is of the same suit as the base card, the ‘athlete‘ pays twice the base amount to the ‘broker‘. If the top card is of the same colour as the base card (but not the same suit), then the ‘athlete‘ pays the base amount to the ‘broker‘. If the top card happens to be of a different colour than the base card, the ‘broker‘ pays the base amount to the ‘athlete‘.

Bosch Babu played the game four times. First time he picked eight of clubs and the ‘broker‘ picked queen of clubs. Second time, he picked ten of hearts and the ‘broker‘ picked two of spades. Next time, Bosch Babu picked six of diamonds and the ‘broker‘ picked ace of hearts. Lastly, he picked eight of spades and the ‘broker‘ picked jack of spades. Answer the following questions based on these four games.

1. The initial money Bosch Babu had (before the beginning of the game sessions) was Rs. X. At no point did he have to borrow any money, the minimum possible value of X

(a) 100

(b) 16

(c) 8

(d) 24

2. If Bosch Babu stopped playing the game on maximizing his gain, what would be the gain?

(a) Rs. 12

(b) Rs. 4

(c) Rs. 20

(d) Rs. 16

3. If the final amount of money that Bosch Babu had with him was Rs. 100, the initial amount he had with him is

(a) 96

(b) 8

(c) 120

(d) 4

S8)

The information given in the question can be presented in the tabular form as under

Game Round	Athlete's Account		Broker's Account	
	Result of game	Result after game	Result of game	Result after game
I	+8 -16= -8	-8	-8 + 16	8
II	+10 +10= 20	-8 + 20= 12	-10 – 10 = -20	+8 – 20= -12
III	+6 - 6=0	12+ 0 =12	-6 + 6= 0	-12 + 0= -12
IV	+8 -16= -8	12 – 8= 4	-8 + 16= 8	-12 + 6= -6

Answer 1: (C)

The minimum amount has to be the negative balance of Rs.8 that was left after the first round.

Answer 2: (A)

The maximum gain of Bosch Babu is Rs. 12 which is same after second and third round of game.

Answer 3: (A)

Let initial amount of money which he had with him be Rs. X. Bosch has gained Rs 4 by playing the game. X + 4 = 100 or X = Rs. 96.

Q9)

Ram Roy was the head of a Steel manufacturing company trying to develop a new drug. Roy and his friend Prashant Agrawal, assessed that such products had mixed success. Roy and Agrawal realized that if a new drug was a success, it may result in sales of 100 crores but if unsuccessful, the sales may be only 20 crores. They further assessed that a new drug was likely to be successful 50% of times. Cost of launching the new drug was likely to be 50 crores. Now, Roy and Agrawal were in a quandary whether company should go ahead and market the drug. They contacted Raj Adhikar, a common friend for advice. Adhikar was of the opinion that given the risky nature of launch, it may be a better idea to test the market. Agrawal and Roy realized test marketing would cost 10 crores. Adhikar told them the previous test marketing results have been favorable 70% of times and success rate of products favorably tested was 80%. Further, when test marketing results were unfavorable; the product had been successful 30% of times.

1. How much profit can the company expect to earn if it launches the new drug (suppose there are no additional costs)?

A) 12 crores
B) 10 crores
C) 10.5 crores

D) 11 crores
E) 11.5 crores
2. If Agrawal and Roy decides to launch the product despite unfavorable test marketing, how much profit can the company expect to earn?
A) 10 crores
B) –13.2 crores
C) –36.8 crores
D) –46.8 crores
E) –16 crores
3. How much profit can the company expect to make if the product is launched after favorable test marketing results.(assume there are no additional costs)?
A. 13.8 crores
B. 11.5 crores
C. 8.5 crores
D. 10 crores
E. 24 crores
4. What is the probability of product failure if Roy and Agrawal decides to test market it?
A. 0.14
B. 0.35
C. 0.28
D. 0.21
E. Cannot be computed

S9)
Answer 1: (B)
The cost of launching the product = 50 crores
The probability of the success of the product is 50%.
Hence, the profit = (50 % of 100 + 50% of 20) – 50 = 10 crores.
Answer 2: (E)
Required profit = (30% of 100 + 70% of 20) – (50 + 10) = –16 crores.
Answer 3: (E)
The correct answer is not present among the options.
The success rate of a product if its results are favorable = 80%
Total cost = 50 + 10 = 60 crores
Sales = 80% of 100 + 20 % of 20 = 84 crore
Hence, the profit = 84 – 60 = 24 crores.
Answer 4: (B)
Required probability = 0.70 × 0.20 + 0.30 × 0.70 = 0.14 + 0.21 =0.35.

5. PUZZLE TEST

Reasoning puzzles are a favorite question type in all aptitude examinations. In the scheme of chapters on reasoning contained in this book, we have created separate chapters for specific kinds of puzzles which are often asked (like arrangements, rankings etc., which constituted the chapters prior to this one). All other categories of puzzles—which cannot be specifically categorized as any of the foregoing chapters—you will get to see and practice in this chapter. As the name suggests, questions on puzzles challenge you to match multiple factors (like name, color of shirt, place of living, car model driven, etc.).The key skills involved in solving questions on puzzles include but are not limited to:

(i) The ability to make a relevant tabular structure for using the clues seamlessly: For example, suppose you have 5 people A, B, C, D, E wearing 5 color of shirts red, yellow, green, blue and white drinking 5 kinds of soft drinks Coke, Pepsi, Mirinda, Thums Up and Seven Up—the solution table structure would look like this:

	Red	Yellow	Green	Blue	White	Coke	Pepsi	Mirinda	Thums Up	Seven Up
A										
B										
C										
D										
E										
Coke										
Pepsi										
Mirinda										
Thums Up										
Seven Up										

In the table above, you can see clearly that there is a direct correlation structure between each of the three ‘variables’ in the problem. More of this you would get to see and experience as you moves through the solved illustrations and the exercises that follow it.

(ii) The ability to order the clues in the correct order of usage (as explained in the theory of logical reasoning): This includes the ability to perceive the direct clues and use them first to set up the ‘framework’ of problem solving.

(iii) The ability to perceive what indirect clues are talking about – and how to use them;

(iv) The ability to convert clues written in language form into visual cues so that you do not need to read the text again and again. Also, converting the language clues to visual cues is critical for the purpose of being able to ‘see’ all the clues at one go.

Illustrated below are the solutions to a few typical questions on puzzles. We would advise you to first have a look at the questions and try to solve the same on your own before looking at the step-by-step process of solving the same.

Example 1)

At a fancy dress party people were asked to dress as an object that represented their professions.

Mr. Abhijit, Mrs. Banerjee, Mrs. Chatterjee, Mr. Dipanjan De and Mr. Elangovan were among the guests. The costumes included a leaf, a pen, a fork, a camera reel, and a stethoscope. The professions included a photographer, a gardener, a compounder, a teacher, and a cook.

1. Mr. Abhijit is a teacher.
2. Neither Mrs. Banerjee nor Mrs. Chatterjee was dressed as a fork.
3. None of the men is a compounder.
4. Mr. Dipanjan De is dressed as a camera reel.
5. Mrs. Chatterjee is a gardener.

1. Which person is dressed as a stethoscope?

A) Abhijit

B) Banerjee

C) Chatterjee

D) Dipanjan

2. What is Elangovan’s profession?

A) Cook

B) Gardener

C) Compounder

D) Teacher

S1)

Putting the direct Clues 1 and 4 in the table we get:

Male	Female	Female	Male	Male
Abhijit	Banerjee	Chatterjee	Dipanjan	De Elangovan
Pen			Camera reel	
Teacher			Photographer	
From Clue 1			From Clue 4	

At this point if we use Clue 2, it is evident that it must be Elangovan who is dressed as a fork. Also using clue 5 (Mrs. Chatterjee is a gardener) completes the solution.

Male	Female	Female	Male	Male
Abhijit	Banerjee	Chatterjee	Dipanjan	De Elangovan
Pen		Leaf	Camera reel	
Teacher		Gardener	Photographer	
From Clue 1		From Clue 5		From Clue 2 once we know the details of Dipanjan and Abhijit.

Obviously, Mrs. Banerjee would be the compounder.

The solutions are: 1. Banerjee Option (b)

2. Elangovan is the cook. Option (a) is correct.

Example 2) Read the information and answer the question.

Four engineers, designated as CE, SE, EE and AE, read a certain number of newspapers early in the morning. One of them reads four newspapers, another read three newspapers, the third reads two newspapers while the fourth one reads one newspaper. Below are some additional facts regarding the names of these officers:

i. Nahiri is not the EE.

ii. Hari is the AE.

iii. Nahiri is not the CE and he reads more number of newspapers than Lahiri.

iv. The one who is the CE reads more number of newspapers than Lahiri.

v. The person who is the SE reads the maximum number of newspapers.

vi. Bahiri does not read two newspapers.

1. Which of the following statements is necessarily true?

(a) Hari is the AE and reads two newspapers.

(b) Lahiri is the EE and reads one newspaper.

(c) Bahiri is the CE and reads three newspapers.

(d) Nahiri is the EE and reads four newspapers.

S2)

From statement (ii), Hari is AE. From Statements (i) and (iii), Nahiri is not the CE or the EE. Hence Nahiri is the SE and reads 4 newspapers as from Statement (v). From Statement (iv), Lahiri is not the CE; this obviously means that Lahiri is the EE and Bahiri is the CE. From Statement (iv) and (vi), Bahiri reads three newspapers. As he reads more than at least one person, we cannot allocate one newspaper to him. Hence Bahiri must read three newspapers. Hence the final distribution is as follows:

Engineer	Name	No. of Newspapers
AE	Hari	2/1
CE	Bahiri	3
SE	Nahiri	4
EE	Lahiri	1/2

Hence, option (c) is the correct answer.

Example 3)

Read the information and answer the questions.

Four brothers Rohan, Sohan, Mohan and Ganesh are at their annual family property fight sitting across a circular table. Their occupations are— author, biologist, chemist and doctor, but not necessarily in that order. Ganesh starts by setting the agenda of the meeting and after him the doctor gives a long discourse of what is right and what is wrong. Rohan is sitting across the doctor and next to the chemist. Mohan is silent throughout the meeting and the chemist speaks only at the very end.

1. The profession of Rohan is

(a) Author

(b) Biologist

(c) Doctor

(d) Cannot be determined

2. Who among the following is the chemist?

(a) Sohan

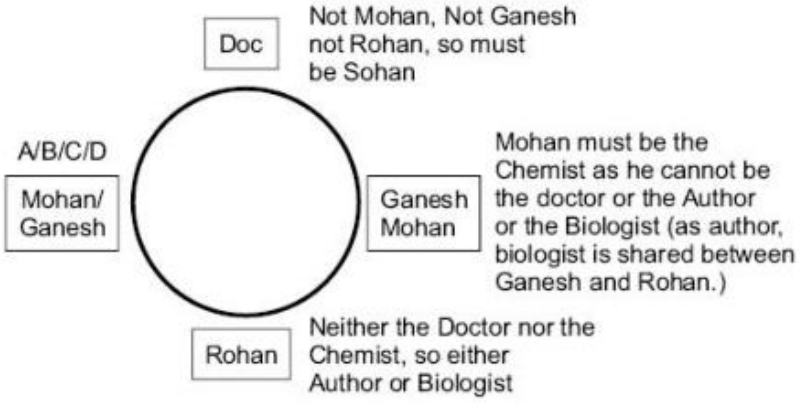
(b) Mohan

(c) Ganesh

(d) Cannot be determined

S3)

The solution of the above question can be seen in the figure below. (The interpretation of Mohan's silence throughout the meeting is that Mohan spoke only at the end.)



Based on the interpretations of the figure above we can answer the questions as:

1. Rohan could be the Author or the biologist. Hence, we cannot determine his profession. Option (d) is correct.

2. Mohan was the Chemist. Option (b) is correct.

Example 4)

For admission into vacant seats of higher secondary classes, most schools prescribe an admission test. There is no uniformity in the nature of these tests, however. Anuja Sinha, Amit Agnihotri, Himesh Reshammiya, Mahima Sharma and Sonit Kala, whose parents had been transferred to Lucknow this summer and who were all seeking admission into Class 9, appeared in entrance tests of different schools. The number of questions in all these tests happened to be different. Amit wrote an entrance test containing exactly twice as many questions as the test that Himesh wrote, Sonit wrote a test that contained 160 questions but it was neither for admission into City Montessori School's nor for Scindia School. Delhi Public School test contained 200 questions but neither Himesh nor Mahima wrote this test. City Montessori School test had 120 questions but it was not written by Himesh. The test that Mahima Sharma wrote had 25% less questions than the test Amit Agnihotri wrote but it was not for City Montessori School and Scindia School. Anuja did not write Millennium School's or Saint Mary's Convent's test. When all the tests are compared with regard to number of questions in them, St. Mary's Convent falls exactly in the middle of the list. Each of the five students wrote exactly one of the tests – the longest one of which contained 200 questions. There is no penalty for wrong answers.

1. Anuja Sinha wrote the test for admission into which school?

(a) Scindia

(b) Sanskar Valley

(c) City Montessori School

(d) Delhi Public School

2. If Amit Agnihotri secured only 50 per cent marks in the test, each question carrying one mark, how many marks did he secure in this test?

(a) 50

(b) 60

(c) 100

(d) None of these

3. If both Sonit and Amit secured equal marks in their respective tests and the test Sonit wrote carried 5 marks per question, how many marks did each question in Amit's test carry? It is given that both Amit and Sonit answered 75% questions of their respective tests correctly. Wrong answers did not fetch any marks.

(a) 1

(b) 2.5

(c) 3

(d) 4

4. Which of the following combinations is true?

(a) Anjua Sinha – City Montessori School – 180 Questions

(b) Amit Agnihotri – Millennium School – 160 Questions

(c) Himesh Reshammiya – Scindia School – 100 Questions

(d) Mahima Sharma – Scindia School – 150 Questions

S4)

Reaction Tracker

With respect to the number of questions in each test it can be inferred that the number of questions are 200,120,160. Two values are unknown and to compensate for them we have been given that the number of questions in the tests of Amit Agnihotri would be (2x), Himesh Reshammiya (x) and Mahima Sharma (1.5x). The starting table for this situation would be:

	Anuja Sinha	Amit Agnihotri (2x)	Himesh R (x)	Mahima Sharma (1.5 x)	Sonit Kapoor	CMS	Scindia	Millenium	St. Marys	DPS
200			NO	NO	NO	NO	NO	NO	NO	YES
160	NO	NO	NO	NO	YES	NO	NO			NO
120			NO		NO	YES	NO	NO	NO	NO
??					NO	NO				NO

??					NO	NO				NO
CMS			NO	NO	NO					
Scindia				NO	NO					
Millennium	NO									
St. Mary's	NO									
DPS			NO	NO	NO					

From the table it is clear that DPS (200) and CMS (120) are shared between Anuja and Amit. This means two possibilities – Anuja 120 and Amit 200 or Amit 120 and Anuja 200. If we take the second possibility into account things would not work out because: If Amit is 120, the missing numbers are 90 (25% less for Mahima) and 60 (Amit's test had twice Himesh's test). Then the number of questions placed in ascending order would be: 60, 90, 120 (CMS), 160, 200 (DPS). This situation contradicts the condition that when all the tests are compared as to number of questions in them, St. Mary's Convent falls exactly in the middle of the list. Thus, we must consider the other option, i.e., Amit had 200 questions and Anuja 120. Then the number of questions is 200, 160, 150 (Mahima), 120, 100. The table would now look as follows:

	Anuja Sinha	Amit Agnihotri (2x)	Himesh R (x)	Mahima Sharma (1.5 x)	Sonit Kapoor	CMS	Scindia	Millenium	St. Marys	DPS
200	NO	YES	NO	NO	NO	NO	NO	NO	NO	YES
160	NO	NO	NO	NO	YES	NO	NO	YES	NO	NO
120	YES	NO	NO	NO	NO	YES	NO	NO	NO	NO
150	NO	NO	NO	YES	NO	NO	NO	NO	YES	NO
100	NO	NO	YES	NO	NO	NO	YES	NO	NO	NO
CMS	YES	NO	NO	NO	NO					
Scindia	NO	NO	YES	NO	NO					
Millennium	NO	NO	NO	NO						
St. Mary's	NO	NO	NO	YES	NO					
DPS	NO	YES	NO	NO	NO					

The solutions are:

1. CMS. Option (c) is correct.

2. He would score 100 marks. Option (c) is correct.

3. Sonit & Amit would score 600 marks out of 800 (The total marks would be 800 because Sonit's test has 160 questions of 5 marks each totaling 800). Hence, Amit's score would also be out of 800 marks—4 marks per question. Option (d) is correct.

4. Option (c) is correct.

Example 5)

Study the following information carefully and answer the questions given below:

Four people were being interviewed for the same job. They were all interviewed on the same day, but in different rooms (Dharti, Prithvi, Akash and Agni), at different times and by different persons. Determine the name of each candidate, which room they were interviewed in and their appointment time and answer the questions.

(i) Aishwarya's appointment was just after Mr. Sharma's, which was just after that of the person in room Prithvi.

(ii) Mr Narurkar's appointment was at least two hours later in the day than Bhagat's.

(iii) Mr Joshi's appointment was just after the person who had an interview in room Agni, who had an appointment just after Chiranjeev.

(iv) Three of the four interviewees were: Dhanush, the one with the interview in room Dharti, and the person who had an appointment at 1 pm.

(v) The four people were interviewed at 11 am, 12 noon, 1 pm and 2 pm.

(vi) Joshi, Narurkar, Zaidi and Sharma were the interviewers whereas Aishwarya, Bhagat, Chiranjeev and Dhanush were the interviewees.

1. Sharma's appointment was with

(a) Aishwarya

(b) Bhagat

(c) Chiranjeev

(d) Can't be determined

2. Dhanush's appointment was in the room

(a) Dharti

(b) Prithvi

(c) Akash

(d) Agni

(e) Can’t be determined

3. Zaidi’s appointment was in the room

(a) Dharti

(b) Prithvi

(c) Akash

(d) Agni

(e) Can’t be determined

S5)

In order to solve this question the starting table should be like this.

			A/B/C/D	A/B/C/D	A/B/C/D	A/B/C/D
			J/N/Z/S	J/N/Z/S	J/N/Z/S	J/N/Z/S
			11 am	12 noon	1 pm	2 pm
A/B/C/D	J/N/Z/S	Dharti				
A/B/C/D	J/N/Z/S	Prithvi				
A/B/C/D	J/N/Z/S	Akash				
A/B/C/D	J/N/Z/S	Agni				

The first clue gives us: Prithvi–Mr. Sharma–Aishwarya. This means that these three can only be at 11,12 and 1 pm or 12,1 or 2 pm.

The table evolves to:

			B/C/D	B/C/D	A/B/C/D	B/C/D
			J/N/Z	J/N/Z/S	J/N/Z/S	J/N/Z
			11 am	12 noon	1 pm	2 pm
A/B/C/D	J/N/Z/S	Dharti				
A/B/C/D	J/N/Z/S	Prithvi			x	x
A/B/C/D	J/N/Z/S	Akash				
A/B/C/D	J/N/Z/S	Agni				

The second clue tells us that Narurkar's appointment is at least 2 hours after Bhagat's appointment. This could only mean that Bhagat is at either 11 am or 12 noon and Narurkar could only be at 1 or 2 PM. The table evolves to:

			B/C/D	B/C/D	A/C/D	C/D
			J/Z	J/Z/S	J/N/Z/S	J/N/Z
			11 am	12 noon	1 pm	2 pm
A/B/C/D	J/N/Z/S	Dharti				
A/B/C/D	J/N/Z/S	Prithvi			x	x
A/B/C/D	J/N/Z/S	Akash				
A/B/C/D	J/N/Z/S	Agni				

Clue (iii) gives us that the sequence Chiranjeev – Agni – Joshi. This again means that this sequence could only be at 11,12 and 1 PM or 12, 1 and 2 PM. Thus, Chiranjeev can only be at 11 or 12, Agni only at 12 or 1 and Joshi only at 1 or 2 pm. Introducing these deductions into the existing table we get:

			B/C/D	B/C/D	A/D	D
			Z	Z/S	J/N/Z/S	J/N/Z
			11 am	12 noon	1 pm	2 pm
A/B/C/D	J/N/Z/S	Dharti				
A/B/C/D	J/N/Z/S	Prithvi			x	x
A/B/C/D	J/N/Z/S	Akash				
A/B/C/D	J/N/Z/S	Agni	x			x

Note: In the above table we are now sure about Zaidi's position at 11 am and Dhanush's position at 2 pm. Thus we can also remove D and Z from the other possibilities. When we do that, notice that at 1 pm Aishwarya remains the only possibility left and for 12 noon Sharma remains the only possibility left. The table then becomes:

			B/C	B/C	A	D
			Z	S	J/N	J/N
			11 am	12 noon	1 pm	2 pm
A/B/C/D	J/N/Z/S	Dharti				
A/B/C/D	J/N/Z/S	Prithvi			x	x
A/B/C/D	J/N/Z/S	Akash				
A/B/C/D	J/N/Z/S	Agni	x			x

Clue (iv) tells us that: Dhanush, the person in Dharti and the 1 pm person are different people. So Dhanush is not in Dharti and Dharti is also not at 1 pm. The table changes to:

			B/C	B/C	A	D
			Z	S	N	J
			11 am	12 noon	1 pm	2 pm
B/C	Z/S	Dharti			x	X
B/C	Z/S	Prithvi			x	X
D	J	Akash	x	x	x	Definite
A	N	Agni	x	x	definite	X

From this point if we go back to Clue (iii) we get: Joshi would be at 2 pm (as he is just after the person who had an interview in Agni— which is fixed at 1 pm). Also, Chiranjeev must be at 12 noon as he is just before Agni. The table evolves to:

			B	C	A	D
			Z	S	N	J
			11 am	12 noon	1 pm	2 pm
B/C	Z/S	Dharti			x	x
B/C	Z/S	Prithvi			x	x
D	J	Akash	x	x	x	definite
A	N	Agni	x	x	definite	x

The only thing left at this stage is: Prithvi was just before Sharma. So Prithvi is at 12 noon.

The table closes to:

			B	C	A	D
			Z	S	N	J
			11 am	12 noon	1 pm	2 pm
C	S	Dharti	x	definite	x	x
B	Z	Prithvi	definite		x	x
D	J	Akash	x	x	x	definite
A	N	Agni	x	x	definite	x

The answers become:

1. Sharma's appointment was with Chiranjeev. Option (c) is correct.

2. Akash. Option (c) is correct.

3. Prithvi. Option (b) is correct.

Example 6)

Six friends A, B, C, D, E, and F work in different companies namely—Pentasoft, Quark, Raymond's, Sunmet, Trump & Gates and Udupi, and each wears different coloured, company-sponsored shirts, viz., blue, green, pink, yellow, purple and red though not necessarily in the same order.

i. The one wearing the blue shirt works in Sunmet and the one wearing a green shirt works in Pentasoft.

ii. F does not work in Raymond's or Trump and Gates.

iii. A wears pink shirt and works in Quark.

iv. D does not work in Trump & Gates and purple coloured shirt is not sponsored by Raymond's.

v. E works in Udupi and neither D nor B works in Sunmet.

vi. Trump & Gates does not sponsor purple or yellow coloured shirts and C works in Pentasoft.

1. Which colour shirt is sponsored by Raymond's?

(a) Yellow (b) Blue

(c) Pink (d) Cannot be determined

2. Which pair is correctly matched?

(a) Red–Raymond's–A

(b) Red–Trump & Gates–B

(c) Green–Raymond's–C

(d) None of these

3. Which of the following is true?

(a) Udupi sponsors green shirts.

(b) D is working in Trump & Gates.

(c) E wears red shirt.

(d) Red shirt is sponsored by Trump & Gates.

4. What is the sequence of companies representing A, B, C, D, E and F?

(a) Quark, Pentasoft, Trump & Gates, Raymond's, Udupi, Sunmet

(b) Quark, Trump & Gates, Pentasoft, Raymond's, Udupi, Sunmet

(c) Quark, Pentasoft, Trump & Gates, Sunmet, Udupi, Raymond's

(d) None of these

5. If Raymond's and Sunmet decide to interchange the colours of sponsored shirts then which two persons will have to interchange their shirts?

(a) D and F (b) A and C

(c) D and E (d) B and D

S6)

The direct clues give you the following linkages—green – Pentasoft, Blue– Sunmet A– Pink – Quark, E – Udupi, C – Pentasoft.

Combining these clues will give you the following table:

A	B	C	D	E	F
Quark		Pentasoft	Udupi		
Pink		Green			

At this point use Clue no. (ii) [about F]. This will give you that the correct pairing for Sunmet is with F. Also use Clue (iv) at this stage.

The table would evolve to:

A	B	C	D	E	F
Quark	Trump & Gates	Pentasoft	Raymonds	Udupi	Summet
Pink	Green				Blue

Now using Clue (vi), you will get that Trump & Gates is red and since purple is not with Raymonds, the final table will be

A	B	C	D	E	F
Quark	Trump & Gates	Pentasoft	Raymonds	Udupi	Summet
Pink	Green	Green	Yellow	Purple	Blue

From this point, you need to just read off the respective answers to the questions asked from the completed table.

The answers are:

1. Yellow. Option (a)

2. Option (b) is correct.

3. Option (d) is correct.

4. Option (c) is correct.

5. Option (a) is correct.

Example 7)

Directions:

i. Seven friends A, B, C, D, E, F and G are in Patna to attend a seminar at Mindworkzz. Five of them have to go back to five different places—Delhi, Chennai, Lucknow, Bangalore, and Kolkata.

ii. Five of them are executives, each specializing in Administration, Human Resource Management (HRM), Marketing, Systems and Finance.

iii. E, an executive is going to Chennai, is neither from Finance nor Marketing. G is a system specialist and is leaving for Delhi. F is an executive but is not going to one of the five places.

iv. B is an executive from HRM but has come at the airport to see off his friends. A is an executive but not from Marketing and is flying to one of the destinations but not to Bangalore or Kolkata.

1. The one who is going to fly to Chennai is:

(a) Not an executive

(b) From Administration

(c) From Systems

(d) From Finance

2. Who among the following specializes in Marketing?

(a) D

(b) A

(c) F

(d) G

3. C has specialized in which field?

(a) Finance

(b) Marketing

(c) Either Finance or Marketing

(d) None

4. Who is flying to Bangalore?

(a) A

(b) C

(c) D

(d) Data inadequate

S7)

From Clues (i), (ii), and (iii):

7 people–A, B, C, D, E, F & G

5 cities – Delhi, Chennai, Lucknow, Bangalore and Kolkata

5 specialisations – Admin, HRM, Marketing, Systems, Finance

From Clues (iii) and (iv) we have:

D	
E- Chennai	X Fin, X Marketing
G- Delhi	Systems
F- No place	?
B- No place	HRM
A- X Bangalore, X Kolkata	X Marketing
C-	

Refining the above table, the following deduction can be made: E must be from Admin. The table evolves to:

D	
E- Chennai	Admin
G- Delhi	Systems
F- No place	Marketing
B- No place	HRM
A- X Bangalore, X Kolkata	Finance
C-	

Hence the answers are:

1. (b)

2. (c)

3. (d)

4. (d)

Example 8)

Amitabh, Bhagyashree, Chunky, Dharmendra, Ekta, Farhan and Govinda are students of a class. Each of them has a different favourite subject, viz., Economics, Commerce, Zoology, Sociology, Statistics, Urdu and Computers but not necessarily in the same order. There are two such students whose one sister each is there in the group. There is no other relation among the students. No boy likes Commerce or Urdu. Dharmendra, who does not like Sociology and

Statistics, is the brother of that student who likes Computers. The student who likes Sociology is the sister of that boy student who likes Economics. F is a boy student, B is sister of A.

1. Which of the following is a pair of brother-sister other than Amitabh and Bhagyashree?

(a) Dharmendra and Govinda

(b) Dharmendra and Chunky

(c) Dharmendra and Ekta

(d) Data Inadequate

2. Which of the following is true?

(a) Dharmendra likes Commerce.

(b) Chunky, Bhagyashree and Dharmendra are girl students.

(c) The number of girls is more than that of the number of boys in the group.

(d) None of these

S8)

Let us tabulate the given information to get a clear picture of the scenario.

The first piece of information that fills the blanks is that no boy likes Commerce or Urdu. Then next information that is useful is that Dharmendra does not prefer Sociology and Statistics and is a boy hence he does not like Urdu and Commerce also. He is also brother of the person who opts for Computers. (Thus, the person opting for computers must be a girl). So he must be having either Economics or Zoology. Now we are given that Amitabh and Bhagyashree are brother and sister and, hence Amitabh and Bhagyashree both will not have computers. Also Farhan is given as a boy student hence, by default from previous information, he does not like Urdu and Commerce, and also since he is a boy he cannot be Dharmendra's sister and thus cannot have computers. Thus, we have covered all the possibilities for boys except for the pair of brother and sister having Economics and Sociology. Clearly since Amitabh and Bhagyashree are the only brother and sister mentioned in the information, they must be the ones having those subjects. Consequently Dharmendra must be having Zoology and Farhan would have Statistics. Commerce, Urdu and Computers would be shared between Chunky, Ekta and Govinda in some random order. Hence, they must be girls. The resultant table can be drawn as follows:

	Brother of Bhagyashree	Sister of Amitabh	Girl	Boy	Girl	Boy	Girl
	Amitabh	Bhagyashree	Chunky	Dharmendra	Ekta	Farhan	Govinda
Economics	√	X	X	X	X	X	X
Commerce	X	X	????	X	????	X	????
Zoology	X	X	X	√	X	X	X
Sociology	X	√	X	X	X	X	X
Statistics	X	X	X	X	X	√	X
Urdu	X	X	????	X	????	X	????
Computers	X	X	????	X	????	X	????

The answers are:

1. Option (d) is the correct answer as we do not know who Dharmendra's sister is.

2. Option (c) is the correct answer as there are 4 girls in the group.

Example9)
Read the following information and answer the questions that follow.

Houses numbered 1A to 4D situated east to west in that order, are each occupied by College of Commerce professors. They all teach different subjects, possess different makes and different models (years) of bikes:

(i) Anurag Kesarwani does not own a Suzuki (owner of which is professor of Gujarati language).

(ii) Anshul has a Honda bike.

(iii) Mr. Khanduja lives in House No. 3.

(iv) Mr. Singh is the professor of Sanskrit language.

(v) 2001 model of bike owned by the Urdu language professor is not of BMW make.

(vi) Vivek is the professor of Bengali language.

(vii) Mr. Saxena is not the owner of 2004 model bike, owner of which lives next to in a house westward of one owning 2002 model bike.

(viii) Siddhartha's is House No. 4.

1. Prof. Singh owns which make and model (year) of bike?

(a) Suzuki of 2001

(b) BMW of 2004

(c) Honda of 2002

(d) BMW of 2003

2. If Urdu and Sanskrit language professors exchange their bikes, who could be the owner of the latest model?

(a) Anurag only

(b) Siddhartha only

(c) Vivek only

(d) Siddhartha or Vivek only

S9)

To solve the question, we have to decode the information step by step and hence interpret the information.

Anshul owns the Honda and Vivek teaches Bengali => Gujarati teacher owning Suzuki – who isn't Anurag Kesarwani – is Siddhartha in House No. 4D (Clues (ii), (vi), (i), (viii)).

Prof. Singh who teaches Sanskrit is Anshul (Clue (iv)). Anurag Kesarwani, who teaches Urdu owns 2001 model bike which is not BMW (Clue (v)), neither is it Suzuki nor Honda – an unknown make.

So, Vivek Khanduja owns BMW in House No. 3C and Siddhartha's surname is Saxena (Clue (iii)).

Since Siddhartha's vehicle is not a 2002 or 2004 model (Clue (vii)) and neither a 2001 model – it is model of an unknown year (which could be 2005 or an earlier year).

Now 2002 model belongs to House No. 2 and 2004 model to House No. 4.

Anshul Singh's is 2002 model and Vivek Khanduja's is 2003 model.

Remaining House No. 1A is Anurag Kesarwani's.

The summary is as under

Anurag – Kesarwani – Urdu –???? – 2001

Anshul – Singh – Sanskrit – Honda – 2002

Vivek – Khanduja – Bengali – BMW – 2003

Siddhartha – Saxena – Gujrati – Suzuki – ????

1. Option (c) is the correct answer.

2. Option (d) is the correct answer.

Example 10)

Read the information and answer the questions that follow.

Five friends—Ramesh, Suresh, Tanveer, Umesh, and Vikram—each present one paper to their class on mathematics, history, biology, chemistry, or dermatology—one day a week, Monday through Friday.

(i) Vikram does not do chemistry and does not give his presentation on Tuesday.

(ii) Suresh makes the dermatology presentation, and does not do it on Monday or Friday.

(iii) The mathematics presentation is made on Thursday.

(iv) Tanveer presents his presentation, which is not on Chemistry, on Wednesday.

(v) The biology presentation is on Friday, and not by Umesh.

(vi) Ramesh makes his presentation on Monday.

1. What day is the Chemistry presentation made?

(a) Friday

(b) Monday

(c) Tuesday

(d) Wednesday

2. What presentation does Vikram do?

(a) Chemistry

(b) Dermatology

(c) Mathematics

(d) Biology

3. What day does Umesh make his presentation on?

(a) Monday

(b) Tuesday

(c) Wednesday

(d) Thursday

S10)

The following solution table would emerge out of these clues.

Day	Monday	Tuesday	Wednesday	Thursday	Friday
Person	Ramesh	Suresh	Tanveer	Umesh	Vikram
Presentation topic	Chemistry	Dermatology	History	Mathematics	Biology

Hence, the answers are:

1. Monday. Option (b)

2. Biology. Option (d)

3. Thursday. Option (d)

Example 11)

A, B, C, D, E, F and G are seven persons who travel to office everyday by a particular train which stops at five stations – I, II, III, IV and V respectively after it leaves base station. Three among them get in the train at the base station. D gets down at the next station at which F gets down. B does not get down either with A or E, G alone gets in at station III and gets down with C after one station. A travels between only two stations and gets down at station V. None of them gets in at station II. C gets in with F but does not get in with either B or D. E gets in with two others and gets down alone after D, B and D work in the same office and they get down together at station III. None of them gets down at station I.

1. At which station does E get down?
(a) II

(b) III
(c) IV
(d) Data inadequate

2. At which station do C and F get in?
(a) I
(b) II
(c) III
(d) Data inadequate

3. At which of the following stations do B and D get in?
(a) I
(b) Base station
(c) III
(d) None of these

4. After how many stations does E get down?
(a) One
(b) Two
(c)Three
(d) Four

5. E gets down after how many stations at which F gets down?
(a) Next station
(b) Two
(c) Three
(d) Four

Example 11)

Person	In	Out
A	IV	V
B	Base or I	III
C	Base or I	V
D	Base or I	III
E	Base	IV
F	Base or I	II
G	III	V

1. **(c)** E gets in with two others and gets down alone after D and we know that D get down at station 3, so option c.

2. **(d)** Since we know that E gets in at station 3 alone and there was nobody who gets in from station 2, therefore D may be get in from station number 1 or base station. So, there is data inadequate so the answer should be d, because C and F get in either at Base station or station I.

3. **(d)** we know that B and D work in the same office and they get down together at station III. It means they cannot get in at station 3 so they also may get it from Base station or Station 1. So none of these is the best answer, so option d.

4. **(c)** We know that E gets in with two others and gets down alone after D, B and D work in the same office and they get down together at station III . So E gets down after three stations. So the best answer for the question is E gets down after three stations (I, II And III).

5. **(b)** We know that D gets down at the next station at which F gets down and E gets in with two others and gets down alone after D, B and D work in the same office and they get down together at station III. From this it is clear that E gets down after two stations at which F gets down.

6. PROPOSITION, ARGUMENTS and SYLLOGISMS

This test used to judge the candidate's ability to arrive at conclusions from a study of the given statements or situations. Ability to grasp relations between ideas or between objects and to analyse and synthesise them on the basis of given statements are also inherent to logical reasoning.

A word gives us a unit of thought. For example, if we say 'Tiger', this word gives us simply a physical image of the animal called tiger, and nothing else, nor even anything about the tiger. But if we say that 'tigers are ferocious animals', we get some idea about the nature of tiger. It can become an unit of an argument. A unit of thought is called a term and a unit of argument is called a proposition, which is also expression of thought. It is also a statement showing some relation between two or more terms analogous to a sentence. Here the example, 'tigers are ferocious animals', is a proposition. Here 'tigers' (subject) is the first term and 'ferocious animals' (predicate) is the second term. The joining word 'are' is called the copula.

I. PROPOSITIONS

Proposition- A proposition is a sentence that makes a statement and gives a relation between two terms. It consists of three parts (a) the subject; (b) the predicate; (c) the relation between the subject and the predicate.

Some examples of propositions are being given below:

(i) All coasts are beaches.

(ii) No students are honest.

(iii) Some documents are secret

(iv) Some cloths are not cotton.

Subject and Predicate- A subject is that part of the proposition about which something is being said. A predicate, on the other hand, is that term of the proposition which is stated about or related to the subject.

Thus, for example, in the four propositions mentioned above, 'coasts', 'students', 'documents' and 'cloths' are subjects while 'beaches', 'honest', 'secret' and 'cotton' are predicates.

Categorical Propositions- A categorical proposition makes a direct assertion. It has no conditions attached with it. For example, "All S are P", "No S are P", "Some S are P" etc are categorical propositions, but "If S, then P" is not a categorical proposition.

Types of Categorical Propositions

(a) Universal Proposition: Universal propositions either fully include the subject or fully exclude it.

Examples are,

All coasts are beaches.

No Students are honest.

Universal propositions are further classified as

(i) Universal Positive Proposition: A proposition of the form "All S are P", for example, "All coasts are beaches", is called a universal positive proposition. And it is usually denoted by a letter "A".

(ii) Universal Negative Proposition: A proposition of the form "No S are P", for example, "No students are honest", is called a universal negative proposition. And it is usually denoted by a letter "E".

(b) Particular Proposition: Particular proposition either only partly include or only partly exclude the subject while making a statement.

Examples are,

Some documents are secret.

Some cloths are not cotton.

Particular propositions are also further classified as

(i) Particular Positive Proposition: A proposition of the form "Some S are P", for example, "Some documents are secret", is called a particular positive proposition and it is denoted by the letter "I".

(ii) Particular Negative Proposition: A proposition of the form "Some S are not P", for example, "Some cloths are not cotton", is called a particular negative proposition. And is usually denoted by the letter "O".

Important Note-

The Four Types of Propositions

Type of Proposition	Universal	Particular
Positive	**A Format**: All S are P Example: All coasts are beaches.	**I Format**: Some S are P. Example: Some documents are secret.
Negative	**E Format**: No S are P Example: No student are honest.	**O Format**: Some S are not P. Example: Some cloths ar

II. ARGUMENTS

An argument is a sequence of two or more propositions, phrases, clauses or statements, of which one is a conclusion that follows from a few other propositions called premises. A proposition by itself is not an argument. Generally, arguments are of two different types—deductive and inductive. In a deductive argument, the premises provide conclusive grounds for the truth of its conclusion or in other words, the conclusion is arrived at by going from the universal to the particular proposition.

Example:

(i) Man is mortal

(ii) King is a man.

Premises

So, king is mortal—Conclusion.

Here, the first premise is a universal statement and the conclusion is particular as it refers to one individual. Whereas, in an inductive argument the premises provide only some support for its conclusion but not conclusive grounds. So, unlike a deductive argument, inductive argument cannot be labeled valid or invalid. Also it may proceed from a particular to a universal proposition.

Example:

(i) All tigers are animals.

(ii) All cats are also animals.

Premises

So, all cats are tigers.—Conclusion

Here the reasoning proceeds from a particular to a universal proposition though it may not be valid.

III. SYLLOGISM

Syllogisms can be defined as a deductive scheme under which a formal argument is made. It is originally a word given by the Greeks, which means ‘inference’ or ‘deduction’. It is a deductive or logical argument in which the conclusion is inferred from two premises.

Syllogism is the process of thinking which aims to arrive at valid conclusions. It means a mediate inference, consisting of two premises and a conclusion. The two premises jointly imply the conclusion, in which a relation between two terms contained in the premises is established through a common term. Syllogism is simply concerned with the validity of the premises and not with material truth. Questions based on Syllogisms always first state the premises and then ask you to derive the conclusion on the basis of the relationship existing between the different elements of the premises.

The Euler's Circles or Venn-Diagrams- There is a pictorial way of representing the propositions, formulated by Euler, an ancient mathematician. Suppose that the proposition is trying to relate the subject (S) with the predicate (P). Then there are four ways in which the relation could be made according to the four propositions shown in the figure I-IV below.

Fig. I clearly represents 'All S are P'. This is denoted by the fact that the whole circle denoting S (denoting 'all S') lies within the circle denoting P. Similarly, Fig. II represents 'No S are P' because the circles denoting S and P do not intersect at all. Fig. III, similarly, represents the proposition "Some S are P" because some part of the circle denoting S as indicated by the shaded area of S (representing 'some S') lies within the circle denoting P. Slightly more attention-seeking is the representation for the O proposition "Some S are not P". For this, take a closer look at Fig. IV. The figure shows that some portion of the circle S has no intersection with the circle P, while the remaining portion of the circle S is left incomplete and it is uncertain whether this portion touches P or not. The verbal interpretation of this figure would be: "there are some S that are definitely not P while there may be some S that might be P or might not be P".

Type-A



Fig. I
All S are P.

Type-E

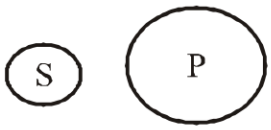


Fig. II
No S are P.

Type-I

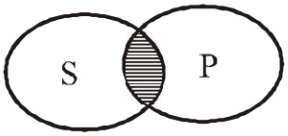


Fig. III
Some S are P.

Type-O

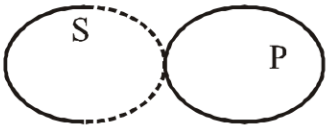
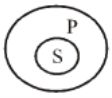
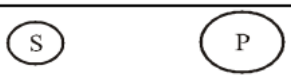
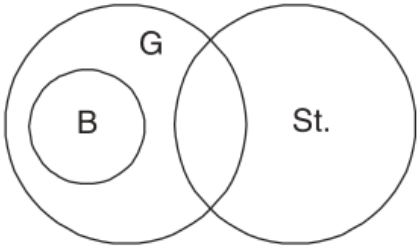


Fig. IV
Some S are not P.

Table- Euler's Circles and representation of the four propositions-

If the Type of the given proposition is	Then its pictorial representation is
A **All S are P**	Always 
E **No S are P**	Always 
I **Some S are P**	Either  Some S are P. [Some S are not P]
	Or,  Some S are P [All S are P]
	Or,  Some S are P [All P are S]
O **Some S are not P**	Either  Some S are not P [Some S are P]
	Or,  Some S are not P [All P are S]
	Or,  Some S are not P [No S are P]

Examples:



1. All boys are girls.

Some girls are students.

Conclusion: All girls are boys.

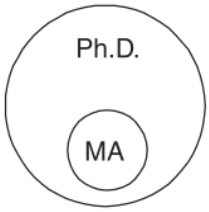
Here, the conclusion is invalid.

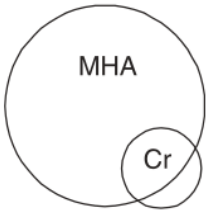
2. All Master of Arts are eligible for Ph.D. course.

A criminal is a master of his art.

Conclusion: A criminal is eligible for Ph.D. course.

Here the conclusion is invalid.





3. All fishes have red blood.

All red blooded animals are quadrupeds.

All quadrupeds are mammals.

Conclusion: All fishes are mammals.

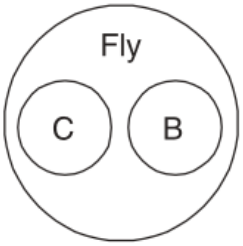
Here, the conclusion is invalid, it contains four propositions.

4. All birds can fly.

All cockroaches can fly.

Conclusion: All cockroaches are birds.

Here, the inference is false, as the middle term ‘fly’ is undistributed in both propositions which are universal affirmatives.

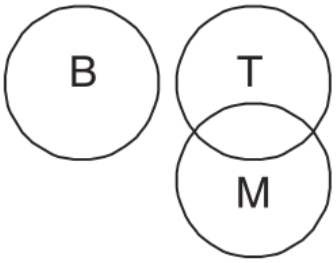


5. No Bengali is Tamil.

Some Tamils are Muslims.

Conclusion: Some Muslims are Bengalis.

Here also, the inference is fallacious.

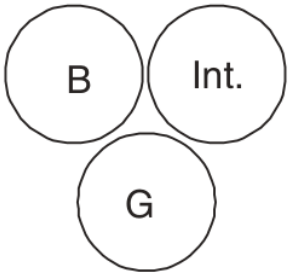


6. No boy is intelligent.

No girl is boy.

Conclusion: No girl is intelligent.

Here, the conclusion is invalid.

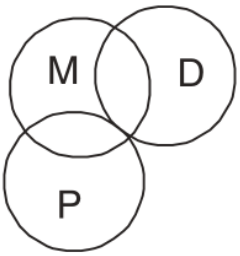


7. Some men are dishonest.

Some men are politicians.

Conclusion: Some politicians are dishonest.

Here, the conclusion is uncertain.

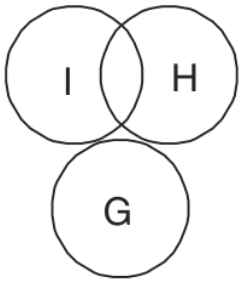


8. Some Indians are Hindus.

No Indian is a Greek.

Conclusion: Some Greeks are Hindus.

Here, the conclusion is invalid.

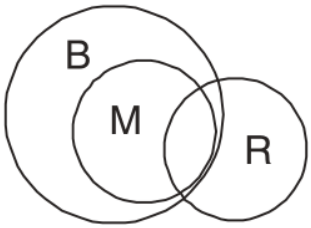


9. Some men are rich.

All men are beautiful.

Conclusion: All beautiful men are rich.

Here also, the inference is invalid.

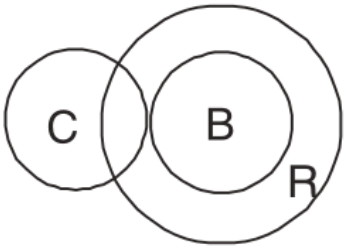


10. No cat is bat.

All bats are rats.

Conclusion: No rat is cat.

Here, the inference is invalid.

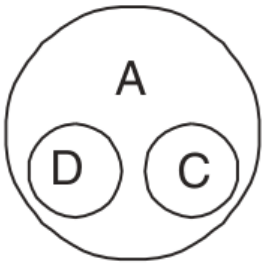


11. All cows are animals.

All dogs are animals.

Conclusion: All dogs are not cows.

Here, the conclusion is invalid.

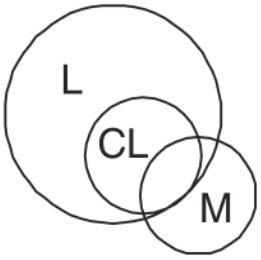


12. All clerks are lazy.

Some men are clerks.

Conclusion: All lazy are men.

Here, the conclusion is invalid.



Directions for examples 13 to 18:

Certain ‘conclusions’ have been drawn on the basis of given facts. Judge whether a conclusion is (A) ‘True’ (correctly drawn from the given facts), (B) ‘False’ (wrongly drawn from the given facts) or (C) ‘Uncertain’ (neither definitely true nor definitely false).

Given facts:

Some athletes are students.

All students are men.

Conclusions:

13. Some men are athletes.

14. All athletes are men.

15. No man is athlete.

16. Some athletes are men.

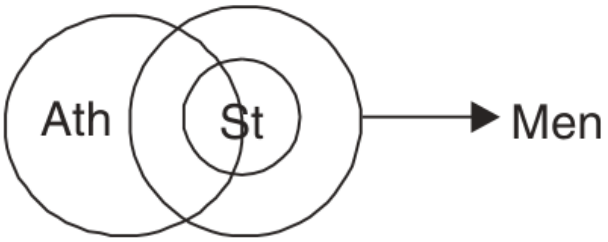
17. Some athletes are not men.

18. Most men are athletes.

Answers and Explanations:

13. (A), 14. (B), 15. (B), 16. (A), 17. (A), 18. (C).

These can be solved easily by using Venn diagrams. According to the given facts in the example “Some athletes are students”—Proposition-I, and “All students are men”—Proposition-II. These proposition according to the Venn diagram should be like this:



Q1) Pick the most logical sequence for the following statements-

A) Some curtains are cloth.

B) All cloth is wood.

C) All that is wood is cloth.

D) All cloth are curtains.

E) All curtains are wood.

F) Some curtains are wood.

(a) BED

(b) BDF

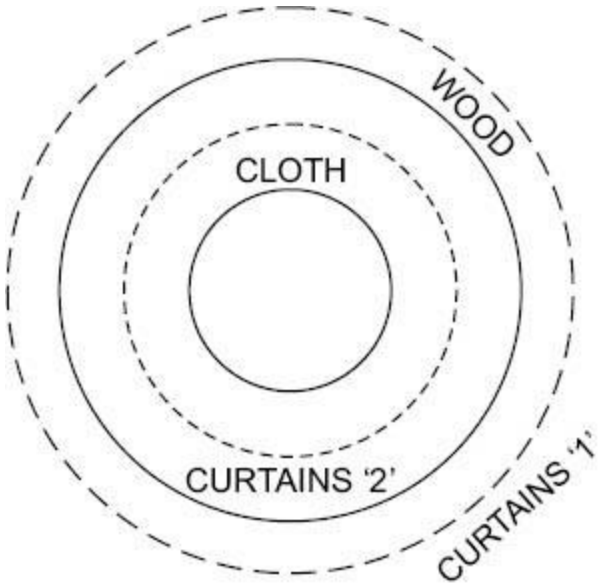
(c) FAB

(d) FBA.

S1)

(b)

BDF is the most logical sequence here since if we take B and D as the premises, then the curtains which are cloth, will also be wood. The Venn Diagram for BD will look like this—



Hence, F is the correct conclusion.

Q2)

Pick the most logical sequence for the following statements-

A) Some stone is bone.

B) All slipper is bone.

C) All bone is stone.

D) No stone is slipper.

E) No bone is stone.

F) Some bone is slipper.

(a) BCA

(b) AFE

(c) DEC

(d) CEA

S2)

(a) is the obvious answer here, since if all bone is stone (C), then obviously, some stone is bone(A).

Q3)

Directions (Q. i-v): In each of the questions below are given two/three statements followed by two conclusions numbered I and II. You have to take the given statements to be true even if they seem to be at variance with commonly known facts. Read all the conclusions and then decide which of the given conclusions logically follows from the given statements, disregarding commonly known facts. Give answer

A) if only conclusion I follows.

B) if only conclusion II follows.

C) if either conclusion I or conclusion II follows.

D) if neither conclusion I nor conclusion II follows.

E) if both conclusion I and conclusion II follows.

i)

Statements: All blades are sharp.

Some blades are blunt.

No blunt is a sword.

Conclusions: I. All sword being blades is a possibility.

II. At least some blades may be swords.

ii)

Statements: All flowers are white.

Some whites are beautiful.

Conclusions: I. All flowers being beautiful is a possibility.

II. At least some white may not be flowers.

iii)

Statements: All fruits are tasty.

No tasty is good.

All goods are yummy.

Conclusions: I. All yummy things being tasty is a possibility.

II. All fruits being yummy is a possibility.

iv)

Statements: All mangoes are grapes.

Some grapes are black.

Conclusions: I. Some mangoes being black is a possibility.

II. There is a possibility that some mangoes are not black.

v)

Statements: No book is a pencil.

All pencils are pens.

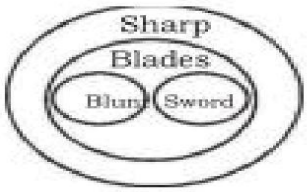
No pen is blue.

Conclusions: I. No blue is a pencil.

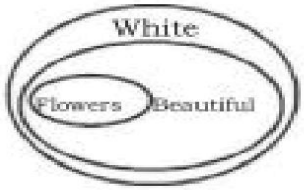
II. All books being pen is a possibility.

S3)

i) E



ii) E



iii) B ;fruit



So, **B** follows.

iv) E ;mango



v) E



Q4).

For this question, you have to base your conclusion on the three premises (viz: a, b and c). Identify the correct answer from amongst the options given to you:

Statements-

a.Some cakes are bikes

b. Some bikes are hills

c. All laptops are bikes

Conclusions I. All laptops are hills

II. Some laptops are cakes

III. Some cakes are hills

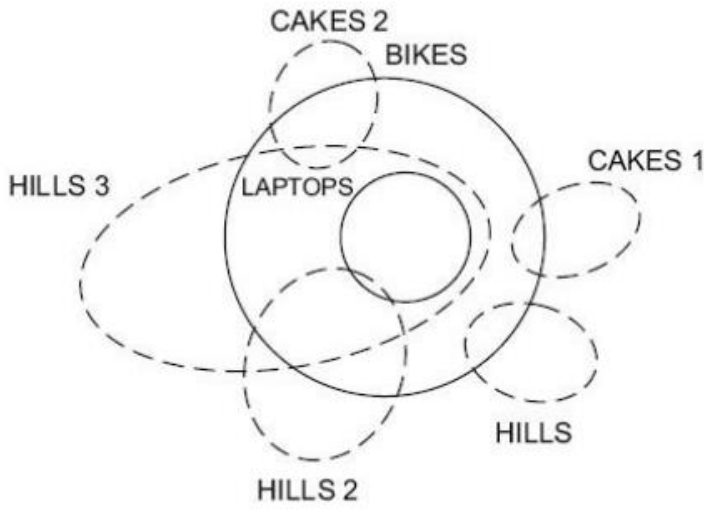
IV. Some laptops are not cakes

A) Only I follows
B) Only either II or IV follows
C) Only I or III follow
D) Only I and IV follows

E) None of these

S4)

For the question above, the following Venn Diagram can be drawn:



It is clear from the figure that the conclusion 'All Laptops are Hills' does not hold. Similarly, some Cakes are hills is also not necessarily true. If we take a look at II & IV, we can conclude that one of them has to be true. Hence, the answer is (B).

Q5)

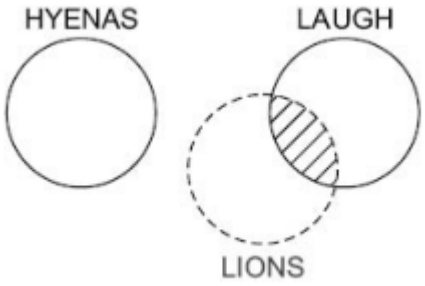
Which of the following conclusion is justified-
(i) No hyenas laugh. Some who laugh are lions. Some lions are not hyenas.
(ii) All ghosts are red. Some ghosts do not dance. Some dancers are not red.
(ii) Australians indulge in swearing. Those who swear are fined. Some who are fined are not Australians.
(iv) Some Europeans are Spanish. All Europeans are British. Some British are Spanish

(A) i and ii
(B) iii only
(C) i and iv
(D) iv only

S5)

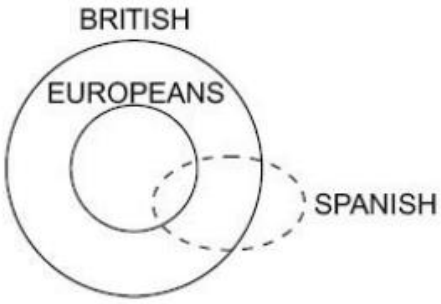
C)

The following Venn Diagram can made for the sequence i:



Obviously, the lions who laugh are not hyenas. Hence, the conclusion is justified.

The following Venn Diagram will be made for the sequence iv:



It is obvious that whatever circle we draw for Spanish (apart from the one shown) it will have to intersect the circle for Europeans. This in effect, means that, it has also to intersect with the British circle. Hence, the conclusion that 'Some British are Spanish' is justified.

Q6)

Directions for Questions i-xii:
Certain 'conclusions' have been drawn on the basis of certain 'given facts'. Judge whether the said conclusion is (A) 'True' (correctly drawn from the given facts) or (B) 'False' (contradicts the given facts) or (C) 'Uncertain' (neither definitely true, nor definitely false).

Given facts:
Sohan is stronger than Rahim.
Rahim is weaker than John.

Conclusions:
i. Rahim is weaker than Sohan.
ii. Rahim is stronger than John.
iii. Sohan is stronger than John.
iv. John is stronger than Rahim.
v. Sohan and John are equally strong.

Given facts:
X is older than Y.
Y is older than Z.
Z is younger than W.

Conclusions:
vi. Z is younger than Y.
vii. Z is older than X.
viii. W and Y are of same age.

Given facts:
A is taller than B.
B is shorter than C.
C is as tall as D.

Conclusions:
ix. D is taller than B.
x. A is as tall as D.
xi. C is taller than A.
xii. D is shorter than B.

S6)
For Questions i-v:
According to the premises we can arrange 'stronger than' as > and 'weaker than' as <, so according to given facts Sohan > Rahim and Rahim < John.
Thus, i. (A), ii. (B), iii. (C), iv. (A), v. (C).

For Questions vi-viii:
As per given facts, we may say > means ‘older than’ and < means ‘younger than’. So X > Y > Z < W.
Thus, vi. (A), vii. (B), viii. (C).

For Questions ix- xii:
As per given facts, we may say > means ‘taller than’, < means ‘shorter than’ and = means ‘as tall as’. So, A > B, B < C, C = D.
Thus, ix. (A), x. (C), xi. (C), xii. (B).

Q7)
Directions for Questions i-xv:
Certain ‘conclusions’ have been drawn on the basis of certain ‘given facts’. Judge whether the said conclusion is (A) ‘True’ (correctly drawn from the given facts) or (B) ‘False’ (contradicts the given facts) or (C) ‘Uncertain’ (neither definitely true, nor definitely false).

Given facts:
No Z is X and
All X is Y.

Conclusions:
i. All Y is Z.
ii. Some Y is not Z.
iii. Some Z is not Y.

Given facts:
All P is Q and
Some P is not R

Conclusions:
iv. All Q is R.
v. Some Q is not R.
vi. No R is Q.

Given facts: No X is Y and
Some Y is Z.

Conclusions:

vii. Some Z is not X.

viii. Some X is Z.

ix. All Z is X.

Given facts: X is larger than Y.

Z is smaller than Y.

W is not smaller than X

Conclusions:

x. X is the largest.

xi. W is larger than Z.

xii. Y is the smallest

xiii. W is larger than X.

xiv. Z is the smallest.

xv. Y is larger than W.

S7)

For Questions i-iii:

On the basis of the given facts, if we draw the following diagrams, we find

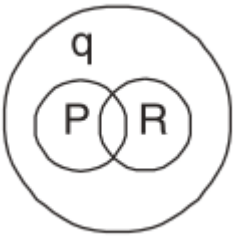


(i) (ii)

Thus, i. (B), ii. (A), iii. (C) → from diagram (i) it is not possible, but in diagram (ii) it is possible, hence 'uncertain' i.e. (C).

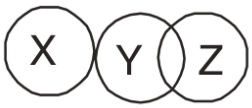
For Questions iv-vi:

On the basis of the given facts, if we draw a diagram, we find



Thus, iv. (B), v. (A), vi. (B).

For Questions vii-ix:
From the given facts two diagrams are possible which are



(i)



(ii)

Thus, vii. (A), viii. (C) [in diagram (i) it is not possible but in (ii) it is possible], ix. (B).

For Questions x-xv:
From the given facts we can arrange like this
X > Y, Z < Y, W < X or W ≥ X, when > means 'greater than', < means 'less than', < means 'not less than' and ≥ means 'greater than or equal to'.
Thus, x. (C), xi. (A), xii. (B), xiii. (C), xiv. (A), xv. (B).

Q8)
Directions for Questions i-vi:
Certain 'conclusion' has been drawn on the basis of 'given facts'. Judge whether the conclusion is (A) true (correctly drawn from the given facts), (B) false (wrongly drawn from the given facts) or (C) uncertain (neither definitely true nor definitely false). Symbols used have the following meanings: > means 'is greater than'
< means 'is less than'
= means 'is equal to'
$\not>$ means 'is not greater than'
$\not<$ means 'is not less than'
≠ means 'is not equal to'.

Given facts:	Conclusions:
i. X > Y and Z > Y	x = z
ii. X $\not<$ Y and Z = Y	z > x
iii. X $\not<$ Y and Y $\not<$ Z	x ≠ z
iv. X $\not>$ Y and Z > Y	x = z
v. X > 3Y and Z < 2Y	z < x
vi. 2X > Y and Z $\not<$ Y	z $\not>$ x

S8)

i. (C) Uncertain.

ii. (B) False. Given that Y is not greater than X and as Z is equal to Y, so, Z cannot be greater than X.

iii. (C) Uncertain. Given that Y is not greater than X and Z is not greater than Y. So it can not be definitely said that X is not equal to Z.

iv. (B) X is not greater than Y and Y is less than Z, so, X cannot be equal to Z.

v. (A) True. X is greater than 3Y and 2Y is greater than Z. So X must greater than Z.

vi. (C) Uncertain. 2X greater than Y and Z is not less than Y. So Z may or may not be greater than X.

8. STATEMENTS

These tests are based on statements. Though these tests may be called as the parts of logical reasoning test, here some statements or assumptions or arguments are given followed by a few conclusions. Students have to say whether the said conclusions are valid or relevant or not. The questions may take any of the following forms: (i) statement-conclusion, (ii) statement-reasoning, (iii) statement-argument (iv) statement and course of action (v) statement and assumptions.

I. STATEMENT-CONCLUSION TYPE

In such questions one or more statements are given. The candidate is required to analyse the given statements, understand their implications and then decide whether the conclusions follow or not.

Examples:

Directions for Questions 1-4:

For each question below some statements are given followed by a conclusion you have to say whether the said conclusions:

(A) necessarily follows from the statements.

(B) is only a long drawn one.

(C) definitely does not follow from the statements.

(D) is doubtful.

1. Statements: (i) Early rising is good for health.
(ii) Rama generally rises early in the morning.

Conclusion: Therefore Rama has good health.

2. Statements: (i) All green apples are sour.
(ii) All the apples of Shimla are green.

Conclusion: All the apples of Shimla are sour.

3. Statements: (i) All boys in a hostel get up at 5 A.M.
(ii) Robin is a boy.

Conclusion: Therefore Robin gets up at 5 A.M.

4. Statements: (i) Last year there was a major train accident in March.
(ii) Year before last there also was a major train accident during March.

Conclusion: So this year also there will be a major train accident in the month of March.

Answers and Explanations:

1. (D) The word ‘generally’ in the 2nd statement makes the conclusion doubtful.
2. (A) The two statements directly lead to the conclusion.
3. (C) Because nowhere it is said that Robin lives in a hostel.
4. (B) Only two examples are not sufficient to come to a general conclusion.

Directions for questions 5- 7: In each of the following questions, a statement is given followed by two conclusions. Give answer

(A) If only conclusion I follows.
(B) If only conclusion II follows.
(C) If both conclusions I and II follow.
(D) If neither I nor II follows.

5. Statement

Anyone who manages an engineering organization like Tata Motors can only be successful if he has the knowledge of the professional work that is carried on in the organization.

Conclusions

I. If you want to run any company, it is essential that you should know the professional work associated with it.

II. In order to manage an engineering organization like Tata Motors, it is essential to have an engineering education background.

6. Statement

According to Baba Ramdev “Values, integrity and peace of mind are essential for a beautiful life.”

Conclusions

I. Baba Ramdev believes that each of values, integrity and peace of mind are essential for a beautiful life.

II. Baba Ramdev believes that materialistic persons cannot have a beautiful life.

7. Statement

India won the second test match against Pakistan by one innings and two hundred and three runs. The greatest contribution was from the bowlers.

Conclusions

I. Most of the Indian players are bowlers.

II. It was India’s biggest win against Pakistan.

Solution

5. The statement particularly talks about engineering organizations like Tata Motors. The first conclusion is generalizing the statement and is hence not valid as a conclusion. The second conclusion follows. Hence option (b) is correct.

6. Conclusion I follows since that is what is meant by the statement. Materialism is generally anti values and integrity and hence we can draw the second conclusion. Hence option (c) is correct.

7. Both conclusions are invalid. Hence, option (d) is correct.

Exercise

Directions for Questions 1-15:

For each question below some statements are given followed by a conclusion. Study the statements and the conclusion carefully and say whether the said conclusion:

(A) necessarily follows from the statements.

(B) is only a long drawn one.

(C) definitely does not follow from the statements.

(D) is doubtful.

1. Statements: (i) Ramesh generally walks in the morning.
(ii) Morning walk is good for health.

Conclusion: Therefore Ramesh has good health.

2. Statements: (i) Clouds are formed in the higher regions.
(ii) Mountains are high.

Conclusion: Therefore there are more rains on the mountains.

3. Statements: (i) People living in cold countries generally have white complexion.
(ii) Germany is a cold country.

Conclusion: All Germans have white complexion.

4. Statements: (i) Gastro-enteritis increases during rainy season.
(ii) July and August are the months of rain.

Conclusion: The number of Gastro-enteritis patients is more during July and August than in other months.

5. Statements: (i) Maximum growth of the brain takes place during the first few years of the child.
(ii) Protein is helpful for the growth of brain.

Conclusion: Mental deficiency occurs during early years of the child.

6. Statements: (i) Metals are generally hard but ductile.
(ii) Stone is also hard but not ductile.
Conclusion: So stone is not a metal.

7. Statements: (i) Students who work hard and attend school regularly pass the examination.
(ii) Namita attends school regularly.
Conclusion: So Namita must have passed the examination.

8. Statements: (i) The sum of the angles of a triangle is 180 degrees.
(ii) The sum of the two sides of a triangle is greater than the third side.
Conclusion: So ABC is a right angled triangle.

9. Statements: (i) All Government offices are closed on Saturday.
(ii) Sudip is a government employee.
(iii) Today is Saturday.
Conclusion: Sudip will not go to office today.

10. Statements: (i) In Kolkata Schools start from 10 A.M.
(ii) Sonia is a school child.
Conclusion: Therefore, Sonia should reach her school at 10 A.M.

11. Statements: (i) Vitamins are good for health.
(ii) Fruits and Vegetables may contain vitamin.
(iii) Apples are fruits and carrots are vegetables.
Conclusion: Therefore those having weak health should take apples and carrots.

12. Statements: (i) Soma is better than Nagen in studies.
(ii) Nagen and Dipak are equal in height.
Conclusion: So Soma is also better than Dipak in studies.

13. Statements: (i) Milk or milk product is good for health.
(ii) Tea is generally prepared with milk.
(iii) Pratap regularly takes tea in the morning.
Conclusion: Therefore Pratap is healthy.

14. Statements: (i) True art has power to move.
(ii) Music often moves listeners.

Conclusion: Music is an art.

15. Statements: (i) Cigarette smoking is injurious to health.
(ii) Cigarette smoking generally causes cancer.
(iii) Subodh is a chain-smoker.
Conclusion: Therefore, Subodh will be attacked with Cancer in future.

16. Statements: (i) Some trees are tall.
(ii) All tall trees are coconut trees.
Conclusion: Therefore all trees are coconut trees.

Directions for Questions 17 to 20: Choose the correct conclusion for the given statement(s).

17. Statement Laughter is the best medicine.
Conclusions (a) All medicines have Benzene enzymes.
(b) Laughter is good for health.
(c) All medicines that make you laugh are good.
(d) Stand-up comedy shows on television are a good source of laughter.

18. Statement 1. Private company's employees work for five working days with 9 working hours every day.
2. IOC's employees work for 6 working days a week and 8 working hours a day.
Conclusions
(a) IOC's employees are better than private company's employees.
(b) IOC's employees work more than Private company's employees.
(c) Private companies employees work more than IOC's employees.
(d) IOC's employees are more efficient than that of private companies.

19. Statement 1. All athletes in the Asian athletics teams are talented.
2. All Indian athletes are now in the Asian athletics team.
Conclusions
(a) Some Indian athletes are talented.
(b) All Indian athletes are talented.
(c) India does not have any athletics team.
(d) India has more cricketers than athletes.

20. Statement 1. Next year, salary of all people working in the Bata Hindicom will increase by Rs 1 lac per year.

2. The current income difference between two employees of Bata Hindicom, Arun and Subhash is Rs 2 lacs.
Conclusions
(a) Arun's salary will be higher after the increment in salary.
(b) Subhash's salary will be higher than Arun's from the next year.
(c) The difference between Arun and Subhash's salary will remain constant.
(d) None conclusion follows.

Directions for Questions 21 to 25: In the following questions there are given some statements followed by conclusions that can be drawn from them. Choose the answer for each question independently.
21. Bombay is more expensive than Delhi. Delhi is more expensive than Calcutta. Bombay is more expensive than Calcutta.
(a) true
(b) probably true
(c) false
(d) can't say

22. Basket ball players are generally tall. Jeevan is a basketball player. So Jeevan is tall.
(a) false
(b) probably false
(c) true
(d) can't say

23. Statement 1. Some actors are dancers.
2. All those, who are dancers, are paid well.
Conclusions Some actors are paid well.
(a) true
(b) probably true
(c) false
(d) can't say

24. Statement Teachers punish students when they do not complete their homework. Ravi (a teacher) punished Soni in school today.
Conclusions Soni did not do her homework.
(a) definitely true
(b) definitely false
(c) can't say

(d) conclusion is irrelevant

25. Statement Smart managers are those who hire smart employees. Conclusions To be a smart manager, I just have to hire smart employees.
(a) true
(b) false
(c) either probably true or probably false
(d) can't say

Answers
1. (D) The word 'generally' makes the conclusion doubtful.
2. (B) The conclusion is a long drawn one. It is only expectation.
3. (D) As in question 1.
4. (A) Both the statements clearly lead to the conclusion.
5. (C) No link between protein and mental deficiency is found from the statements.
6. (D) The word generally makes the conclusion doubtful.
7. (D) Here Namita shows only one quality, which is not sufficient to come to a conclusion. So, it is doubtful.
8. (C) Conclusion does not tally with the statements.
9. (A) Conclusion directly drawn from the statements.
10. (C) Nowhere it is written that Sonia lives in Kolkata.
11. (B) Apples and carrots are one type of fruit and vegetable. So, no conclusion can be drawn from this single type.
12. (C) No relation between studies and height.
13. (D) The word 'generally' makes the conclusion doubtful.
14. (A) The conclusion clearly follows the statements.
15. (D) The word 'generally' makes the conclusion doubtful.
16. (C) Some trees, not all trees are tall, so it contradicts the statements.
17. A medicine is something that is beneficial for health. Hence, we can conclude that conclusion (b) follows from the statement. Options (a), (c) and (d) are irrelevant conclusions.
18. Options (a) and (d) do not follow as we do not know anything about who is better. From the two statements it is easy to conclude that IOC's employees work more than Private company's employees. Hence, option (b) is correct.
19. Obviously conclusion (b) follows.
20. Obviously conclusion (c) is correct. We cannot say anything about whose salary is higher.
21. Clearly the conclusion holds true. Hence, option (a) is correct.
22. Clearly we cannot claim that every basket ball player is tall. However, we cannot also say for sure that Jeevan is not tall. Hence, option (d) is correct.

23. Definitely true. Option (a) is correct.
24. From the given statement it is not clear as to why Ravi punished Soni. While her not doing her homework could have been a reason it could equally well not have been the reason too. Hence, we cannot say anything about this. Option (c) is correct.
25. We cannot say for sure whether this conclusion is true or false. It might be true and at the same time might be false if there are more things that are needed to be done for being a smart manager. Hence, option (d) is correct.

II. STATEMENT-REASONING TYPE

In making logical conclusions from some statements it is desirable to judge whether they logically follow from the given statements, disregarding commonly known facts. Because, sometimes, the statements, given to be true, even they seem to be at variance from commonly known facts.
Examples:
Directions for Questions 1-2:
1. Statements: All goats are cows but all cows are not goats.
Conclusion: I. Some cows are goats.
II. Some goats are not cows.

2. Statements: At high altitudes there is less oxygen in the air.
Oxygen is necessary for life.
Conclusion: I. Oxygen is necessary for burning.
II. Mountaineers need supplementing oxygen supply for their expedition.

Answers and Explanations:
1. (A) Only I follows the statements.
2. (B) Only conclusion II is correct as conclusion I does not have any reference in the statements.

Directions for Questions 1-15:
Each question below consists of some statements followed by two conclusions numbered I and II. You have to decide whether the conclusions.
(A) only I follows or,
(B) only II follows or,
(C) both I and II follow or
(D) either I or II follows or

(E) neither I nor II follows from the statements.

1. Statements: If a body is moving it will continue moving in the same direction unless some force is applied to stop it. If someone brakes suddenly while driving a scooter moving fast.
Conclusions: I. He will stop abruptly, without falling.
II. He will fall back.

2. Statements: His brother is a good football player.
He is a good cricketer.
His sister is a good singer.
Conclusions: I. They are all talented.
II. They are all sportsmen.

3. Statements: Wood catches fire when heated.
Metals expand on heat.
Conclusions: I. Wood may be a metal.
II. Wood cannot be a metal.

4. Statements: Robin is an early riser and has a habit of morning walk.
Probir also gets up early in the morning but does not go for a walk.
Conclusions: I. In one respect they are similar.
II. In the other they are dissimilar.

5. Statements: A is the brother of B.
C is the brother of A.
Conclusions: I. B is a girl.
II. B is a boy.

6. Statements: Fish is a protein food and good for health.
Prawns are not fish.
Conclusions: I. Prawn is not good for health.
II. Chicken has got protein.

7. Statements: Oxygen and Hydrogen are gases.
Water contains 2 parts of Hydrogen and 1 part of Oxygen.
Conclusions: I. Water is also a gas.
II. Water is a compound of two gases.

8. Statements: X is a friend of Y.
Y is a friend of Z.
Conclusions: I. X is a friend of Z.
II. X, Y and Z are friends.

9. Statements: The metropolis does not represent the real India.
Conclusions: I. People in New Delhi live in the western style.
II. The real India can be seen only in the rural areas.

10. Statements: All the years that are divided by four, are leap years.
Conclusions: I. 1964 was a leap year.
II. 3000 will not be a leap year.

11. Statements: Treatment of Aids is very expensive.
Conclusions: I. People with low incomes do not suffer from Aids.
II. People prefer expensive treatment.

12. Statements: Dipak is studying in a co-educational college.
In his class the students sit on dual desks, two on one desk.
Conclusions: I. Dipak sits by the side of a boy.
II. A girl sits by the side of Dipak.

13. Statements: Half of the people in a locality are Hindus. One fourth are Muslims.
Conclusions: I. Hindus are in majority.
II. One fourth people are neither Hindus nor Muslims.

14. Statements: Water, when heated changes into steam but when cooled changes into ice.
Conclusions: I. Water is a gas.
II. Water is a solid.

15. Statements: All the goats are tigers but all tigers are not goats.
Conclusions: I. Some tigers are goats.
II. Some goats are not tigers.

Answers and Explanations:
1. (E) None of the conclusions follows from the statements.
2. (A) Only conclusion I follows, conclusion II is wrong.
3. (E)

4. (C) Both the conclusions are possible.
5. (D) Either conclusion I or II is possible, both not possible at the same time.
6. (E) None of the conclusions follows from the statements.
7. (B) Here only II follows, I is not correct.
8. (E) None of the conclusions follows from the statements.
9. (B) Only conclusion II follows.
10. (A) Only I is true.
11. (E) Both the conclusions are not relevant to the statements.
12. (D) Both cannot follow. Only one follows, either I or II.
13. (C) Both the conclusions can follow.
14. (E) Neither I or nor II follows from the statement.
15. (A) Only conclusion I follows from the statement.

III. STATEMENT-ARGUMENTS

Sometime while making or justifying any decision we are to deal with certain arguments which may be 'strong' or 'weak'. 'Strong' arguments are those which are directly related to the decision and are very important too. Whereas 'weak' arguments may or may not directly related to the decision and have either minor importance or no importance at all.
In this type of questions, a statement is given, followed by two arguments. The candidate is required to analyze the statement first, then the arguments in context of the statement and then decide which of the arguments holds strong as regards the statement.

Examples:
Directions for Questions 1 and 2:
Each question below is a statement followed by two arguments numbered I and II. You have to decide which of the arguments is 'strong' and which is 'weak' and give answer accordingly if (A) only I is strong (B) only II is strong (C) both I and II are strong (D) either I or II is strong (E) neither I nor II is strong.

1. A stitch in time saves nine.
Arguments:
I. Yes, it is true that if we rectify the fault immediately, it does not magnify.
II. No, one should not worry unnecessarily.
2. Degrees should not be compulsory for jobs.
Arguments:
I. No, they are compulsory; otherwise it will become very difficult to access the eligibility of candidates for a particular job.

II. No, otherwise there would be a lack of motivation among the students towards education.

Answers and Explanations:
1. (A) Only Argument I is strong, the second one is hardly any argument in favour.
2. (C) Here both the arguments are equally strong.

Directions for Questions 1-15:
Each question below is a statement followed by two arguments numbered I and II. You have to decide which of the arguments is 'strong' or 'weak' and give answer accordingly if, either (A) only I is strong, or (B) only II is strong, or (C) both I and II are strong, or (D) either I or II is strong or (E) neither I nor II is strong.

1. Nuclear families are better than joint families.
Arguments:
I. Yes, members of nuclear families enjoy greater freedom.
II. No, in joint families there is greater security.

2. Sanskrit should be the national language of India.
Arguments:
I. Yes, it is our cultural heritage.
II. Yes, because every Indians, more or less, knows the language.

3. Should railway travelling without a valid ticket be encouraged?
Arguments:
I. Yes, railway fares are excessively high.
II. No, it is antinational to do so.

4. Should smoking be prohibited?
Arguments:
I. Yes, smoking needs to be abolished because it is injurious to health.
II. No, otherwise it will throw thousands of workers in the tobacco industry out of employment.

5. English language should be made compulsory in primary education in our country.
Arguments:
I. Yes, with only the knowledge of English language, students could be enriched in education.
II. No, it would be an extra burden of language for the little children, because only mother tongue can help the student to enrich in education.

6. Honesty always pays.
Arguments:
I. Amongst dishonest people it may not pay immediately.
II. Amongst honest people it does pay immediately.

7. Begging should be banned.
Arguments:
I. Begging lowers the self esteem of a person.
II. Begging is not good.

8. Inter-caste marriages should be encouraged.
Arguments:
I. Yes, this will lead to National integration.
II. No, this will lead to a lot of maladjustment of married couples in adjusting with the rituals and dogmas of different castes.

9. Plain living and high thinking is a good principle of life.
Arguments:
I. This makes one a great person.
II. No, one cannot think high with simple living.

10. Married life is better than unmarried life.
Arguments:
I. Married life enables one to adjust better emotionally.
II. Unmarried persons can contribute more to the welfare of the society.

11. Should one follow a more or less rigid programme daily?
Arguments:
I. Yes, it will lead to better discipline.
II. No, it will kill creativity in individual.

12. Should all property belong to State?
Arguments:
I. Yes, it will ensure equitable distribution of necessities of life to all.
II. No, it will kill private endeavor.

13. Traditional culture is better than the new culture.
Arguments:

I. The new culture is ruining the society.
II. Both have their good points as well as weak points.

14. The drop-outs in elementary schools can be checked.
Arguments:
I. By increasing awareness among the parents.
II. With the improvement of economic condition of the families.

15. Should students have their unions?
Arguments:
I. Yes, it will help fighting for the cause of the students.
II. No, it will hamper their study.

Answers and Explanations:
1. (C) Both the arguments are equally strong.
2. (E) None of the arguments puts forth any support to the proposition.
3. (B) Here only argument II is strong, the first one is hardly any argument in favour.
4. (A) Here argument I is much stronger than the second argument.
5. (D) Here either of the arguments may be strong but not both at the same time.
6. (C) Here both the arguments are equally strong.
7. (A) Here only argument I is strong, argument II is not at all an argument.
8. (A) Here also argument I is much relevant for the proposition.
9. (A) Here only argument I is relevant.
10. (C) Here both the arguments are equally strong for the proposition.
11. (D) Though both the arguments are strong, they can't go together.
12. (C) Both arguments are weighty.
13. (B) Here only argument II is strong.
14. (C) Both arguments are essential for the proposition.
15. (C) Both arguments are strong.

IV. STATEMENT AND COURSES OF ACTION

In this type of questions some report or statement is given followed by two or three suggested courses of actions. You are to find out the appropriate course or courses of action assuming the problem or policy being talked about in the statement.
Examples:

1. Statement: There will be a summit talk between India and Pakistan in the month of July this year.
Courses of action:
I. They will discuss only the main issue, i.e. Kashmir.
II. They should discuss and resolve all the issues including Kashmir.

2. Statement: Six infants died on the same day at the medicine ward of a hospital.
Courses of action:
I. All the attending physicians of the medicine ward are to be dismissed immediately.
II. An enquiry is to be done immediately and possible preventive measures are to be taken.

3. Statement: The death toll from consuming mushrooms in Darjeeling has risen to ten.
Courses of action:
I. An immediate enquiry in this regard should be done and in order to find out the actual causes of death and preventive measures to be taken accordingly.
II. Production and consumption of mushrooms are to be stopped immediately by an ordinance.

4. Statement: The army has been alerted in the coastal areas of Gujarat following a meteorological warning of severe cyclonic storm within 48 hours.
Courses of action:
I. People of coastal areas should be shifted to the safer places.
II. Relief to these people should be arranged.
III. Adequate medical facilities should be arranged.

Answers and Explanations:
1. There are many problems between India and Pakistan, of which Kashmir is one of these. All these problems are to be solved mutually through bilateral discussion between the heads of two neighboring countries. So, only II course of action is important here and must follow.
2. The death of so many infants at a same time in a hospital is a very serious issue. An immediate enquiry and preventive measure should be taken immediately. So, only courses of action II follows here.
3. So many deaths from mushroom consumption is also a serious issue. An immediate enquiry is to be done to find out whether these mushrooms were poisonous or not or any other cause for such death and preventive measures are to be taken accordingly. So course of action I will follow.
4. Since severe damage may be caused by cyclone, so all the courses of actions are to be taken.

Directions for Questions 1-15:
In each question below is given a Statement followed by two Courses of action numbered I and II. Assuming the problem given in the statement you have to decide which course of action is to be taken or logically follows and answer in the following way:
(A) If only I follows;
(B) If only II follows;
(C) If either I or II follows;
(D) If neither I nor II follows;
(E) If both I and II follows.
1. Statement: A devastating earthquake has ravaged the city of Bhuj killing thousands of people and rendering many more homeless.
Courses of action:
I. The affected people should immediately be shifted to the alternate temporary housing in a safer place.
II. Government and civic administration should immediately arrange for relief, food, medicine for the victims.

2. Statement: Chinese missiles can strike major Indian cities.
Courses of action:
I. India should ask China to stop its missile production.
II. India should take immediate preventive measures for its defence.

3. Statement: The already high prices of potatoes in the state will increase further owing to a mismatch in demand and supply.
Courses of action:
I. State Government should take immediate steps to stop transporting potatoes outside the state by the businessmen and buy potatoes from elsewhere so that supply in the state can be maintained.
II. People should be asked to stop consuming potatoes.

4. Statement: Youngsters are often found staring at obscene posters.
Courses of action:
I. Children should be punished and penalized if they are found doing so.
II. Any display of such material should be banned.

5. Statement: Since its launching in 1981, Vayudoot has so far accumulated losses amounting Rs. 153 crore during the last ten years.
Courses of action:

I. Vayudoot should be directed to reduce wasteful expenditure and to increase passenger fare.
II. An amount of about Rs. 300 crore should be provided to Vayudoot to make the airliner economically viable.

6. Statement: There are more than 200 villagers in the hill area of Uttar Pradesh which are severely damaged due to cyclone and it causes an extra burden of Rs. 200 crore on State Government for relief and rehabilitation work.
Courses of action:
I. People of hill area should be shifted to other safer places.
II. State Government should ask more financial support from Central Government.

7. Statement: One of the problems facing the food processing industry is the irregular supply of raw material. The producers of raw material are not getting the reasonable price.
Courses of action:
I. The Government should regulate the supply of raw material to other industries also.
II.The Government should announce an attractive package to ensure regular supply of raw material for food processing industry.

8. Statement: If the retired Professors of the same Institute are also invited to deliberate on restructuring of the organization, their contribution may be beneficial to the Institute.
Courses of action:
I. Management may seek opinion of the employees before calling retired professors.
II. Management should involve experienced people for the systematic restructuring of the organization.

9. Statement: Doordarshan is concerned about the quality of its programmes particularly in view of stiff competition it is facing from STAR and other satellite T.V. channels and is contemplating various measures to attract talent for its programmes.
Courses of action:
I. In an effort to attract talent, the Doordarshan has decided to revise its fee structure for the artists.
II. The fee structure should not be revised until other electronic media also revise it.

10. Statement: Universities of Eastern regions are facing severe financial crisis even to run their regular courses.
Courses of action:
I. These Universities should close down some of its regular courses immediately till the situation improves.

II. The Universities should curtail the staff strength at least by 50% immediately.

11. Statement: The minister said that the teachers are still not familiarized with the need, importance and meaning of population education in the higher education system. They are not even clearly aware about their role and responsibilities in the population education programme.
Courses of action:
I. Population education programme should be included in the college curriculum.
II. Orientation programme should be conducted for teachers on population education.

12. Statement: The Government has decided not to provide financial support to voluntary organization from next five year plan and has communicated that all such organizations should raise funds to meet their financial needs.
Courses of action:
I. Voluntary organizations should explore other sources of financial support.
II. They should collaborate with foreign agencies.

13. Statement: Exporters in the capital are alleging that commercial banks are violating a Reserve Bank of India directive to operate a post shipment export credit denominated in foreign currency at international interest rates from January this year.
Courses of action:
I. The officers concerned in the commercial banks are to be suspended.
II. The RBI should be asked to stop giving such directives to commercial banks.

14. Statement: The sale of Ambassador car has gone down considerably in recent years causing great concern to the company.
Courses of action:
I. The company should make a proper study of other rival cars in the market.
II. The price of the ambassador car should be reduced much to attract the prospective buyers.

15. Statement: India's performance in the recent Olympic games was very poor not even a single medal could be bagged by the players. Government has spent Rs. 5 crores in training and deputing a team of players to participate in the Olympic games.
Courses of action:
I. India should stop sending players to the future Olympic games.
II. Government should immediately set up an enquiry commission to find out the reason for India's dismal performance.

16. Statement: A leading U.S. multinational engineering and construction firm is keen to invest in India in a variety of sectors ranging from power to land management.
Courses of action:
I. Such multinational companies should not be allowed to operate in India.
II. India should encourage multinational companies from other developed countries to invest in power sectors to bring in competitive climate.

17. Statement: The chairman stressed the need for making education system more flexible and regretted that the curriculum has not been revised in keeping with the pace of the changes taking place.
Courses of action:
I. Curriculum should be reviewed and revised periodically.
II. System of education should be made more flexible.

18. Statement: The Asian Development Bank has approved a $285 million loan to finance a project to construct coal ports by Paradip and Madras Port Trusts.
Courses of action:
I. India should use financial assistance from other International financial organisations to develop such ports in other places.
II. India should not seek such financial assistance from the International financial agencies.

19. Statement: The Government could consider the possibility of increasing the software budget from the current Rs. 20 crore to Rs. 100 crore in the Eighth Plan, provided there are concrete suggestions for the utilisation of the funds.
Courses of action:
I. The Government should consult the trade unions in this regard.
II. Software companies should submit detailed proposals to the Government.

20. Statement: There is a serious constitutional crisis in the State of Manipur after its two years old elected government lost its majority in the State Assembly due to a large scale defection.
Courses of action:
I. Opposition parties should be invited by the Governor to form the Government.
II. President's rule should be imposed in the State immediately after dissolving the State Assembly.

Directions for Questions 21-35:
In each question below a statement is given followed by three courses of action numbered I, II and

III. Analyze the statement and decide which of the three courses of action logically follows and answer according to the alternative answers given along with each question.

21. Statement: If the faculty members also join the strikes there is going to be a serious problem.
Courses of action:
I. The faculty members should be persuaded not to go on strike.
II. Those faculty members who join the strike should be suspended.
III. The management should not worry about such small things.
(A) None follows
(B) Only I follows
(C) Only I & II follow
(D) Only II & III follow
(E) All follow.

22. Statement: Higher disposal costs encourage those who produce waste to look for cheaper ways
to get rid of it. (Bank P.O. 1993)
Courses of action:
I. The disposal costs should be made higher.
II. The disposal costs should be brought down.
III. A committee should be set up to study the details in this respect.
(A) All follow
(B) Only I follows
(C) Only II follows
(D) Either I & II follow
(E) Only I & III follow.

23. Statement: The army has been alerted in the district following floods triggered by incessant rains.
Courses of action:
I. Relief to flood affected people should be arranged.
II. Supply of food articles should be arranged.
III. Adequate medical facilities should be arranged.
(A) None follows
(B) only I follows
(C) only II follows
(D) Only I & III follow
(E) All follow.

24. Statement: Faced with a serious resource crunch and a depressing overall economic scenario, Orissa is unlikely to achieve the targeted percent compound annual growth rate during the 8th plan.
Courses of action:
I. The target growth should be reduced for the next year.
II. The reasons for the failure should be studied.
III. Orissa's performance should be compared with that of other States.
(A) None follows
(B) Only I follows
(C) Only II & III follow
(D) Only I & III follow
(E) All follow.

25. Statement: Over 27,000 bonded laborers identified and freed are still awaiting rehabilitation.
Courses of action:
I. More cases of bonded laborers should be identified.
II. Till the proper rehabilitation facilities are available, the bonded laborers should not be freed.
III. The impediments in the way of speedy and proper rehabilitation of bonded laborers should be removed.
(A) None follows
(B) Only I follows
(C) Only II follows
(D) Only III follows
(E) Only II and III follow.

Answers and Explanations:
1. (E) Here both the actions logically follow as in natural calamities like earthquakes such actions should immediately be taken.
2. (B) In a situation a possible threat to our defense, India should strengthen its defense first, hence only II follows logically. Action I is irrelevant here.
3. (A) Here also action II is irrelevant. Action I is the most appropriate step to check the price rise of potatoes.
4. (B) Bad things often attract youngsters, but often they are not quite aware of those, but punishing them should not be the remedy, they should be made aware or rather convinced in this regard. So, best way is to stop display of such type of posters.
5. (A) Only way to reduce the loss is to reduce the wasteful expenditure, at the same time for economic gain, passenger fare may be increased, hence I follows.

6. (E) In natural calamities like cyclone, top priority to be given to action I and for relief work, huge amount needed must be met by the central government, so both the actions should follow here.
7. (B) For a food processing industry a regular supply of raw material is a must, hence action II is must here.
8. (B) For restructuring of any organization top priority to be given to get the help of experienced people, hence only II follows. For this opinion of all the employees will do no good.
9. (A) In a competitive market Doordarshan, as a national channel should take upper hand in attracting talents, without waiting for the other rival groups. So, action I logically follows here.
10. (D) None of the actions is the remedial measure for a fund crunch University, only Government grant and reduction of wasteful expenditure will help, hence none of the courses follows here.
11. (B) In order to make awareness and knowledge to the teachers regarding population programme, orientation course is most helpful, which should be taken immediately, hence only II follows here.
12. (A) Voluntary organizations should not depend entirely on government grant it should explore other sources also, hence only I follows here.
13. (D) Here the allegation against the commercial banks is the violation of R.B.I., but none of the actions has taken any step against this, hence none follows.
Statement Tests 197
14. (A) The cause of diminished sale of any product in the market is that it lost to attract people, hence, to assess that a proper study of other rival cars whose sale have increased, and then proper action may be taken, hence only I follows.
15. (B) In a competition first step to find out is weakness of the team and then take remedial measures, hence only II follows here.
16. (B) Such arrangement should always be welcomed for the development of the country, hence only II follows here.
17. (E) With the change of time periodical review and revision of curriculum should be done at the same time education should be made more flexible, hence both courses of action follow.
18. (A) For the development of a country funding from outside agencies is always welcome, hence the action I follows.
19. (B) Government funds to any projects should be properly utilised and for this detailed proposal should be submitted to the government in advance; hence only action II follows here.
20. (C) In such constitutional crisis either there should be an alternative government or President's rule, no other option is there. But both cannot be done at the same time, hence either I or II follows.
21. (B) In this case somehow faculty members are to be refrained from joining the strike, hence only I follows.

22. (E) Waste disposal is a must and at the same time its cost should not be less, otherwise the waste producers would be encouraged, hence only I and III will follow.
23. (E) For the flood affected people all the three courses of action are equally important, hence all follow.
24. (C) Under this circumstances Orissa's performance during the 8th plan should be studied in comparison with other states and its reasons behind the failure also be detected, hence II and III follow.
25. (D) All freed bonded laborers are to rehabilitate quickly, hence only III follows.

V. STATEMENT AND ASSUMPTION

Most questions on Statement and Assumptions give you a statement and follow it up with two further statements which have to be tested for whether they are implied in the original statement or not. In order to identify whether an assumption is implied in a statement the best way to do so is to look at the original statement in the question by negating the assumption which is being tested. In case the assumption being tested is implied, negating it will have the effect of weakening the argument of the original statement.
Directions: In each question below is given a statement followed by two assumptions numbered I and II. Consider the statement and decide which of the given assumption is implicit.
Given answer
(a) If only assumption I is implicit.
(b) If only assumption II is implicit.
(c) If both I and II are implicit.
(d) If neither I nor II is implicit.

Example 1
Statement
Strong programming skills cannot be the only criteria for a successful career in the field of Information Technology.
Assumptions
1. Bright careers in Information Technology are not linked only with strong programming skills.
II. If a person has excellent communication skills as well as strong programming skills, only then would he be successful.

Solution
Clearly, there should be some other skills such as interpersonal skills, communication skills, negotiation skill, etc., for a successful career. Hence, assumption I is implicit. Assumption II is not

implicit because it talks only about communication and technical skills. Hence option (a) is correct.
Note: Try the negation test in this question. If Assumption 1 is not true, i.e. “Bright careers in Information Technology are not linked only with strong programming skills” is untrue, then the argument contained in the statement “Strong programming skills cannot be the only criteria for a successful career in the field of Information Technology” loses its strength. Hence, this assumption is implicit in the statement.

Example 2
Statement: The maximum IAS exam selection will be from our institute—A director of a coaching institution.
Assumptions:
I. Most of the students who study in this institute are genius.
II. Teachers of the coaching institution are preparing their students thoroughly.
Solution
Assumption I is irrelevant because there might be some other genius students who are studying in other coaching institutions. Assumption II is implicit. A coaching institution can only claim such statements if it is preparing its students thoroughly. Hence, option (b) is correct.

Exercise

Directions:
(a) If only assumption I is implicit.
(b) If only assumption II is implicit.
(c) If both I and II are implicit.
(d) If neither I nor II is implicit.

1. Statement: UPSC is tougher than MPPSC. So I will appear in MPPSC.
Assumptions:
I. Every graduate should appear in MPPSC.
II. One should avoid tougher things because one might fail in them and failure might lead to a loss of time. On the other hand if something is easy the chances of succeeding are high.

2. Statement: Rahim is a tall boy
Assumptions:
I. Rahim’s father’s height is 6 feet 7 inches.
II. Any height above 6 feet is considered to be tall.

3. Statement: What is the distance of my home from my office?
Assumptions:
I. My home is at the centre of the city.
II. My office is 10 km away from the park, which is 1 km awayfrom my home.

4. Statement: A says to B “Refer to the Cambridge dictionary instead of the Oxford dictionary to be good at English.”
Assumptions:
I. B wants to be good at English.
II. A is a good advisor.

5. Statement: Narayan the CEO of the company says to his employee, “Achieve the targets we have set for you or I will fire you.”
Assumptions:
I. With Narayan’s statement, the employee will work hard to achieve his targets.
II. All bosses are in the habit of intimidating their employees.

6. Statement: In the last year, Lucknow University has launched a number of management courses for the better future of its students.
Assumptions:
I. These courses may attract students.
II. Lucknow University is conscious about the future of students.

7. Statement: “We should take permission from the director of the college in order to organize the annual fest.” one student said to another.
Assumptions:
I. Without the director’s permission we can’t organize the annual fest.
II. The director does not want to celebrate annual fests.

8. Statement: “If you want to specialize in Human Resource Management in India, join XLRI.”
Assumptions:
I. XLRI provides specialization in HRM.
II. XLRI has the best HRM course in India.

9. Statement: Most rural folks are uneducated and therefore superstitious.
Assumptions:
I. Education increases rational thinking.
II. Rural folks don’t go to school.

10. Statement: Since the exam is held on the third Sunday of November it is going to be held on 18th November this year.
Assumptions:
I. This year it will not be held on 15th, 19th, 20th or 21st of November.
II. This year it will not be held on 16th or 17th November.

11. Statement: "See it first only on our channel", an advertisement of a news channel.
Assumptions:
I. Advertising is the first and most important thing for every channel.
II. Other news channels do not want to show things first.

12. Statement: To keep my knowledge about world affairs strong I always read The Economist magazine.
Assumptions:
I. There is no other magazine except The Economist for world affairs.
II. The Economist is a cheap magazine.

13. Statement: The present condition of law and order is the worst of all time.
Assumptions:
I. The present government does not pay heed to the requirements of maintaining law and order.
II. In earlier days, the law and order condition was better.

14. Statement: India's contribution to the growing global market of management consultancy may touch $5 billion by 2030, which is 10 per cent of the current export size.
Assumptions:
I. The current export size is $50 billion.
II. The current year is 2010.

15. Statement: A Harvard University study raises fresh concerns about the anti-allergy drug Crocin which is prescribed to patients following antibiotic medication. Compared with other two anti-allergy drugs (one of the three must be prescribed), this study shows that Crocin significantly raises the risk of adverse side effects following antibiotic medication.
Assumptions:
I. There is no risk if one of the two other anti-allergy drugs are administered.
II. It is necessary to take an anti-allergy drug after antibiotic medication.

Solution

1. Only Assumption II is relevant here as it justifies the argument in the statement of going for easier things rather than tougher things.

2. Both assumptions are invalid and not implied. Hence, option (d) is correct.

3. Neither assumption is valid. Hence, option (d) is correct.

4. This advice will only be given if Assumption I is true. We cannot assume anything about A's advisory skills hence assumption II is not necessary for the statement.

5. Assumption I is implied as the boss would want (and assumes) this reaction from the employee. The second assumption is invalid.

6. Assumption I is an assumption since the University has launched the courses with this intention. Assumption II is also valid since the launching of these courses shows that the Lucknow University is conscious about the future of its students.

7. Assumption I is obvious while Assumption II is something that we know nothing about.

8. Both assumptions are implied.

9. Assumption I supports the basic conclusion that is drawn in the statement and is hence implied but Assumption II is something that we cannot make a judgment about since it is possible to be uneducated even if you have gone to school.

10. Both Assumptions are implied.

11. The first Assumption is irrelevant while the second is also not implied in the statement since we cannot judge about the intentions of every other news channel on the basis of the statement.

12. Neither Assumption is implied.

13. If the present condition is the worst ever, obviously law and order being the responsibility of the government it must be the case that the present government does not pay heed to the requirements of maintaining law and order. Besides, we can also assume that the condition must have been better earlier.

14. We cannot say anything about the current year.

15. Since one of the three must be prescribed, Assumption II must be implied.

9. SET THEORY, VENN DIAGRAMS AND NETWORK DIAGRAMS

9.1 SET THEORY AND VENN DIAGRAMS

Venn diagrams are an efficient way of representing and analyzing sets and performing set operations. As such, the usage of Venn diagrams is just the elaboration of a solving technique.
Let us first take a look at some standard theoretical inputs related to set theory. Look at the following diagrams:
Figure 1: Refers to the situation where there are two attributes A and B. (Let's say A refers to people who passed in Physics and B refers to people who passed in Chemistry.) Then the shaded area shows the people who passed both Physics and Chemistry.

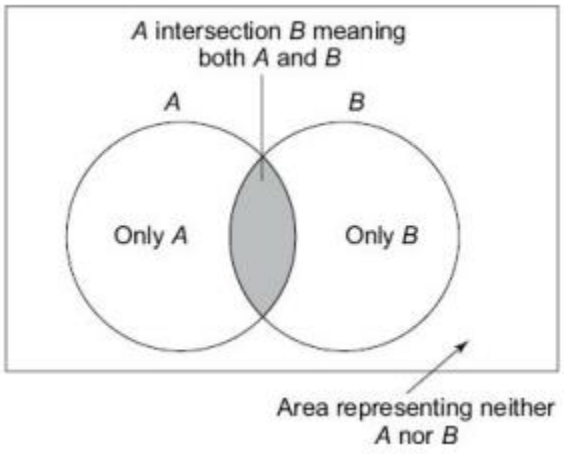
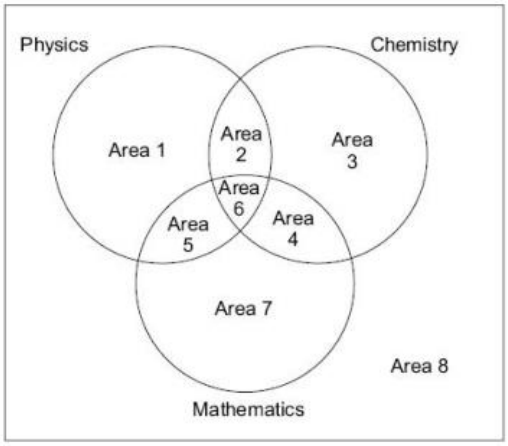


Figure 1.
In mathematical terms, the situation is represented as:
Total number of people who passed at least 1 subject = A + B – A « B
(A « B means A∩B, or in words A intersection B, while A » B means A∪B, or in words A union B)

Figure 2: Refers to the situation where there are three attributes being measured. In the figure below, we are talking about people who passed Physics, Chemistry and/or Mathematics.



In the above figure, the following explain the respective areas:
Area 1: People who passed in Physics only
Area 2: People who passed in Physics and Chemistry only (in other words—people who passed Physics and Chemistry but not Mathematics)
Area 3: People who passed Chemistry only.
Area 4: People who passed Chemistry and Mathematics only (also, can be described as people who passed Chemistry and Mathematics but not Physics).
Area 5: People who passed Physics and Mathematics only (also, can be described as people who passed Physics and Mathematics but not Chemistry).
Area 6: People who passed Physics, Chemistry and Mathematics.
Area 7: People who passed Mathematics only.
Area 8: People who passed in no subjects.
Also take note of the following language which there is normally confusion about:
People passing Physics and Chemistry—Represented by the sum of areas 2 and 6
People passing Physics and Maths—Represented by the sum of areas 5 and 6
People passing Chemistry and Maths—Represented by the sum of areas 4 and 6
People passing Physics—Represented by the sum of the areas 1, 2, 5 and 6
In mathematical terms, this means:
Total number of people who passed at least 1 subject =
P + C + M – P « C – P « M – C « M + P « C « M

9.2 NETWORK DIAGRAMS

A network diagram is made up of a set of nodes and lines that connect those nodes. In questions involving networks, a complex route diagram between two particular cities or places is given and the student is required to identify the total number of distinct routes or the total amount of a particular product flow from one city to other through pipelines. Other variations include complex questions that involve calculation of the shortest distance, the time taken to reach based on speed calculations etc.

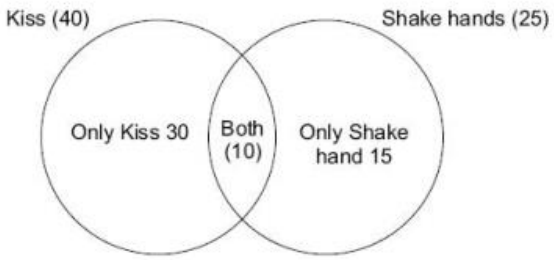
Strategies**:**
For any given network, wherever it is logically permitted, divide the network into parts. If there are two parts of an operation and these two parts can be performed in m and n ways, then that operation can be performed in m x n ways.
If using one tool an operation can be performed in x ways, using another tool in y ways, and using yet another tool in z ways, the overall operation can be performed in x + y z ways.
In case of product flow related questions, a proper understanding of the slack concept can be helpful. Slack = the total capacity of the pipeline – the amount of product flow within that pipeline.

Examples
1)

At the birthday party of Sherry, a baby boy, 40 persons chose to kiss him and 25 chose to shake hands with him. 10 persons chose to both kiss him and shake hands with him.

How many persons turned out at the party?
(a) 35
(b) 75
(c) 55
(d) 25

Answer-
From the figure, it is clear that the number of people at the party were 30 + 10 +15 = 55.
We can of course solve this mathematically as below:
Let n(A) = No. of persons who kissed Sherry = 40
n(B) = No. of persons who shake hands with Sherry = 25 and n(A « B) = No. of persons
who shook hands with Sherry and kissed him both = 10



Then using the formula, n(A » B) = n(A) + n(B) – n(A « B)
n(A » B) = 40 + 25 – 10 = 55

2.

Directions for Questions i to iv: Refer to the data below and answer the questions that follow:
In an examination 43% passed in Math, 52% passed in Physics and 52% passed in Chemistry. Only 8% students passed in all the three. 14% passed in Math and Physics and 21% passed in Math and Chemistry and 20% passed in Physics and Chemistry. Number of students who took the exam is 200. Let Set P, Set C and Set M denotes the students who passed in Physics, Chemistry and Math respectively. Then,

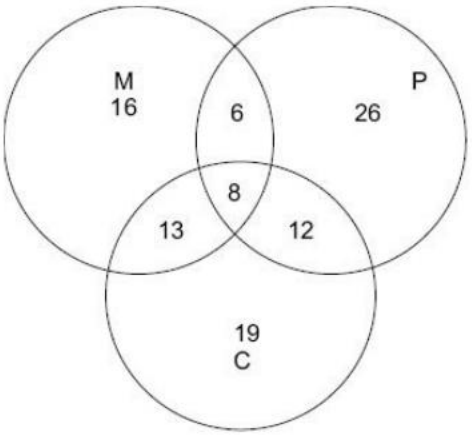
i. How many students passed in Math only?
(a) 16
(b) 32
(c) 48
(d) 80

ii. Find the ratio of students passing in Math only to the students passing in Chemistry only?
(a) 16:37
(b) 29:32
(c) 16:19
(d) 31:49

iii. What is the ratio of the number of students passing in Physics only to the students passing in either Physics or Chemistry or both?
(a) 34/46
(b) 26/84
(c) 49/32
(d) None of these

iv. A student is declared pass in the exam only if he/she clears at least two subjects. The number of students who were declared passed in this exam is?
(a) 33
(b) 66
(c) 39
(d) 78

Answer
Let P denote Physics, C denote Chemistry and M denote Maths.
% of students who passed in P and C only is given by
% of students who passed in P and C – % of students who passed all three = 20% – 8% = 12%
% of students who passed in P and M only is given by % of students who passed in P and M – % of students who passed all three = 14% – 8% = 6%
% of students who passed in M and C only is: % of students who passed in C and M – % of students who passed all three = 21% – 8% = 13%
So, % of students who passed in P only is given by:
Total no. passing in P – No. Passing in P & C only – No. Passing P & M only – No. Passing in all threeÆ 52% – 12% – 6% – 8%- = 26%
% of students who passed in M only is:
Total no. passing in M – No. Passing in M & C only – No. Passing P & M only – No. Passing in all threeÆ 43% – 13% – 6% – 8%- = 16%
% of students who passed in Chemistry only is
Total no. passing in C – No. Passing in P & C only – No. Passing C & M only – No. Passing in all three Æ 52% – 12% – 13% – 8%– = 19%



The answers are:

i. Only Math =16% = 32 people. Option (b) is correct.
ii. Ratio of Only Math to Only Chemistry = 16:19. Option (c) is correct.
iii. 26:84 is the required ratio. Option (b) is correct.
iv. 39 % or 78 people. Option (d) is correct.

3.

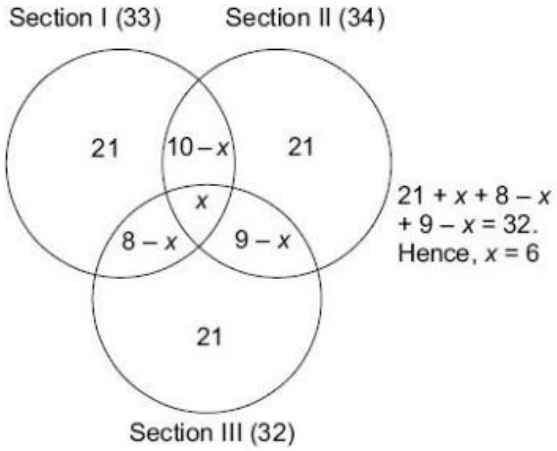
Directions for Questions i to iii: Refer to the data below and answer the questions that follow.
Last year, there were 3 sections in the Catalyst, a mock CAT paper. Out of them 33 students cleared the cut-off in Section 1, 34 students cleared the cut-off in Section 2 and 32 cleared the cut-off in Section 3. 10 students cleared the cut-off in Section 1 and Section 2, 9 cleared the cut-off in Section 2 and Section 3, 8 cleared the cut-off in Section 1 and Section 3. The number of people who cleared each section alone was equal and was 21 for each section.

i. How many cleared all the three sections?
(a) 3
(b) 6
(c) 5
(d) 7

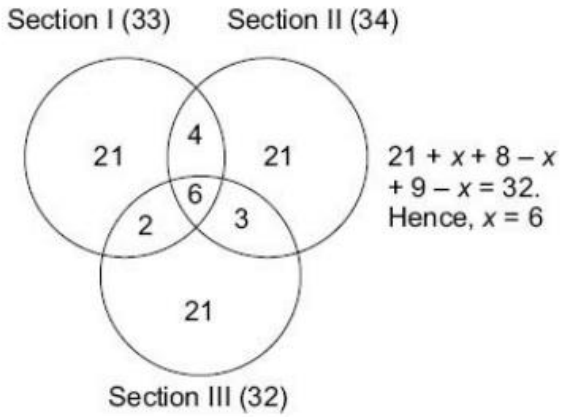
ii. How many cleared only one of the three sections?
(a) 21
(b) 63
(c) 42
(d) 52

iii. The ratio of the number of students clearing the cut-off in one or more of the sections to the number of students clearing the cutoff in Section 1 alone is?
(a) 78/21
(b) 3
(c) 73/21
(d) None of these

Answer



Since, x = 6, the figure becomes:



The answers would be:

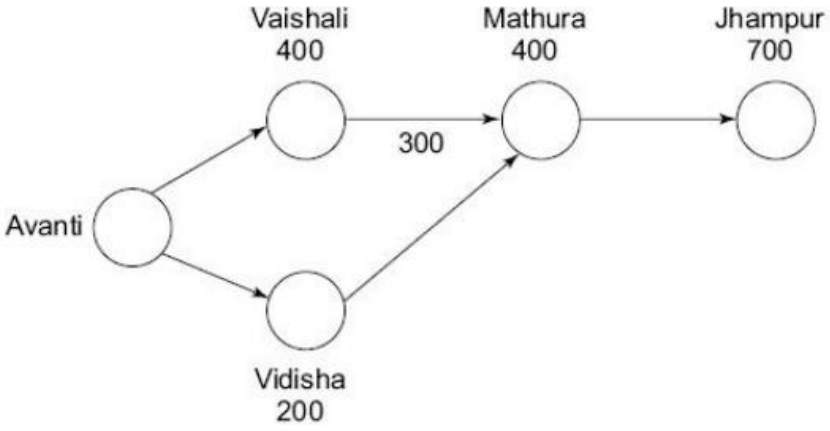
i. 6. Option (b) is correct.

ii. 21 + 21 + 21= 63. Option (b) is correct.

iii. (21 + 21 + 21 + 6 + 4 + 3 + 2)/21 = 78/21. Option (a) is correct.

4.

Directions for Questions i to iii: Answer these questions based on the pipeline diagram below.



The following sketch shows the pipelines carrying material from one location to another. Each location has a demand for material. The demand at Vaishali is 400, at Mathura is 400, at Jhampur is 700 and at Vidisha is 200. Each arrow indicates the direction of material flow through the pipeline. The flow from Vaishali to Mathura is 300. The quantity of material flow is such that the demands at all these locations are exactly met. The capacity of each pipeline is 1000.

i. What is the free capacity available in the Avanti-Vidisha pipeline?
(a) 300
(b) 200
(c) 100
(d) 0

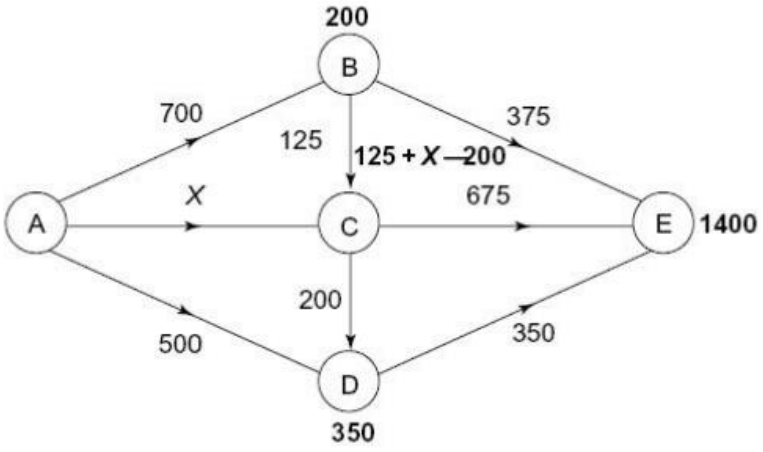
ii. What is the free capacity available from Avanti to Vaishali?
(a) 0
(b) 100
(c) 200
(d) 300

iii. The quantity moved from Avanti to Vidisha is
(a) 200
(b) 800
(c) 700
(d) 1000

Answer-
Since 700 is required at Jhampur, the requirement at Mathura must be 1100, which has to be supplied from the two pipelines coming into Mathura. It is clear that since Vaishali to Mathura is only 300, the Vidisha-Mathura pipeline should carry 800. Hence, Avanti Vidisha should have 1000.

i. There is no free capacity in the Avanti Vidisha pipeline. Option (d) is correct.
ii. Avanti-Vaishali flow should be 700 and hence the free capacity is 300. Option (d) is correct.
iii. 1000. Option (d) is correct.

5.
Directions for Questions i to iii: In the network diagram above, the figures represent the flow of natural gas through pipelines between major cities A, B, C, D & E (in suitable units). Assume that supply equals demand in the network (although not on individual nodes).



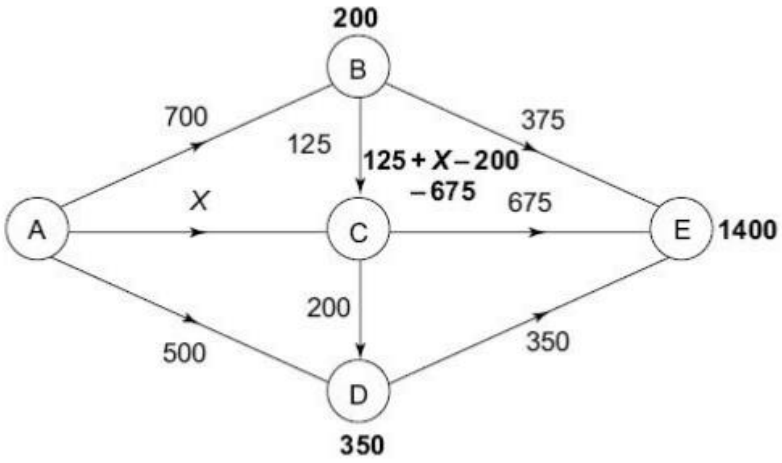
In the network diagram above, the figures represent the flow of natural gas through pipelines between major cities A, B, C, D & E (in suitable units). Assume that supply equals demand in the network (although not on individual nodes).

i. What is the number of units demanded at B?
(a) 175
(b) 200
(c) 225
(d) 250

ii. If the number of units demanded in C is 225, what is the value of x?
(a) 975
(b) 875
(c) 775
(d) 950

iii. What is the demand in D?
(a) 300
(b) 350
(c) 375
(d) 450

Sol. Refer to the following figure for the solution:



i. From the figure it would be 700 – (375 + 125) = 200. Option (b) is correct.
ii. 125 + X – 200 – 675 = 225 Æ X = 975. Option (a) is correct.
iii. 500 + 200 – 350 = 350. Option (b) is correct.

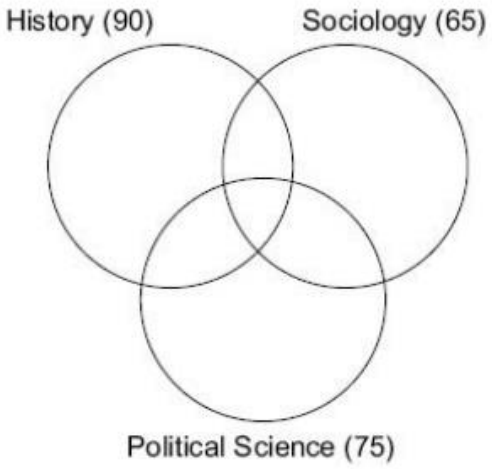
Exercise

Q1)
Directions for Questions i and ii: Refer to the data below and answer the questions that follow.
In a test in which 120 students appeared, 90 passed in History, 65 passed in Sociology and 75 passed in Political Science. 30 students passed in only one subject and 55 students in only two. 5 students passed no subjects.

i. How many students passed in all the three subjects?
(a) 25
(b) 30
(c) 35
(d) Data insufficient

ii. Find the number of students who passed in at least two subjects.
(a) 85
(b) 95
(c) 90
(d) Data insufficient

Answer-The given situation can be read as follows:



115 students are being counted 75 + 65 + 90 = 230 times.
This means that there is an extra count of 115. This extra count of 115 can be created in 2 ways.
A. By putting people in the 'passed exactly two subjects' category. In such a case each person would get counted 2 times (double counted), i.e., an extra count of 1.
B. By putting people in the 'all three' category, each person put there would be triple counted. 1 person counted 3 times – meaning an extra count of 2 per person.
The problem tells us that there are 55 students who passed exactly two subjects. This means an extra count of 55 would be accounted for. This would leave an extra count of 115–55 = 60 more

to be accounted for by ‘passed all three’ category. This can be done by using 30 people in the ‘all 3’ category. Hence, the answers are:
i. Option (b)
ii. Option (a)

Q2)
Directions for Questions i to vii: Refer to the data below and answer the questions that follow.
In a survey among students at all the IIMs, it was found that 48% preferred coffee, 54% liked tea and 64% smoked. Of the total, 28% liked coffee and tea, 32% smoked and drank tea and 30% smoked and drank coffee. Only 6% did none of these. If the total number of students is 2000 then find

i. The ratio of the number of students who like only coffee to the number who like only tea is
(a) 5:3
(b) 8:9
(c) 2:3
(d) 3:2

ii. Number of students who like coffee and smoking but not tea is
(a) 600
(b) 240
(c) 280
(d) 360

iii. The percentage of those who like coffee or tea but not smoking among those who like at least one of these is
(a) more than 30
(b) less than 30
(c) less than 25
(d) none of these

iv. The percentage of those who like at least one of these is
(a) 100
(b) 90
(c) Nil
(d) 94

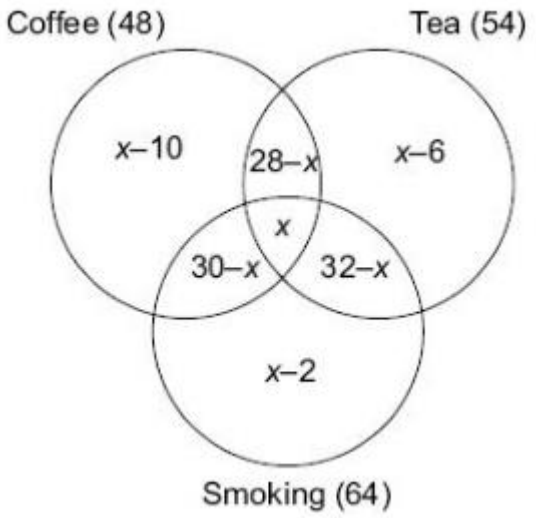
v. The two items having the ratio 1:2 are
(a) Tea only and tea and smoking only.
(b) Coffee and smoking only and tea only.
(c) Coffee and tea but not smoking and smoking but not coffee and tea.
(d) None of these

vi. The number of persons who like coffee and smoking only and the number who like tea only bear a ratio
(a) 1:2
(b) 1:1
(c) 5:1
(d) 2:1

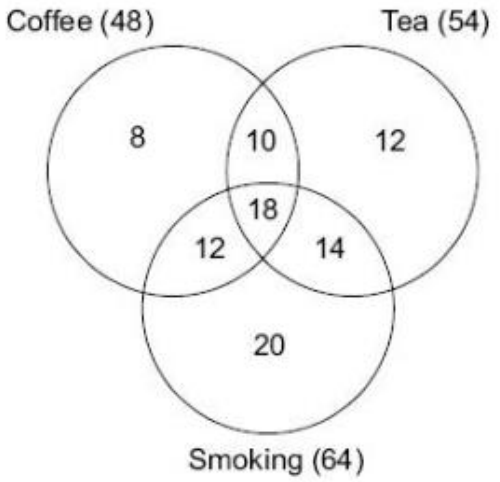
vii. Percentage of those who like tea and smoking but not coffee is
(a) 14
(b) 14.9
(c) less than 14
(d) more than 15

S2)

If you try to draw a figure for this question, the figure would be something like:



We can then solve this as:
x – 10 + 28 – x + x + 30 – x + x + 2 + 32 – x + x – 6 = 94 Æ x + 76 = 94 Æ x = 18.
Note: In this question, since all the values for the use of the set theory formula are given, we can find the missing value of students who liked all three as follows:
94 = 48 + 54 + 64 – 28 – 32 – 30 + All three Æ All three = 18
As you can see this is a much more convenient way of solving this question, and the learning you take away for the 3 circle situation is that whenever you have all the values known and the only unknown value is the center value – it is wiser and more efficient to solve for the unknown using the formula rather than trying to solve through a venn diagram. Based on this value of x we get the diagram completed as:



The answers then are:

i. 8:12 = 2:3 Æ Option (c) is correct.
ii. 12 % of 2000 = 240. Option (b) is correct.
iii. 30/94 Æ more than 30%. Option (a) is correct.
iv. 94%. Option (d) is correct.
v. Option (c) is correct as the ratio turns out to be 10:20 in that case.
vi. 12:12 = 1:1 Æ Option (b) is correct.
vii. 14%. Option (a) is correct.

Q3)

Directions for Questions i to v: Refer to the data below and answer the questions that follow.
In the McGraw Hill Mindworkzz Quiz held last year, participants were free to choose their respective areas from which they were asked questions. Out of 880 participants, 224 chose Mythology, 240 chose Science and 336 chose Sports, 64 chose both Sports and Science, 80 chose Mythology and Sports, 40 chose Mythology and Science and 24 chose all the three areas.

i. The percentage of participants who did not choose any area is
(a) 23.59%
(b) 30.25%
(c) 37.46%
(d) 27.27%

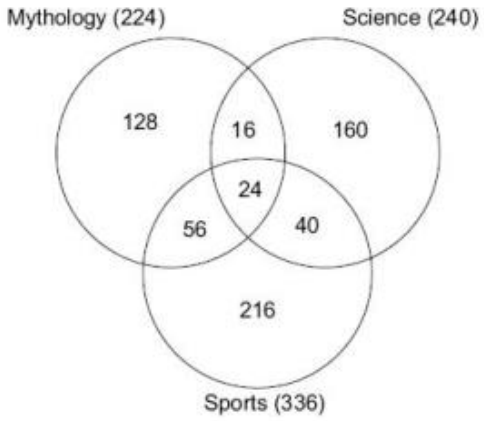
ii. Of those participating, the percentage who choose only one area is
(a) 60%
(b) more than 60%
(c) less than 60%
(d) more than 75%

iii. Number of participants who chose at least two areas is
(a) 112
(b) 24
(c) 136
(d) None of these

iv. Which of the following areas shows a ratio of 1:8?
(a) Mythology & Science but not Sports: Mythology only
(b) Mythology & Sports but not Science: Science only
(c) Science: Sports
(d) None of these

v. The ratio of students choosing Sports & Science but not Mythology to Science but not Mythology & Sports is
(a) 2:5
(b) 1:4
(c) 1:5
(d) 1:2
S3)

The following figure would emerge on using all the information in the question:



The answers would then be:
i. 240/880 = 27.27%. Option (d) is correct.
ii. 504/880 = 57.27%. Hence, less than 60. Option (c) is correct.
iii. 40 + 16 + 56 + 24 = 136. Option (c) is correct.
iv. Option a gives us 16:128 = 1:8. Option (a) is hence correct.
v. 40:160 Æ 1:4. Option (b) is correct.

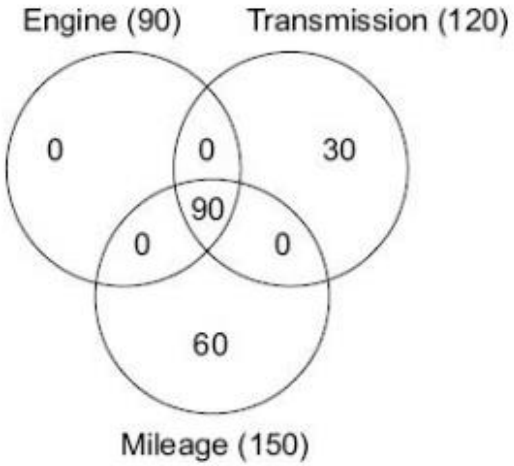
Q4)
Answer the questions based on the following information.
The following data was observed from a study of car complaints received from 180 respondents at Colonel Verma's car care workshop, viz., engine problem, transmission problem or mileage problem. Of those surveyed, there was no one who faced exactly two of these problems. There were 90 respondents who faced engine problems, 120 who faced transmission problems and 150 who faced mileage problems.

i. How many of them faced all the three problems?
(a) 45
(b) 60
(c) 90
(d) 20

ii. How many of them faced either transmission problems or engine problems?
(a) 30
(b) 60
(c) 90
(d) 40

S4)

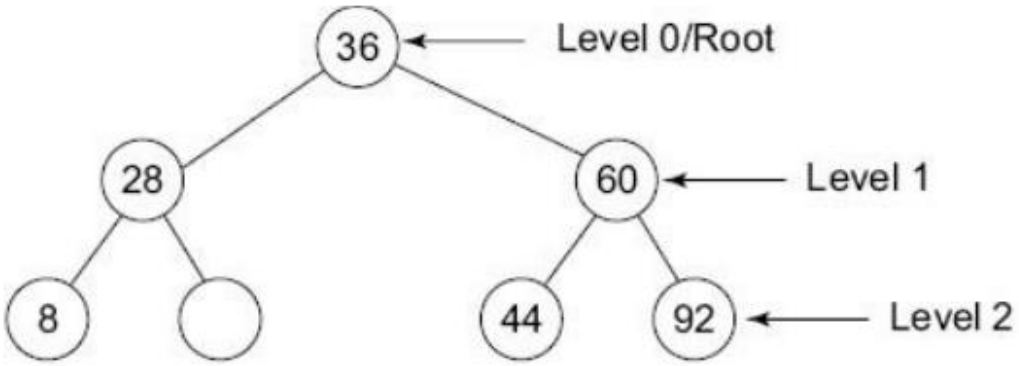
i. Option (c) is correct.
ii.There are 30 such people. Option (b) is correct.



Q5)

Direction for Questions i to iii: In the ancient game of "placing the numbers on a tree branch" a sequence of numbers is provided to an individual participant and he is then asked to place the numbers in the branches of a hanging tree with the first number forming the top most node of the tree (and this node is called the root of the tree) and the subsequent numbers are placed in the left sub tree if the number is smaller than the number at the root or in the right sub tree, if

the number is greater than the number in the root. A node is denoted by an oval and has to include a number in it. Each node at a level can be viewed as a tree itself and the same rule applies to it as well. For every node, if a number is greater than the number at the node, then it must be in the nodes to its right and if the number be smaller than the number at the node, then it must be in the nodes to its left. New numbers must obey the rule of all nodes above itself. Numbers shall not repeat themselves in the nodes and in case a number appears which has already appeared earlier in the tree it has to be placed on the right most of its possible positions. E.g, 36, 28, 60, 92, 8, 44

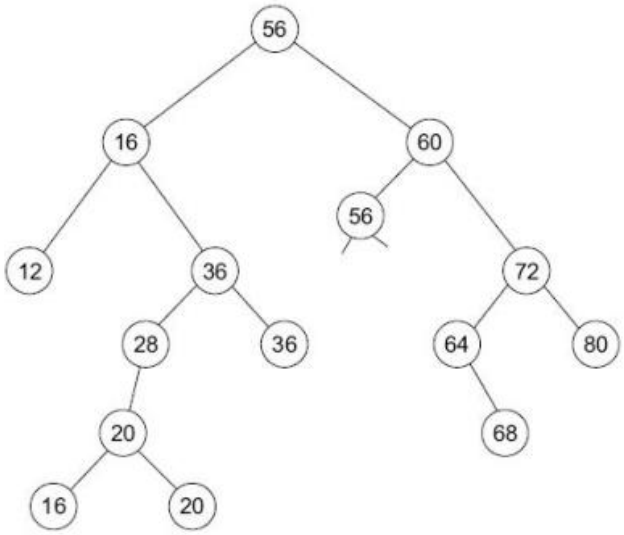


Given is a sequence of numbers and an empty tree. Fill the numbers in the appropriate nodes. The sequence is given in order and starts from 56
56, 60, 16, 36, 28, 72, 12, 20, 64, 16, 80, 68, 36, 56, 20

i. What are the numbers at level 2?
(a) 12, 20, 28
(b) 12, 36, 72
(c) 12, 36, 56, 72
(d) 16, 36, 64, 80

ii. What is the sum of the numbers at the level 3?
(a) 172
(b) 128
(c) 76
(d) 208

iii. How many nodes are at the 4th level?
(a) 7
(b) 3
(c) 9
(d) 2
S5)



The answers can be read off the above grid.

i. Option (c) is correct.

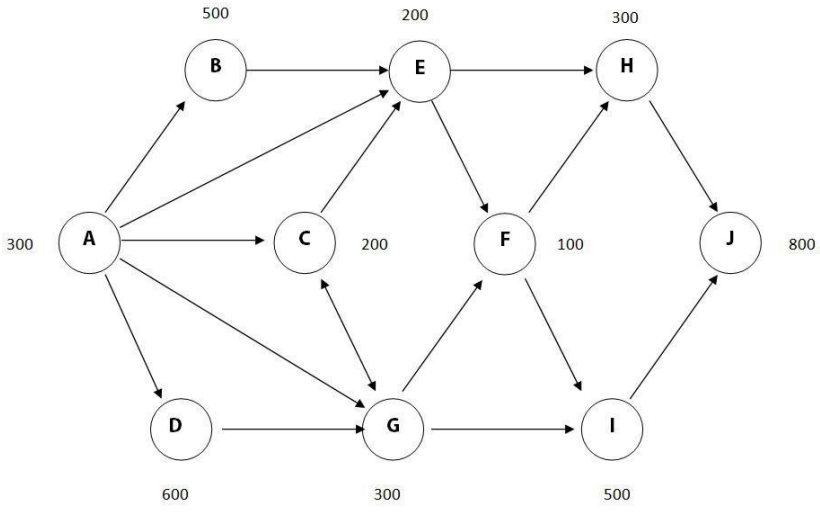
ii. The sum is 208 = 80 + 64 + 36 + 28. Option (d) is correct.

iii. There are 2 nodes at the 4th level. Option (d) is correct.

Q6)

Direction for Questions i to iii:

Given, a network of pipelines carrying oil for distribution among ten towns A, B, C, D, E, F, G, H, I and J.



The maximum capacity of every pipeline shown is 1000 units.

The values given next to a town represents the consumption of oil at that town.

Given,

a) The oil flows only in the directions shown by the arrows.

b) The slack in any pipeline is the extra oil that can flow through it to bring the pipeline to full capacity.

c) At every town, the requirements are exactly met.

The slack details of some of the pipelines are given as follows:

Pipeline	Slack
AB	200
AC	200
FH	900
CE	500
DG	700
EF	800
GI	600

i. The difference between the slack in the pipeline connecting A and G and the slack in the pipeline connecting F and I is

(1) 300 units
(2) 600 units
(3) 400 units
(4) 500 units
(5) Cannot be determined

ii. If the flow in the pipeline connecting I and J is 300 units, then the slack in the pipeline connecting A and E is
(1) 700 units
(2) 200 units
(3) 500 units
(4) 600 units
(5) Cannot be determined

iii. If the slack in the pipeline that supplies to town A is 400 units, then the capacity of that pipeline (assuming that it is the only pipeline supplying to town A) is
(1) 3600 units
(2) 2800 units
(3) 4200 units

(4) 4400 units
(5) 4900 units

iv. The flow in the pipeline from C to G is

(1) 100
(2) 400
(3) 700
(4) 200
(5) Cannot be determined

S6)

The basic approach here is to balance the inflows and out flows plus consumption at each city.

i. Consider city F. Inflow = outflow + consumption EF + GF = FH + FI + Consumption at F

200 + GF = 100 + FI + 100, GF = FI ……..(a)

Again at city G,

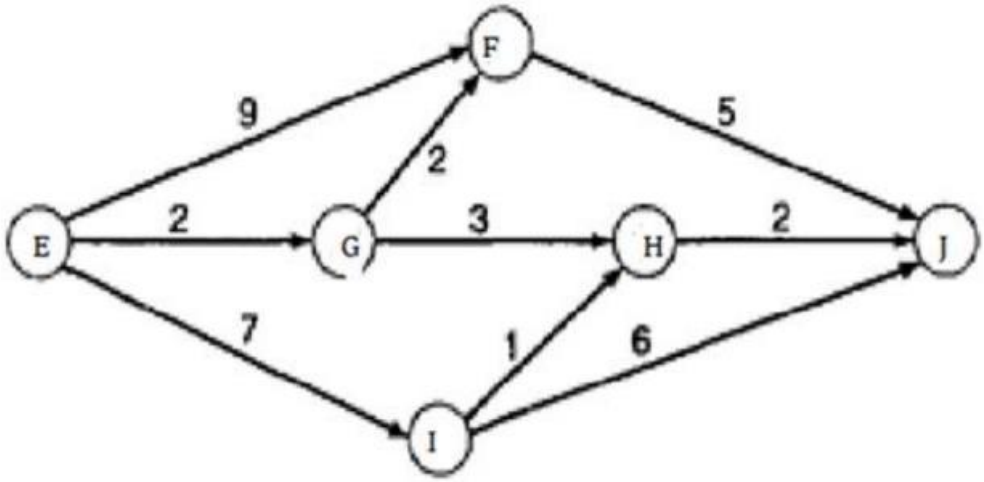
CG + AG + DG = GF + GI + Consumption at G
Now CG = AC – 200 – 500 = 100
Hence, 100 + AG + 300 = GF + 400 + 300, AG – GF = 300 ……..(b)
From (a) and (b)
AG – FI = 300 (Note: Difference in flows = difference in slacks)

ii. Using the information given that IJ = 300 units, we can solve for HJ as follows
HJ – 800 – 300 = 500 (a) (at city J)
From (a) EH = (300 + 500) – 100 = 700 (b) (at city H)
Now consider city E
AE = 300 + 500 = 200 + 700 + 200 à AE = 300 (BE = 800 – 500 = 300)
Hence, slack = 1000 – 300 = 700.

iii. The total inflow into the city A has to be equal to the total requirements of all the cities together i.e. Inflow into A = 300 + 500 + 200 + 600 + 200 + 100 + 300 + 300 + 500 + 800 = 3800. If slack is 400, then the capacity = 3800 + 400 = 4200.

iv. Given that the flow from AC= 800 and at C we have consumption = 200 Flow in CE = 500
Hence, C to G = 800 – (500 + 200) = 100

Q7)
A significant amount of traffic flows from point E to point J in the one-way street network shown below. Points F, G, H and I are junctions in the network, and the arrows mark the direction of traffic flow. The fuel cost in rupees for travelling along a street is indicated by the number adjacent to the arrow representing the street.



Motorists travelling from point E to point J would obviously take the route for which the total cost of travelling is the minimum. If two or more routes have the same least travel cost, then motorists are indifferent between them. Hence, the traffic gets evenly distributed among all the least cost routes. The government can control the flow of traffic only by levying appropriate toll at each junction. For example, if a motorist takes the route E-F-J (using junction F alone), then total cost of travel would be Rs.14. (i.e., Rs. 9 + Rs 5) plus the toll charged at junction F.

Question 1: If the government wants to ensure that the traffic at S gets evenly distributed along streets from E to F, from E to G, and from E to I, then a feasible set of toll charged (in rupees) at junctions F,G,H and I, respectively to achieve this goal is
(a) 0, 4, 3, 2
(b) 1, 5, 3, 3
(c) 0, 5, 2, 2
(d) 1, 5, 3, 2
(e) 0, 5, 4, 1

Question 2 : If the government wants to ensure that all motorists travelling from E to J pay the same amount (fuel costs and toll combined) regardless of the route they choose and the street from G to H is under repairs (and hence unusable), then a feasible set of toll charged (in rupees) at junctions F,G, H and I respectively to achieve this goal is :
(a) 2, 5, 3, 2
(b) 0, 5, 3, 1
(c) 1, 5, 3, 0
(d) 2, 3, 5, 1
(e) 1, 3, 5, 1

Question 3: If the government wants to ensure that all routes from E to J get the same amount of traffic, then a feasible set of toll charged (in rupees) at junctions F,G,H and Irespectively to achieve this goal is
(a) 0, 5, 4, 1
(b) 0, 5, 2, 2
(c) 1, 5, 4, 2
(d) 1, 5, 3, 2
(e) 1, 5, 3, 3

Question 4: If the government wants to ensure that no traffic flows on the street from I to J, while equal amount of traffic flows through junctions F and H, then a feasible set of toll charged (in rupees) at junctions F,G,H and I respectively to achieve this goal is
(a) 1, 5, 3, 3
(b) 0, 5, 2, 3
(c) 1, 5, 4, 2
(d) 1, 4, 4, 3
(e) 0, 5, 2, 2

Question 5: The government wants to devise a toll policy such that the total cost to the commuters per trip is minimized. The policy should also ensure that not more than 70% of the total traffic passes through junction G. The cost incurred by the commuter travelling from point E to point J under this policy will be
(a) Rs. 7
(b) Rs. 9
(c) Rs. 10
(d) Rs. 13
(e) Rs. 14

S7)
Answer 1: (d) Traffic among the different paths will be evenly distributed if the cost of travel (fuel cost plus toll charges) is same through all the given routes. Option (d) gives the cost of travel through all the routes as same.
F= 1, G=5, H = 3, I = 2
E ⇒ F: EFJ = 9 + 1 +5 = Rs.15
E⇒G: EGFJ = 2 + 5 + 2 + 1 +5 = Rs.15
EGHJ=2 + 5+3+3 + 2 = Rs.15
E⇒I: EIHJ = 7 + 2 + 1 +3 + 2 = Rs.15
EIJ= 7 + 2 + 6 = Rs.15

Answer 2: (b) It is given that path GH is under repair, hence, available paths to reach from E to J are EGHJ, EFJ, EIHJ and EIJ. Now, using options, we find that option (b) gives the same cost of travel from S to T.
F=0, G = 5, H = 3, I = 1
EGFJ = 2 + 5 + 2 + 0 + 5 = Rs.14

EFJ = 9 + 0 + 5=Rs.14
EIHJ = 7 + l + l + 3 + 2 = Rs.14
EIJ = 7 + l + 6 = Rs.14

Answer 3: (d) For option (d), we have F= 1, G=5, H = 3, I = 2
EFJ = 9 + l + 5=Rs.15
EGFJ = 2 + 5 + 2 + 1 +5 = Rs.15
EIJ = 7 + 2 + 6 = Rs.15
EIHJ = 7 + 2 + l + 3 + 2 = Rs.15
EGHJ = 2 + 5 + 3+3 + 2 = Rs.15

Answer 4: (e) Traffic from I to J is restricted, hence, available routes through F and H are EFJ,EGFJ,EGHJ,EIHJ. Option (e) gives the same cost of travel.
EFJ = 9 + 0 + 5=Rs.14
EGFJ = 2 + 5 + 2+0 + 5 = Rs.14
EGHJ =2 + 5 + 3+2 + 2 = Rs.14
EIHJ = 7 + 2 + l + 2 + 2 = Rs.14

Answer 5: (c) Option (a) is ruled out because cost incurred from E to J will be Rs 7 only through EGHJ
assuming toll charges as G = 0 and H = 0, but in this case all traffic will be move through junction G. Option (b) is ruled out because cost incurred from E to J will be Rs 9 when toll at G = 0 and H = 2. In this case, all traffic will be diverted through junction G. Using option (c), we find that if toll charges at G = 3 and H = 0, then travel cost from E to J through G will be Rs 10.
EGHJ = 2 + 3+3 + 0 + 2 = Rs.10 and the same cost will be incurred through EIHJ = 7 + 0 + l + 0 + 2 = Rs.10. Since, Rs 10 is the least, hence, options (d) and (e) are ruled out.

10. BINARY LOGIC

These question types are called binary logic questions simply because each question contains two logic streams that have to be matched, in order to get to the correct answer. In a binary logic problem, we have people who either speak a true statement or a false statement. These people are divided into three categories:

1. Truth-teller: This person will always speak the truth. All the statements made by this person are true.
2. Liar: This person will always tell a lie. All statements made by this person are false.
3. Alternator: This person always alternates between the truth and the lie. If first statement of this person is true, then second will be false, third will be true and so on. Similarly, if first statement made by this person is false, then second will be true, third will be false and so on. There is no particular number of true statements or false statements made by this person but order is always TRUE-FALSE-TRUE- or FALSE-TRUE-FALSE.

Mainly, there are two types of questions as below:

Type 1

In this question type, three persons speak two statements each—one of which is true, the other is false. The two logic streams to be considered are:

1. The logic of the Statements, i.e., the logic of what is said within the statements.

2. The logic of the Basic Conditions, i.e., the logic of the fact that if one sentence is taken as true, the other will be false automatically.

The process of solving these questions is best illustrated through an example. Gauri Islands is the name of an island. The inhabitants of this island always answer any question with two sentences. One of which is always true and the other is always false. Milly, Silly and Dilly are the three daughters of the chief whip of this island. Out of them, two are minor and one is of a marriageable age. You have been caught as an intruder on the island and you have two options given by the chief whip: identify his daughter who is of marriageable age. If you do so, you can have the privilege of marrying her and becoming the new chief whip in the future. On the other hand, if you cannot, you will be executed. Only Silly has dentures in her teeth. On questioning the three daughters, these are the answers you get: Milly: “I am shorter than Silly. The girl of marriageable age has dentures in her teeth.”

Silly: “I am shorter than Milly. Dilly is the one who is of a marriageable age.”

Dilly: The girl of marriageable age is amongst the three of us. I am of a marriageable age.”

Who is the girl of marriageable age?

(a)Milly

(b) Dolly

(c) Silly

(d) Can’t say

In the above question, you should see that, the first statement of Dilly has to be correct (By Logic of the statement—If you evaluate what the statement is saying, it is clear that it has to be true. It can be easily understood that the girl of marriageable age is amongst the three of the girls.) In this case, if this statement is true, then Dilly’s second statement is automatically false. Further, since Dilly’s second statement is false, Silly’s second statement will also be false. (By evaluating the sentence logic—as both of these statements are saying the same thing.) Hence, Silly’s first statement will be true (Basic Condition logic) and hence further, Milly’s first statement will be false (It is saying the opposite of Silly’s true first statement—Statement Logic). Hence, Milly’s second statement has to be true. Hence, Silly has to be the one of marriageable age.

Type 2

In this type of questions, we have two types of people: those who always speak the truth, and those who always lie. These questions are also classified as Binary Logic questions since they are solved on the basis of two logical streams that run parallel to each other, viz: Basic Condition Logic (BCL) and Logic of the Statements (LoS). The Basic Condition Logic flows from the fact that if a person always speaks the truth, then whatever statement he says must be true. However, in some cases, a clash might arise between the Basic Condition Logic and the Statement Logic, if it can be seen that the statement is obviously false. The best way to understand this question type is by looking at an example:

You meet three inhabitants—Rohit, Mohit and Sohit—standing together. You ask Rohit

“Are you a type X or type Y?” He mumbles something, which you cannot catch, so you ask Mohit, “What did Rohit say?” Mohit replies “Rohit said that he is type Y.” You look at Sohit and he says, “Do not believe Mohit, he is lying.”

Which of the following is true?

1. Rohit is a type X.

2. Mohit is a type Y.

3. Sohit is a type Y.

4. All of these.

By BCL, Sohit could be Type X or Type Y:

If Sohit is Type Y, his statement must be false. That means Mohit is not lying. Hence, Mohit should be Type X and hence his statement must be true, i.e., Rohit must have said that he is Type Y. However, if you look at Rohit, he could not have made this statement in either case. If we assume that Rohit is Type X, he would only speak the truth and he would then say that he is Type X, not that he is Type Y. On the other hand, if we were to assume that Rohit is a Type Y, he would only speak false. But then he could not have said that he is Type Y, since that statement would be true.

Q1)

Directions for questions i-iv: Toya-Toya is an island in the Indian Ocean. A peculiar quality of all the inhabitants of Toya-Toya is that they always answer any question with two sentences only out of which one is always true and the other always false.

(i) You are walking on a road and come to a village. You ask the inhabitants Ramu, Lala and Lalit, —Which road will take me to the village?

Ramu says, —I never speak to strangers, I am new to these parts.

Lala says, —I am married to Lalit. Take the left road.

Lalit says, —I am married to Ramu. He is not new to this place.

Which of the following is true?

(a) Left road takes you to the village

(b) Right road takes you to the village

(c) Lalit is married to Lala

(d) None of these

(ii) You find that your boat is stolen. You question three inhabitants of the island and they reply as follows: Jerry says, —I didn‘t do it. Maya didn‘t do it.

Maya says, —I didn‘t do it. Kanishka didn‘t do it.

Kanishka says, —I didn‘t do it. I don‘t know who did it.

Who stole your boat?

(a) Jerry

(b) Maya

(c) Kanishka

(d) None of them

(iii)You want to speak to the chief of the village. You question three inhabitants, Amardeep, Bhanu and Chandu. Only Bhanu is wearing red shirt.

1. Amardeep says, —I am not Bhanu‘s son. The chief wears a red shirt.

2. Bhanu says, —I am Amardeep father, Chandu is the chief.

3. Chandu says, —The chief is one among us. I am the chief.

Who is the chief?

(a) Amardeep

(b) Bhanu

(c) Chandu

(d) None of them

(iv) There is only one pilot on the island. You interview three men-Kundan, Lucy and Mona. You also notice that Kundan is wearing a cap.

1. Mona says, —Lucy‘s father is the pilot. Lucy is not the priest’s son.

2. Kundan says, —I am the priest. On this island, only priests can wear caps.

3. Lucy says, —I am the priest’s son. Kundan is not the priest.

Which statement is true?

(a) Lucy is not Kundan’s son

(b) Kundan is the pilot

(c) Mona is the pilot

(d) Lucy is the priest

S1)

(i) : (c)

Lalit is married to Lala is the correct statement. To solve this one needs to proceed further after considering one of the statements to be true and then validating all the other statements spoken by the other inhabitants. The statement which does not contradict with any other statement becomes the true answer.

(ii) : (b)

Let us assume first part of the first statement to be true. Then second part will be false. If we relate these results with second and then third statements assuming one part to be true and other false, we will find that answer is Maya. On the contrary if we assume first half of the first statement to be false and second to be true, then there will be two answers; Jerry and Kanishka, which cannot be true. Hence the correct answer is Mathew.

(iii): (b)

It is clear that Bhanu is the chief. All other options go contrary to the information.

(iv) : (b)

Option (b) is the only correct statement. To solve this one needs to proceed further after considering one of the statements to be true and then validating all the other statements spoken by the other inhabitants. The statement which does not contradict with any of the other statements becomes the true answer.

Q10)

Directions for questions i- ii: Consider the following statements and answer the questions that follow. Three criminals were arrested for shop lifting. However, when interrogated, only one of them told the truth in both his statements, while the other two each told one true statement and one lie. The statements were:

	Ti-Ti:	Ki-Ki	Chi-Chi
Statement 1	Chi-chi passed the goods.	Ti-Ti passed the goods.	I took the goods out of the shop.
Statement 2	Ki- Ki created the diversion.	I created the diversion.	Ki-Ki passed goods.

i. Who created the diversion?

A) Ti-Ti
B) Chi-chi
C) Ki-Ki
D) Either 1 or 2

ii. Which of these statements is correct?

A) Chi-Chi created the diversion.
B) Ti-Ti took the goods out of the shop.
C) Ti-Ti passed the goods.
D) Chi-Chi passed the goods.

S10)

For (I), let us assume that Ti – Ti speaks the truth. Then as his both statements are true, so we get the arrangement as Chi -Chi passed the goods, Ki-Ki created diversion and Ti-Ti took goods out of shop.
But when this arrangement is validated as per the statement, we find that there is one Truth Teller, one Liar and one alternator. This scenario violates the given condition that there is one truth teller and two alternators. So our assumption that Ti – Ti is the truth teller is wrong. Assume that Ki-Ki speaks the truth. Then we get the following table:

	Ti-Ti:	Ki-Ki	Chi-Chi
Statement 1	Chi-chi passed the goods. **FALSE**	Ti-Ti passed the goods. **TRUE**	I took the goods out of the shop. **TRUE**
Statement 2	Ki- Ki created the diversion. **TRUE**	I created the diversion. **TRUE**	Ki-Ki passed goods. **FALSE**

The possibility which is mentioned above satisfies the conditions. So, Ti-Ti passed the goods, Ki-Ki created diversion and Chi-Chi took goods out of shop. So answer for (i) is option C.

And for (ii) is option D.

Q12)

Raj, Rahul and Vicky answered the question, “Who among you is a Doctor” in the following manner: We know exactly one of these boys is a Doctor, one is a Painter and one is an Athlete. Further, one always speaks the truth, one always lies and one alternates between the truth and the lie.

	Raj	Rahul	Vicky
Statement 1	*I am a Doctor*	*Vicky is an Athlete*	*I am not a Painter*
Statement 2	*Rahul is a Painter*	*Raj is not a Doctor*	*Rahul is not a painter*
Statement 3	*I am an alternator*	*I am a liar*	*Raj is a liar*

i. Who among them is a doctor?

A) Raj

B) Rahul

C) Vicky

D) Could not be interpreted

ii. Who among them is a painter?

A) Raj

B) Rahul

C) Vicky

D) Could not be interpreted

iii. Who among them is an athlete?

A) Raj

B) Rahul

C) Vicky

D) Could not be interpreted

S12)

In our example question, Rahul makes the statement, “I am a liar”, which clearly tells us that Rahul is the alternator. Also, Raj said that he is an alternator, which means that he can be either a liar or an alternator. Now, since Rahul is the alternator, Raj is the liar. This leaves us with Vicky as the truth-teller. Since Rahul is the alternator, so the statements proceeding/preceding the statement, “I am a liar” are true, we can say that Raj is not a Doctor. Also as Vicky is a truth-teller, according to his statements, Rahul is not a painter and he himself is not a painter, thus Raj is the painter. Now, Rahul’s first statement is a lie (as the order for Rahul’s statements is 1st-false, 2nd-

true,3rd-false), this means Vicky is not an athlete, which leaves us with the only option for Vicky as Doctor and Rahul will then be an athlete.

Summary:

Raj	**Rahul**	**Vicky**
Liar	Alternator	Truth-teller
Painter	Athlete	Doctor

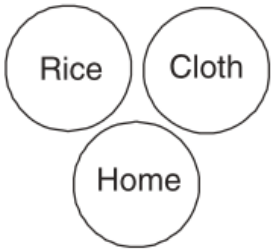
So the answer for (i) is Vicky, option C.

Answer for (ii) is Raj, option A.

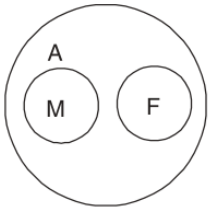
Answer for (iii) is Rahul, option B.

11. LOGICAL VENN DIAGRAMS

In logical Venn diagrams relation between certain groups of items are analyzed. Suppose, you are to relate between three items, e.g. Rice, Cloth and House. As these three items bear no relationship to each other, the possible Venn diagrams for this is



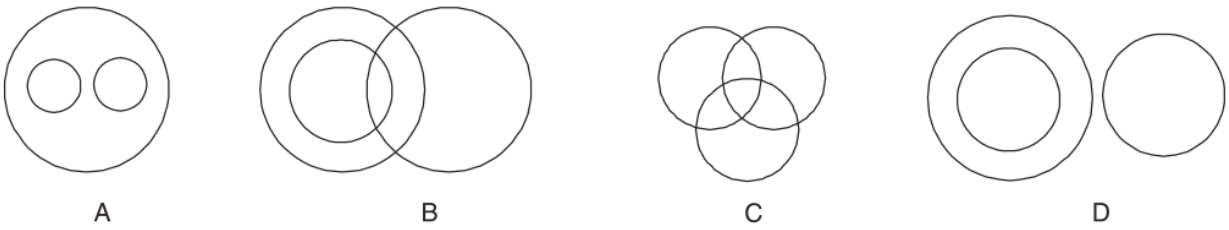
Suppose you are given another three items, Audience, Males, Females. Here the diagram would be-



Clearly males and females are two separate classes, though both of them are audience.

Examples-

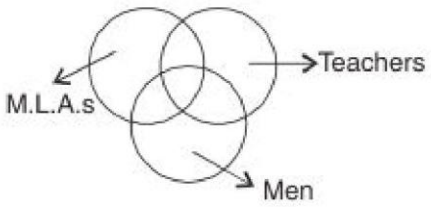
1.



I. M.L.A.s, Teachers, Men
II. Dolphins, Mammals, Fishes
III. Mango, Orange, Fruits
IV. Wives, Widows, Women
V. Boys, Students, Players.

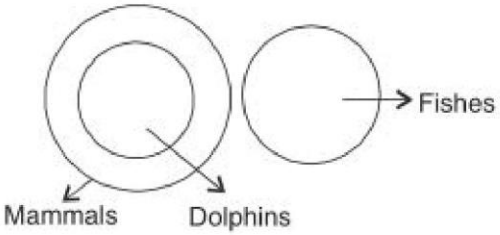
Answer and explanation

I. (C)



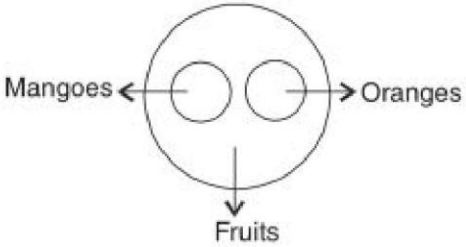
Some teachers can be M.L.A.s, some teachers can be men, so M.L.A.s can be men.

II. (D)



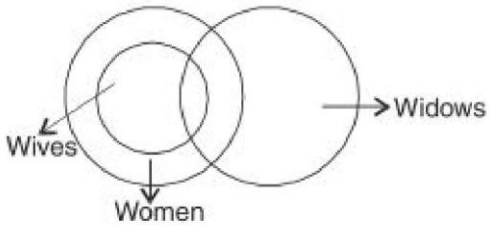
All dolphins are mammals, but fishes are not mammals.

III. (A)



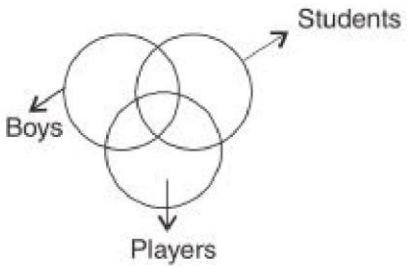
Mangoes and oranges are two different types of fruits.

IV. (B)



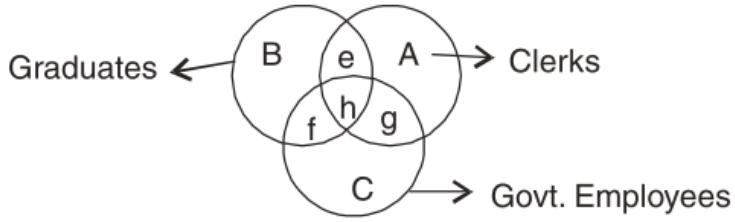
All wives are women. Some wives are widows, So some women can be widows.

V. (C)



5. (C) Some boys are students, some students are players. So some boys are players.

2.
In the following diagrams three intersecting circles each representing graduates, clerks, government employees are given. The circles are marked A, B and C, and the intersections are marked e, f, g and h. Which part best represents the statements in questions I-V?



I. Some clerks are graduates.
(A) e
(B) h
(C) g
(D) f

II. Who are clerks as well as government employees?
(A) e
(B) f
(C) g
(D) h

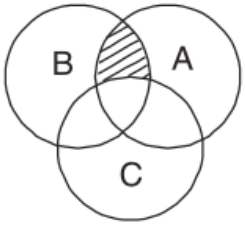
III. Who are graduate govt. employees but not clerks?
(A) e
(B) f
(C) g
(D) h

IV. Who are graduate govt. employees working as clerks?
(A) e
(B) f
(C) g
(D) h

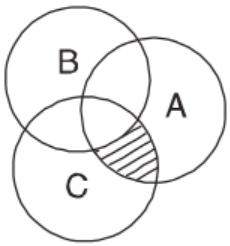
V. Who are non-govt. graduate clerks?
(A) e
(B) f
(C) g
(D) h

Answer and explanation

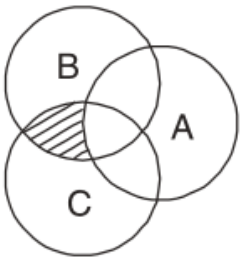
I. (A) Here the shaded portion, between A and B in figure represents the graduate clerks.



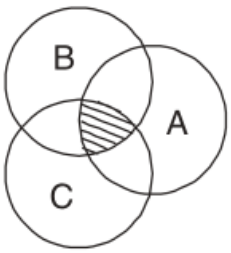
II. (C) Here the shaded portion between A and C in figure is the answer.



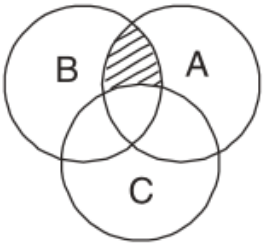
III. (B) Here the shaded portion in Fig. 3 represents the graduate govt. employees.



IV. (D) Here the shaded portion between A, B and C in figure represents the graduate govt. employees working as clerks.



V. (A) The non-govt. graduate clerks are represented in the shaded portion mentioned in figure.

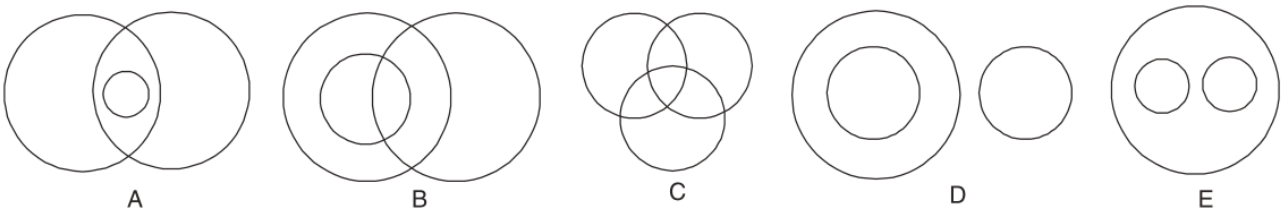


Exercise

Q1)

Directions for Questions i-x:

Below, five diagrams are given followed by a few questions. Select the diagram that best

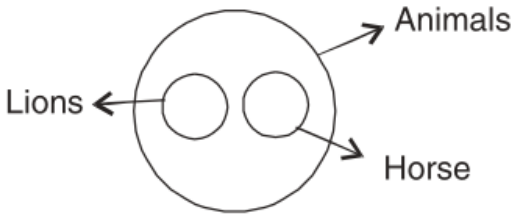


represents the relation of terms given in the questions that follow:

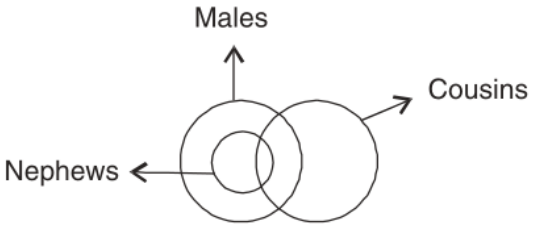
i. Lions, Horse, Animals.
ii. Cousins, Males, Nephews.
iii. Doctors, Surgeons, Musicians.
iv. Medical entrance candidates, Males, Students.
v. Bedroom, Drawing room, Dwelling.
vi. Chemical engineers, Mechanical engineers, Engineers.
vii. People, Professors, Cats.
viii. Young creatures, Colts, Horses.
ix. Men, Story teller, Liers.
x. Mothers, Females, Nurses.

S1)

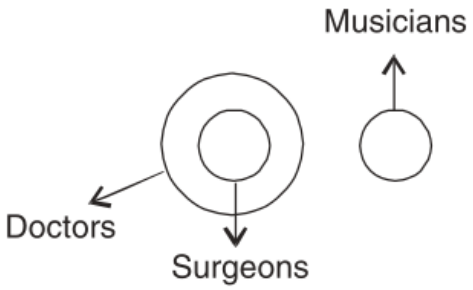
i. (E) Lions and Horses both are animals but of different species.



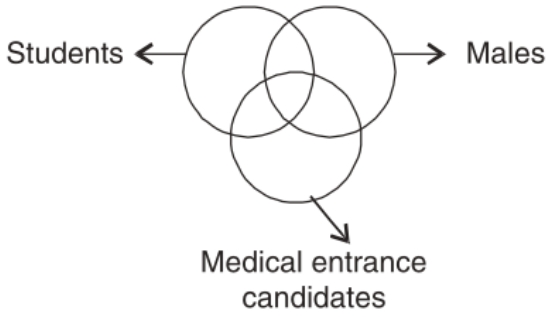
ii. (B) All Nephews are males, but all cousins are not males. Some cousins may be nephews, and also may be males.



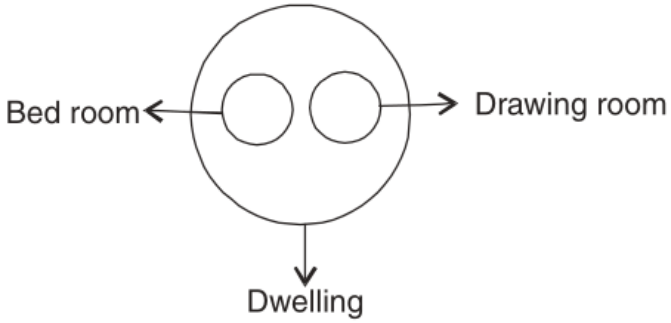
iii. (D) All surgeons are doctors, but musicians are different entities.



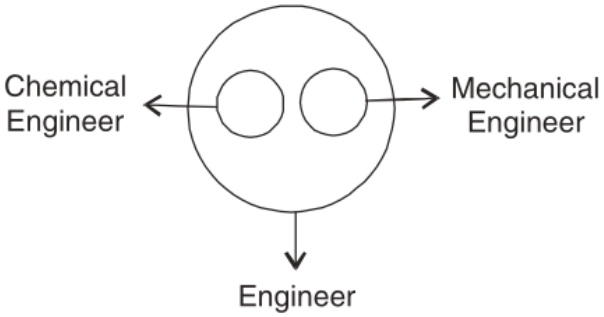
iv. (C) All these are overlapping. Medical-entrance candidates may be students and also males. Students may also be males.



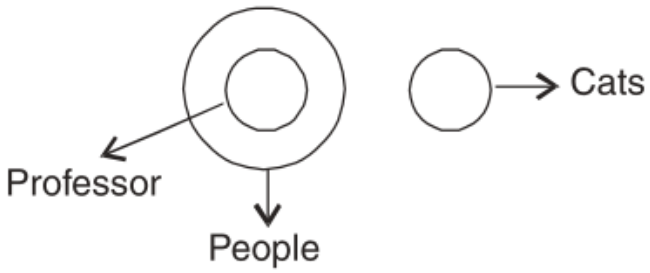
v. (E) Both bedroom and drawing room are within a dwelling house, but each has a separate entity.



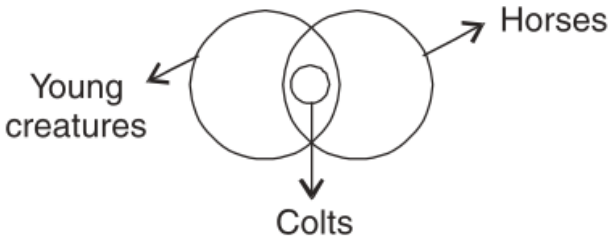
vi. (E) Both Chemical and Mechanical are different branches of Engineering.



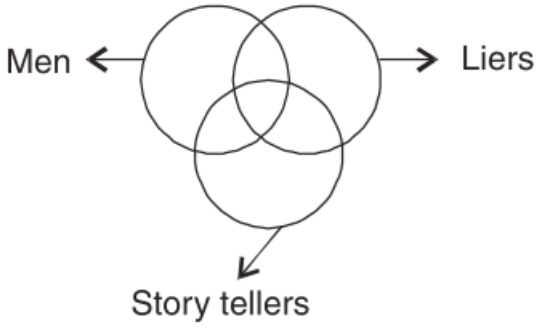
vii. (D) All professors are people, but cat has different entity.



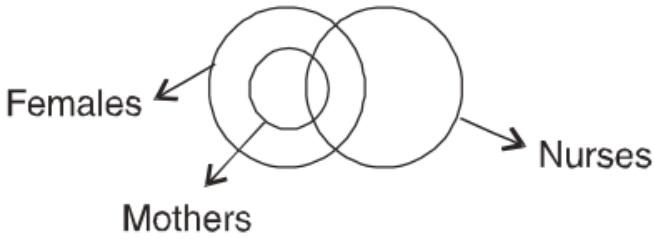
viii. (A) All colts are young creatures, all colts are also horses. Some horses are also young creatures.



ix. (C) Some men are story tellers. Some story tellers are liers. So, some men are also liers. All are overlapping each other.

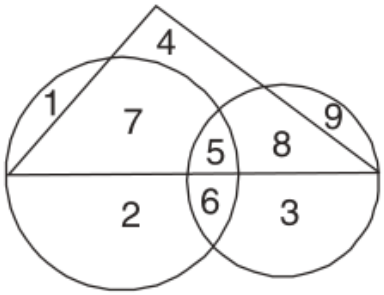


x. (B) All mothers are females. Some nurses are also females. Some nurses are also mothers.



Q2)
Directions for Questions i-v:

The following five questions are based on the following diagram in which the triangle represents female graduates, small circle represents self employed females and the big circle represents self employed females with bank loan facility. Numbers are shown in the different sections of the diagram.



On the basis of these numbers answer the following.

i. Who are graduate self employed females without using bank loan facility?
(A) 4
(B) 8
(C) 9
(D) 3

ii. Who are unemployed female graduates?
(A) 7
(B) 5
(C) 4
(D) None of these.

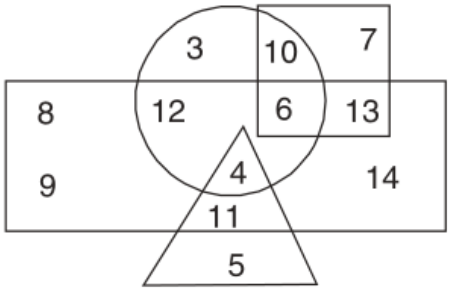
iii. Who are non-graduate self-employed females with and without bank loan facility?
(A) 6
(B) 5
(C) 2
(D) 3

iv. Who are graduate, self-employed females with bank loan facility?
(A) 1
(B) 7
(C) 2
(D) 8

v. Who are graduate, self-employed females with and without bank loan facility?
(A) 2
(B) 6
(C) 7
(D) 5

S2)
i. (B) The required region is common to the triangle and small circle only.
ii. (C) The required region is only within the triangle.
iii. (A) The required region is within both the circles but not in the triangle.
iv. (B) The required region is common to the triangle and the big circle but not within small circle.
v. (D) The required region is within all the three figures.

Q3)
Directions for Questions i-viii:
The following questions are based on the diagram given below:
(a) Rectangle represents Males.
(b) Triangle represents educated.
(c) Circle represents urban.
(d) Square represents civil servants.



i. Who among the following is an educated male who is not an urban resident?
(A) 4
(B) 5
(C) 9
(D) 11

ii. Who among the following is neither a civil servant, nor educated but is urban and not a male?
(A) 2
(B) 3
(C) 6
(D) 10

iii. Who among the following is a female, urban resident and also a civil servant?
(A) 6
(B) 7
(C) 10
(D) 13

iv. Who among the following is an educated male who hails from urban area?
(A) 4
(B) 2
(C) 11
(D) 5

v. Who among the following is uneducated and also an urban male?
(A) 2
(B) 3
(C) 11
(D) 12

vi. Who among the following is only a civil servant but not a male, nor urban oriented and uneducated?
(A) 7
(B) 8
(C) 9
(D) 14

vii. Who among the following is a male, urban oriented, and also a civil servant but not educated?
(A) 13
(B) 12
(C) 6
(D) 10

viii. Who among the following is a male civil servant, who is neither educated nor belongs to urban area?
(A) 7
(B) 13
(C) 4
(D) 1

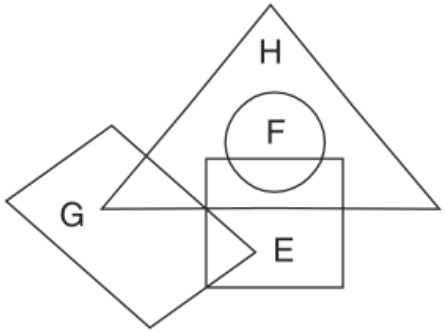
S3)

i. (D) The required region is common to rectangle and triangle only.
ii. (B) The required area is only within the circle.
iii. (A) The region is within circle and square only.

iv. (A) The region is within triangle, circle and rectangle, but not in the square.
v. (D) The area is common to circle and rectangle only.
vi. (A) The area is only within the square.
vii. (C) The area is common to square, rectangle and circle, but not in the triangle.
viii. (B) The area common to square and rectangle only.

Q4)
Directions (Questions i-iii):
These questions are based on the following:

(a) The triangle stands for Hindi people.
(b) Circle for French-speaking people.
(c) Square for English speaking people.
(d) Rectangle for German-speaking people.



i. In the above diagram, which one of the following statements is true?
(A) All French speaking people speak German.
(B) All French speaking people speak English.
(C) All German speaking people speak English and Hindi.
(D) All French speaking people speak Hindi also.

ii. In the diagram which one of the following statements is true?
(A) There are some people who speak all four languages.
(B) Some German speaking people can speak either Hindi or English.
(C) Some English speaking people can speak all the other languages.
(D) All Hindi speaking people speak French but not German.

iii. In the above diagram, which one of the following is not true?
(A) German-speaking people cannot speak French.
(B) No French-speaking people can speak German.
(C) Some Hindi-speaking people can speak English, French and German as well.
(D) Some French-speaking people can speak Hindi and English but not German.

S3)

i. (D) As the circle is completely within the triangle. No other alternatives are possible.
ii. (B) Rectangle (German-speaking) overlaps triangle (Hindi-speaking) and square (English-speaking).
iii. (C) If this is true then all the figures including the circle are to be overlapped each other.

12. CRITICAL REASONING

The critical reasoning tests are designed to test the ability to evaluate an assumption, paragraph, inference or argument. This test is also basically a type of reasoning test, but it measures the following abilities:

(a) to comprehend
(b) to draw inference from a given statement or passage
(c) to recognize assumptions underlying the given statement
(d) to examine the validity of an assumption in the light of given subsequent statements
(e) to evaluate the arguments and counter arguments.

Each question consists of a short passage/paragraph/proverb/statement/argument followed by a question or an assumption or conclusion about the said passage/statement, etc. The principal object of the critical reasoning test is to test skills in constructing and evaluating arguments. An argument is a sequence of two or more phrases, clauses, sentences or statements followed by a conclusion.

Example :

Sentence:

He has got a high fever, he must have been suffering from malaria. Here the first part of the sentence, 'He has got a high fever', is a premise and the second part of the sentence, i.e., 'he must have been suffering from malaria', is the conclusion of the sentence based on the premise. Now, the question is, whether the said conclusion is valid or not. Here, the conclusion is not valid. Because, 'high fever' may be the cause for a variety of diseases, so, no definite conclusion can be drawn here.

There may be a few types of critical reasoning tests, e.g., (i) paragraph type, (ii) proverb/assumption type, (iii) inference type, (iv) flaws type and (v) statements of facts.

I. PARAGRAPH TYPE

This type consists of a paragraph followed by a few conclusions drawn from the paragraph. Your task will be to study the paragraph carefully and assess the conclusions whether they are:

A) Definitely true
B) Probably true
C) Can't say
D) Probably false
E) Definitely false.

Example:

Paragraph:

Blood flow in human or animal body is indeed complex due to very complicated structure and function of blood and blood vessels. However the structure and function of blood as well as blood vessels including the various aspects of blood flow are widely used for the diagnosis of pathological pattern in human or animal physiology.

Conclusions:

1. Blood is composed by many chemicals.
2. The composition of blood helps in the diagnosis of various ailments.
3. In man the blood flow system is very simple.
4. Pathological pattern of a disease is very much understood from the examination of blood.
5. Various aspects of blood flow are important for diagnosis.
6. Diagnosis of a disease in human being is not always possible only by the blood test.

Answers and Explanations:

1. (C) No such information given in the paragraph.
2. (A) It is clearly described in the paragraph.
3. (E) It contradicts the paragraph.
4. (B) It may be true.
5. (A) It is clearly written in the paragraph.
6. (D) It seems to be a vague statement and probably false.

Exercise

Paragraph:

In the sphere of education, in order to foster awareness and understanding of the impact of ecological interdependence, environment education should be made compulsory in the curricula of colleges and schools, starting from the primary level. This education, while catering to students should also be imparted to adults and professionals whose activities are directly concerned with the preservation, improvement and quality of the environment.

Conclusions:

1. Environment education should be included in the curriculum of the primary education.
2. In the environment there is interdependence of human life and animal life.
3. Education should lead to character building.
4. Environment education should be given to all adults.
5. There are certain things in nature which do not fall within the ecological interdependence.

6. Environmental awareness should be encouraged among the students.
7. Quality of environment is generally improved with the environment education.

Paragraph:
At the time of introduction of computer system in India there was an abrupt objection to its introduction by the leaders of various Trade Unions. It was argued that this technical device will certainly minimize the chances of employment. It was described as anti-labor. Indian Statistical Institute, Calcutta, got the first computer system in India. But the results are just the reverse. It has, not only created better opportunities for fresh jobs, but has also entrusted the employees with more creative jobs, motivation and job satisfaction with qualitative and quantitative advancement in the routine work. In fact such technological changes should always be welcomed with enthusiasm, leaving aside the wrong notions.

Conclusions:
8. Trade Unionist do not believe in the nation's progress.
9. Computer system has proved very useful for the workers.
10. Laborers have wrong notions about the various schemes launched by the government.
11. Shallow knowledge is the cause of various industrial disputes.
12. Technological advancement brings satisfaction among the working class.
13. India's first computer system was introduced in Calcutta.
14. There was labor unrest in the Indian Statistical Institute for the introduction of computer system.
15. Computer system can also create jobs.
16. Employment opportunity is not at all minimized with the introduction of computer system.
17. The notion of the Trade Union leaders about the computer system in India was found perfect.

Paragraph:
Growth indicates improved motor performances with age. Growth is more than enlargement. Parts of the body change in relative size, glands and their secretion take on new functions and innumerable changes take place in the body's microanatomy and biochemistry. Change in height and weight is most striking in early adolescence. Adolescence sees great increase in height and weight, change from childish to adult body proportions, change in the sexual organs, change in the glandular functions and change in voice.

Conclusions:
18. Growth means obesity.
19. Physical changes of human body generally takes place during puberty.
20. Change in height and weight usually occurs during adolescence.

21. Glands stop secretion with the advancement of adolescence.
22. Biological changes of the body extend over several years.
23. Change of voice does not depend on the glandular function.
24. In adolescence, the child undergoes some behavioral changes also.
25. Girls enter the adolescence ahead of boys.

Answers and Explanations:

1. (A) It directly follows the paragraph.
2. (C) It is nowhere mentioned in the paragraph.
3. (E) It does not follow the paragraph at all.
4. (B) Though it is not directly mentioned in the paragraph there are some hints, so it is probably true.
5. (D) It seems to be a vague statement and probably false.
6. (A) It directly follows from the paragraph.
7. (B) Though it is not written directly in the paragraph it may be assessed from the paragraph.
8. (E) It is nowhere mentioned in the paragraph so, definitely false.
9. (A) It has direct reference in the paragraph.
10. (C) Can't say.
11. (B) To some extent it is mentioned in the paragraph.
12. (E) It contradicts the paragraph.
13. (A) It is mentioned in the paragraph.
14. (C) Can't say.
15. (B) It has some reference in the paragraph.
16. (A) It directly follows from the paragraph.
17. (E) It contradicts the paragraph.
18. (D) It has no direct reference in the paragraph.
19. (C) It cannot be said from the paragraph.
20. (A) It directly follows from the paragraph.
21. (E) It is nowhere written in the paragraph.
22. (B) Though it is not directly mentioned in the paragraph, it may be assumed from the statements.
23. (D) The statement is vague, and probably false.
24. (C) It is not mentioned in the paragraph.
25. (D) It is vague conclusion, hence, may be false.

II. PROVERB/ASSUMPTION TYPE

Proverbs are generally explained in different ways. In this type you will have to find out the most logical explanation of a proverb from a number of such explanations.

Examples:

In each question below some proverbs/statements/assumptions are given at the top followed by certain explanations marked A, B, C and D. You are to choose from these explanations the best logical explanation of the said proverb/statement or assumption.

1. Proverb: Brevity is the soul of wit.

(A) All brave people are witty.

(B) 'Wit' expressed in a few wards is the best.

(C) Wit is the soul of humor.

(D) If expressed in short, wit losses its soul.

2. Proverb: Man proposes God disposes.

(A) God does not allow us to make plans.

(B) God has always been against man.

(C) Man made plans are often shattered by God's will.

(D) Man does not know the actions of God.

Answers and Explanations:

1. (B) If one knows the literary meaning of the words of the proverb, he can answer this easily. Here brevity means briefness hence B is the appropriate answer.

2. (C) All the other three alternatives, A, B, and D, are not reflecting the idea of the proverb.

Exercise

1. Proverb: Pen is mightier than sword.

(A) The tip of the pen is much pointed than that of the sword.

(B) The character assault through writings can harm a person much more than wounds caused by a sword.

(C) Swords are cheaper than pen.

(D) Pen is harder than sword, it can't break easily.

2. Proverb: Rome was not built in a day.

(A) The Romans were a lazy lot, they could not work in time.

(B) It took about twelve years to build Rome.

(C) The Romans did not have sophisticated equipments and infrastructure to build a city like Rome.

(D) It took years of patience, hard work and determination to construct a city like Rome.

3. Proverb: Faint heart never won fair lady.
(A) One should be bold enough to achieve some extraordinary thing.
(B) Those who faint easily cannot get beautiful wives.
(C) Ladies do not like men who have heart trouble.
(D) Strong men always have some weakness towards pretty women.

4. Proverb: A king has no relations.
(A) Those who are alone and got no relatives are made kings.
(B) A king is so powerful that no one dares to have any relationship with him.
(C) A king should not be partial to his relatives.
(D) A king has no diplomatic relations with other countries.

5. Proverb: A liar needs a good memory.
(A) A liar can memorize easily.
(B) Because he can make controversial statements.
(C) Because with a good memory one can take it lying down.
(D) Because he can try to speak the truth.

6. Proverb: Beggar may sing before a pick-pocket.
(A) A beggar has nothing to lose.
(B) Beggars always sing while begging.
(C) Beggars are good singers.
(D) A pick-pocket always helps the beggars.

7. Proverb: A bad workman quarrels with his tools.
(A) Bad workmen are always quarrelsome.
(B) Workmen who are quarrelsome are not good.
(C) A bad workman cannot utilize their tools properly.
(D) An inefficient man complains of circumstances.

8. Proverb: Fair words butter no parsnips.
(A) Parsnips are no choice for butter.
(B) Parsnips cannot grow only by word.
(C) Empty promise will not help anybody in distress.
(D) Fair words are necessary to get some work to be done.

9. Proverb: Barking dogs seldom bite.
(A) Angry words and threats lead to nothing worse.
(B) Beware of dogs.
(C) Dogs always bark before bite.
(D) Tame dogs do not bite.

10. Proverb: Those who live in glass-houses should not pelt stones at others.
(A) Those who live in glass houses are always anxious.
(B) Those who deserves criticism should criticize others.
(C) Pelting stones towards glass-houses are harmful.
(D) Make sure of your position before you fall foul of others.

11. Proverb: A broken hand works but not a broken heart.
(A) If heart is broken, the person dies instantly.
(B) If a person loses interest he cannot work.
(C) Lovers with broken heart are unable to do any work.
(D) Broken hand can be replaced but not the broken heart.

12. Proverb: Blood is thicker than water.
(A) The relative density of blood is more than water.
(B) Blood clots easily while water does not.
(C) A person feels more attached to his close relatives than to others.
(D) Blood stain cannot be wiped out by water.

13. Proverb: Ailing marriages often need first-aid.
(A) A marriage on breaking point can be saved by timely advice, help and understanding.
(B) It is better to avoid uneven marriages.
(C) Every married couple should keep a first-aid box handy because ailments are inevitable.
(D) Every ailment needs first aid.

14. Proverb: He who loses honor has nothing else to lose.
(A) Person having honor has nothing else.
(B) Careless persons lose honor.
(C) People only lose honor when they have lost everything else.
(D) Nothing is more valuable than honor.

Directions for Questions 15-20:
In each question below there is an assumption followed by four implications marked as A, B, C and D. Choose the most appropriate implication and mark accordingly. If all the implications are equally appropriate mark E on the answer sheet.

15. Assumption: The doctor was surprised to see that the patient was talking to his friends.
(A) The patient was dumb.
(B) The patient did not talk to anyone.
(C) The patient was under anesthesia.
(D) The doctor had declared the patient dead.

16. Assumption: Arun said to Bimal, “So you have come”!
(A) Arun had been waiting for Bimal.
(B) Arun had been waiting for someone else, who sent Bimal instead.
(C) Bimal had said that he would not be able to come.
(D) Arun had given up all hopes of his coming.

17. Assumption: X to Y, “Tell me your father’s office address”.
(A) X had some official work with Y’s father.
(B) X did not want to go to Y’s home.
(C) X was transferred to Y’s father’s office.
(D) Y’s father had asked X to come to his office.

18. Assumption: P to Q, “Here is the dictionary!”
(A) P and Q wanted to find out the meaning of some word.
(B) P had challenged Q about the spellings or use of certain word.
(C) The dictionary had been misplaced.
(D) P had challenged Q about some word.

19. Assumption: I like Byron too.
(A) I like Byron too much.
(B) I am a bit like Byron.
(C) I have not read much about Byron, still I like him.
(D) I like some other poet and Byron is my second choice.

20. Assumption: Mr. Jalan provides no medical facilities to the workers.
(A) Mr. Jalan’s workers are never ill.
(B) It is a criticism of Mr. Jalan’s attitude towards his employees.

(C) Mr. Jalan himself gives the medicines.
(D) Only the family members of Mr. Jalan get such facilities.

Answers and Explanations:
1. (B) All the other three alternatives are not appropriate.
2. (D) No. great work can be done easily.
3. (A), 4. (C) King should be impartial.
5. (B), 6. (A) Beggars are not afraid of pick-pocket.
7. (D), 8. (C) (People in distress need concrete help, not more promise).
9. (A),
10. (D),
11. (B) Every work needs proper interest.
12. (C) Here 'blood' means blood relation.
13. (A),
14. (D),
15. (B) Self explanatory.
16. (E) All the choices are probable,
17. (A),
18. (E), All the choices are possible,
19. (D),
20. (B).

III. INFERENCE TYPE

In each question some statement or fact is given. You are to identify a conclusion or inference or claim which follows from it.
Examples:
1. Monopoly is characterized by an absence of or decline in competition, Maruti-Suzuki company realizes that its cars are in competitive industries.
Which of the following conclusions may be inferred from the above statement?
(A) Monopoly is defined as one seller in the market.
(B) Maruti-Suzuki company has no domestic competitors.
(C) Maruti-Suzuki's market is not monopolistic.
(D) Maruti-Suzuki has enjoyed the largest government share.

2. A government press note quoted as saying that the rate of inflation has gone up because of a recent increase in the price of fruits and vegetables.
Which of the following cannot be inferred from the statement?

(A) Consumers have decreased their consumption of fruits and vegetables.
(B) The costs of fruits and vegetables is a major item in the price index.
(C) The cost of fruits and vegetables has risen sharply.
(D) Other items that can make up the inflation rate have not significantly decreased in price.

Answers and Explanations:
1. (C) Since Maruti-Suzuki cars are in competitive industry, there can be no monopoly in its market. So, 'C' is the correct answer.
2. (A) It is not possible to infer whether consumers have decreased their consumption of fruits and vegetables with the increase of price of these products. While the other alternative answers are quite possible. So 'A' is the correct answer.

IV. FLAWS TYPE

In this type of test, the best or correct answer would be the one which, if true, would mean that either the original statement had a flaw or was weak.
Example:
The United States gives billions of dollars in foreign aid to India. But the leaders at New Delhi resent foreign aid, particularly from the U.S. The United States should discontinue direct foreign aid to developing countries.
Which of the following statements, if true, would weaken the above argument?
(A) India's economy is growing.
(B) India is not a developing country.
(C) India has got a democratic government.
(D) India does not need any foreign help.

Answers and Explanations:
1. (B) From the above statement it is clear that U.S. aid is only for the developing countries.
But, if it proved that India is not a developing country, then only the premise and the conclusion can be said to be invalid. So, 'B' is the correct answer.

V. STATEMENT OF FACTS

In these type of questions you are to find out the answer that best agrees with, summarizes or completes the statement.
Example:
1. Which of the following statements best completes the passage below?
Passage:

In department stores the placing of aisles, display stands and shelves, is designed to direct a customer's movements. Different sorts of products are also situated to ensure that a customer will move through different parts of the store. More expensive items are usually placed in the upper floors, while cheaper items are usually placed near the entrance and also adjacent to the cashier.

This sort of planning is intended to —

(A) keep customer flowing in the store as smoothly as possible.

(B) utilize the sales persons as efficiently as possible.

(C) prevent long queues and undue waiting time at cash counter.

(D) maximize sales by exposing customers to as many products as possible.

Answers and Explanations:

1. (D) Here, though all the alternative answers seem to be associated with the idea expressed in the passage, the best among them is 'D', where both the customers and the store authority are benefited.

TACTICS FOR ANSWERING CRITICAL REASONING TESTS

The following are a few tips of how to answer the questions of critical reasoning tests. These tips are all equally useful in dealing with the questions of all the types of logical reasoning tests also. If you follow the tips carefully you will find all the logical reasoning tests, critical reasoning tests, analytical reasoning tests, etc., are quite easy to answer.

1. First read the passage/statement/argument carefully. Then read all the possible answers carefully, in order to identify the question type.

2. Now, if you recognize just what a given question is asking for, i.e., assumption, or inference or conclusion, or central point, or support, or weaken, or summarizes or completes the statement.

3. Then pinpoint the conclusion of the statements/arguments and grounds on which it is based, e.g., an argument or statement is based upon certain assumptions made by its author. If an argument's/statement's basic premises are sound, the argument/statement is strengthened, but if they are flawed, the statement/argument is weakened. Pinpoint what the argument/statement assumes, then choose the best answer from the given alternatives.

Exercise

Directions for Questions 1-20:

For each question below some statement/argument/report/passage is given, followed by a few alternative answer choices marked A, B, C, D and E, of which only one would be the best possible answer. Read the passage/statement/argument carefully and choose the best possible answer from the alternatives given.

Questions:

1. Some experts have concluded that the rise in price of pepper means that the switch by some growers from pepper to cocoa left those growers no better off than if none of them had switched. This conclusion, however, is unwarranted because it can be inferred likely that

(A) Those growers could not have foreseen how high the price of pepper would go.

(B) Supplies of pepper would not be as low as they are if those growers had not switched crops.

(C) As more growers turn to growing cocoa, cocoa supplies will increase and the price of cocoa will fall precipitously.

(D) The initial cost involved in switching from pepper to cocoa is substantial.

(E) Cocoa crops are not so susceptible to being affected by bad weather as are pepper crops.

2. It is important to each student to use computers effectively. Therefore, students should be taught computer programming in school. Which of the following, if true, most weakens the argument above?

(A) Only people who use computers effectively are skilled at computer programming.

(B) Only people skilled at computer programming use computer effectively.

(C) Some people who use computers effectively cannot write computer programmes.

(D) School students can learn computer programming easily than the elders.

(E) Some schools teach computer programming more effectively than others.

3. Four years ago, the government introduced a youth training programme to guarantee, to teenagers leaving school, an alternative to dole. Today over 1,50,000, 16 and 17 year-olds are still signing on for unemployment benefits.

Each of the following statements, if true, could account for the above, except —

(A) The number of 16 and 17 year old youths has increased over the past 4 years.

(B) The programme provides uninteresting work.

(C) It is difficult to find work for all the programme's graduates.

(D) Unemployment benefits are known while future salaries are not.

(E) Youths are unaware of the programme's benefits.

4. Many people are murdered by killers whose homicidal tendencies are triggered by an official execution. There was a murder-rate increase of at least 66% of executions since 1977. In each of the 1,788 death row prisoners were to be executed up to 7,152 addition murders would be one of the results. Which of the following, if true, would weaken the above argument?

(A) The rate of murders to executions is 1 to 1.66.

(B) Executions result from the higher incidence of violent crime.

(C) The death penalty will be abolished.

(D) There is no relation between executions and murders.

(E) Not all death row prisoners will be executed.

5. Sales campaigns aimed at strengthening the faltering Personal Computer (PC) market have strongly emphasized ease-of-use, called user-friendliness. This emphasis is oddly premature and irrelevant in the eyes of most potential buyers who are trying to address the logically prior issue of whether

Which of the following best completes the above passage?

(A) User-friendliness also implies that owners can service their own computers.

(B) The more user-friendly the PC, the more it costs.

(C) Currently available models are user-friendly enough to suit them.

(D) The people promoting PCs use them in their own homes.

(E) They have enough sensible uses for a PC to justify the expense of buying one.

6. Literary historians today, have rejected conventional analyses of the development of English Renaissance drama. They no longer accept the idea that the sudden achievement of the Elizabethan playwrights was a historical anomaly, a sort of magical rediscovery of ancient Greek dramatic form applied to contemporary English subject matter. Instead, most students of the theater now view Elizabethan drama as being organically related to traditional local drama, particularly medieval morality plays.

Which of the following is not consistent with the passage above?

(A) England had a dramatic tradition before the Renaissance period.

(B) Elizabethan drama, once thought to be a sudden blossoming forth of creativity, is now seen as part of a historical continuum.

(C) Although English Renaissance drama treats English as subject matter, its source of form and method is classical Greek drama.

(D) Historians' views of the antecedents of English Renaissance drama have changed considerably.

(E) Current scholarship applies an evolutionary model to English Renaissance drama.

7. A study of illusionistic painting inevitably begins with the Greek painter, Zeuxis. In an early work, which is the basis for his fame, he painted a bowl of grapes that was so lifelike that birds pecked at the fruit. In an attempt to expand his achievement to encompass human figures, he painted a boy carrying a bunch of grapes. When birds immediately came to peck at the fruit, Zeuxis judged that he had failed. This judgment of Zeuxis was based on an assumption. Which of the following could have been that assumption?

(A) People are more easily fooled by illusionistic techniques than the birds.

(B) Birds are not likely to peck at fruits when they see that a human being is present.

(C) The use of illusionistic techniques in painting had become commonplace by the time Zeuxis had completed his later work.
(D) The grapes in the later painting were even more realistic than the ones in the earlier work.
(E) After the success of his early work, Zeuxis was unable to live up to the expectations of the general public.

8. I'm afraid that Rahim will never be an outstanding football player again. Last year he injured his knee and the doctors had to remove some of the cartilage.
The above argument is based on which of the following assumptions?
I. One must have healthy knees to play football.
II. How well one plays football may be influenced by the condition of one's knees.
III. Healthy knees are necessary for a professional football career.
(A) only I (B) I and II only (C) I, II and III (D) only II (E) II and III only

9. In an examination system of an institution, 100 students have been graded (from A - D in descending order) on the basis of the marks they received in the three terminal examinations, in which the pass-marks were 50%. Examiners are instructed to follow the following criteria:
I. All students who scored between 90% and 100% in any two examinations could receive an A.
II. Students who came in the top overall were to be awarded an A.
III. Notwithstanding I and II, if any student failed a paper, the highest he or she could get was a B.
IV. The top 20 students in the whole year, when the overall examination percentages were averaged, could receive an A.
On the basis of above criteria, which of the following would definitely not be permissible?
(A) Bikash, who got 95% in Chemistry and 92% in Biology, received a B.
(B) Suparna, who stood first in Physics and got 96% in Mathematics, received a B.
(C) Amitava failed in English, but because he ranked 9th overall out of 100 students was awarded an A grade.
(D) Bandana was given an A because she came 20th though she had failed to get above 90% in any of the three examinations.
(E) Bipin though failed in Mathematics, but topped his other two examinations and was awarded a B.

10. The only unemployment problem we have is not that people can't find work, but that they won't work. Thousands of jobs go begging everyday, but the unemployed are too lazy to go out and find them.
The above argument would be more persuasive if it were established that
(A) Most unemployed persons do not seek work until their unemployment benefits expire.
(B) The majority of available jobs require unusually high levels of skill or experience or both.

(C) Unemployed persons tend to be geographically clustered in regions distant from available jobs.
(D) Most unemployed persons are back at work within 6 months.
(E) A high unemployment rate has been fostered by the government in order to control inflation.

11. My father, three of my uncles and both my grandfathers became bald within 5 years of starting practice of law. I don't want to lose my hair so I'm going to become a doctor. Which of the following most closely resembles the reasoning used in the argument above?
(A) Every time I drink coffee after dinner, I have a disturbed sloop. But I want to sleep well tonight, so I am going to take a sleeping pill.
(B) Each time train accident occurs only on Thursdays. So I don't want to travel by train on Thursdays.
(C) Everyone, except I, got transferred out of our department within three years of starting. I don't want to work in another department, so I am going to start working harder.
(D) Each of the other operators on my shift was seriously injured on the job within a week of eating at the factory's canteen. I want to maintain my safety record, so I am bringing my food from home.
(E) Each of the three other men who are members of our tea-club became irritable after they quit smoking. I want to be pleasant, so I am going to quit my tea club.

12. The new PXB model of our automatic washing machine is the best washing machine you will ever own. It comes in eight decorator-colors. It is so quiet that you will find yourself checking to see if it is really on. And best of all, it comes in different heights and widths so that finding one that fits into the space you have won't be a problem.
The argument above is most weakened by its failure to mention
(A) The terms of warranty.
(B) How well the machine washes clothes.
(C) The specific sizes available.
(D) How much electricity the machine uses.
(E) How many clothes the machine can hold at a time.

13. The 1999 Hollywood film, Titanic (winner of 10 Oscar awards), is clearly the best foreign film of the year since its box-office breaks the record of Ben Hur, which also won 10 Oscar a few years ago.
The argument above is based on which of the following assumptions?
(A) Only a film based on a true story and having good photography can be judged to be best.
(B) Foreign films should be judged by standards different from those used for Indian films.
(C) Foreign films should be judged by the same standards used for Indian films.

(D) The criteria for judging whether the film is the best one should be the number of Oscars won.
(E) The quality of a film can be measured by the number of people who go to see it.

14. Which of the following contradicts the view that only the smart become rich?
(A) Amar is stupid, yet he amassed a large fortune by the age of 30.
(B) Biren was smart, yet he was poor his whole life.
(C) Smart and rich are relative terms.
(D) Different people are smart in different ways.
(E) Some smart people do not desire to become rich.

15. Television convinces viewers that the likelihood of their becoming the victims of violent crime is extremely high; at the same time by its very nature, television persuades viewers to passively accept whatever happens to them. The argument above leads most logically to the conclusion that
(A) People should not watch television.
(B) Television promotes a feeling of helpless vulnerability in its viewers.
(C) Television-viewers are more likely to be victimized than other persons.
(D) The contents of television programmes should be changed to avoid fostering the attitudes mentioned.
(E) Television viewing promotes criminal behavior.
16. Over the last 20 years the rate of increase in total production in India has been second to none in the world. However, the growth is more modest when calculated per capita of total population. Over the last ten years progress has been much slower.
If the information above is accurate, which of the following must be true?
(A) India has a very large population.
(B) Total production has increased faster than population growth.
(C) Productivity per capita has not grown as fast during the past ten years.
(D) The birth rate has declined.
(E) The per capita production rate has not declined.

17. Lavoisier, the 18th century scientist, became more influential and famous than most of his contemporaries in his field, because not only did he discover and isolate many of the chemical elements but he also gave them names which both described the element in terms of its power and function, and which came to be accepted by other scientists in subsequent generation.
Which of the following can be inferred from the above passage?
(A) Lavoisier was a great scientist of 18th century.
(B) Lavoisier discovered many chemical elements.

(C) Discovery of Lavoisier had been largely accepted by the other scientists in subsequent generations.
(D) The elements which Lavoisier isolated were given names which described their properties.
(E) Lavoisier was the most influential scientist of 18th century.

18. If Robin is on the steering committee, then he is on the central committee. This statement can be logically deduced from which of the following statement?
(A) Some members of the central committee are on the steering committee.
(B) All members of the central committee are on the steering committee.
(C) Robin is on either the central committee or on the steering committee.
(D) No central committee members are members of steering committee.
(E) Everyone who is on the steering committee is also on the central committee.

19. Vinod must be a football player, he is wearing a football jersey. The conclusion above is valid only if it is true that
(A) Only football players wear football jerseys.
(B) Football players often wear football jerseys.
(C) All football players wear football jerseys.
(D) Football players never use any other garments except football jerseys.
(E) Football players are required to wear football jerseys.

20. A garden in springtime is beautiful, even if no one is there to appreciate it.
The statement above would be a logical rebuttal to which of the following claims?
(A) People will see only what they want to see.
(B) Beauty exists only to be appreciated.
(C) Beauty exists only in the eye of the beholder.
(D) There is no accounting for taste.
(E) The greatest pleasures available to mankind is the contemplation of beauty.

Answers and Explanations:
1. (B) Here the assumption is that those growers, if they had not switched crops would have continued to grow pepper. But if suppliers of pepper were not as low as they are, the price of pepper would not have raised enough either to equal or above the price of cocoa, and the growers in question would not have done financially, as they actually did as a result of switching to cocoa. Therefore ‘B’ is the best answer here, Though better foresight mentioned in option ‘A’, but it is irrelevant here like ‘C’, ‘D’ and ‘E’.
2. (C) In the passage it is stated that to use computers effectively, knowledge of computer programming is a must. But choice (C) contradicts this statement.

3. (A) It is stated that the number of 16 and 17 year-olds has increased, but it is not stated that they do not join the training. It is possible that all of them be absorbed into the programme and the increase in population may not account for the increase in number of youths on dole.
4. (D) In the statement we find a positive correlation between execution and homicides (murders). Among the alternatives only 'D' contradicts the argument; hence 'D' is the answer.
5. (E) Only answer choice 'E' should be appropriate answer as it directly questions the logic of emphasizing the user-friendliness in campaigns. Choice 'A' just describes the meaning of user-friendliness in campaigns. Choice 'A' just describes the meaning of user-friendliness and 'B', 'C' and 'D' are irrelevant.
6. (C) Here the task is to find an answer choice not consistent with the statement, and in only 'C' we find this inconsistency. So, 'C' is the best answer choice.
7. (B) His conclusion was that he failed to paint the human figure with the illusion of life.
8. (D) A person can play football and even be a professional player even if he has weak knees. Only to be an outstanding player one must have healthy knees. So, the health of one's knees affects the quality of playing.
9. (C) Bikash and Suparna may have failed in any other subject, so, it is possible they get B grades. Amitava, however, could not have got an A as he failed in one subject.
10. (A) The argument says that the only cause of the unemployment problem is the laziness of workers. This is stated in choice A. Choice D tends to weaken the cause and B, C and E are irrelevant.
11. (D) In the argument we find an example of poor reasoning. It is reasoned that practicing law causes baldness; the heredity factor which might have been the probable cause has been overlooked. In D, we find similar faulty reasoning where unsafe working conditions have been overlooked.
12. (B) The advertisement states that the said washing machine is the best but only gives details of its luxury features, not how well it functions. This is stated in (B) and thus it is the reason that weakens the advertisement.
13. (E) The Titanic is judged the best foreign film because of its better box-office record. So, the quality of a film is measured by the number of people who watch it.
14. (A) As per (A), a stupid person, Amar, amassed a fortune. Thus the statement is contradicted. In B, Biren could have been rich, so the statement is not contradicted. The other sentences are irrelevant.
15. (B) The statements suggest that it is very possible that people be victimized, and that when they are victimized they should accept it. Thus B is a logical conclusion. Choices 'A' and 'D' are incorrect as they are statements of fact and not conclusions.
16. (C) From the statement it is clear that increase in production has not been at par with the increase in population, i.e., population has increased at a faster rate than productivity.

17. (D) Choices A, B, C and E refer only a part about Lavoisier, whereas the choice ‘D’ describes about this main performance for which he became a famous scientist of 18th century. Hence ‘D’ is the best answer choice.
18. (E) The statement given is true only if all members of the steering committee belong to the central committee.
19. (A) If anyone other than a football player wears a football jersey, the conclusion would not be valid. So it is valid if only choice A is true.
20. (C) The original statement says that the beauty has an existence independent of the persons perceiving it. D says the opposite; something is beautiful only if a person looking at it finds it beautiful.

13. SEQUENTIAL INPUT-OUTPUT TRACING

In this type of questions some inputs are given as a message comprising of randomized words followed by steps of rearrangement to give sequential outputs. You are to trace out the pattern of the rearrangement and determine the actual output asked in the questions.

Example:

Study the following information to answer the given questions:

A word arrangement machine, when given an input line of words, rearranges them following a particular rule in each step. The following is an illustration of input and the steps of rearrangement:

Input : before its court seeking permission to carry out a test.
Step I : it's before court seeking permission to carry out a test.
Step II : it's before carryout court seeking permission to a test.
Step III : it's before carryout court permission seeking to a test.
Step IV : a it's before carryout court permission seeking to test.
Step V : a it's before carryout court permission seeking test to.

As per the rules followed in the above steps, find out in the given questions the appropriate step for the given input.

Questions:

1. Which step is the last step for the given input?
Input : in this volume provides the
(A) I (B) II (C) III (D) IV (E) None of these.

2. Input : these many other clinicians and which of the following steps would be – and clinicians many these others?
(A) I (B) III (C) IV (D) II (E) None of these.

3. Step III of a certain line of words is like this – in play role schooling vital, which of the following will definitely be the input?
(A) in play schooling vital role
(B) schooling play vital role in
(C) in schooling play vital role
(D) Cannot be determined
(E) None of these.

4. Which of the following will be Step II for an input – modern man is becoming a chronic victim?
(A) a becoming modern man is chronic victim.
(B) a becoming chronic modern man is victim.
(C) a modern man is becoming chronic victim.
(D) a becoming chronic is man modern victim.
(E) None of these.

Answers and Explanations:
From the information and the arrangements in steps it is clear that the words have been arranged in alphabetical sequence altering the position of only one word at a time in each step. Following this rule if we rearrange the words given in the question we find.
1. (C) Input : in this volume Provides the.
Step I : in volume provides the this
Step II : in provides volume the this
Step III : in provides the this volume, so step III is last.
2. (B) Input : these many other clinicians and
Step I : and these many other clinicians.
Step II : and clinicians these many others.
Step III : and clinicians many these others. So, it is step III.
3. (B) Tracing the output steps for an input we have Step III as given in the question, we must start with that first and going back like this.
Step III : in play role schooling vital.
Step II : in play schooling vital role.
Step I : in schooling play vital role.
Input : Schooling play vital role in.
4. (A) Input : modern man is becoming a chronic victim.
Step I : a modern man is becoming chronic victim.
Step II : a becoming modern man is chronic victim. so, (A) is the correct answer.

Exercise-

Directions for Questions 1-5:
Study the following information and answer the questions given follow it:
The admission ticket for an exhibition bears a password which is changed after every clock hour based on set of words chosen for each day. The following is an illustration of the code and steps of rearrangement for subsequent clock hours. The time is 9 a.m. to 3 p.m.
Day's first password: First Batch – 9 a.m. to 10 a.m. is not ready cloth simple harmony burning.
Second Batch – 10 a.m. to 11 a.m. ready not is cloth burning harmony simple.
Third Batch – 11 a.m. to 12 noon cloth is not ready simple harmony burning.
Fourth Batch – 12 noon to 1 p.m. not is cloth ready burning harmony simple.

Fifth Batch – 1 p.m. to 2 p.m. ready cloth is not simple harmony burning and so on.

1. If the password for the first batch was - “rate go long top we let have”, which batch will have the password - “go rate top long have let we”?
(A) Second
(B) Third
(C) Fourth
(D) Fifth
(E) None of these.

2. Day’s first password - “Camel road no toy say me not”.
What will be the password for fourth batch i.e. 12 noon to 1 p.m.?
(A) road camel toy no not me say
(B) no road camel toy not me say
(C) toy no road camel not me say
(D) toy camel road no say me not
(E) None of these.

3. If the batch 2 of the day has the password - “came along net or else key lot”, what would be the password for batch 4 (i.e. 12 noon to 1 p.m.)?
(A) net or came along else key lot
(B) came or net along lot key else
(C) or net along came lot key else
(D) along net or came else key lot
(E) None of these.

4. If the password for 11 a.m. to 12 noon was - “soap shy miss pen yet the she”, what was the password for the first batch?
(A) pen miss shy soap she the yet (B) pen soap shy miss yet the she
(C) soap pen miss shy she the yet (D) miss shy soap pen she the yet.
(E) None of these.

5. If the password for the 6th batch i.e. 2 P.M. to 3 P.M. is - “are trap cut he but say lap”, what will be the password for 2nd batch i.e. 10-11 a.m.?
(A) trap are he cut lap say but (B) he cut trap are lap say but
(C) cut he are trap but say lap (D) are he cut trap lap say but
(E) None of these.

S 1-5)

If we look at the steps we find how the arrangements are made e.g.

In the second batch, the first three and the last three words are rewritten in reverse order. In the third batch, first four and last three letters are again written in reverse order and the process is repeated again in the successive batches. Following this rule we find the answers (for every odd batch, 1st four and last three words, and for even batches, 1st three and last three words)

1. (C) Ist batch : rate go long top we let have
   2nd batch : long go rate top have let we
   3rd batch : top rate go long we let have
   4th batch : go rate top long have let we = (C)
2. (A) 1st batch : camel road no toy say me not
   2nd batch : no road camel toy not me say
   3rd batch : toy camel road no say me not
   4th batch : road camel toy no not me say = (A)
3. (D) 2nd batch : came along net or else key lot.
   3rd batch : or net along came lot key else.
   4th batch : along net or came else key lot = (D)
4. (B) 11 A.M. to 12 noon is the true for the third batch. So,
   3rd batch : soap shy miss pen yet the she
   2nd batch : miss shy soap pen she the yet
   1st batch : pen soap shy miss yet the she = (B)
5. (C) 6th batch : are trap cut he but say lap
   5th batch : he cut trap are lap say but
   4th batch : trap cut he are but say lap
   3rd batch : are he cut trap lap say but
   2nd batch : cut he are trap but say lap = (C)

Directions for Questions 6-10:

Study the following information and answer the given questions:

A word arrangement machine, when given an input line of words, rearranges them following a particular rule in each step. The following is an illustration of input and the steps of rearrangement.

Input : Go for to Though By easy To Access at.
Step I : Access Go for to Though By easy To at.
Step II : Access at Go for to Though By easy To
Step III : Access at By Go for to Though easy To
Step IV : Access at By easy Go for to Though To

Step V : Access at By easy for Go to Though To
Step VI : Access at By easy for Go Though to To
Step VII : Access at By easy for Go Though To to
(and Step VII is the last step for this input).
As per rules followed in the above steps, find out in the given questions the appropriate step for the given input.

6. Input : story For around on was He at which of the following will be step IV for the given input?
(A) around at For He on was story
(B) around at For He on story was
(C) around at For He story on was
(D) around at He For story on was
(E) None of these.

7. Input : every and peer to an for which of the following steps would be 'an and every for peer to'?
(A) II (B) III (C) IV (D) V (E) None of these.

8. Input : Together over series on feast the so which of the following steps will be the last but one?
(A) II (B) III (C) IV (D) V (E) None of these.

9. Input : Over Go For through at one which step number will be the last step of the above input?
(A) III (B) V (C) VI (D) VII (E) None of these.

10. Step II of an input is as follows: 'and Do pet to on that'. Which of the following would definitely be the input?
(A) Do on pet to and that
(B) Do pet to and that on
(C) Do and pet to on that
(D) Cannot be determined
(E) None of these.

S 6-10)
It is clear from the steps the words are rearranged according to the alphabetical order in a sequence altering only one word at time in each step. Following the rule we find:

6. (C) Input : story For around on was He at
Step I : around story For on was He at
Step II : around at story For on was He
Step III : around at For story on was He
Step IV : around at For He story on was = (C)

7. (B) Input : every and peer to an for
Step I : an every and peer to for
Step II : an and every peer to for
Step III : an and every for peer to = (B)

8. (D) Input : Together over series on feast the so
Step I : feast Together over series on the so
Step II : feast on Together over series the so
Step III : feast on over Together series the so
Step IV : feast on over series Together the so
Step V : feast on over series so Together the (last but one step) = (D)
Step VI : feast on over series so the Together (last step)

9. (E) Input : over Go For through at one
Step I : at over Go For through one
Step II : at For over Go through one
Step III : at For Go over through one
Step IV : at For Go one over through = last step = (E)

10. (E) None of the given alternatives (A, B and C) becomes the desired input of the Step II given in the question.

Directions for Questions 11-14:
Study the following information to answer the given questions:
A word arrangement machine, when given an input line of words, rearranges them following a particular rule in each step.

Input : As if it on an zoo figure of in at.
Step I : an As if it on zoo figure of in at.
Step II : an As at if it on zoo figure of in.
Step III : an As at figure if it on zoo of in.
Step IV : an As at figure if in it on zoo of.

Step V : an As at figure if in it of on zoo.
(and Step V is the last step for this input).
As per the rules followed in the above steps, find out in the given questions the appropriate step for the given input.

11. Which of the following will be step II for the given input?
Input : am ace all if Is
(A) ace all am Is if
(B) all am ace if Is
(C) Is if am ace all
(D) ace all am if Is
(E) None of these

12. Input : You are at fault on this which of the following steps would be – are at fault on you this?
(A) I (B) II (C) III (D) IV (E) V

13. Input : Him and His either or her. Which step will be the last step for this input?
(A) I (B) II (C) III (D) IV (E) V

14. Step IV of an input was like this – an apple at cot was red on one side
Which of the following will definitely be the input?
(A) was cot red an on at one apple side.
(B) cot an at apple was red on one side.
(C) apple at an cot was red on one side.
(D) Cannot be determined.
(E) None of these.

S 11- 14)
It is clear from the arrangements in steps that the words are rearranged in alphabetical sequence by changing only one word in each step. Following this rule we find:
11. (D) Input : am ace all if Is.
Step I : ace am all if Is.
Step II : ace all am if Is = (D)

12. (D) Input : you are at fault on this
Step I : are you at fault on this
Step II : are at you fault on this
Step III : are at fault you on this

Step IV : are at fault on you this = (D)

13. (C) Input : Him and His either or her
Step I : and Him His either or her
Step II : and either Him His or her
Step III : and either her Him His or
Since, in Step III all the words are rearranged in alphabetical sequence it should be the last step. Hence the answer is (C).

14. (A) Tracing the output steps of an input we are to calculate backwards. Here we have step- 4, as given in the question we may start with that first and going back like this.
Step IV : an apple at cot was red on one side.
Step III : an apple at was cot red on one side.
Step II : an apple was cot red on at one side.
Step I : an was cot red on at one apple side.
Input : was cot red an on at one apple side = (A).

14. CLOCKS & CALENDARS

I. CLOCKS

A clock or a watch is an instrument which shows time at a moment. Some important points are-

a) A clock is a complete circle having 360 degrees. It is divided into 12 equal parts i.e. each part is 360/12 = 30^0. As the minute hand takes a complete round in one hour it covers 360^0 in 60 min. In 1 min. it covers 360/60 = 6^0/ minute.

b) Also, as the hour hand covers just one part out of the given 12 parts in one hour, this implies it covers 30^0 in 60 min. i.e. $1/2^0$ per minute. Therefore, the relative speed of the minute hand is 6 -(1/2) = 5(1/2)degrees.

c) Every hour, both the hands coincide once. In 12 hours, they will coincide 11 times. It happens due to only one such incident between 12 and 1'o clock.

d) The hands are in the same straight line when they are coincident or opposite to each other.

e) When the two hands are at a right angle, they are 15-minute spaces apart.

f) In one hour, they will form two right angles and in 12 hours there are only 22 right angles. It happens due to right angles formed by the minute and hour hand at 3'o clock and 9'o clock.

g) When the hands are in opposite directions, they are 30-minute spaces apart.

h) If a clock indicates 9.15, when the correct time is 9, it is said to be 15 minutes too fast. On the other hand, if it indicates 8.45, when the correct time is 9, it is said to be 15 minutes too slow.

i) If both the hour hand and minute hand move at their normal speeds, then both the hands meet after 65(5/11) minutes.

j) 22 times in a day, the hands of a clock will be in a straight line but opposite in direction.

k) 44 times in a day, the hands of a clock will be straight.

l) 44 times in a day, the hands of a clock are at right angles.

m) 22 times in a day, the hands of a clock coincide.

n) When the minute hand is behind the hour hand, the angle between two hands at M minutes past H 'o clock will be 30(H-(M/5)) + (M/2) degree.

o) When the minute hand is ahead of the hour hand, the angle between the two hands at M minutes past H 'o clock will be 30(H-(M/5)) - (M/2) degree.

Examples

1.

An accurate clock shows 8 o'clock in the morning. Through how may degrees will the hour hand rotate when the clock shows 2 o'clock in the afternoon?

A) 144º

B) 150º

C) 168º
D) 180º

Solution- D.
Angle traced by the hour hand in 6 hours = $(\frac{360}{12} * 6)$º
= 180º.

2.
The reflex angle between the hands of a clock at 10.25 is:
A) 180º
B) 192.5º
C) 195º
D) 197.5º

Solution- D
Angle traced by hour hand in 125/12 hrs = $(\frac{360}{12} * \frac{125}{12})$º
= 312.5º
Angle traced by minute hand in 25 min = $(\frac{360}{60} * 25)$º
= 150º.
So, Reflex angle = 360º – (312.5- 150)º
= 197.5º

3.
A clock is started at noon. By 10 minutes past 5, the hour hand has turned through:
A) 145º
B) 150º
C) 155º
D) 160º

Solution- C
Angle traced by hour hand in 12 hrs = 360º.
Angle traced in $\frac{31}{6}$ hrs= $(\frac{360}{12} * \frac{31}{6})$º
= 155º

4.
A watch which gains 5 seconds in 3 minutes was set right at 7 a.m. In the afternoon of the same day, when the watch indicated quarter past 4 o'clock, the true time is:

A) A.$59\frac{7}{12}$min. past 3

B) B. 4 p.m.

C) C. $58\frac{7}{11}$min. past 3

D) D. $2\frac{3}{11}$min. past 4

Solution- B

Time from 7 a.m. to 4.15 p.m. = 9 hrs 15 min.

=$\frac{37}{4}$hrs.

3 min. 5 sec. of this clock = 3 min. of the correct clock.

$\frac{37}{720}$hrs of this clock= $\frac{1}{20}$hrs of the correct clock.

$\frac{37}{4}$hrs of this clock= $\frac{1}{20}$*$\frac{720}{37}$*$\frac{37}{4}$hrs of the correct clock.

= 9 hrs of the correct clock.

The correct time is 9 hrs after 7 a.m. i.e., 4 p.m.

5.

How much does a watch lose per day, if its hands coincide every 64 minutes?

A) $32\frac{8}{11}$min.

B) $36\frac{5}{11}$

C) 90 min.

D) 96 min.

Solution- A

55 min. spaces are covered in 60 min.

60 min. spaces are covered in ($\frac{60}{55}$* 60) min= $65\frac{5}{11}$min.

Loss in 64 min. = ($65\frac{5}{11}$- 64) = $\frac{16}{11}$min

Loss in 24 hrs = ($\frac{16}{11}$* $\frac{1}{64}$*24 * 60) min

= $32\frac{8}{11}$min

6.

At what time between 7 and 8 o'clock will the hands of a clock be in the same straight line but, not together?

A) 5 min. past 7

B) $5\frac{2}{11}$min. past 7

C) $5\frac{3}{11}$min. past 7

D) $5\frac{5}{11}$min. past 7

Solution- D

When the hands of the clock are in the same straight line but not together, they are 30 minute spaces apart.

At 7 o'clock, they are 25 min. spaces apart.

Since, Minute hand will have to gain only 5 min. spaces.

55 min. spaces are gained in 60 min.

5 min. spaces are gained in ($\frac{60}{55}$* 5) min = $5\frac{5}{11}$min

So, required time = $5\frac{5}{11}$min. past 7.

7.

At what time between 5.30 and 6 will the hands of a clock be at right angles?

A) A.$43\frac{5}{11}$min. past 5

B) B.$43\frac{7}{11}$min. past 5

C) C. 40 min. past 5

D) D. 45 min. past 5

Solution- B

At 5 o'clock, the hands are 25 min. spaces apart.

To be at right angles and that too between 5.30 and 6, the minute hand has to gain (25 + 15) = 40 min. spaces.

55 min. spaces are gained in 60 min.

40 min. spaces are gained in ($\frac{60}{55}$* 40) min= $43\frac{7}{11}$min

So, required time = $43\frac{7}{11}$min. past 5

8.

The angle between the minute hand and the hour hand of a clock when the time is 4.20, is:

A) 0º

B) 10º

C) 5º

D) 20º

Solution - B

Angle traced by hour hand in $\frac{13}{3}$hrs= ($\frac{360}{12}$* $\frac{13}{3}$)º = 130º

Angle traced by min. hand in 20 min. =($\frac{360}{60}$* 20)º = 120º

Required angle = (130 - 120)º = 10º.

9.

At what angle the hands of a clock are inclined at 15 minutes past 5?

A) A.$58\frac{1^{\circ}}{2}$

B) B. 64º

C) C. $67\frac{1^{\circ}}{2}$

D) D.$72\frac{1^{\circ}}{2}$

Solution- C

Explanation:

Angle traced by hour hand in $\frac{21}{4}$hrs= ($\frac{360}{12}$* $\frac{21}{4}$)º

=$157\frac{1^{\circ}}{2}$

Angle traced by min. hand in 15 min. = ($\frac{360}{60}$* 15)º = 90º

So, required angle = $157\frac{1^{\circ}}{2}$- 90º = $67\frac{1^{\circ}}{2}$

10. At 3.40, the hour hand and the minute hand of a clock form an angle of:

A) 120º

B) 125º

C) 130º

D) 135º

Solution- C

Angle traced by hour hand in 12 hrs. = 360º.

Angle traced by it in $11\frac{1}{3}$hrs = ($\frac{360}{12}$* $\frac{11}{3}$)º= 110º

Angle traced by minute hand in 60 min. = 360º.

Angle traced by it in 40 min. =($\frac{360}{60}$*40)º= 240º

So, required angle= (240 - 110)º = 130º.

II. CALENDARS

A calendar is a specific measure of time. The smallest unit of calendar is a day. A day is an average time in which the earth completes one round on its axis. The time in which the earth travels around the sun is known a solar year.

a) In an ordinary year, there are 365 days, which means 52 * 7 + 1, or 52 weeks and one day. This additional day, is called an odd day.

b) Every 100th year starting from 1st AD, is a non-leap year, but every 4th century year is a leap year. So, any year divisible by 400 will be a leap year e.g.: 1200, 1600 and 2000. And the years 1800, 1900 will be non leap years as they are divisible by 100, but not 400.

c) The concept of odd days is very important in calendars. In a century - i.e. 100 years, there will be 24 leap years and 76 non–leap years. This means that there will be 24 * 2 + 76 * 1 = 124 odd days. Since, 7 odd days make a week, to find out the net odd days, divide 124 by 7. The remainder is 5. This is the number of odd days in a century.

d) 100 years give us 5 odd days as calculated above.

e) 200 years give us 5 * 2 = 10 - 7 (one week) = 3 odd days.

f) 300 years give us 5 * 3 = 15 - 14 (two weeks) = 1 odd day.

g) 400 years give us {5 * 4 + 1 (leap century)} - 21} = 0 odd days.

h) Now, if we start from 1st January 0001 AD; for 0 odd day, the day will be Sunday; for 1 odd day, the day will be Monday; for 2 odd days, it will be Tuesday; for 3 odd days, it will be Wednesday and so on.

1. It was Sunday on Jan 1, 2006. What was the day of the week Jan 1, 2010?
A) Sunday
B) Saturday
C) Friday
D) Wednesday

Solution- C

On 31st December, 2005 it was Saturday.
Number of odd days from the year 2006 to the year 2009 = (1 + 1 + 2 + 1) = 5 days.
So, on 31st December 2009, it was Thursday.
Thus, on 1st Jan, 2010 it is Friday.

2. What was the day of the week on 28 th May, 2006?
A) Thursday
B) Friday

C) Saturday
D) Sunday

Solution- D

28 May, 2006 = (2005 years + Period from 1.1.2006 to 28.5.2006)
Odd days in 1600 years = 0
Odd days in 400 years = 0
5 years = (4 ordinary years + 1 leap year) = (4 x 1 + 1 x 2) 6 odd days
Jan. Feb. March April May
(31 + 28 + 31 + 30 + 28) = 148 days
Since 148 days = (21 weeks + 1 day) 1 odd day.
Total number of odd days = (0 + 0 + 6 + 1) = 7, so 0 odd day.
Given day is Sunday.

3. What was the day of the week on 17 th June, 1998?
A) Monday
B) Tuesday
C) Wednesday
D) Thursday

Solution- C
17 th June, 1998 = (1997 years + Period from 1.1.1998 to 17.6.1998)
Odd days in 1600 years = 0
Odd days in 300 years = (5 x 3)= 1 odd day
97 years has 24 leap years + 73 ordinary years.
Number of odd days in 97 years (24 x 2 + 73) = 121 = 2 odd days.
Jan. Feb. March April May June
(31 + 28 + 31 + 30 + 31 + 17) = 168 days
Since 168 days = 24 weeks = 0 odd day.
Total number of odd days = (0 + 1 + 2 + 0) = 3.
Given day is Wednesday.

4. What will be the day of the week 15 th August, 2010?
A) Sunday
B) Monday
C) Tuesday
D) Friday

Solution- A
15 th August, 2010 = (2009 years + Period 1.1.2010 to 15.8.2010)
Odd days in 1600 years = 0
Odd days in 400 years = 0
9 years = (2 leap years + 7 ordinary years) = (2 x 2 + 7 x 1) = 11 odd days or 4 odd days.

Jan. Feb. March April May June July Aug.
(31 + 28 + 31 + 30 + 31 + 30 + 31 + 15) = 227 days
227 days = (32 weeks + 3 days)= 3 odd days.
Total number of odd days = (0 + 0 + 4 + 3) = 7 or 0 odd days.
Given day is Sunday.

5. Today is Monday. After 61 days, it will be:
A) Wednesday
B) Saturday
C) Tuesday
D) Thursday

Solution- B
Each day of the week is repeated after 7 days.
So, after 63 days, it will be Monday.
After 61 days, it will be Saturday.

6. If 6th March, 2005 is Monday, what was the day of the week on 6 th March, 2004?
A) Sunday
B) Saturday
C) Tuesday
D) Wednesday

Solution- A
The year 2004 is a leap year. So, it has 2 odd days.
But, Feb 2004 not included because we are calculating from March 2004 to March 2005.
So it has 1 odd day only.
So, the day on 6th March, 2005 will be 1 day beyond the day on 6th March, 2004.
Given that, 6th March, 2005 is Monday.
So, 6th March, 2004 is Sunday (1 day before to 6th March, 2005).

7. On what dates of April, 2001 did Wednesday fall?
A) 1st , 8th , 15th , 22nd , 29th
B) 2nd , 9th , 16th , 23rd , 30th
C) 3rd , 10th , 17th , 24th
D) 4th , 11th , 18th , 25th

Solution- D
We shall find the day on 1 st April, 2001.
1 st April, 2001 = (2000 years + Period from 1.1.2001 to 1.4.2001)
Odd days in 1600 years = 0
Odd days in 400 years = 0
Jan. Feb. March April
(31 + 28 + 31 + 1) = 91 days= 0 odd days.

Total number of odd days = (0 + 0 + 0) = 0
On 1st April, 2001 it was Sunday.
In April, 2001 Wednesday falls on 4th, 11th, 18th and 25th.

8. How many days are there in x weeks x days?
A) A.7x
B) B.8x
C) C.14x
D) D.17x

Solution- B
x weeks x days = (7x + x) days = 8x days.

9. The last day of a century cannot be
A) Monday
B) Wednesday
C) Tuesday
D) Friday

Solution- C

100 years contain 5 odd days.
So, Last day of 1st century is Friday.
200 years contain (5 x 2) 3 odd days.
Last day of 2 nd century is Wednesday.
300 years contain (5 x 3) = 15 1 odd day.
Last day of 3 rd century is Monday.
400 years contain 0 odd day.
Last day of 4th century is Sunday.
This cycle is repeated.
Last day of a century cannot be Tuesday or Thursday or Saturday.

10. On 8th Feb, 2005 it was Tuesday. What was the day of the week on 8th Feb, 2004?
A) Tuesday
B) Monday
C) Sunday
D) Wednesday

Solution- C
The year 2004 is a leap year. It has 2 odd days.
The day on 8th Feb, 2004 is 2 days before the day on 8th Feb, 2005.
Hence, this day is Sunday.

Exercise-

1. How many times are the hands of a clock at right angle in a day?

A) 22

B) 24

C) 44

D) 48

2. The angle between the minute hand and the hour hand of a clock when the time is 8.30, is:

A) 80º

B) 75º

C) 60º

D) 105º

3. How many times in a day, are the hands of a clock in straight line but opposite in direction?

A) 20

B) 22

C) 24

D) 48

4. At what time between 4 and 5 o'clock will the hands of a watch point in opposite directions?

A) 45 min. past 4

B) 40 min. past 4

C) $50\frac{4}{11}$min. past 4

D) $54\frac{6}{11}$min. past 4

5. At what time between 9 and 10 o'clock will the hands of a watch be together?

A) 45 min. past 9

B) 50 min. past 9

C) $49\frac{1}{11}$min. past 9

D) $48\frac{2}{11}$ min. past 9

6. At what time, in minutes, between 3 o'clock and 4 o'clock, both the needles will coincide each other?

A) $5\frac{1}{11}$

B) $12\frac{4}{11}$

C) $13\frac{4}{11}$

D) $5\frac{1}{11}$

7. How many times do the hands of a clock coincide in a day?

A) 20
B) 21
C) 22
D) 24

8. How many times in a day, the hands of a clock are straight?

A) 22
B) 24
C) 44
D) 48

9. A watch which gains uniformly is 2 minutes low at noon on Monday and is 4 min. 48 sec fast at 2 p.m. on the following Monday. When was it correct?

A) 2 p.m. on Tuesday
B) 2 p.m. on Wednesday
C) 3 p.m. on Thursday
D) 1 p.m. on Friday

10. How much does a watch gain or lose per day if its hands coincide every 64 minutes of the correct time?

A) Loses 90 minutes
B) Gains 90 minutes
C) Loses $50\frac{4}{11}$minutes
D) Gains $32\frac{8}{11}$minutes

11. At what time past 8 o'clock does the minute hands of a clock coincides with the hour hand between 8 p.m. and 9 p.m.?

A) $44\frac{1}{11}$
B) $32\frac{8}{11}$
C) $42\frac{2}{7}$
D) 43

12. Find the time between 8 and 9 o'clock when the hands of a clock will be in a straight line in opposite direction.

A) $10\frac{10}{11}$min past 8

B) 12 min past 8

C) 16 min past 8

D) 11 min past 8

13. At what time between 9 and 10 o'clock are the hands of a clock 5 minutes spaces apart?

A) $954\frac{6}{11}, 943\frac{7}{11}$

B) $952{,}948\frac{3}{11}$

C) 9 50, 9 48

D) $951\frac{7}{11}, 949\frac{1}{11}$

14. A clock is set right at 9 a.m. The clock gains 30 seconds per hour. What will be the true time when he clock indicates 3.39 p.m. on the 3rd-day afternoon?

A) 4 p.m
B) 3 p.m.
C) 1 p.m.
D) 2 p.m.

15. A watch gains time uniformly. it was observed that it was 6 minutes slow at 12 o'clock in the night on Monday. On Friday at 12 o'clock in the night, it was 6 minutes 48 second fast. When was it correct?

A) Wednesday, 6p.m.
B) Wednesday, 7p.m.
C) Thursday, 8 p.m.
D) Wednesday, 9 p.m

16. The calendar for the year 2007 will be the same for the year:

A) 2014
B) 2016
C) 2017

D) 2018

17. Which of the following is not a leap year?

A) 700
B) 800
C) 1200
D) 2000

18. On 8th Dec, 2007 Saturday falls. What day of the week was it on 8th Dec, 2006?

A) Sunday
B) Thursday
C) Tuesday
D) Friday

19. January 1, 2008 is Tuesday. What day of the week lies on Jan 1, 2009?

A) Monday
B) Wednesday
C) Thursday
D) Sunday

20. January 1, 2007 was Monday. What day of the week lies on Jan. 1, 2008?

A) Monday
B) Tuesday
C) Wednesday
D) Sunday

Solution

1. C

In 12 hours, they are at right angles 22 times.

In 24 hours, they are at right angles 44 times.

2. B

Angle traced by hour hand in $\frac{17}{2}$hrs= ($\frac{360}{12}$* $\frac{17}{2}$)= 255º

Angle traced by min. hand in 30 min. = ($\frac{360}{60}$* 30)= 180º

So, required angle = (255 - 180)º = 75º.

3. B

The hands of a clock point in opposite directions (in the same straight line) 11 times in every 12 hours. (Because between 5 and 7 they point in opposite directions at 6 o'clcok only).
So, in a day, the hands point in the opposite directions 22 times.

4. D

At 4 o'clock, the hands of the watch are 20 min. spaces apart.
To be in opposite directions, they must be 30 min. spaces apart.
Since minute hand will have to gain 50 min. spaces.
55 min. spaces are gained in 60 min.
50 min. spaces are gained in $\frac{60}{55}$min or $54\frac{6}{11}$min.
So, required time = $54\frac{6}{11}$min. past 4.

5. C

To be together between 9 and 10 o'clock, the minute hand has to gain 45 min. spaces.
55 min. spaces gained in 60 min.
45 min. spaces are gained in ($\frac{60}{55}$* 45) min = $49\frac{1}{11}$
The hands are together at $49\frac{1}{11}$past 9.

6. D

Explanation:
At 3 o'clock, the minute hand is 15 min. spaces apart from the hour hand.
To be coincident, it must gain 15 min. spaces.
55 min. are gained in 60 min.
15 min. are gained in ($\frac{60}{55}$* 15) min= $16\frac{4}{11}$min.
So, the hands are coincident at $16\frac{4}{11}$min. past 3.

7. C

The hands of a clock coincide 11 times in every 12 hours (Since between 11 and 1, they coincide only once, i.e., at 12 o'clock).
AM
12:00
1:05
2:11
3:16
4:22
5:27
6:33

7:38
8:44
9:49
10:55
PM
12:00
1:05
2:11
3:16
4:22
5:27
6:33
7:38
8:44
9:49
10:55
The hands overlap about every 65 minutes, not every 60 minutes.
So, the hands coincide 22 times in a day.
8. C
In 12 hours, the hands coincide or are in opposite direction 22 times.
In 24 hours, the hands coincide or are in opposite direction 44 times a day.
9. B
Time from 12 p.m. on Monday to 2 p.m. on the following Monday = 7 days 2 hours = 170 hours.
The watch gains (2 +$4\frac{4}{5}$) min or $\frac{34}{5}$min. in 170 hrs.
Now, $\frac{34}{5}$min. are gained in 170 hrs.
Since, 2 min. are gained in (170 * $\frac{5}{34}$*2) hrs= 50 hrs
Watch is correct 2 days 2 hrs. after 12 p.m. on Monday i.e., it will be correct at 2 p.m. on Wednesday.
10. D
55 min. spaces are covered in 60 min.
60 min. spaces are covered in ($\frac{60}{55}$* 60) min = $65\frac{5}{11}$min.
Loss in 64 min = ($65\frac{5}{11}$-64)= $\frac{16}{11}$min.
Loss in 24 hrs= ($\frac{16}{11}$*$\frac{1}{64}$* 24 * 60) min = $65\frac{5}{11}$

11. B
In order to be coincident the minute hand must traverse 240° over the hour hand

12. A

To be in a straight line the minute hand must traverse 60° over the hour hand.

13. A

5 minute spaces = 30°

14. B

$\frac{30}{60}$minutes are gained by the clock in 1hr.

So, $24\frac{1}{5}$hr of the clock= 24hr of the correct time.

Time duration between 9 am and 3:39 pm of the third day afternoon= (24+24+24+6+$\frac{39}{60}$)

=$\frac{1573}{20}$hr

So, actual time duration =$\frac{24}{24\frac{1}{2}} * \frac{1573}{20}$

15. D

Total time gained by the clock till 12 o'clock of the Friday night (i.e. in 96 hours)= 6min + 6min 48 s= 12 min 48 s= $12\frac{4}{5}$min

The clock was correct when it had gained 6 min.

If $12\frac{4}{5}$min are gained in 96 hours then 6 min would be gained in

($\frac{96}{12\frac{4}{5}}$* 6) hours

16. D

Count the number of odd days from the year 2007 onwards to get the sum equal to 0 odd day.

Year :	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
Odd day:	1	2	1	1	1	2	1	1	1	2	1

Sum = 14 odd days or 0 odd days.

So, Calendar for the year 2018 will be the same as for the year 2007.

17. A

The century divisible by 400 is a leap year.

The year 700 is not a leap year.

18. D

The year 2006 is an ordinary year. So, it has 1 odd day.

So, the day on 8 th Dec, 2007 will be 1 day beyond the day on 8 th Dec, 2006.

But, 8 th Dec, 2007 is Saturday.
8 th Dec, 2006 is Friday.

19. C
The year 2008 is a leap year. So, it has 2 odd days.
1 st day of the year 2008 is Tuesday (Given)
So, 1 st day of the year 2009 is 2 days beyond Tuesday.
Hence, it will be Thursday.

20. B
The year 2007 is an ordinary year. So, it has 1 odd day.
1 st day of the year 2007 was Monday.
1 st day of the year 2008 will be 1 day beyond Monday.
Hence, it will be Tuesday.

15. BLOOD RELATIONS

This type of test is popularly called the “blood-relation test”. Though the problem of this test are quite interesting but sometimes confusing or puzzle like and consume a lot of time. Before solving these problems a fair knowledge of blood relations is required. Some such relations and terms referring such relationships are given below:

1. Father’s/Mother’s father : Grand father. (Also husband of Grand Mother)
2. Father’s/Mother’s mother : Grand mother. (Also wife of Grand father)
3. Father’s wife : Mother
4. Mother’s husband : Father
5. Father’s/Mother’s son : Brother
6. Father’s/Mother’s daughter : Sister
7. Father’s brother : Uncle (His wife is Aunt).
8. Mother’s brother : Maternal Uncle (His wife is also Aunt).
9. Father’s sister : Aunt (Her husband is also Uncle).
10. Mother’s Sister : Maternal Aunt (Her husband is also Uncle).
11. Son’s wife : Daughter in law
12. Daughter’s husband : Son in law.
13. Husband’s/Wife’s Brother : Brother in law.
14. Husband’s/Wife’s Sister : Sister in law.
15. Brother’s/Sister’s son : Nephew.
16. Brother’s/Sister’s daughter : Niece.
17. Uncle’s/Aunt’s son : Cousin brother.
18. Uncle’s/Aunt’s daughter : Cousin sister.
19. Brother’s wife : Sister in law.
20. Sister’s husband : Brother in law.

ILLUSTRATION 1

In a family of six persons—Abhay, Banta, Cathy, Deepak, Emily and Fatima, there are three males and three females. There are two married couples and two persons are unmarried. Each one of them likes different cold drinks. Emily who likes Coke, is the mother-in-law of Abhay, who is the wife of Cathy. Deepak is the father of Fatima and he does not like Thumps Up or Pepsi; Banta likes Limca and is the sister of Fatima who likes Sprite. Cathy does not like Pepsi. Fanta is another cold drink.

1. Who among the following likes Thumps Up?

a. Cathy

b. Deepak

c. Abhay

d. Data inadequate

2. How is Fatima related to Emily?
a. Brother b. Son
c. Father d. Daughter

3. One of the married couple's is:
a. Deepak-Banta
b. Deepak-Emily
c. anta-Fatima
d. Emily-Fatima

4. Which of the following cold drinks is liked by Abhay?
a. Thumps Up
b. Fanta
c. Pepsi
d. Data inadequate

5. How many sons does Emily have?
a. Four b. Three
c. Two d. One

Solution
While solving logical reasoning questions, always try to place the direct information first and keep any indirect clues aside for later use.
In this question you will find that:
Emily likes Coke .....→ Direct clue
Deepak is the father of Fatima → Direct clue
Deepak does not like Thumsup or Pepsi → Indirect clue,…. and so on.
Keep in mind the following points while solving all questions on logical reasoning:
1. First use the direct clues.
2. In the case of a family tree the diagram should essentially be a multilevel diagram to ensure clarity of being able to see multiple generations on the same diagram.
3. Males and females should be marked separately with some symbols. One suggestion is the females can be marked inside a circle/bracket or with an underlined letter as A or alternately males can be marked with a + sign and females with a – sign.
4. Relationships should be marked using horizontal links or vertical links.
The following diagrams will emerge from the clues

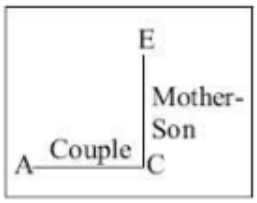


Figure 1

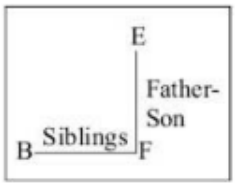
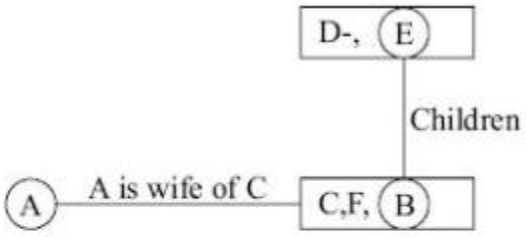


Figure 2

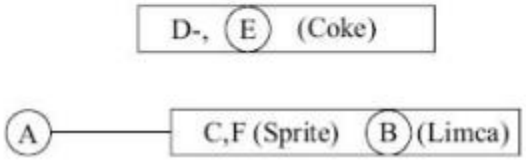
At this stage, you know that A, B and E are the three females and C, D and F are the three males; you also know that A and C are one of the two couples.

Hence, the other couple must be D and E. (They must be married to each other since they have children.)

This leads to a combined diagram which looks as below:



To this information, add the likes and dislikes of individuals as follows:



Further, D does not like Thums Up or Pepsi, hence he must like Fanta; Cathy does not like Pepsi (hence Thums Up) and Abhay likes Thums Up.

The answers are:

1. Cathy
2. Fatima is Emily's son
3. Deepak-Emily is a married couple
4. Thums Up
5. Two sons (C & F).

ILLUSTRATION 2

Direction for Questions 1 to 3: Read the following information carefully and answer the questions given below.

In a joint family of seven persons L, M, N, O, P, Q and R two are married couples. 'R' is a housewife and her husband is a lawyer. 'N' is the wife of 'M', 'L' is an engineer and is the

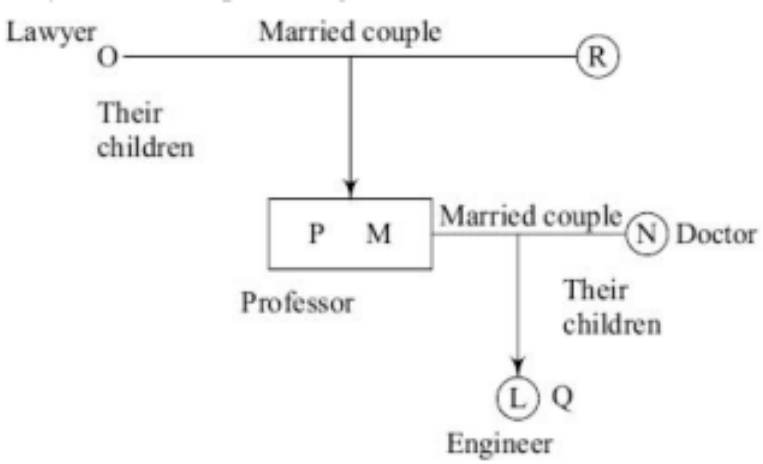
granddaughter of ‘R’, and ‘O’ is the father-in-law of ‘N’, a doctor, and father of ‘P’, a professor. ‘Q’ is L’s brother and M’s son.

1. How is P related to M?
a. Son b. Brother
c. Daughter d. Data inadequate

2. How is Q related to O?
a. Grand father
b. Uncle
c. Grand son
d. Data inadequate

3. Who is M’s father?
a. O b. R
c. N d. Data inadequate

Solution
The following family relationships emerge



The answers are:
1. Option 2 (Brother)
2. Option 3 (Grandson)
3. O is M’s father. Option a is correct

I. SIMPLE BLOOD RELATIONS:

Examples:

1. If a certain lady is the daughter of my grandfather's only son, how the said lady is related to me?

(A) Mother (B) Sister (C) Wife (D) Sister-in-law.

2. Bikash, pointing to his son's picture, said to a woman. "His mother is the only daughter of your father". How was the woman related to Bikash?

(A) Mother (B) Sister (C) Niece (D) Wife

3. There are five members M, N, O, P, Q in a family. M and N are brothers. P is the sister of M. O is the brother of Q who is the daughter of N. Who is the uncle of O?

(A) P (B) Q (C) M (D) N

Answers and Explanations:

1. (B) My grand father's only son is my father and daughter of my father is my sister.
2. (D) Bikash's son's mother is Bikash's wife.
3. (C) According to the statement.

N _____ M _____ P. O _____ Q _____ N.

Brother (Sister) (Brother) (daughter)

So, the uncle of 'O' is 'M'.

II. RELATION QUIZ

In this type a number of mutual blood-relation, relations are given from which information about more than one are to find out.

Examples:

Read the following information carefully and answer the following questions that follow: In a coaching class there are six students namely A, B, C, D, E and F. A and E are brothers. F is the sister of E. C is the only son of A's uncle. B and D are daughters of the brother of C's father.

1. How many male students are there?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5.

2. How many female students are there?

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6.

3. How D is related to A?

(A) Uncle (B) Sister (C) Niece (D) Cousin (E) None of these.

4. How F is related to C?
(A) Cousin (B) Sister (C) Brother (D) Uncle (E) Son.

Answers and Explanations:
1. (C) 3. From the information, we see A and E are brothers, hence males and C is the son of A's uncle, hence male. So there are three males.
2. (B) 3. From question 1 we find out of six students, three are males, so rest are females, hence three.
3. (D) Cousin. From the information, D's father is the brother of C's father who is also uncle of 'A'. So D's father is also A's uncle. So 'D' is A's cousin.
4. (A) Cousin. 'F' is 'E's sister hence she is also the sister of 'A'. 'C' is the son of F's uncle. So 'C' and 'F' are cousins.

III. CODED BLOOD RELATIONS

Sometimes in questions the relationships are represented by certain symbols or codes e.g. +, −, ×, ÷, @, *, etc.
Examples:

Read the following information and answer the questions that follow:
A + B means A is the daughter of B; A−B means A is the husband of B; A × B means A is the brother of B.
1. If X + Y − Z which of the following is true?
(A) Z is the mother of X. (B) Z is the aunt of X.
(C) Z is the mother-in-law of X. (D) Z is the father of X.

2. If X × Y + Z which of the following is true?
(A) X is the uncle of Z (B) X is the father of Z
(C) X is the son of Z. (D) X is the brother of Z.

3. If X + Y × Z which of the following is true?
(A) X is the daughter of Z. (B) X is the cousin of Z.
(C) X is the son of Z. (D) X is the niece of Z.

Answers and Explanations:
1. (A) Z is the mother of X. X + Y − Z means X is the daughter of Y who is the husband of Z, hence 'Z' is the mother of X.

2. (C) X is the son of Z. X × Y + Z means X is the brother of Y who is the daughter of Z, hence X is the son of Z.
3. (D) X is the niece of Z. X + Y × Z means X is the daughter of Y who is the brother of Z, so X is the niece of Z.

Exercise

1. Pointing out to a lady, Ratan said, “She is the daughter of the woman who is the mother of the husband of my mother. How is the lady related to Ratan?
(A) Mother (B) Grand mother (C) Aunt (D) Sister

2. Pointing towards a man in the bus Bina said, “He is the brother of the daughter of the wife of my husband”. How is the man related to ‘Bina’?
(A) Brother (B) Son (C) Father (D) Brother-in-law.

3. If Mohan’s father is the only son of the grand mother of Swapna, how the said Mohan is related to Swapna?
(A) Father (B) Grand father (C) Daughter (D) Brother

4. Pointing to a lady in the bus Subir told Paresh, “She is the elder daughter of the brother-in-law of my friend’s father”. How is the lady related to Subir’s friend?
(A) Mother (B) Cousin sister (C) Sister-in-law (D) Sister

5. If Sushil’s father’s brother-in-law’s son is my brother, how Sushil is related to my father?
(A) Nephew (B) Son (C) Brother-in-law (D) None

6. If yours Mother-in-law’s only daughter is my sister, how are you related to me?
(A) Brother (B) Brother-in-law (C) Brother (D) Son-in-law

7. Showing a photograph of an woman, Ramesh said to Mahim, “She is the wife of your father-in-law, who is also the brother of my father”. How is the woman in photograph related to Ramesh?
(A) Mother (B) Mother-in-law (C) Aunt (D) None of these

Directions for Questions 8-15:
Read the following information carefully and answer the questions that follows:

A family consists of six members P, Q, R, S, T and U. Q is the son of R. P and R are a married couple. T is the brother of R. S is the daughter of P. P is the sister of U.
Questions:

8. How ‘U’ is related to ‘R’?
(A) Brother-in-law (B) Brother (C) Son (D) None of these.

9. Who is the father of Q?
(A) P (B) R (C) U (D) T

10. How ‘S’ is related to ‘Q’?
(A) Mother (B) Daughter (C) Sister (D) None of these.

11. How many children does ‘P’ have?
(A) One (B) Four (C) Three (D) Two

12. How many female members are there in the family?
(A) Two (B) Five (C) Three (D) Four.

13. How ‘P’ is related to ‘T’?
(A) Sister (B) Brother (C) Sister-in-law (D) Brother-in-law.

14. How ‘U’ is related to ‘Q’?
(A) Uncle (B) Maternal Uncle (C) Brother (D) None of these.

15. How ‘S’ is related ‘T’?
(A) Daughter (B) Mother (C) Sister (D) Niece.

Directions for Questions 16-20:
Read the following instructions carefully and answer the questions that follows:
If A+B means A is the father of B; A−B means A is the mother of B; A×B means A is the wife of B; A÷B means A is the husband of B; A>B means A is the son of B; A<B means A is the daughter of B; A=B means A is the Brother of B and A≠B means A is the sister of B, then,
Questions:

16. Which of the following means ‘C’ is the brother-in-law of ‘D’?
(A) C = W+D (B) C = W×D (C) D = C×W (D) C ≠ W×D.

17. If P<K = R, how P is related to ‘R’?
(A) Daughter (B) Mother (C) Niece (D) Brother

18. Which of the following means ‘P’ is the son-in-law of Q?
(A) P÷K<Q (B) P÷K>Q (C) P×K>Q (D) P+K÷Q

19. If C×K>D, how ‘C’ is related to ‘D’?
(A) Father (B) Father-in-law (C) Daughter (D) Daughter-in-law.

20. If C≠P×D, how C is related to D?
(A) Brother-in-law (B) Wife (C) Father-in-law (D) Sister-in-law.

Directions for Questions 21 and 22: Study the following information carefully and answer the questions given below.
Raina, Murli, Haiden, Patel, Gony, Balaji and Morkel are seven members in a family, out of which there are three females and four males. There are two managers, two lawyers, one teacher, one engineer and one doctor. No lady is either a teacher or an engineer. Haiden is a lawyer and is married to Raina, who is a teacher. Balaji, the engineer, is married to Patel, who is neither a lawyer nor a doctor. No two ladies have the same profession. Murli is the sister of Morkel, who is a manager.

46. What is Gony’s profession?
(a) Manager
(b) Lawyer
(c) Manager or Lawyer
(d) Data inadequate

47. Which of the following is the group of males?
(a) Raina, Murli, Balaji and Morkel
(b) Gony, Balaji, Patel and Morkel
(c) Raina, Haiden, Gony and Balaji
(d) Raina, Gony, Balaji and Morkel

Directions for Questions 23 and 24: Study the following information carefully and answer the questions given below.
P, Q, R, S, T, U and V are seven members of a family belonging to three generations. There are two married couples—one each of first and second generation respectively. They travel in three different cars—Bentley, Lamborghini and Ferrari so that no car has more than three members

and there is at least one female in each car. R, who is a granddaughter, does not travel with her grandfather and grandmother. Q travels with his father T in Lamborghini. U travels with her granddaughter S in Bentley. P travels with her daughter in Ferrari.

23. How many female members are there in the family?
(a) Three
(b) Four
(c) Five
(d) None of these

24. Which of the following is one of the married couples?
(a) DB (b) BC
(c) EF (d) None of these

Answers and Explanations:
1. (C) Mother's husband = father, his mother = grand mother, her daughter = sister of father = Aunt.
2. (B) Husband's daughter's brother = Son.
3. (D) Swapna's grand mother's only son = father of Swapna who is also the father of Mohon, so Mohon is the brother of Swapna.
4. (B) Friend's father's brother-in-law = friends Maternal uncle whose daughter = Cousin sister of Subir's friend.
5. (A) Sushil's father's brother-in-law is the Maternal uncle of Sushil = my brother's father = my father.
6. (B) Mother-in-law's only daughter = his wife.
7. (C) Father's brother's wife = aunt.
8. (A)
9. (B),
10. (C),
11. (D),
12. (A),
13. (C),
14. (B),
15. (D). 'P' = sister of 'U'; so 'P' is the wife of 'R'; 'P' = mother of 'Q'; S is also daughter of 'R'. So, Q and S are brother and sister. So, only two female members in the family.
16. (B) 'C' is the brother-in-law of 'D' means 'C' is the brother of the husband/wife (say 'W') of 'D'. So C = W×D.
17. (C)

18. (A),
19. (D),
20. (D).
Solutions 21 and 22

From the clues we can see that no lady is either a teacher or an engineer. Hence these have to be males. Now since Raina is a male, Haiden is a female because Haiden is a lawyer and is married to Raina. Balaji the engineer is married to Patel. Hence Balaji is a male and Patel is a female so Patel cannot be teacher or engineer. It is also given that she is neither a lawyer or a doctor. Thus, she is a manager. Again, Murli is sister of Morkel who is the manager. Morkel is a male because no two females have the same profession and Patel is a female and a manager. On the same grounds, Murli cannot be lawyer hence she is a doctor. By elimination of clues Gony is a lawyer and a male.
Now let us tabulate all the conclusions in a table:

Person	Profession	Gender
Raina	Teacher	Male
Murli	Doctor	Female
Haiden	Lawyer	Female
Patel	Manager	Female
Gony	Lawyer	Male
Balaji	Engineer	Male
Morkel	Manager	Male

Now we can answer the questions.
21. Option (b)
22. Option (d)
Solutions to 23 and 24
On the basis of the given clues we get the following information:

Car	Persons travelling in car
Bentley	U(F), S(F)
Lamborghini	Q(M), T(M)
Ferrari	P(F), ?(F)

Besides this we came to know that R is a female but we still do not know the gender of V. But it is given that there is at least one female in each car. The above table shows that Lamborghini has

a vacant seat for a female passenger. Hence, the seventh person must be a female to fulfill the condition. Hence, V is a female. But we still do not know in which car R and V are travelling knowing that both are not in the same car. Now we need to construct a family tree:

$$[T(M)\leftrightarrow U(F)] \dashrightarrow [Q(M)\leftrightarrow P(F)] \dashrightarrow [R(F)+S(F)]$$

FIRST SECOND THIRD

Since R does not travel with her grandfather and grandmother, R is not in Bentley, hence R is in Lamborghini. We are still not aware of V's position in the family tree.

23. Option (c)

24. Option (c)

16. DIRECTION TEST

I. Simple Direction Test

In such tests, questions involve a direction puzzle. You are expected to analyse the given information pertaining to the movement of any person or vehicle in a particular direction and to ascertain the final direction or position of the person or vehicle. To solve these problems you must have a knowledge about the main directions as shown in the following Fig. 1.

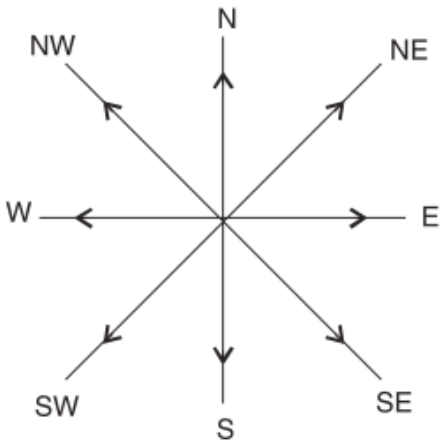


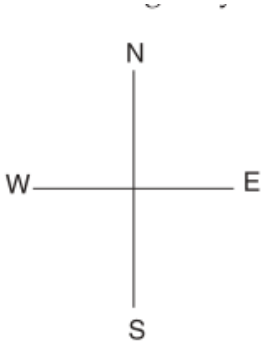
Fig. 1

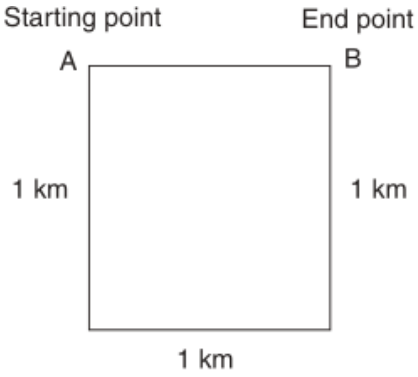
Examples:
Rahim starts from point 'A' goes one kilometre south then one kilometre east. Again he turns left and goes one kilometre north.
Questions:

1. How far Rahim away from the starting point 'A'?
(A) 3 km (B) 2 km (C) 1 km (D) 4 km.

2. How much did Rahim walk?
(A) 3 km (B) 5 km (C) 4 km (D) 1 km.

Answers and Explanations:
1. (C),
2. (A) Here you have to remember the directions and turns given in the question and the distance covered. For easy solution the situation should be quickly sketched in the following way:

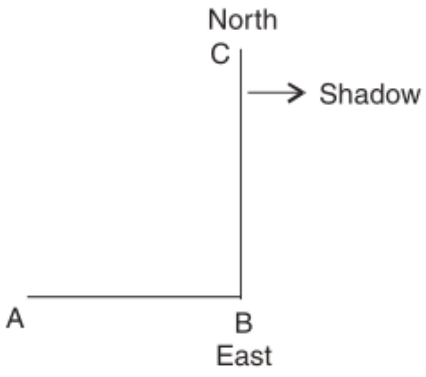




Here, 'A' = Starting point
B = Final or finishing point
Distance between A and B = 1 km.
Position of 'B' from 'A' is 'East'.

Example:
3. You are walking straight in the afternoon, then turned left. While proceeding farther you noticed your shadow in your right, in which direction you started walking in the beginning?
(A) Towards the north
(B) Towards the east
(C) Towards the south
(D) Towards the west

Answer and Explanation:
3. (B)



Consider, AB = initial direction and BC = final direction.

Here, afternoon sun is in the ‘west’, so if we travel ‘north’ only then our shadow would be in the ‘right’. So at the final stage he was walking towards north. But before that he made a turn left, so initially he was travelling towards east as shown in the sketch.

II. Direction Sense Test

These tests usually measure the sense of direction of the candidates. A Fig. is given below which shows four main directions e.g. North, South, East and West and also the four angular direction showing the four cardinal points, e.g. North-East, North-West, South-East and South West.

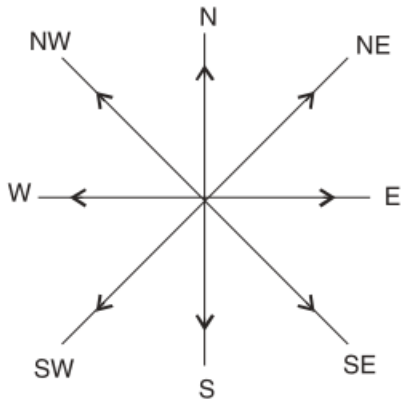


Fig. 1

Examples:

1. A man is facing East. He turns 120° in the clockwise direction and then 210° in the anticlockwise direction. Which direction is he facing now?
(A) South (B) South West (C) North (D) North East.

2. A man is facing North-East. He turned 90° in the clockwise direction, then 180° in the anticlockwise and then another 90° in the same direction. Which direction is he facing now?
(A) North (B) South West (C) North-West (D) South-East.

3. Ramesh starts walking from point a walks 12 kms towards North from there the turns right and walks 4 km, then he again turns right and walks 9 km. How far and which direction he is from his starting point?
(A) 13 km. East (B) 13 km. South (C) 5 km. North (D) 5 km. North-East.

Answers and Explanations:

1. (C) North

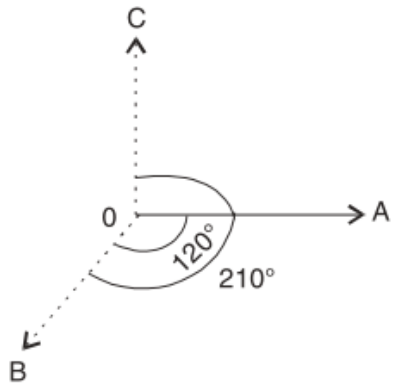


Fig. 1

As shown in the Fig. 1 the man initially faces OA direction. On moving 120° clockwise he faces OB direction. On further moving 210° anti-clockwise he faces OC direction, which is North.

2. (B) South-West

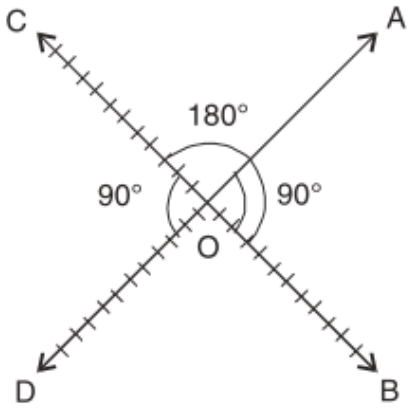


Fig. 2

As shown in the Fig. 2 the man initially faces North-East direction, i.e. OA. On moving 90° clockwise he faces OB direction. On further moving 180° anticlockwise he faces OC direction and at last after moving 90° in the anticlockwise he finally faces OD direction, which is South-West.

3. (D) 5 km. North-East

It is clear from the Fig. 3 that Ramesh moves from A 12 km Northwards upto B, then turns right (i.e. towards East) 4 km upto C then moves right (towards South) and walks 9 km upto D.

Then AE = AB – BE or AB – CD = 12 – 9 = 3 km.
and ED = BC = 4 Km.

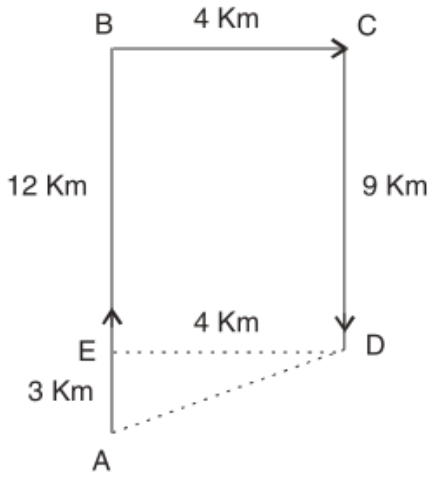


Fig. 3

$$\therefore \quad AD = \sqrt{(AE)^2 + (DE)^2} = \sqrt{3^2 + 4^2}$$

$$= \sqrt{9+16} = \sqrt{25} = 5 \text{ Km.}$$

and D is North-East of A. So, answer is (D).

Exercise

1. A man is facing North-West. He turns 90° in the clockwise direction and then 135° in the anticlockwise direction. Which direction he is facing now?
(A) East (B) West (C) North (D) South

2. A man is facing south. He turns 135° in the anticlockwise direction and then 180° in the clockwise direction. Which direction he is facing now?
(A) North-East (B) North-West (C) South-East (D) South-West.

3. A man is facing north-west. He turns 90° in the clockwise direction then 180° in the anticlockwise direction and then another 90° in the same direction. Which direction he is facing now?
(A) South East (B) West (C) South West (D) South.

4. A man is facing west. He turns 45° in the clockwise direction and then another 180° in the same direction and then 270° in the anticlockwise direction. Which direction he is facing now?
(A) South (B) North West (C) South West (D) West.

5. I am facing east. I turn 100° in the clockwise direction and then 145° in the anticlockwise direction. Which direction I am facing now?
(A) East (B) North East (C) North (D) South West

6. Radha moves towards South-East a distance of 7 km, then she moves towards west and travels a distance of 14 km. From here, she moves towards North-West a distance of 7 km and finally she moves a distance of 4 km towards East and stood at a point. How far is the starting point from where she stood?
(A) 3 km (B) 4 km (C) 10 km (D) 11 km

7. A child is looking for his father. He went 90 metres in the east before turning to his right. He went 20 metres before turning to his right again to look for his father at his uncle's place 30 metres from this point. His father was not there. From there he went 100 metres to his North before meeting his father in a street. How far did the son meet his father from the starting point?
(A) 100 metres (B) 80 metres (C) 140 metres (D) 260 metres

8. Kishenkant walks 10 km towards north. From there he walks 6 km towards south. Then he walks 3 km towards east. How far and in which direction he is with reference to his starting point?
(A) 5 km West (B) 5 km North-East (C) 7 km. East (D) 7 km West

9. A man walks 1 km towards East and then he turns to south and walks 5 km again he turns to east and walks 2 km, after this he turns to north and walks 9 km. Now, how far he is from his starting point?
(A) 3 km (B) 4 km (C) 5 km (D) 7 km

10. Rohit walked 25 metres towards south. Then he turned to his left and walked 20 metres. He then turn to his left and walked 25 metres. He again turn to his right and walked 15 metres. At what distance he is from the starting point and in which direction?
(A) 35 metres east (B) 35 metres north (C) 40 metres east (D) 60 metres east.

Solution

1. (B) West. It is clear from the Fig. 1 that the man initially faces in the direction OA. Then he moves 90° clockwise and faces OB. Then again moving 135° anticlockwise, he faces in the direction OC, which is west.

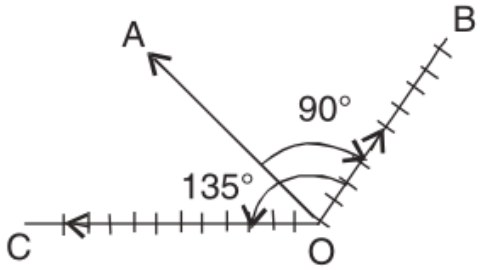


Fig. 1

2. (D) South west. From the Fig. 2 it is clear that originally the man was facing south in the direction OA. Then he moves 135° anticlockwise and facing in the direction OB. Then he again moves 180° clockwise and facing in OC direction which is south west.

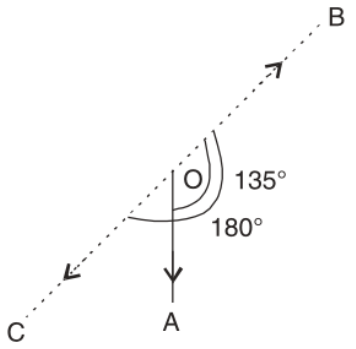


Fig. 2

3. (A) South east. From Fig. 3, it is clear that the man initially was facing north west in the direction OA, then he turns 90° clockwise and faces OB direction, then he turns 180° anti-clockwise and faces OC direction and finally he turns 90° anticlockwise and faces OD direction which is south east.

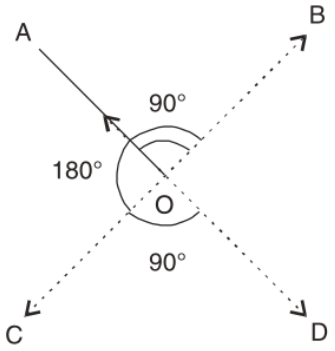


Fig. 3

4. (C) South west. From Fig. 4 it is clear that the initially the man was facing west in OA direction; then he moves OB direction which is 45° clockwise from his original position, then he moves 180° clockwise and facing OC direction, then he moves 270° anticlockwise and facing OD direction which is south west.

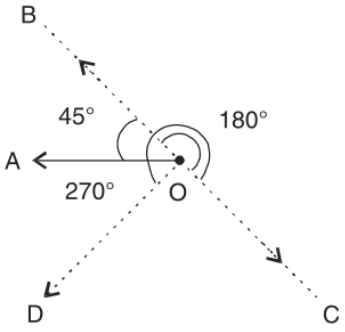


Fig. 4

5. (B) North east. From the Fig. 5 it is clear that initially I was facing east in OA direction. Then I turn 100° clockwise and facing OB direction, then I turn 145° anticlockwise direction and facing OC which is north east.

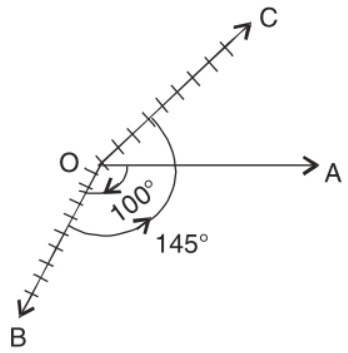


Fig. 5

6. (C) From the Fig. 6 Radha starts from O goes 7 km south-east reaches A, then goes 14 km west reaches B, then goes 7 km north-west reaches C and finally goes 4 km east reaches D. So her distance
from the starting point is OD which is OD = (OC − CD) = (14 − 4) = 10 kms [AB = OC]

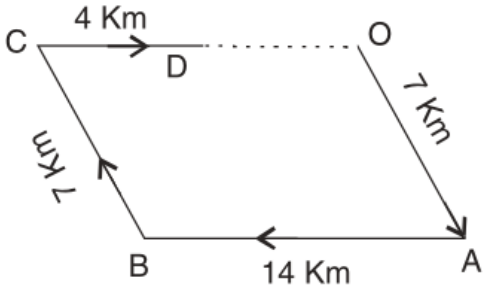


Fig. 6

7. (A) 100 metres. From the Fig. 7 it is clear that the child moves from O to 90 metres east reaches A, from there 20 metres towards south reaches B,
then turns right, i.e. east was 30 metres reaches C, then moves northwards 100 metres and reaches D.

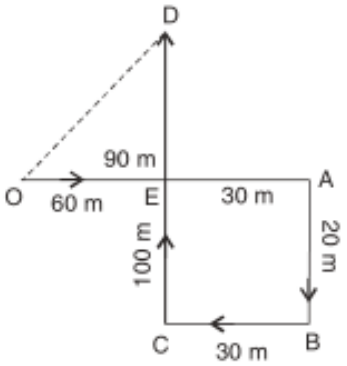


Fig. 7

So, OA = 90 m. AB = 20 m. BC = 30 m. CD = 100 m.
CD = (CE + ED)
ED = (CD − CE) = (100 − 20) = 80 m (CE = AB = 20m)
OA = (OE + AE) or OE = (OA − AE) = (90 − 30)
= 60 m. (AE = BC = 30 m)
∴ child's distance from the starting point is OD, which is

$$OD = \sqrt{(OE)^2 + (DE)^2}$$

$$= \sqrt{60^2 + 80^2} = \sqrt{3600 + 6400} = \sqrt{10000} = 100\,\text{m}.$$

8. (B) 5 km. North east

From the Fig. 8, it is clear that Kishenkant starts from O goes to A 10 km north and back to B 6 km south and finally goes to C which is 3 km east. Clearly 'C' is north east of 'O' his starting point. Now the distance of

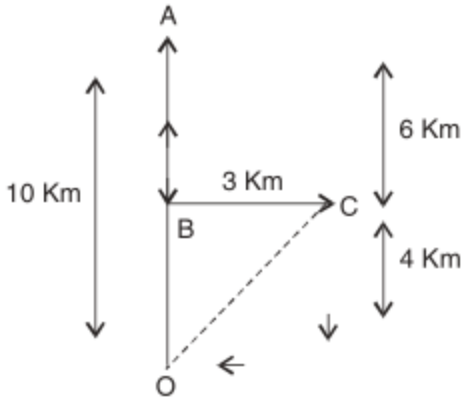


Fig. 8

9. (C) 5 km

10. (A) 35 metres east

The movements of Rohit are shown in Fig. 10. Rohit starts from O and finally reaches D. So OD is the distance from his starting point. OD = OC + CD = AB + CD

= 20 + 15 = 35 metres

[∵ OC = AB = 20 m]

Also D is east of O.

17. DIAGRAMMATIC REASONING

1) Matrices include a group of figures in a single panel. Each panel follows a certain rule or pattern. You will determine what this pattern is and choice the answer that fits best.

2) Next in the Series are questions that have a row of images that follow a pattern. Identify the pattern to determine the next in the sequence.

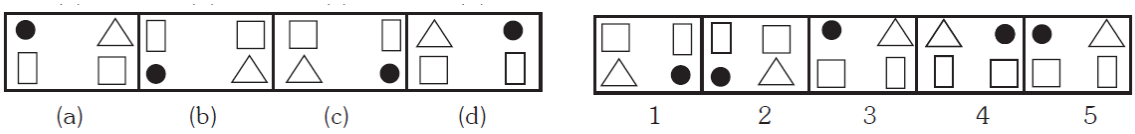
3) Odd one Out present you with several images in an order that follow a particular pattern. Find the images that do not fit the pattern.

4) Analogies begin with two pairs of shapes that share a relationship. Your task is to find and locate the relationship and implement it into the one of the given answers.

I. Series Questions: Concept Builders

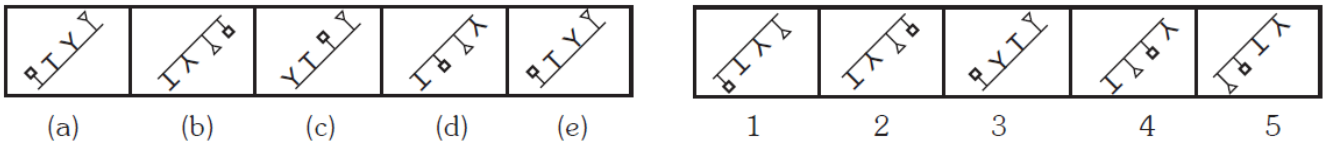
In each of the following questions you are given five series of questions you have to find out the next series from the answer figures that follows the sequence of the questions figures.

Q1.



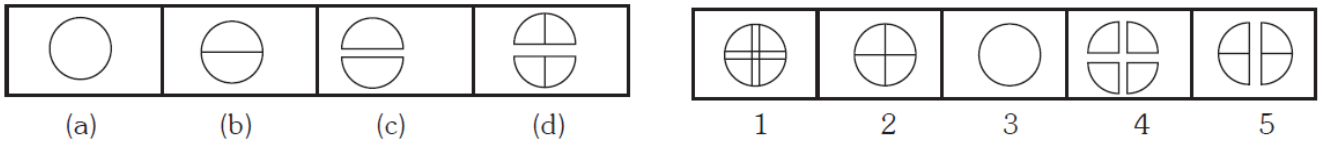
Solution: (5)-Circle (Shaded circle) moves anti- clockwise and rest figures interchanged their position with the Opposite figure.

Q2.



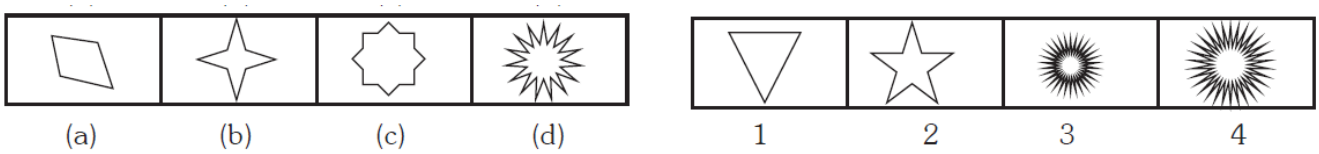
Solution : (2)

Q3.



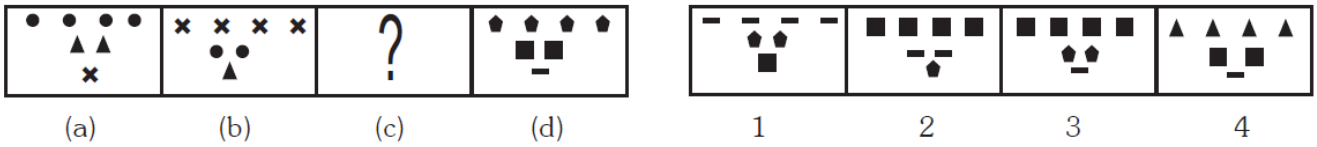
Solution: (4)-In one stop, the figure gets a dividing mark or line than in the very next step it is separated and this pattern repeated alternatively.

Q4.



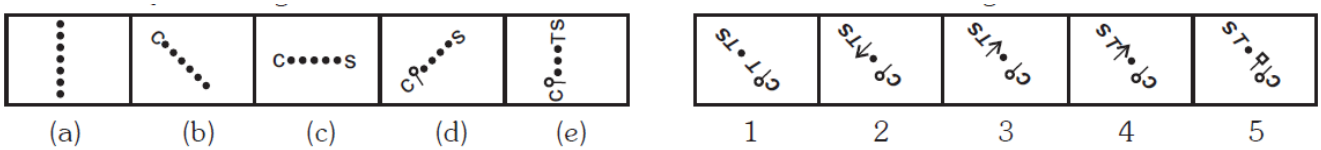
Solution: (4)-The number of projections doubles in each step.

Q5.



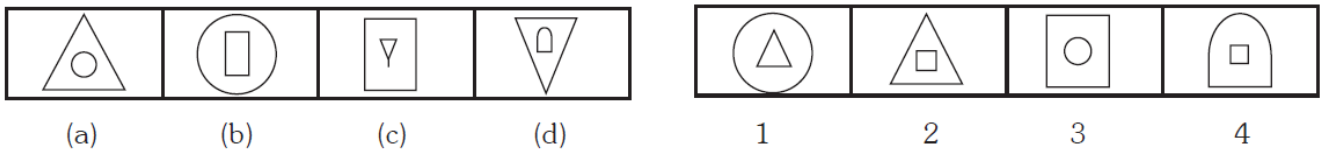
Solution: (2) All the elements of the top row moves to second row , second row elements move to third row and third row elements move to first row.

Q6.



Solution: (1) one dot is replaced by a symbol and the replacement takes place on either end. And rotates anti clock wise at 45 degree in each step.

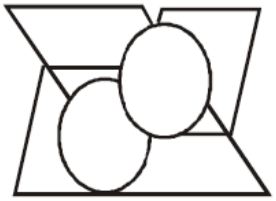
Q7.

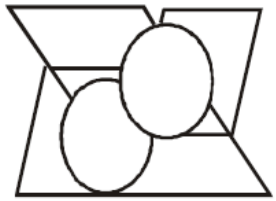


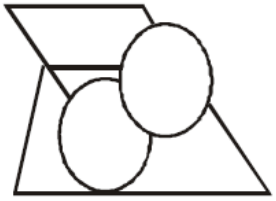
Solution: (4)-Every time the inner figure enlarges and become the outer figure and the row inner figure changes to new element.

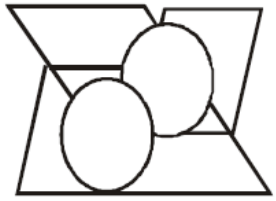
II. Find The Odd One Out

Q1.

(a) 

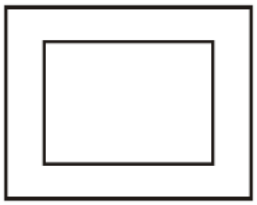
(b) 

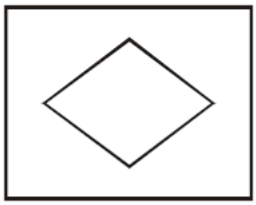
(c) 

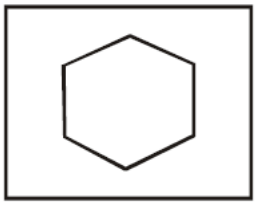
(d) 

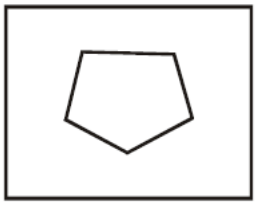
Solution: (c)

Q2.

(a) 

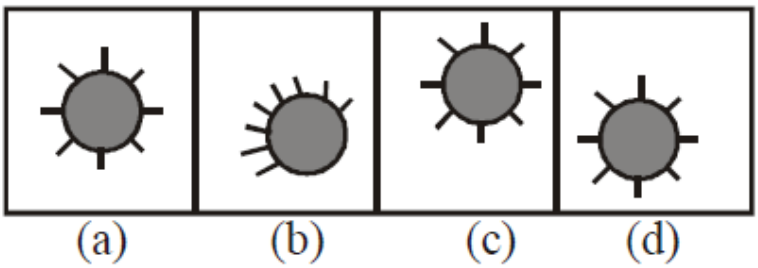
(b) 

(c) 

(d) 

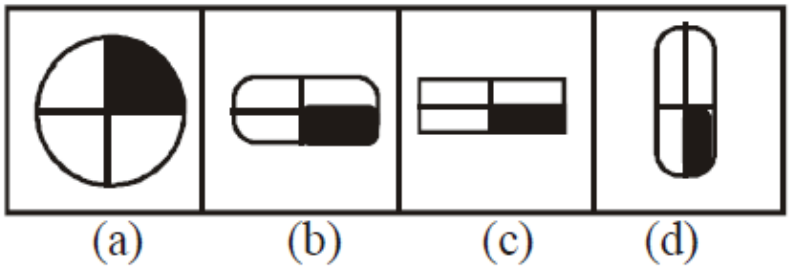
Solution: (d)

Q3.



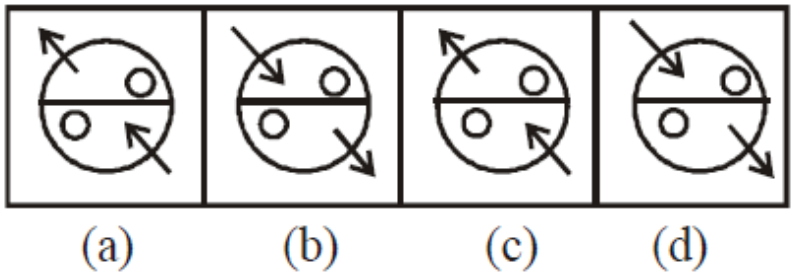
Solution: (c)

Q4.



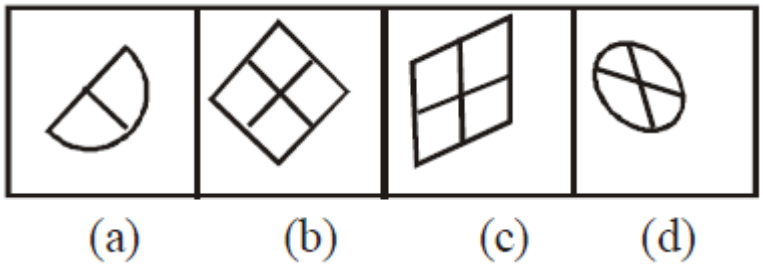
Solution: (a) In all other figures bottom right corners have been shaded.

Q5.



Solution: (b) Arrows are not in the same direction.

Q6.

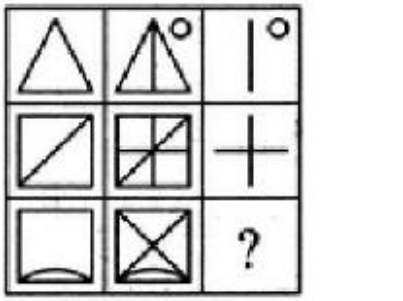


Solution: (a) All others have four parts.

III. Matrix Completion Problems:

In each of the following questions find out which of the answer figures (1), (2), (3) and (4) completes the figure matrix.

Q1.



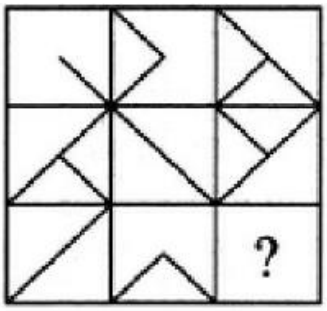


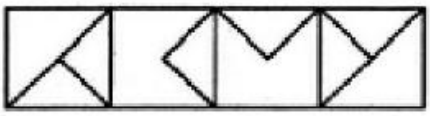
(1) (2) (3) (4)

(a) 1 (b) 2 (c) 3 (d) 4

Solution: (d) In each row, the first unit is deleted in the third unit.

Q2.



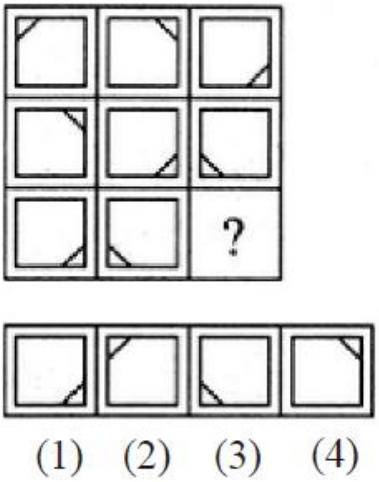


(1) (2) (3) (4)

(a) 1 (b) 2 (c) 3 (d) 4

Solution: (d) Answer figure (d) will complete the matrix.

Q3.



(a) 1 (b) 2 (c) 3 (d) 4

Solution: (b) Answer figure (b) will complete matrix.

IV. Logical Figures

Q1.

1- Choose the image that completes the pattern:



Which of the shapes below continues the sequence:

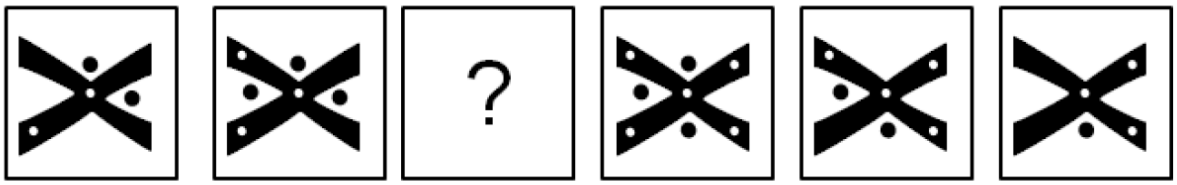


Solution:
All triangles “moves” slightly counter-clockwise and outside. A good solving tip would be to try and disassemble the complete figure to its elements (triangles) and to focus each time on one of the elements. Thus, the answer is:



Q2.

2- Choose the image that completes the pattern:

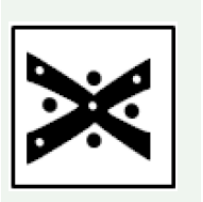


Which of the shapes below continues the sequence:



Solution:

The logic: An X shape is dotted with black and white dots. Both ssets of dots are independent and follow a similar pattern. In each frame, a black dot is added counter-clockwise in the angles of the X shape, until all the angles are occupied. Then a dot is reduced, also counter-clockwise. The same pattern occurs with the white dots, only in a clockwise manner.
Examining the changes before and after, the “question mark” figure should look the same as in frame 2, only with an additional black dot (making all four black dots present) and an additional white dot in the upper right corner, as determined by the pattern.



Q3.

5- Choose the image that completes the pattern:



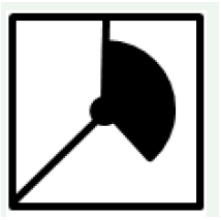
Which of the shapes below continues the sequence:



Solution:

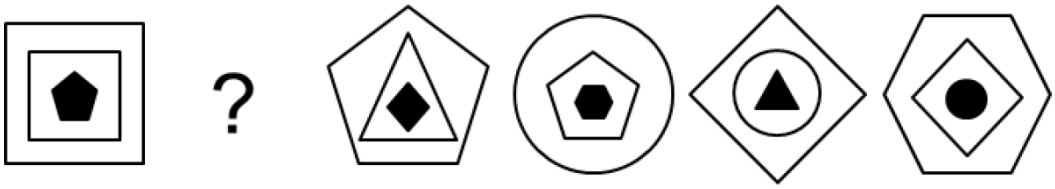
1) Each step. 45 degrees of the black circle is painted in white, counter clockwise.

2) Each step, a line drawn from the centre of the circle to the edge of the frame turns 45 degrees clockwise.



Q4.

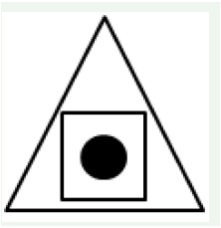
6- Choose the image that completes the pattern:



Which of the shapes below replaces the missing box above?

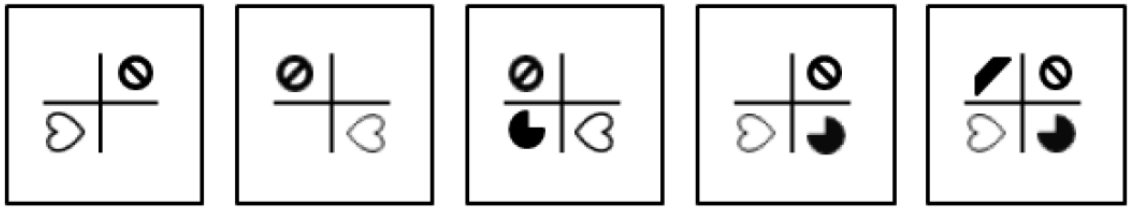


Solution:
Each object in the series consists of 3 shapes. The logic of the sequence is defined by the relations between these shapes.
The external shape appears as the middle shape in the next frame; the middle shape disappears in the next frame; the inner shape determines the external shape two steps forward, e.g. the inner pentagon in the first frame appeared as the external shape in the third frame.

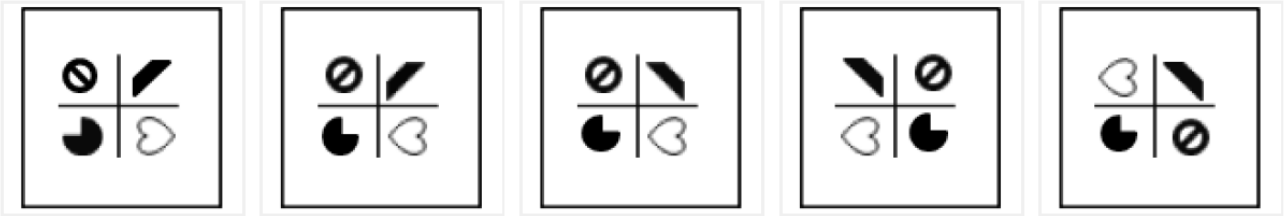


Q5.

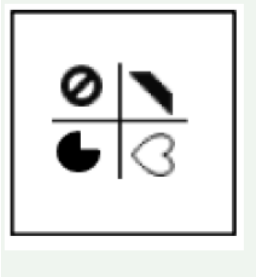
3- Choose the image that completes the pattern:



Which of the shapes below continues the sequence:

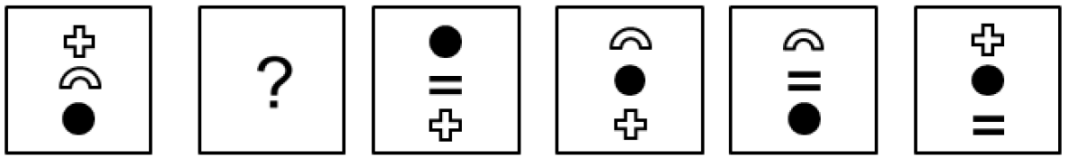


Solution:
The logic: The next frame after each step portrays a mirror image of the previous frame. In addition, every two steps a shape is added to the frame. Answer choice 3 is a mirrored version of frame 5 and is therefore the correct answer.



Q6.

7- Choose the image that completes the pattern:



Which of the shapes below replaces the missing box above?



Solution:

Each square contains 3 shapes. The black circle is the only shape that constantly appears. It "travels" along the column up and down, one step at a time. The other shapes appear in two consecutive squares, and then do not appear in the next square. The shape that remains in the frame (in regards to the last step from the left) maintains its relative position to the other shape (not the circle)

